

Tzuchien

Gauge Theory

on Four-Manifolds

Preface

These notes grew out of our seminar on gauge theory on four-manifolds. Their central subject is the construction of the Seiberg–Witten invariants and their use in studying smooth four-dimensional geometry. The question guiding the exposition is how equations for connections and spinors can detect information that the intersection form alone does not determine.

The first chapter recalls the topological data used throughout the book: intersection forms, characteristic classes, and the Hodge decomposition of two-forms. Chapter 2 develops connections, curvature, Chern–Weil theory, and gauge transformations. Abelian gauge fixing provides the estimates needed later, while Uhlenbeck compactness introduces the different behavior of non-Abelian equations, where curvature can concentrate. Chapter 3 constructs the four-dimensional spinor representations, Spin^c structures, their connections, and the Dirac operator, culminating in the Weitzenböck formula.

Chapter 4 brings these ingredients together. Starting from the Seiberg–Witten equations and their gauge symmetry, it develops the local deformation problem, computes the expected dimension, and establishes the regularity, compactness, and orientation results needed to define the invariant. The discussion includes both signed counts in dimension zero and cohomology insertions in higher dimensions. Chapter 5 turns to geometric consequences: obstructions to positive scalar curvature, the blow-up formula, genus bounds from the adjunction inequality, Taubes’ results for symplectic manifolds, and applications to smooth structures.

Much of the detail is devoted to the steps between these constructions. Local slices give coordinates on the irreducible gauge quotient; a choice of gauge need not select a canonical representative; an index becomes the actual dimension only under appropriate regularity assumptions. Similarly, compactness requires estimates on representatives and a convergence argument, while invariance requires a compact oriented cobordism. Keeping these distinctions explicit makes it possible to follow how the analytic and topological parts of the argument fit together. Local formulas are derived with consistent conventions for the determinant connection, Clifford multiplication, and orientations.

The intended reader has some familiarity with smooth manifolds, differential forms, vector bundles, and basic algebraic topology. The analytic appendix collects the Sobolev, elliptic, and Fredholm theory used in the main text, including finite-dimensional reduction and the Sard–Smale theorem. It can be read alongside the relevant arguments rather than in advance. The notes do not give complete proofs of every theorem they use: the index theorem and the deeper gluing and symplectic results supply inputs whose roles and hypotheses are explained where they enter.

The references include Donaldson and Kronheimer’s *The Geometry of Four-Manifolds*, together with treatments by Haydys, Nicolaescu, and Perutz. These notes form the gauge-theoretic volume of a pair; surfaces in four-manifolds and Kirby calculus are treated separately in *4-Manifolds & Kirby Calculus*. The present volume retains the topological preliminaries needed for its own arguments. It remains a set of seminar notes, and corrections and suggestions are welcome.

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Chapter 1

Invariants of 4-manifolds

1.1 The infeasibility of a complete classification of 4-manifolds

Before studying 4-manifolds, the most natural question is whether one can completely classify all closed smooth 4-manifolds.

In general, this goal is algorithmically impossible. The underlying reason is that the fundamental group of a 4-manifold can be as complicated as an arbitrary finitely presented group.

Theorem 1.1.1 (Adyan–Rubin, 1955). There is no algorithm that can decide, for an arbitrary finite presentation of a group

$$G = \langle g_1, \dots, g_n \mid r_1, \dots, r_m \rangle$$

whether the group G it defines is the trivial group.

On the other hand, every finitely presented group can be realized by a 4-manifold.

Theorem 1.1.2 (Markov). For every finitely presented group G there exists a closed smooth 4-manifold X_G such that

$$\pi_1(X_G) \cong G.$$

Hence the fundamental group of a closed smooth 4-manifold is at least as complicated as all finitely presented groups. Since the fundamental group is a homotopy invariant, any method that effectively classifies all closed smooth 4-manifolds up to homotopy type must be able to handle the triviality problem for arbitrary finitely presented groups. This contradicts the undecidability theorem above.

Therefore no algorithmic complete classification can classify all closed smooth 4-manifolds up to homotopy. Still less can it classify them up to diffeomorphism.

In other words, the classification problem for general closed smooth 4-manifolds already contains the undecidable problem for finitely presented groups, so from the outset one cannot expect a complete algorithmic classification of the kind provided by the classification theorem for surfaces.

The idea behind the construction in Markov’s theorem Let

$$G = \langle g_1, \dots, g_n \mid r_1, \dots, r_m \rangle$$

be a finitely presented group. First consider

$$N = \#^n(S^1 \times S^3).$$

By the Seifert–van Kampen theorem,

$$\pi_1(N) \cong \underbrace{\mathbb{Z} * \cdots * \mathbb{Z}}_n = F_n = \langle g_1, \dots, g_n \rangle.$$

For each relation r_j , choose an embedded closed curve

$$\gamma_j \subset N$$

representing the corresponding element of the fundamental group. Since these curves are 1-dimensional and

$$1 + 1 < 4,$$

we may assume by general position that the γ_j are pairwise disjoint.

Fix one of these curves γ , representing the relation r . It has a tubular neighborhood

$$S^1 \times B^3.$$

Perform surgery on γ , that is, remove $S^1 \times B^3$, and then glue in $B^2 \times S^2$ along the common boundary

$$S^1 \times S^2$$

to obtain

$$\widetilde{N} = (N \setminus (S^1 \times B^3)) \cup_{S^1 \times S^2} (B^2 \times S^2).$$

Since $B^2 \times S^2$ is simply connected, applying the Seifert–van Kampen theorem once more gives

$$\pi_1(\widetilde{N}) \cong \pi_1(N) / \langle\langle r \rangle\rangle,$$

where $\langle\langle r \rangle\rangle$ denotes the normal closure generated by r .

Performing the above surgery for all relations $r_1, \dots, r_m$ simultaneously, we finally obtain a closed smooth 4-manifold X_G satisfying

$$\pi_1(X_G) \cong F_n / \langle\langle r_1, \dots, r_m \rangle\rangle \cong G.$$

Thus every finitely presented group can occur as the fundamental group of some closed smooth 4-manifold.

Remark 1.1.3. The above difficulty comes mainly from the fundamental group. Hence, in order to obtain a tractable classification theory, we shall subsequently restrict attention to closed orientable 4-manifolds with

$$\pi_1(X) = 0$$

the closed orientable 4-manifolds.

In the simply connected case this undecidable complexity of the fundamental group is eliminated, and the intersection form

$$Q_X : H^2(X; \mathbb{Z}) \times H^2(X; \mathbb{Z}) \longrightarrow \mathbb{Z}$$

begins to become the central algebraic invariant governing the homotopy type and the topological type of a 4-manifold.

In what follows in this chapter, unless otherwise stated, X will always denote a simply connected, closed, orientable 4-manifold.

1.2 Intersection forms: first steps

1.2.1 The intersection form

Let X be a simply connected, closed, orientable 4-manifold. Fix an orientation on X , and denote the corresponding fundamental class by

$$[X] \in H_4(X; \mathbb{Z}).$$

Since X is simply connected, we have $\pi_1(X) = 0$, hence $H_1(X; \mathbb{Z}) = 0$. By the universal coefficient theorem,

$$H^2(X; \mathbb{Z}) \cong \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}).$$

In particular, $H^2(X; \mathbb{Z})$ is a free Abelian group. On the other hand, by Poincaré duality,

$$PD : H^2(X; \mathbb{Z}) \xrightarrow{\cong} H_2(X; \mathbb{Z}).$$

Hence $H_2(X; \mathbb{Z})$ is a free Abelian group as well.

Definition 1.2.1. Define the **intersection form** of X to be the symmetric bilinear form

$$\begin{aligned} Q_X : H^2(X; \mathbb{Z}) \times H^2(X; \mathbb{Z}) &\longrightarrow \mathbb{Z} \\ (a, b) &\longmapsto \langle a \smile b, [X] \rangle. \end{aligned}$$

Here $\smile$ denotes the cup product in cohomology.

For $a, b \in H^2(X; \mathbb{Z})$, the graded commutativity of the cup product gives

$$a \smile b = (-1)^{2 \cdot 2} b \smile a = b \smile a.$$

Hence $Q_X(a, b) = Q_X(b, a)$, so Q_X is a symmetric bilinear form.

Via Poincaré duality, we may also regard Q_X as a bilinear form defined on $H_2(X; \mathbb{Z})$:

$$Q_X : H_2(X; \mathbb{Z}) \times H_2(X; \mathbb{Z}) \longrightarrow \mathbb{Z}.$$

Concretely, for $\alpha, \beta \in H_2(X; \mathbb{Z})$, define

$$Q_X(\alpha, \beta) = \langle PD^{-1}(\alpha) \smile PD^{-1}(\beta), [X] \rangle.$$

When no confusion can arise, we write both the cohomological and the homological version as Q_X . Since $H_2(X; \mathbb{Z})$ is a finitely generated free Abelian group, it can be written as

$$H_2(X; \mathbb{Z}) \cong \mathbb{Z}^{\oplus k}.$$

Take a basis $\{e_1, \dots, e_k\}$; then the matrix of Q_X with respect to this basis is

$$M = \left(Q_X(e_i, e_j) \right)_{1 \leq i, j \leq k} \in M_k(\mathbb{Z}).$$

Since Q_X is a symmetric bilinear form, $M^T = M$, and if $\{f_1, \dots, f_k\}$ is another basis with

$$(f_1, \dots, f_k) = (e_1, \dots, e_k)C,$$

where

$$C \in GL_k(\mathbb{Z}),$$

then the matrix of Q_X with respect to the new basis is $M' = C^T M C$. Hence

$$\det M' = \det(C^T) \det(M) \det(C) = (\det C)^2 \det M.$$

Since $C \in GL_k(\mathbb{Z})$, we have $\det C = \pm 1$. Therefore $\det M' = \det M$, so we may define $\det Q_X := \det M$, and this definition is independent of the choice of basis.

Definition 1.2.2 (unimodular). Let $L \cong \mathbb{Z}^{\oplus k}$ be a finitely generated free Abelian group, and let

$$Q : L \times L \longrightarrow \mathbb{Z}$$

be a bilinear form. Define the homomorphism

$$\begin{aligned} \Phi_Q : L &\longrightarrow L^* := \text{Hom}(L, \mathbb{Z}) \\ x &\longmapsto Q(x, -). \end{aligned}$$

If Φ_Q is an isomorphism, then Q is called **unimodular**.

Take a basis $\{e_1, \dots, e_k\}$ of L , and denote the matrix of Q with respect to this basis by

$$A = (Q(e_i, e_j)) \in M_k(\mathbb{Z}).$$

Take moreover the dual basis $\{e^1, \dots, e^k\}$, that is, $e^i(e_j) = \delta_{ij}$. Then

$$\Phi_Q(e_i) = \sum_j a_{ij} e^j.$$

Hence the matrix of Φ_Q with respect to the basis $\{e_i\}$ and the dual basis $\{e^j\}$ is precisely A . Therefore

$$\Phi_Q \text{ is an isomorphism} \iff A \in GL_k(\mathbb{Z}) \iff \det A = \pm 1.$$

In other words, Q is unimodular if and only if the determinant of its matrix is ± 1 .

Proposition 1.2.3. Let X be a simply connected, closed, orientable 4-manifold. Then its intersection form Q_X is unimodular.

Proof. By the universal coefficient theorem, there is a short exact sequence

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}^1(H_1(X; \mathbb{Z}), \mathbb{Z}) \longrightarrow H^2(X; \mathbb{Z}) \longrightarrow \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}) \longrightarrow 0.$$

Since X is simply connected, $H_1(X; \mathbb{Z}) = 0$, hence

$$H^2(X; \mathbb{Z}) \cong \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}).$$

This isomorphism is precisely the evaluation map

$$\begin{aligned} \text{ev} : H^2(X; \mathbb{Z}) &\longrightarrow \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}) \\ a &\longmapsto (\beta \mapsto \langle a, \beta \rangle). \end{aligned}$$

On the other hand, by Poincaré duality,

$$PD : H^2(X; \mathbb{Z}) \xrightarrow{\cong} H_2(X; \mathbb{Z}).$$

Hence we may consider the composite isomorphism

$$H_2(X; \mathbb{Z}) \xrightarrow{PD^{-1}} H^2(X; \mathbb{Z}) \xrightarrow{\text{ev}} \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}).$$

For any $\alpha, \beta \in H_2(X; \mathbb{Z})$, we have

$$\text{ev}(PD^{-1}(\alpha))(\beta) = \langle PD^{-1}(\alpha), \beta \rangle = Q_X(\alpha, \beta).$$

Thus the above composite isomorphism is exactly

$$\Phi_{Q_X} : H_2(X; \mathbb{Z}) \longrightarrow \text{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z}), \quad \alpha \longmapsto Q_X(\alpha, -).$$

Therefore Φ_{Q_X} is an isomorphism, that is, Q_X is unimodular. Consequently

$$\det Q_X = \pm 1.$$

□

Finally we explain the relation between the intersection form and the orientation. Denote by $\bar{X}$ the manifold obtained by giving X the opposite orientation. Then

$$[\bar{X}] = -[X].$$

Hence for any $a, b \in H^2(X; \mathbb{Z})$,

$$Q_{\bar{X}}(a, b) = \langle a \smile b, [\bar{X}] \rangle = -\langle a \smile b, [X] \rangle = -Q_X(a, b).$$

Therefore

$$Q_{\bar{X}} = -Q_X.$$

Remark 1.2.4. The above definition 1.2.1 was given in the case where X is simply connected, closed and orientable. These three conditions do not play the same role.

First, simple connectivity is not needed in order to define the intersection form. If X is merely a closed, orientable 4-manifold, we can still define

$$Q_X(a, b) = \langle a \smile b, [X] \rangle, \quad a, b \in H^2(X; \mathbb{Z}).$$

In this case $H^2(X; \mathbb{Z})$ may contain torsion. If a is a torsion element, then for any $b \in H^2(X; \mathbb{Z})$, the element $a \smile b \in H^4(X; \mathbb{Z}) \cong \mathbb{Z}$ is also torsion, hence must be 0. Thus Q_X vanishes identically on the torsion part, and induces

$$Q_X : H^2(X; \mathbb{Z}) / \text{Tor} \times H^2(X; \mathbb{Z}) / \text{Tor} \longrightarrow \mathbb{Z}.$$

Equivalently, via Poincaré duality, we may regard it as

$$Q_X : H_2(X; \mathbb{Z}) / \text{Tor} \times H_2(X; \mathbb{Z}) / \text{Tor} \longrightarrow \mathbb{Z}.$$

For any closed, orientable 4-manifold, this intersection form on the free part is still unimodular. The role of the simple connectivity condition is mainly to ensure that $H_1(X; \mathbb{Z}) = 0$, so that $H_2(X; \mathbb{Z})$ is itself a free Abelian group and there is no need to remove torsion.

If X is compact and orientable but has non-empty boundary ∂X , then the fundamental class no longer lies in $H_4(X; \mathbb{Z})$, but rather is the relative fundamental class

$$[X, \partial X] \in H_4(X, \partial X; \mathbb{Z}).$$

In this case one can use the relative cup product to define

$$Q_X(a, b) = \langle a \smile b, [X, \partial X] \rangle, \quad a, b \in H^2(X, \partial X; \mathbb{Z}).$$

By Poincaré–Lefschetz duality,

$$H^2(X, \partial X; \mathbb{Z}) \cong H_2(X; \mathbb{Z}),$$

so this gives an intersection form on $H_2(X; \mathbb{Z})$. Unlike the case of a closed manifold, this intersection form may in general be degenerate. The perfect pairing corresponding to it is

$$H^2(X; \mathbb{Z}) \times H^2(X, \partial X; \mathbb{Z}) \longrightarrow \mathbb{Z}, \quad (a, b) \longmapsto \langle a \smile b, [X, \partial X] \rangle.$$

Finally, if X is non-orientable, then no fundamental class $[X] \in H_4(X; \mathbb{Z})$ with integer coefficients exists, so one cannot define an integer-valued intersection form directly in the manner of definition 1.2.1. The simplest remedy is to switch to $\mathbb{Z}_2$ coefficients. In that case a fundamental class always exists,

$$[X]_2 \in H_4(X; \mathbb{Z}_2),$$

and one can define the mod 2 intersection form

$$Q_X^{(2)}(a, b) = \langle a \smile b, [X]_2 \rangle, \quad a, b \in H^2(X; \mathbb{Z}_2).$$

If one wishes to retain integer coefficients, then one must use the orientation local coefficient system $\mathcal{O}_X$. In this case Poincaré duality has to be formulated with local coefficients, and correspondingly what one obtains is a pairing with local coefficients, no longer a symmetric bilinear form defined on a single ordinary integral homology group.

1.2.2 Parity, definiteness and signature of the intersection form

Let

$$Q : L \times L \longrightarrow \mathbb{Z}$$

be a symmetric bilinear form on a finitely generated free Abelian group $L \cong \mathbb{Z}^n$.

Definition 1.2.5. If for every $x \in L$ we have

$$Q(x, x) \in 2\mathbb{Z},$$

then Q is called **even**.

If there exists $x \in L$ such that $Q(x, x)$ is odd, then Q is called **odd**.

If we choose a basis $\{e_1, \dots, e_n\}$ of L and denote by

$$A = (Q(e_i, e_j)),$$

the matrix of Q with respect to this basis, then Q is even if and only if all diagonal entries of A are even. Hence parity is independent of the choice of basis.

For a closed oriented 4-manifold X , we correspondingly call its intersection form Q_X even or odd.

In order to define definiteness and signature, we extend Q to the real vector space

$$L_{\mathbb{R}} := L \otimes_{\mathbb{Z}} \mathbb{R},$$

obtaining a real symmetric bilinear form

$$Q_{\mathbb{R}} : L_{\mathbb{R}} \times L_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

When no confusion can arise, we still denote it by Q .

Definition 1.2.6. If for every non-zero $x \in L_{\mathbb{R}}$ we have $Q(x, x) > 0$, then Q is called **positive definite**.

If for every non-zero $x \in L_{\mathbb{R}}$ we have $Q(x, x) < 0$, then Q is called **negative definite**.

If there exist non-zero vectors $x, y \in L_{\mathbb{R}}$ such that

$$Q(x, x) > 0, \quad Q(y, y) < 0,$$

then Q is called **indefinite**.

By Sylvester's law of inertia, there is a basis of $L_{\mathbb{R}}$ with respect to which the matrix of Q can be written as

$$\text{diag}(\underbrace{1, \dots, 1}_{b^+}, \underbrace{-1, \dots, -1}_{b^-}, \underbrace{0, \dots, 0}_{b^0}).$$

Here b^+ , b^- , b^0 are independent of the choice of basis and are called the **positive index of inertia**, the **negative index of inertia** and the **nullity** of Q , respectively.

If Q is non-degenerate, then $b^0 = 0$, so that

$$\text{rank } Q = b^+ + b^-.$$

For a closed oriented 4-manifold X , the intersection form Q_X is unimodular, hence in particular non-degenerate. We write

$$b_2^+(X) := b^+(Q_X), \quad b_2^-(X) := b^-(Q_X).$$

Thus

$$b_2(X) = b_2^+(X) + b_2^-(X).$$

Definition 1.2.7. The **signature** of a symmetric bilinear form Q is defined by

$$\sigma(Q) := b^+ - b^-.$$

For a closed oriented 4-manifold X , we define

$$\sigma(X) := \sigma(Q_X) = b_2^+(X) - b_2^-(X).$$

Consequently,

$$b_2^+(X) = \frac{b_2(X) + \sigma(X)}{2}, \quad b_2^-(X) = \frac{b_2(X) - \sigma(X)}{2}.$$

In particular, Q_X is positive definite if and only if $b_2^-(X) = 0$, in which case

$$\sigma(X) = b_2(X).$$

Similarly, Q_X is negative definite if and only if $b_2^+(X) = 0$, in which case

$$\sigma(X) = -b_2(X).$$

And Q_X is indefinite if and only if

$$b_2^+(X) > 0, \quad b_2^-(X) > 0.$$

Finally, if we reverse the orientation of X , then

$$Q_{\bar{X}} = -Q_X.$$

Hence

$$b_2^+(\bar{X}) = b_2^-(X), \quad b_2^-(\bar{X}) = b_2^+(X),$$

and therefore

$$\sigma(\bar{X}) = -\sigma(X).$$

1.2.3 The intersection form and characteristic classes

Let X be a simply connected, closed, orientable smooth 4-manifold. Beyond the homology groups themselves, a smooth manifold carries characteristic classes coming from its tangent bundle TX . In dimension 4, one mainly considers the Stiefel–Whitney classes

$$w_i(TX) \in H^i(X; \mathbb{Z}_2),$$

as well as the Euler class and the first Pontryagin class

$$e(TX), p_1(TX) \in H^4(X; \mathbb{Z}).$$

For a simply connected closed oriented 4-manifold, these characteristic classes are in fact all determined by the intersection form Q_X .

The second Stiefel–Whitney class First consider $w_2(TX)$. By the Wu formula, for every $x \in H^2(X; \mathbb{Z}_2)$ we have

$$\langle w_2(TX) \smile x, [X] \rangle = \langle x \smile x, [X] \rangle \pmod{2}. \quad (1.2.E1)$$

The right-hand side is precisely the self-intersection number of the intersection form reduced modulo 2. Hence $w_2(TX)$ is the class uniquely determined by Q_X modulo 2.

In other words, $w_2(TX)$ is a characteristic element for the intersection form modulo 2, that is, it satisfies

$$Q_X(x, x) \equiv Q_X(w_2(TX), x) \pmod{2}.$$

Here $w_2(TX)$ is identified, via Poincaré duality, with an element of homology with $\mathbb{Z}_2$ coefficients.

In particular,

$$Q_X \text{ is even} \iff w_2(TX) = 0.$$

Therefore, for a simply connected closed oriented 4-manifold,

$$Q_X \text{ is even} \iff X \text{ is spin.}$$

This can also be understood from the point of view of embedded surfaces. Let $\alpha \in H_2(X; \mathbb{Z}_2)$; since X is simply connected, α can be represented by some closed oriented embedded surface $\Sigma \subset X$. On Σ we have

$$TX|_{\Sigma} \cong T\Sigma \oplus \nu_{\Sigma},$$

where ν_{Σ} is the normal bundle of Σ in X . By the Whitney product formula,

$$w_2(TX|_{\Sigma}) = w_2(T\Sigma) + w_2(\nu_{\Sigma}),$$

and since both $T\Sigma$ and ν_{Σ} are oriented, their first Stiefel–Whitney classes are both 0.

For an oriented real 2-dimensional vector bundle one has $w_2 \equiv e \pmod{2}$. Hence

$$\langle w_2(T\Sigma), [\Sigma] \rangle \equiv \chi(\Sigma) \equiv 0 \pmod{2},$$

whereas

$$\langle w_2(\nu_{\Sigma}), [\Sigma] \rangle \equiv \langle e(\nu_{\Sigma}), [\Sigma] \rangle \equiv \Sigma \cdot \Sigma \pmod{2}.$$

Therefore

$$\langle w_2(TX), [\Sigma] \rangle \equiv \Sigma \cdot \Sigma \pmod{2},$$

which is precisely the geometric manifestation of the Wu formula (1.2.E1) above.

The remaining Stiefel–Whitney classes Since X is orientable, we have $w_1(TX) = 0$. And since X is simply connected, Poincaré duality gives $H^3(X; \mathbb{Z}_2) = 0$, hence

$$w_3(TX) = 0.$$

The top Stiefel–Whitney class satisfies

$$w_4(TX) \equiv e(TX) \pmod{2}.$$

Thus once the Euler class is determined, $w_4(TX)$ is determined as well.

The Euler class Since X is simply connected,

$$b_0 = b_4 = 1, \quad b_1 = b_3 = 0.$$

Hence the Euler characteristic is

$$\chi(X) = \sum_{i=0}^4 (-1)^i b_i(X) = 2 + b_2(X).$$

On the other hand,

$$\langle e(TX), [X] \rangle = \chi(X).$$

Since $H^4(X; \mathbb{Z}) \cong \mathbb{Z}$, choosing a generator $\mu \in H^4(X; \mathbb{Z})$ with $\langle \mu, [X] \rangle = 1$, we get

$$e(TX) = (2 + b_2(X))\mu.$$

And $b_2(X) = \text{rank } Q_X$, so the Euler class is completely determined by the rank of the intersection form.

The first Pontryagin class The first Pontryagin class is governed by the Hirzebruch signature theorem. In dimension 4,

$$\langle p_1(TX), [X] \rangle = 3\sigma(X).$$

Here

$$\sigma(X) = \sigma(Q_X) = b_2^+(X) - b_2^-(X).$$

Again since $H^4(X; \mathbb{Z}) \cong \mathbb{Z}$, if we still take $\langle \mu, [X] \rangle = 1$, then

$$p_1(TX) = 3\sigma(X)\mu.$$

Hence the first Pontryagin class is completely determined by the signature of the intersection form.

To sum up, for a simply connected closed oriented smooth 4-manifold X , the intersection form Q_X determines the main characteristic class data of the tangent bundle. Concretely,

$$\begin{aligned} w_1(TX) &= 0, \\ w_2(TX) &\text{ is determined by } Q_X \text{ mod } 2, \\ w_3(TX) &= 0, \\ w_4(TX) &\equiv e(TX) \pmod{2}, \\ e(TX) &= (2 + \text{rank } Q_X)\mu, \\ p_1(TX) &= 3\sigma(Q_X)\mu. \end{aligned}$$

Thus, in the case of a simply connected closed oriented 4-manifold, the intersection form determines not only the integral cohomology ring, but also the Stiefel–Whitney classes, the Euler class and the first Pontryagin class of the tangent bundle.

1.2.4 Geometric interpretation of the intersection form

In what follows we give a geometric interpretation of the intersection form. First we explain that every homology class in $H_2(X; \mathbb{Z})$ can be represented by an embedded oriented surface.

Proposition 1.2.8. For any $\alpha \in H_2(X; \mathbb{Z})$, there exists a closed oriented surface Σ_α and a smooth embedding

$$\Sigma_\alpha \hookrightarrow X$$

such that

$$[\Sigma_\alpha] = \alpha \in H_2(X; \mathbb{Z}).$$

Proof. By Poincaré duality, α corresponds to a cohomology class

$$PD^{-1}(\alpha) \in H^2(X; \mathbb{Z}).$$

Since a complex line bundle is classified by its first Chern class, we can take a complex line bundle

$$L \longrightarrow X$$

such that

$$c_1(L) = PD^{-1}(\alpha).$$

Take a smooth section $s : X \rightarrow L$ of L and perturb it slightly so that s is transverse to the zero section. Then its zero set

$$\Sigma_\alpha := s^{-1}(0)$$

is a smooth closed submanifold of X . Since L has real rank 2, Σ_α has codimension 2, and hence Σ_α is a closed surface.

The complex line bundle L carries a natural orientation, so the normal bundle of Σ_α also carries a natural orientation. Combined with the orientation of X , this endows Σ_α with an orientation. By the relation between the first Chern class and the zero set,

$$PD[\Sigma_\alpha] = c_1(L) = PD^{-1}(\alpha).$$

Therefore

$$[\Sigma_\alpha] = \alpha.$$

□

Now take two homology classes $\alpha, \beta \in H_2(X; \mathbb{Z})$ and choose closed oriented embedded surfaces Σ_α and Σ_β representing them.

After a small perturbation, we may assume that Σ_α and Σ_β intersect transversely. Since

$$\dim \Sigma_\alpha + \dim \Sigma_\beta = 2 + 2 = 4 = \dim X,$$

their intersection $\Sigma_\alpha \cap \Sigma_\beta$ is a finite set of points.

Let

$$p \in \Sigma_\alpha \cap \Sigma_\beta.$$

Since the intersection is transverse, we have

$$T_p X = T_p \Sigma_\alpha \oplus T_p \Sigma_\beta.$$

The orientations of the two surfaces induce an orientation on the direct sum

$$T_p \Sigma_\alpha \oplus T_p \Sigma_\beta$$

carries a natural orientation, namely that induced by the orientations of $T_p \Sigma_\alpha$ and $T_p \Sigma_\beta$. We define the sign of the intersection point p by comparing this orientation with the given orientation of $T_p X$:

$$\varepsilon(p) = \begin{cases} +1, & T_p \Sigma_\alpha \oplus T_p \Sigma_\beta \text{ has the same orientation as } T_p X, \\ -1, & T_p \Sigma_\alpha \oplus T_p \Sigma_\beta \text{ has the opposite orientation to } T_p X. \end{cases}$$

Definition 1.2.9 (Intersection number). The quantity

$$\Sigma_\alpha \cdot \Sigma_\beta := \sum_{p \in \Sigma_\alpha \cap \Sigma_\beta} \varepsilon(p)$$

is called the **algebraic intersection number** of Σ_α and Σ_β .

Note that what is computed here is not the number of geometric intersection points, but the signed number of intersection points. Hence intersection points of opposite signs may cancel one another.

Proposition 1.2.10. Let $\alpha, \beta \in H_2(X; \mathbb{Z})$ and represent them by transversely intersecting oriented embedded surfaces Σ_α and Σ_β , respectively. Then

$$Q_X(\alpha, \beta) = \Sigma_\alpha \cdot \Sigma_\beta = \sum_{p \in \Sigma_\alpha \cap \Sigma_\beta} \varepsilon(p).$$

Thus the intersection form Q_X can be understood entirely through the geometric intersection of surfaces. For $\alpha, \beta \in H_2(X; \mathbb{Z})$, the number $Q_X(\alpha, \beta)$ is precisely the signed count of intersection points obtained by intersecting two oriented surfaces representing α and β transversely.

In particular, when $\alpha = \beta$, the number

$$Q_X(\alpha, \alpha)$$

is called the **self-intersection number** of α . Geometrically, one can first take the surface Σ_α representing α and then perturb it a sufficiently small amount in the normal direction to obtain Σ'_α . Then

$$Q_X(\alpha, \alpha) = \Sigma_\alpha \cdot \Sigma'_\alpha.$$

1.2.5 The de Rham description

In what follows we give another description of the intersection form in terms of differential forms.

Let $a, b \in H^2(X; \mathbb{Z})$ and, via the change of coefficients, regard them as cohomology classes in $H^2(X; \mathbb{R})$. Write

$$\alpha = PD(a), \quad \beta = PD(b),$$

and take closed oriented embedded surfaces $\Sigma_\alpha, \Sigma_\beta \subset X$ representing α, β , respectively. After a small perturbation, we may assume that Σ_α and Σ_β intersect transversely.

By the de Rham theorem, there is an isomorphism

$$DR : H_{\text{dR}}^k(X) \xrightarrow{\cong} H^k(X; \mathbb{R}).$$

Take closed 2-forms ω_1, ω_2 such that

$$[\omega_1] = DR^{-1}(a), \quad [\omega_2] = DR^{-1}(b).$$

Proposition 1.2.11. In the above notation,

$$Q_X(a, b) = \langle a \smile b, [X] \rangle = \sum_{p \in \Sigma_\alpha \cap \Sigma_\beta} \varepsilon(p) = \int_X \omega_1 \wedge \omega_2.$$

We first explain that

$$\langle a \smile b, [X] \rangle = \int_X \omega_1 \wedge \omega_2.$$

The de Rham isomorphism is not merely an isomorphism of vector spaces; it is compatible with the multiplicative structure. More precisely, for closed differential forms ω, η we have

$$DR([\omega \wedge \eta]) = DR([\omega]) \smile DR([\eta]).$$

Hence DR is a graded ring isomorphism.

From $DR([\omega_1]) = a$ and $DR([\omega_2]) = b$, we obtain

$$DR([\omega_1 \wedge \omega_2]) = a \smile b.$$

Thus

$$\langle a \smile b, [X] \rangle = \int_X \omega_1 \wedge \omega_2.$$

Next we explain that

$$\sum_{p \in \Sigma_\alpha \cap \Sigma_\beta} \varepsilon(p) = \int_X \omega_1 \wedge \omega_2.$$

We can choose the representatives ω_1, ω_2 so that $\text{supp } \omega_1$ and $\text{supp } \omega_2$ are contained in sufficiently small tubular neighborhoods of Σ_α and Σ_β , respectively. Consequently the support of $\omega_1 \wedge \omega_2$ occurs only near the individual intersection points of $\Sigma_\alpha \cap \Sigma_\beta$, so that the whole integral decomposes as a sum of local contributions near the individual intersection points.

Take any $p \in \Sigma_\alpha \cap \Sigma_\beta$. Since Σ_α and Σ_β intersect transversely at p , we can choose local coordinates (x, y, u, v) in a neighborhood of p such that

$$\Sigma_\alpha = \{x = y = 0\}, \quad \Sigma_\beta = \{u = v = 0\}.$$

Take a bump function f on $\mathbb{R}^2$ with sufficiently small support satisfying

$$\int_{\mathbb{R}^2} f(x, y) dx \wedge dy = 1.$$

In the local neighborhood of the intersection point p , we can take

$$\omega_1 = f(x, y) dx \wedge dy, \quad \omega_2 = f(u, v) du \wedge dv.$$

Then

$$\omega_1 \wedge \omega_2 = f(x, y) f(u, v) dx \wedge dy \wedge du \wedge dv.$$

If

$$\left(\frac{\partial}{\partial u}, \frac{\partial}{\partial v}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right)$$

gives the same orientation as $T_p X$, then $\varepsilon(p) = +1$, and in this case the local contribution of the intersection point p to the integral is

$$\int f(x, y) f(u, v) dx \wedge dy \wedge du \wedge dv = 1.$$

If that orientation is opposite to the orientation of $T_p X$, then $\varepsilon(p) = -1$, and the corresponding local contribution is -1 .

Hence each transverse intersection point p contributes exactly $\varepsilon(p)$ to the integral $\int_X \omega_1 \wedge \omega_2$. Adding up the local contributions of all intersection points, we obtain

$$\int_X \omega_1 \wedge \omega_2 = \sum_{p \in \Sigma_\alpha \cap \Sigma_\beta} \varepsilon(p).$$

In summary,

$$Q_X(a, b) = \langle a \smile b, [X] \rangle = \Sigma_\alpha \cdot \Sigma_\beta = \int_X \omega_1 \wedge \omega_2.$$

Thus the intersection form can be understood from three equivalent points of view: the cup product gives its algebraic definition, the signed count of intersection points of transverse surfaces gives its geometric interpretation, and the integral of the wedge product of differential forms gives its de Rham description.

1.2.6 Some basic examples

In what follows we compute the intersection forms of some common 4-manifolds.

Example 1.2.12 (S^4). Since

$$H_2(S^4; \mathbb{Z}) = 0,$$

there are no non-trivial 2-dimensional homology classes on S^4 . Hence the intersection form of S^4 is trivial, that is,

$$Q_{S^4} = 0.$$

Example 1.2.13 ($\mathbb{CP}^2$). Take the complex orientation of $\mathbb{CP}^2$. We have

$$H_2(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}.$$

Take a complex projective line

$$L = \mathbb{CP}^1 \subset \mathbb{CP}^2,$$

Then $[L]$ generates $H_2(\mathbb{CP}^2; \mathbb{Z})$.

Take two complex projective lines L, L' in general position. They intersect transversely in exactly one point, and since the orientation induced by the complex structure is compatible with the complex orientation of $\mathbb{CP}^2$, the sign of this intersection point is +1. Hence

$$[L] \cdot [L] = 1.$$

Thus the matrix of the intersection form with respect to the basis $\{[L]\}$ is

$$Q_{\mathbb{CP}^2} = (1).$$

Example 1.2.14 ($\overline{\mathbb{CP}}^2$). Denote by $\overline{\mathbb{CP}}^2$ the manifold obtained from $\mathbb{CP}^2$ by reversing the orientation. From

$$Q_{\overline{X}} = -Q_X,$$

we immediately obtain

$$Q_{\overline{\mathbb{CP}}^2} = (-1).$$

Example 1.2.15 ($S^2 \times S^2$). We have

$$H_2(S^2 \times S^2; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Take

$$A = [S^2 \times \{p\}], \quad B = [\{q\} \times S^2],$$

where $p, q \in S^2$.

First we compute the self-intersection numbers. We may perturb $S^2 \times \{p\}$ slightly in the direction of the second factor, obtaining $S^2 \times \{p'\}$, where $p' \neq p$. Thus

$$(S^2 \times \{p\}) \cap (S^2 \times \{p'\}) = \emptyset,$$

hence $A \cdot A = 0$. Similarly $B \cdot B = 0$.

On the other hand,

$$(S^2 \times \{p\}) \cap (\{q\} \times S^2) = \{(q, p)\}.$$

This intersection point is transverse, and with respect to the product orientation its sign is +1. Hence

$$A \cdot B = 1.$$

Since the intersection form is symmetric, we also have $B \cdot A = 1$.

Thus with respect to the basis $\{A, B\}$ the matrix of the intersection form is

$$Q_{S^2 \times S^2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

One usually writes

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and calls it the hyperbolic form. Hence

$$Q_{S^2 \times S^2} = H.$$

Proposition 1.2.16 (The connected sum property of the intersection form). Let X, Y be closed oriented 4-manifolds. Then

$$Q_{X \# Y} \cong Q_X \oplus Q_Y.$$

This is because

$$H_2(X \# Y; \mathbb{Z}) \cong H_2(X; \mathbb{Z}) \oplus H_2(Y; \mathbb{Z}),$$

and the 2-dimensional homology classes coming from the two sides X and Y can be represented by surfaces chosen so as to lie on the two sides of the neck of the connected sum. Thus the intersection number between classes from different direct summands is 0, so the intersection form splits as an orthogonal direct sum.

For example,

$$Q_{\#^m \mathbb{CP}^2 \#^n \overline{\mathbb{CP}}^2} \cong m(1) \oplus n(-1).$$

Example 1.2.17 (The K3 surface). A **K3 surface** is a compact complex surface X whose canonical bundle

$$K_X = \Lambda^{2,0} T^* X$$

is trivial, equivalently $c_1(X) = 0$, and whose first Betti number vanishes; such a surface is automatically simply connected. The standard examples are the smooth quartic surfaces $X = \{z \in \mathbb{CP}^3 : f(z) = 0\}$, where f is a homogeneous polynomial of degree 4 in the variables z_0, z_1, z_2, z_3 (for instance the Fermat quartic $z_0^4 + z_1^4 + z_2^4 + z_3^4 = 0$), and the Kummer surface, obtained by resolving the 16 ordinary double points of the quotient $T^4/\{\pm 1\}$. For a smooth hypersurface of degree d in $\mathbb{CP}^3$ the adjunction formula gives $c_1 = (4 - d)h$, where h denotes the hyperplane class, so for $d = 4$ the canonical bundle is trivial, while the Lefschetz hyperplane theorem gives $\pi_1 = 0$; hence every smooth quartic is a K3 surface. All K3 surfaces are diffeomorphic, so in the smooth category one speaks simply of *the* K3 surface.

The intersection form of a K3 surface is

$$Q_{K3} \cong 2(-E_8) \oplus 3H,$$

where E_8 denotes the positive definite E_8 lattice of rank 8, that is,

$$E_8 = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

H (see Example 1.2.15) denotes the hyperbolic form

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Hence the rank of Q_{K3} is

$$2 \cdot 8 + 3 \cdot 2 = 22,$$

its signature is -16 and its positive index is $b_2^+ = 3$.

Example 1.2.18 (T^4). Although T^4 is not simply connected, its intersection form is straightforward to compute, so it also often serves as a basic example.

Write

$$T^4 = S^1 \times S^1 \times S^1 \times S^1.$$

Take standard coordinates x^1, x^2, x^3, x^4 and take the orientation

$$dx^1 \wedge dx^2 \wedge dx^3 \wedge dx^4 > 0.$$

By the Künneth formula,

$$H^2(T^4; \mathbb{Z}) \cong \Lambda^2 \mathbb{Z}^4,$$

hence $\text{rank } H^2(T^4; \mathbb{Z}) = 6$.

We may take the basis

$$e_{12} = [dx^1 \wedge dx^2], \quad e_{13} = [dx^1 \wedge dx^3], \quad e_{14} = [dx^1 \wedge dx^4],$$

and

$$e_{34} = [dx^3 \wedge dx^4], \quad -e_{24} = -[dx^2 \wedge dx^4], \quad e_{23} = [dx^2 \wedge dx^3].$$

From

$$\int_{T^4} dx^1 \wedge dx^2 \wedge dx^3 \wedge dx^4 = 1,$$

we obtain

$$e_{12} \cdot e_{34} = 1, \quad e_{13} \cdot (-e_{24}) = 1, \quad e_{14} \cdot e_{23} = 1.$$

Apart from these, all remaining pairwise intersection numbers among the basis elements above are 0. Hence

$$Q_{T^4} \cong H \oplus H \oplus H = 3H.$$

1.3 The intersection form and topological classification

Let X be a simply connected, closed, oriented 4-manifold. We have

$$H_0(X; \mathbb{Z}) \cong \mathbb{Z}, \quad H_2(X; \mathbb{Z}) \cong \mathbb{Z}^{b_2}, \quad H_4(X; \mathbb{Z}) \cong \mathbb{Z},$$

while the homology groups in all remaining degrees are 0.

Correspondingly, the cohomology groups satisfy

$$H^0(X; \mathbb{Z}) \cong \mathbb{Z}, \quad H^2(X; \mathbb{Z}) \cong \mathbb{Z}^{b_2}, \quad H^4(X; \mathbb{Z}) \cong \mathbb{Z},$$

and $H^1(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0$.

Thus, if we consider the cohomology ring $H^*(X; \mathbb{Z})$, then apart from the unit element the only cup product that can possibly be non-trivial is

$$H^2(X; \mathbb{Z}) \times H^2(X; \mathbb{Z}) \longrightarrow H^4(X; \mathbb{Z}) \cong \mathbb{Z}.$$

and this multiplication is precisely the one recorded by the intersection form

$$Q_X(a, b) = \langle a \smile b, [X] \rangle$$

In other words, for a simply connected closed oriented 4-manifold, the intersection form Q_X determines the multiplicative structure of its integral cohomology ring.

In what follows we explain, from the point of view of CW complexes, why the intersection form can further control the homotopy type of X .

Since X is simply connected, we may choose a suitable CW decomposition such that the 2-skeleton of X is homotopy equivalent to a wedge sum of b_2 copies of S^2 ,

$$\bigvee_{i=1}^{b_2} S_i^2.$$

Hence X can be written as

$$X \simeq \left(\bigvee_{i=1}^{b_2} S_i^2 \right) \cup_f D^4,$$

where

$$f : S^3 \longrightarrow \bigvee_{i=1}^{b_2} S_i^2$$

is the attaching map of the unique 4-cell.

Denote by

$$\iota_i : S_i^2 \longrightarrow \bigvee_{j=1}^{b_2} S_j^2$$

the inclusion of the i -th sphere. Let $\eta : S^3 \rightarrow S^2$ be the Hopf map, and write

$$\eta_i := \iota_i \circ \eta.$$

Moreover, for $i \neq j$ there is also the Whitehead product

$$[\iota_i, \iota_j] \in \pi_3 \left(\bigvee_{k=1}^{b_2} S_k^2 \right).$$

These elements generate

$$\pi_3 \left(\bigvee_{i=1}^{b_2} S_i^2 \right),$$

and the coefficients of the individual Hopf maps and Whitehead products in the attaching map f are controlled precisely by the matrix entries of the intersection form.

If $\{e_1, \dots, e_{b_2}\}$ is a basis of $H_2(X; \mathbb{Z})$, then roughly speaking, f can be written as

$$f \simeq \sum_i Q_X(e_i, e_i) \eta_i + \sum_{i < j} Q_X(e_i, e_j) [\iota_i, \iota_j].$$

Thus the diagonal terms $Q_X(e_i, e_i)$ of the intersection form control the coefficients of the Hopf maps, while the off-diagonal terms $Q_X(e_i, e_j)$ control the coefficients of the Whitehead products. In other words, Q_X records how the 4-cell is attached to the 2-skeleton.

Theorem 1.3.1 (Whitehead–Milnor). Let X, Y be simply connected, closed, oriented 4-manifolds. Then

$$Q_X \cong Q_Y$$

if and only if X and Y have the same orientation-preserving homotopy type.

Hence, in the case of simply connected closed oriented 4-manifolds, the intersection form determines not only the cohomology ring but also the orientation-preserving homotopy type of the manifold.

Thus the next question arises naturally: if

$$Q_X \cong Q_Y,$$

can we further conclude that

$$X \cong_{\text{Top}} Y?$$

This is precisely the question studied by Freedman theory.

1.3.1 Freedman theory and topological classification

We have seen above that for a simply connected closed oriented 4-manifold the intersection form determines its orientation-preserving homotopy type (Theorem 1.3.1). Next we consider a stronger question: given an abstract unimodular symmetric integral bilinear form, is it necessarily realized by some simply connected closed oriented 4-manifold?

Theorem 1.3.2 (Freedman). Let

$$Q : \mathbb{Z}^n \times \mathbb{Z}^n \longrightarrow \mathbb{Z}$$

be a unimodular symmetric integral bilinear form. Then there exists a simply connected closed oriented topological 4-manifold X_Q such that

$$Q_{X_Q} \cong Q.$$

Hence, in the topological category, any unimodular symmetric integral bilinear form can occur as the intersection form of some simply connected closed oriented 4-manifold.

For example, E_8 (see Example 1.2.17) is an even, positive definite, unimodular symmetric integral bilinear form. Hence by the Freedman theorem, Theorem 1.3.2, there exists a simply connected closed oriented topological 4-manifold X_{E_8} satisfying

$$Q_{X_{E_8}} \cong E_8.$$

However, the intersection form alone is in general still not enough to determine the topological homeomorphism type completely. To obtain a complete topological classification one must also introduce the Kirby–Siebenmann invariant.

Definition 1.3.3. For a closed topological 4-manifold X , its Kirby–Siebenmann invariant is denoted by

$$ks(X) \in H^4(X; \mathbb{Z}_2).$$

In the present simply connected closed oriented situation,

$$H^4(X; \mathbb{Z}_2) \cong \mathbb{Z}_2,$$

so we may regard $ks(X)$ as an element of $\mathbb{Z}_2$.

The Kirby–Siebenmann invariant measures the obstruction for a topological manifold to admit a PL or smooth structure.

Theorem 1.3.4 (Freedman). Let X, Y be simply connected closed oriented topological 4-manifolds. Then

$$X \cong_{\text{Top}} Y$$

if and only if

$$Q_X \cong Q_Y$$

and

$$ks(X) = ks(Y).$$

Hence the topological homeomorphism type of a simply connected closed oriented topological 4-manifold is determined jointly by the intersection form Q_X and the Kirby–Siebenmann invariant $ks(X)$.

When Q_X is even, the Kirby–Siebenmann invariant is already determined by the intersection form. Specifically,

$$ks(X) \equiv \frac{\sigma(Q_X)}{8} \pmod{2}. \quad (1.3.E1)$$

Here $\sigma(Q_X)$ denotes the signature of the intersection form (Definition 1.2.7).

Hence, for a given even unimodular form Q , there is exactly one orientation-preserving topological homeomorphism type corresponding to it.

On the other hand, when Q is odd, there are in general two topological homeomorphism types with the same intersection form, corresponding respectively to

$$ks = 0 \quad \text{and} \quad ks = 1.$$

In other words, in the topological category even and odd intersection forms behave slightly differently. For an even unimodular form the intersection form itself already determines the Kirby–Siebenmann invariant, see (1.3.E1), and hence determines the topological homeomorphism type. For an odd unimodular form one must specify ks in addition.

1.3.2 Constraints in the smooth category

In what follows we consider the smooth category further. If X is a smooth closed 4-manifold, then its Kirby–Siebenmann invariant (Definition 1.3.3) must satisfy

$$ks(X) = 0. \quad (1.3.E2)$$

Hence, if X, Y are both simply connected closed oriented smooth 4-manifolds and $Q_X \cong Q_Y$, then by the Freedman theorem, Theorem 1.3.4, it follows that

$$X \cong_{\text{Top}} Y.$$

However, in general one cannot further conclude that

$$X \cong_{\text{Diff}} Y.$$

In other words, the intersection form has very strong classification power in the topological category, but it does not determine the smooth structure.

Freedman theory (Theorem 1.3.2, Theorem 1.3.4) tells us which unimodular symmetric integral bilinear forms can be realized in the topological category, whereas Donaldson theory (Theorem 1.3.5) imposes very strong restrictions on intersection forms in the smooth category.

We have seen above that, by the Freedman theorem, Theorem 1.3.2, there exists a simply connected closed oriented topological 4-manifold X_{E_8} such that

$$Q_{X_{E_8}} \cong E_8.$$

Since E_8 is even, if X_{E_8} admitted a smooth structure, then X_{E_8} would be a spin 4-manifold. Moreover, since

$$\sigma(X_{E_8}) = \sigma(E_8) = 8,$$

by the Rokhlin theorem, Theorem 1.3.7, for any smooth closed spin 4-manifold X we have

$$\sigma(X) \equiv 0 \pmod{16}.$$

This contradicts $\sigma(X_{E_8}) = 8$. Hence X_{E_8} cannot admit a smooth structure.

That is, E_8 can occur as the intersection form of some simply connected closed oriented topological 4-manifold, but not as the intersection form of a simply connected closed oriented smooth 4-manifold.

This shows that in dimension 4 there is an essential difference between the topological category and the smooth category.

1.3.3 Cases of intersection forms of smooth manifolds

In what follows let X be a simply connected, closed, oriented smooth 4-manifold, and let us discuss the cases according to the parity of the intersection form Q_X and to whether it is definite. The conclusions in the various cases are collected in Table 1.

(1) The odd indefinite case If Q_X is an odd indefinite unimodular form, then the classification of odd indefinite unimodular forms over the integers tells us that

$$Q_X \cong p\langle 1 \rangle \oplus q\langle -1 \rangle,$$

where $p, q \geq 1$. Here $\langle 1 \rangle$ and $\langle -1 \rangle$ denote the rank 1 bilinear forms defined by the matrices (1) and (-1) , respectively.

On the other hand,

$$Q_{\mathbb{CP}^2} = \langle 1 \rangle, \quad Q_{\overline{\mathbb{CP}}^2} = \langle -1 \rangle,$$

while connected sums satisfy

$$Q_{X\#Y} \cong Q_X \oplus Q_Y.$$

Hence

$$Q_{\#^p \mathbb{CP}^2 \#^q \overline{\mathbb{CP}}^2} \cong p\langle 1 \rangle \oplus q\langle -1 \rangle.$$

Since X is smooth, (1.3.E2) gives $ks(X) = 0$. At the same time $\#^p \mathbb{CP}^2 \#^q \overline{\mathbb{CP}}^2$ is also smooth, so its Kirby–Siebenmann invariant likewise vanishes. By Freedman’s theorem,

$$X \cong_{\text{Top}} \#^p \mathbb{CP}^2 \#^q \overline{\mathbb{CP}}^2.$$

Therefore, for a smooth simply connected closed oriented 4-manifold, the odd indefinite case is in the topological category completely determined by p and q .

(2) The even indefinite case If Q_X is an even indefinite unimodular form, then by the classification of even indefinite unimodular forms, Q_X can be written as an orthogonal direct sum of E_8 -type summands and hyperbolic summands H .

Following the notation of these notes, we may write

$$Q_X \cong kE_8 \oplus lH,$$

where the sign of k records the orientation of the E_8 part. More precisely, $k > 0$ means k copies of E_8 , while $k < 0$ means $|k|$ copies of $-E_8$.

Since

$$\sigma(E_8) = 8, \quad \sigma(H) = 0,$$

we have

$$k = \frac{\sigma(X)}{8}.$$

On the other hand, $\text{rank}(E_8) = 8$ and $\text{rank}(H) = 2$, so

$$b_2(X) = 8|k| + 2l.$$

Hence

$$l = \frac{b_2(X) - |\sigma(X)|}{2}.$$

Thus the isomorphism type of an even indefinite unimodular form is determined by $b_2(X)$ and $\sigma(X)$.

In the smooth category this case is subject to a further restriction coming from Rokhlin’s theorem 1.3.7, which we discuss later.

(3) The definite case The classification of definite unimodular integral forms in general is far more difficult. If however we restrict to the smooth category, Donaldson’s theorem 1.3.5 gives a very strong restriction.

Theorem 1.3.5 (Donaldson). Let X be a simply connected, closed, oriented smooth 4-manifold. If Q_X is definite, then Q_X can be diagonalized over the integers.

Therefore, if Q_X is positive definite, then

$$Q_X \cong \langle 1 \rangle^{\oplus b_2(X)}.$$

If Q_X is negative definite, then

$$Q_X \cong \langle -1 \rangle^{\oplus b_2(X)}.$$

By Freedman's theorem 1.3.4, in the positive definite case

$$X \cong_{\text{Top}} \#^{b_2(X)} \mathbb{CP}^2,$$

while in the negative definite case

$$X \cong_{\text{Top}} \#^{b_2(X)} \overline{\mathbb{CP}}^2.$$

Corollary 1.3.6. A non-zero even definite unimodular form cannot be the intersection form of a simply connected, closed, oriented smooth 4-manifold.

In fact, if Q_X is definite, then by Donaldson's theorem 1.3.5 we may choose an integral basis $\{e_1, \dots, e_{b_2}\}$ such that

$$Q_X(e_i, e_i) = \pm 1.$$

Hence Q_X must be odd. This contradicts the fact that Q_X is even.

In particular, E_8 is an even, positive definite, unimodular form, so by Corollary 1.3.6 it cannot be the intersection form of any simply connected closed oriented smooth 4-manifold.

Further restrictions in the even indefinite case We now consider further the case where X is a simply connected, closed, oriented smooth 4-manifold and Q_X is even indefinite.

Since $\pi_1(X) = 0$ and Q_X is even, Wu's formula (1.2.E1) shows that

$$w_2(X) = 0.$$

Hence X is a spin 4-manifold.

Theorem 1.3.7 (Rokhlin). Let X be a smooth closed spin 4-manifold. Then

$$\sigma(X) \equiv 0 \pmod{16}.$$

On the other hand, an even indefinite unimodular form can be written as

$$Q_X \cong kE_8 \oplus lH,$$

where $k \in \mathbb{Z}$ and $l \geq 1$. Since $\sigma(E_8) = 8$ and $\sigma(H) = 0$, we have

$$\sigma(X) = 8k.$$

By Rokhlin's theorem 1.3.7, $\sigma(X)$ is divisible by 16, so k must be even.

Thus we may write

$$k = 2m,$$

and therefore

$$Q_X \cong 2mE_8 \oplus lH.$$

By reversing the orientation of X if necessary, we may assume that $\sigma(X) \leq 0$. Note that after reversing the orientation we have $Q_{\bar{X}} = -Q_X$, and

$$-H \cong H.$$

Hence we may as well write the intersection form as

$$Q_X \cong 2m(-E_8) \oplus lH, \quad m \geq 0.$$

In this notation,

$$\sigma(X) = -16m, \quad b_2(X) = 16m + 2l.$$

On the other hand, the intersection form of the K3 surface is

$$Q_{K3} \cong 2(-E_8) \oplus 3H.$$

Hence

$$Q_{\#^m K3} \cong 2m(-E_8) \oplus 3mH.$$

Since

$$Q_{S^2 \times S^2} = H,$$

whenever $l \geq 3m$ we have

$$Q_X \cong Q_{\#^m K3} \oplus (l - 3m)H.$$

That is,

$$Q_X \cong Q_{\#^m K3 \# \#^{l-3m}(S^2 \times S^2)}.$$

Since these manifolds are all smooth, their Kirby–Siebenmann invariants are all 0. Hence, by Freedman’s theorem 1.3.4,

$$X \cong_{\text{Top}} \#^m K3 \# \#^{l-3m}(S^2 \times S^2),$$

as long as $l \geq 3m$.

Thus, in the even indefinite case, a natural question is: for an arbitrary simply connected closed smooth spin 4-manifold, do we always have

$$l \geq 3m?$$

This is precisely the 11/8-conjecture 1.3.8.

Conjecture 1.3.8 (11/8-conjecture). Let X be a smooth closed spin 4-manifold. Then

$$b_2(X) \geq \frac{11}{8}|\sigma(X)|.$$

For

$$Q_X \cong 2m(-E_8) \oplus lH,$$

since $b_2(X) = 16m + 2l$ and $|\sigma(X)| = 16m$, the above inequality is equivalent to

$$16m + 2l \geq \frac{11}{8} \cdot 16m = 22m.$$

Hence

$$l \geq 3m.$$

Without reversing the orientation beforehand, this can be stated uniformly as

$$l \geq 3|m|.$$

Therefore the 11/8-conjecture 1.3.8 is equivalent to the assertion that for an even indefinite unimodular form

$$Q_X \cong 2m(\pm E_8) \oplus lH$$

arising from a smooth closed spin 4-manifold, its H part must satisfy

$$l \geq 3|m|.$$

The 11/8-conjecture and known results We have seen above that for a simply connected, closed, oriented smooth spin 4-manifold X , if

$$Q_X \cong 2m(-E_8) \oplus lH,$$

then the 11/8-conjecture 1.3.8 is equivalent to

$$l \geq 3m.$$

Hence, if the 11/8-conjecture 1.3.8 holds, then for every even indefinite unimodular form

$$Q \cong 2m(\pm E_8) \oplus lH$$

satisfying

$$l < 3|m|,$$

there is no simply connected closed smooth 4-manifold X with

$$Q_X \cong Q.$$

In other words, the 11/8-conjecture 1.3.8 gives a very strong restriction on whether an even indefinite unimodular form can be realized in the smooth category.

At present the 11/8-conjecture has not been completely resolved. One important known result is Furuta's 10/8-theorem, namely Theorem 1.3.9.

Theorem 1.3.9 (Furuta). Let X be a smooth closed spin 4-manifold. Then

$$b_2(X) \geq \frac{10}{8}|\sigma(X)| + 2.$$

For

$$Q_X \cong 2m(-E_8) \oplus lH,$$

we have

$$b_2(X) = 16m + 2l, \quad |\sigma(X)| = 16m.$$

Hence Furuta's theorem 1.3.9 gives

$$16m + 2l \geq \frac{10}{8} \cdot 16m + 2 = 20m + 2.$$

Thus

$$l \geq 2m + 1.$$

Without fixing the sign of $\sigma(X)$ beforehand, this can be written as

$$l \geq 2|m| + 1.$$

Therefore, although Furuta's 10/8-theorem does not yet reach the bound

$$l \geq 3|m|$$

required by the 11/8-conjecture 1.3.8, it is already much stronger than the restriction obtained from Rokhlin's theorem 1.3.7 alone.

Definiteness	Parity	Intersection form Q_X	Topological type and conditions
—	even	(0)	S^4 , $b_2(X) = 0$
definite	odd	$\langle 1 \rangle^{\oplus b_2(X)}$	$\#^{b_2(X)} \mathbb{CP}^2$
		$\langle -1 \rangle^{\oplus b_2(X)}$	$\#^{b_2(X)} \overline{\mathbb{CP}}^2$
indefinite	odd	$p\langle 1 \rangle \oplus q\langle -1 \rangle$	$\#^p \mathbb{CP}^2 \#^q \overline{\mathbb{CP}}^2$, $p, q \geq 1$
indefinite	even	$2m(-E_8) \oplus lH$	$\#^m K3 \#^{l-3m} (S^2 \times S^2)$
			$l \geq 3m$
			smooth realization is not yet completely determined by known results $2m + 1 \leq l < 3m$ no corresponding simply connected closed smooth 4-manifold exists $1 \leq l \leq 2m$

Table 1: Overview of the intersection forms and topological types of simply connected closed oriented smooth 4-manifolds.

1.4 Self-duality and Hodge theory

1.4.1 The Hodge star and the decomposition of 2-forms

Let X be a closed, orientable smooth 4-manifold, and choose a Riemannian metric g on X .

The metric g together with the orientation of X allow us to define the Hodge star operator

$$* : \Omega^k(X) \longrightarrow \Omega^{4-k}(X).$$

In particular, when $k = 2$,

$$* : \Omega^2(X) \longrightarrow \Omega^2(X).$$

In dimension 4, the Hodge star satisfies $*^2 = 1$ on 2-forms. Hence the only eigenvalues of $*$ on $\Omega^2(X)$ are $+1$ and -1 .

Define

$$\Omega^+(X) := \{\alpha \in \Omega^2(X) \mid *\alpha = \alpha\},$$

and

$$\Omega^-(X) := \{\alpha \in \Omega^2(X) \mid *\alpha = -\alpha\}.$$

We call the elements of $\Omega^+(X)$ **self-dual 2-forms**, and the elements of $\Omega^-(X)$ **anti-self-dual 2-forms**.

Since $*^2 = 1$, we have the direct sum decomposition

$$\Omega^2(X) = \Omega^+(X) \oplus \Omega^-(X).$$

Once harmonic forms have been defined below, we write

$$\mathcal{H}^2(X; g) = \{\alpha \in \Omega^2(X) \mid \Delta\alpha = 0\}$$

for the space of harmonic 2-forms. Since the Hodge star preserves harmonic 2-forms, we can further define

$$\mathcal{H}^+(X; g) := \{\alpha \in \mathcal{H}^2(X; g) \mid *\alpha = \alpha\},$$

and

$$\mathcal{H}^-(X; g) := \{\alpha \in \mathcal{H}^2(X; g) \mid *\alpha = -\alpha\}.$$

Thus

$$\mathcal{H}^2(X; g) = \mathcal{H}^+(X; g) \oplus \mathcal{H}^-(X; g).$$

By the Hodge theorem,

$$\mathcal{H}^2(X; g) \cong H_{\text{dR}}^2(X) \cong H^2(X; \mathbb{R}).$$

Hence

$$H^2(X; \mathbb{R}) \cong \mathcal{H}^+(X; g) \oplus \mathcal{H}^-(X; g).$$

Note that the subspaces $\mathcal{H}^+(X; g)$ and $\mathcal{H}^-(X; g)$ themselves depend on the chosen Riemannian metric g . However, we will prove later that their dimensions are independent of g , and that

$$b_2^+(X) = \dim \mathcal{H}^+(X; g), \quad b_2^-(X) = \dim \mathcal{H}^-(X; g).$$

If $[\alpha], [\beta] \in H^2(X; \mathbb{R})$, then in de Rham cohomology the intersection form is expressed as

$$Q_X([\alpha], [\beta]) = \int_X \alpha \wedge \beta.$$

We will prove later that Q_X is positive definite on $\mathcal{H}^+(X; g)$ and negative definite on $\mathcal{H}^-(X; g)$. Hence

$$\sigma(X) = b_2^+(X) - b_2^-(X) = \dim \mathcal{H}^+(X; g) - \dim \mathcal{H}^-(X; g).$$

1.4.2 Inner products on differential forms and formal adjoints

Let (X, g) be an n -dimensional Riemannian manifold. For any $p \in X$, the Riemannian metric g gives an inner product on the tangent space $T_p X$, and hence induces a dual inner product on the cotangent space $T_p^* X$.

Furthermore, this inner product naturally induces an inner product on the exterior power

$$\Lambda^k T_p^* X$$

For decomposable k -forms $\xi_1 \wedge \cdots \wedge \xi_k$ and $\eta_1 \wedge \cdots \wedge \eta_k$, define

$$\langle \xi_1 \wedge \cdots \wedge \xi_k, \eta_1 \wedge \cdots \wedge \eta_k \rangle := \det(\langle \xi_i, \eta_j \rangle).$$

For example, let $\{e_1, \dots, e_n\}$ be an orthonormal basis of $T_p X$, and let $\{e^1, \dots, e^n\}$ be the corresponding dual basis. For a multi-index

$$I = (i_1 < \cdots < i_k),$$

write

$$e^I := e^{i_1} \wedge \cdots \wedge e^{i_k}.$$

Then the forms e^I all together constitute an orthonormal basis of $\Lambda^k T_p^* X$.

If

$$\alpha = \sum_{|I|=k} a_I e^I, \quad \beta = \sum_{|I|=k} b_I e^I,$$

then

$$\langle \alpha, \beta \rangle = \sum_{|I|=k} a_I b_I,$$

in particular,

$$|\alpha|^2 = \langle \alpha, \alpha \rangle = \sum_{|I|=k} a_I^2.$$

The Hodge star can be characterized by this inner product. For any $\alpha, \beta \in \Omega^k(X)$, we have

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle d\mu_g,$$

where $d\mu_g$ is the volume form determined by g and the orientation of X .

We can therefore define the L^2 inner product on $\Omega^k(X)$ by

$$(\alpha, \beta)_{L^2} := \int_X \alpha \wedge * \beta = \int_X \langle \alpha, \beta \rangle d\mu_g.$$

Correspondingly,

$$\|\alpha\|_{L^2}^2 = (\alpha, \alpha)_{L^2} = \int_X |\alpha|^2 d\mu_g.$$

Formal adjoints Consider the exterior derivative

$$d : \Omega^k(X) \longrightarrow \Omega^{k+1}(X).$$

We wish to find an operator

$$d^* : \Omega^{k+1}(X) \longrightarrow \Omega^k(X),$$

such that for any $\alpha \in \Omega^k(X)$ and $\beta \in \Omega^{k+1}(X)$, we have

$$(d\alpha, \beta)_{L^2} = (\alpha, d^*\beta)_{L^2}.$$

We call d^* the **formal adjoint** of d with respect to the L^2 inner product.

In what follows we assume that X is a closed manifold. By definition,

$$(d\alpha, \beta)_{L^2} = \int_X d\alpha \wedge *\beta.$$

On the other hand, by the Leibniz formula,

$$d(\alpha \wedge *\beta) = d\alpha \wedge *\beta + (-1)^k \alpha \wedge d(*\beta).$$

Since X is a closed manifold, Stokes' theorem gives

$$\int_X d(\alpha \wedge *\beta) = 0.$$

Therefore

$$\int_X d\alpha \wedge *\beta = (-1)^{k+1} \int_X \alpha \wedge d(*\beta).$$

Combining this with the identity

$$*^2 = (-1)^{r(n-r)}$$

satisfied by the Hodge star on r -forms, we obtain the expression

$$d^* = (-1)^{n(r+1)+1} * d * \quad \text{on } \Omega^r(X).$$

In particular, when $\dim X = 4$, the above formula becomes

$$\boxed{d^* = - * d * .}$$

Remark 1.4.1. The formal adjoint d^* is unique. Indeed, if

$$D_1, D_2 : \Omega^{k+1}(X) \longrightarrow \Omega^k(X)$$

both satisfy

$$(d\alpha, \beta)_{L^2} = (\alpha, D_i\beta)_{L^2},$$

then for every α we have

$$(\alpha, (D_1 - D_2)\beta)_{L^2} = 0.$$

Taking $\alpha = (D_1 - D_2)\beta$, we obtain

$$\|(D_1 - D_2)\beta\|_{L^2}^2 = 0.$$

Hence $D_1 = D_2$.

1.4.3 The Hodge Laplacian and harmonic forms

Define the Hodge Laplacian

$$\Delta := dd^* + d^*d.$$

Definition 1.4.2. If $\alpha \in \Omega^k(X)$ satisfies

$$\Delta\alpha = 0,$$

then we call α a **harmonic k -form**.

We write the space of all harmonic k -forms as

$$\mathcal{H}^k(X; g) := \ker \left(\Delta : \Omega^k(X) \longrightarrow \Omega^k(X) \right).$$

For a closed Riemannian manifold, the harmonic condition can equivalently be written as

$$\Delta\alpha = 0 \iff d\alpha = 0 \quad \text{and} \quad d^*\alpha = 0.$$

Indeed,

$$(\Delta\alpha, \alpha)_{L^2} = (dd^*\alpha, \alpha)_{L^2} + (d^*d\alpha, \alpha)_{L^2}.$$

Since d^* is the formal adjoint of d ,

$$(\Delta\alpha, \alpha)_{L^2} = \|d^*\alpha\|_{L^2}^2 + \|d\alpha\|_{L^2}^2.$$

Hence if $\Delta\alpha = 0$, then

$$d\alpha = 0, \quad d^*\alpha = 0.$$

The converse is clear.

We now assume that $\dim X = 4$ and consider harmonic 2-forms. Since

$$* : \Omega^2(X) \longrightarrow \Omega^2(X), \quad *^2 = 1,$$

we show below that $*$ preserves harmonic 2-forms.

Let $\alpha \in \mathcal{H}^2(X; g)$. Then

$$d\alpha = 0, \quad d^*\alpha = 0.$$

In dimension 4 we have $d^* = - * d *$, and therefore

$$d^*\alpha = 0 \implies d(*\alpha) = 0.$$

On the other hand,

$$d^*(*\alpha) = - * d * (*\alpha) = - * d\alpha = 0.$$

Hence

$$\Delta(*\alpha) = 0.$$

Therefore

$$* : \mathcal{H}^2(X; g) \longrightarrow \mathcal{H}^2(X; g).$$

Since $*^2 = 1$, we can define

$$\mathcal{H}^+(X; g) := \{\alpha \in \mathcal{H}^2(X; g) \mid *\alpha = \alpha\},$$

and

$$\mathcal{H}^-(X; g) := \{\alpha \in \mathcal{H}^2(X; g) \mid *\alpha = -\alpha\}.$$

Thus we have the direct sum decomposition

$$\boxed{\mathcal{H}^2(X; g) = \mathcal{H}^+(X; g) \oplus \mathcal{H}^-(X; g).}$$

1.4.4 The Hodge theorem and the intersection form

By the Hodge theorem, every de Rham cohomology class has a unique harmonic representative. Hence there is a natural isomorphism

$$H_{\text{dR}}^2(X) \cong \mathcal{H}^2(X; g).$$

Then, by the de Rham theorem,

$$H^2(X; \mathbb{R}) \cong H_{\text{dR}}^2(X),$$

and thus

$$H^2(X; \mathbb{R}) \cong \mathcal{H}^2(X; g) = \mathcal{H}^+(X; g) \oplus \mathcal{H}^-(X; g).$$

Note that $\mathcal{H}^+(X; g)$ and $\mathcal{H}^-(X; g)$ depend on the chosen Riemannian metric g . Strictly speaking, therefore, one should write them as

$$\mathcal{H}_g^+(X), \quad \mathcal{H}_g^-(X).$$

We now describe the signs of the intersection form on these two subspaces.

Proposition 1.4.3. The intersection form Q_X is positive definite on $\mathcal{H}^+(X; g)$ and negative definite on $\mathcal{H}^-(X; g)$.

Proof. Let $0 \neq \alpha \in \mathcal{H}^+(X; g)$. Since $*\alpha = \alpha$, we have

$$Q_X([\alpha], [\alpha]) = \int_X \alpha \wedge \alpha = \int_X \alpha \wedge *\alpha = \int_X |\alpha|^2 d\mu_g > 0.$$

Hence Q_X is positive definite on $\mathcal{H}^+(X; g)$.

Similarly, if $0 \neq \alpha \in \mathcal{H}^-(X; g)$, then $*\alpha = -\alpha$, and hence

$$Q_X([\alpha], [\alpha]) = \int_X \alpha \wedge \alpha = - \int_X \alpha \wedge *\alpha = - \int_X |\alpha|^2 d\mu_g < 0.$$

Hence Q_X is negative definite on $\mathcal{H}^-(X; g)$. □

It follows that

$$b_2^+(X) = \dim \mathcal{H}^+(X; g), \quad b_2^-(X) = \dim \mathcal{H}^-(X; g).$$

Therefore

$$\sigma(X) = b_2^+(X) - b_2^-(X) = \dim \mathcal{H}^+(X; g) - \dim \mathcal{H}^-(X; g).$$

Although the subspaces $\mathcal{H}^\pm(X; g)$ themselves depend on the metric g , their dimensions equal the positive and negative indices of inertia of the intersection form Q_X , respectively; hence $b_2^\pm(X)$ and $\sigma(X)$ are both independent of the choice of g .

1.4.5 The operator d^+ and the self-dual complex

Denote by

$$\pi_+ : \Omega^2(X) \longrightarrow \Omega^+(X)$$

the projection onto the self-dual part. Since

$$\Omega^2(X) = \Omega^+(X) \oplus \Omega^-(X),$$

and $*|_{\Omega^+} = 1$, $*|_{\Omega^-} = -1$, we have

$$\pi_+ = \frac{1}{2}(1 + *).$$

Define

$$d^+ := \pi_+ \circ d : \Omega^1(X) \longrightarrow \Omega^+(X).$$

Thus, for any $\alpha \in \Omega^1(X)$,

$$d^+\alpha = (d\alpha)^+ = \frac{1}{2}(d\alpha + *d\alpha).$$

If $f \in \Omega^0(X)$, then

$$d^+df = \pi_+(d^2f) = 0.$$

Hence we obtain the complex

$$0 \longrightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d^+} \Omega^+(X) \longrightarrow 0.$$

Write the cohomology groups of this complex as $H_{d^+}^0$, $H_{d^+}^1$ and $H_{d^+}^2$.

Proposition 1.4.4. The cohomology of the above complex satisfies

$$H_{d^+}^0 \cong H^0(X; \mathbb{R}), \quad H_{d^+}^1 \cong H^1(X; \mathbb{R}), \quad H_{d^+}^2 \cong \mathcal{H}^+(X; g).$$

Proof. First,

$$H_{d^+}^0 = \ker(d : \Omega^0(X) \longrightarrow \Omega^1(X)).$$

If $df = 0$, then f is constant on each connected component. Therefore

$$H_{d^+}^0 \cong H^0(X; \mathbb{R}).$$

We now compute $H_{d^+}^1$. The key point is to prove that

$$\ker d^+ = \ker d.$$

Clearly $\ker d \subseteq \ker d^+$. Conversely, let $\alpha \in \Omega^1(X)$ satisfy $d^+\alpha = 0$. Decompose $d\alpha$ as

$$d\alpha = (d\alpha)^+ + (d\alpha)^-.$$

Since $\Omega^+(X)$ and $\Omega^-(X)$ are orthogonal with respect to the L^2 inner product, we have

$$\int_X d\alpha \wedge d\alpha = \|(d\alpha)^+\|_{L^2}^2 - \|(d\alpha)^-\|_{L^2}^2.$$

On the other hand,

$$d\alpha \wedge d\alpha = d(\alpha \wedge d\alpha).$$

Since X is a closed manifold, Stokes' theorem gives

$$\int_X d\alpha \wedge d\alpha = 0.$$

Hence

$$\|(d\alpha)^+\|_{L^2}^2 = \|(d\alpha)^-\|_{L^2}^2.$$

Since $d^+\alpha = (d\alpha)^+ = 0$, it follows that

$$(d\alpha)^- = 0.$$

Therefore $d\alpha = 0$, and thus

$$\ker d^+ = \ker d.$$

Consequently,

$$H_{d^+}^1 = \frac{\ker d^+}{\operatorname{im} d} = \frac{\ker d}{\operatorname{im} d} = H_{\text{dR}}^1(X) \cong H^1(X; \mathbb{R}).$$

Finally we compute

$$H_{d^+}^2 = \frac{\Omega^+(X)}{\operatorname{im} d^+}.$$

We first show that

$$\mathcal{H}^+(X; g) \perp \operatorname{im} d^+.$$

Indeed, take $h \in \mathcal{H}^+(X; g)$ and $\alpha \in \Omega^1(X)$. Since h is self-dual while $(d\alpha)^- \in \Omega^-(X)$, we have

$$\begin{aligned} (h, d^+\alpha)_{L^2} &= (h, (d\alpha)^+)_{L^2} \\ &= (h, d\alpha)_{L^2} \\ &= (d^*h, \alpha)_{L^2} \\ &= 0. \end{aligned}$$

The last equality holds because h is harmonic, so that $d^*h = 0$.

Therefore the natural map

$$\mathcal{H}^+(X; g) \longrightarrow \frac{\Omega^+(X)}{\operatorname{im} d^+}$$

is injective.

We next prove that it is surjective. Take any $\omega^+ \in \Omega^+(X)$. By the Hodge decomposition, it can be written as

$$\omega^+ = h + d\alpha + d^*\gamma,$$

where $h \in \mathcal{H}^2(X; g)$, $\alpha \in \Omega^1(X)$, $\gamma \in \Omega^3(X)$.

In dimension 4, $d^* = - * d *$, so setting

$$\beta := - * \gamma \in \Omega^1(X),$$

we get

$$d^*\gamma = *d\beta.$$

Hence

$$\omega^+ = h + d\alpha + *d\beta.$$

Since ω^+ is self-dual, we have $*\omega^+ = \omega^+$. Thus

$$\omega^+ = *h + *d\alpha + d\beta.$$

By the uniqueness of the Hodge decomposition, the harmonic part, the exact part and the coexact part are respectively equal. Therefore

$$*h = h, \quad d\alpha = d\beta.$$

In particular, $h \in \mathcal{H}^+(X; g)$.

Consequently,

$$\begin{aligned} \omega^+ &= h + d\alpha + *d\alpha \\ &= h + 2(d\alpha)^+ \\ &= h + 2d^+\alpha. \end{aligned}$$

Hence in the quotient space,

$$[\omega^+] = [h].$$

So the natural map

$$\mathcal{H}^+(X; g) \longrightarrow \frac{\Omega^+(X)}{\text{im } d^+}$$

is surjective.

Putting everything together,

$$H_{d^+}^2 = \frac{\Omega^+(X)}{\text{im } d^+} \cong \mathcal{H}^+(X; g).$$

□

The identification of the cokernel with $\mathcal{H}^+(X; g)$ has a concrete application in Proposition 2.4.3: it is exactly the obstruction to prescribing the self-dual curvature of a line-bundle connection.

1.4.6 The self-dual complex and its relation to gauge theory

Above we studied the complex

$$0 \longrightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d^+} \Omega^+(X) \longrightarrow 0.$$

This complex is by no means merely a formal construction within Hodge theory. In 4-dimensional gauge theory it is precisely the basic model that emerges after linearizing the anti-self-dual equations.

Why dimension 4 is particularly natural Let $P \rightarrow X$ be a principal G -bundle and A a connection on it, with curvature denoted by

$$F_A \in \Omega^2(X; \text{ad } P).$$

Since X is a 4-dimensional oriented Riemannian manifold, the Hodge star yields a decomposition

$$\Omega^2(X; \text{ad } P) = \Omega^+(X; \text{ad } P) \oplus \Omega^-(X; \text{ad } P).$$

Thus the curvature likewise decomposes uniquely as

$$F_A = F_A^+ + F_A^-.$$

In Donaldson theory one studies mainly the connections satisfying

$$F_A^+ = 0,$$

which are called **anti-self-dual connections**, or ASD connections for short.

Therefore the projection defined earlier

$$\pi_+ : \Omega^2 \longrightarrow \Omega^+$$

together with the operator

$$d^+ = \pi_+ \circ d$$

are exactly the objects that arise naturally when linearizing the ASD equations.

Linearization of the ASD equations Fix a connection A and consider a small perturbation

$$A + a, \quad a \in \Omega^1(X; \text{ad } P).$$

The curvature satisfies

$$F_{A+a} = F_A + d_A a + a \wedge a.$$

Hence its self-dual part is

$$F_{A+a}^+ = F_A^+ + d_A^+ a + (a \wedge a)^+,$$

where

$$d_A^+ := \pi_+ \circ d_A : \Omega^1(X; \text{ad } P) \longrightarrow \Omega^+(X; \text{ad } P).$$

If A itself satisfies the ASD equation $F_A^+ = 0$, then after discarding the second-order term $(a \wedge a)^+$, the first-order linearization of the ASD equation is

$$d_A^+ a = 0.$$

So, in the case of the trivial connection, this linearized operator is precisely the one studied above,

$$d^+ : \Omega^1(X) \longrightarrow \Omega^+(X).$$

Gauge symmetry and the first differential The connection itself, however, is not gauge invariant. The infinitesimal action of the gauge group is given by

$$d_A : \Omega^0(X; \text{ad } P) \longrightarrow \Omega^1(X; \text{ad } P)$$

That is, if

$$\phi \in \Omega^0(X; \text{ad } P),$$

then $d_A \phi$ represents an infinitesimal change along the gauge orbit.

When A is an ASD connection,

$$d_A^+ d_A \phi = [F_A^+, \phi] = 0.$$

Hence we obtain the deformation complex

$$0 \longrightarrow \Omega^0(X; \text{ad } P) \xrightarrow{d_A} \Omega^1(X; \text{ad } P) \xrightarrow{d_A^+} \Omega^+(X; \text{ad } P) \longrightarrow 0.$$

This is precisely the twisted version of the self-dual complex above.

Geometric meaning of the three cohomology groups Write the cohomology groups of the above complex as

$$H_A^0, \quad H_A^1, \quad H_A^2.$$

They each carry a direct gauge theoretic meaning.

First,

$$H_A^0 = \ker d_A$$

describes the infinitesimal gauge transformations that leave the connection A fixed. Thus H_A^0 is related to the stabilizer of A . For an irreducible connection one usually has

$$H_A^0 = 0.$$

Second,

$$H_A^1 = \frac{\ker d_A^+}{\text{im } d_A}.$$

The numerator $\ker d_A^+$ describes the first-order solutions of the ASD equation, while the denominator $\text{im } d_A$ describes the changes coming purely from gauge transformations. Thus H_A^1 is exactly the formal tangent space to the ASD moduli space at $[A]$.

In other words,

$$T_{[A]} \mathcal{M}_{\text{ASD}} \cong H_A^1$$

holds when the moduli space is smooth and A is irreducible.

Finally,

$$H_A^2 = \frac{\Omega^+(X; \text{ad } P)}{\text{im } d_A^+}$$

is the obstruction space in deformation theory. If

$$H_A^2 = 0,$$

then the ASD equation satisfies the required transversality near A , and in this case the moduli space has the expected smooth structure near $[A]$.

Thus the three groups may be summarized as

$$H_A^0 \sim \text{infinitesimal stabilizer}, \quad H_A^1 \sim \text{deformations}, \quad H_A^2 \sim \text{obstructions}.$$

Gauge fixing and the operator $d_A^* \oplus d_A^+$ To genuinely analyze the ASD equation, we still need to eliminate the gauge symmetry. A standard choice is the Coulomb gauge

$$d_A^* a = 0.$$

Hence the linearized ASD equation and the gauge fixing condition can be combined into a single operator

$$D_A := d_A^* \oplus d_A^+ : \Omega^1(X; \text{ad } P) \longrightarrow \Omega^0(X; \text{ad } P) \oplus \Omega^+(X; \text{ad } P).$$

This operator is elliptic. Consequently its kernel and cokernel are both finite dimensional, and the Fredholm theory of elliptic operators applies.

In the untwisted case, the kernel and cokernel of the corresponding operator

$$D = d^* \oplus d^+ : \Omega^1(X) \longrightarrow \Omega^0(X) \oplus \Omega^+(X)$$

can be described by the cohomology of the self-dual complex computed above.

For connected X , the Euler characteristic of this complex is

$$\chi_{\text{SD}} = 1 - b_1(X) + b_2^+(X),$$

while the Fredholm index of the operator $d^* \oplus d^+$ is

$$\text{ind}(d^* \oplus d^+) = b_1(X) - 1 - b_2^+(X).$$

Thus the quantities appearing earlier,

$$b_1(X) \quad \text{and} \quad b_2^+(X)$$

are not merely topological invariants: they also enter directly into the index of the gauge theoretic deformation operator.

Relation to Donaldson theory The basic object of Donaldson theory is the ASD moduli space

$$\mathcal{M}_{\text{ASD}} = \{A \mid F_A^+ = 0\} / \mathcal{G},$$

where $\mathcal{G}$ is the gauge group.

Near an ASD connection A , the local structure of this nonlinear equation is governed precisely by the deformation complex

$$0 \longrightarrow \Omega^0(\text{ad } P) \xrightarrow{d_A} \Omega^1(\text{ad } P) \xrightarrow{d_A^+} \Omega^+(\text{ad } P) \longrightarrow 0$$

Therefore the complex studied above,

$$0 \longrightarrow \Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d^+} \Omega^+ \longrightarrow 0$$

may be regarded as the most basic Abelian model for the ASD deformation complex in Donaldson theory.

Relation to Seiberg–Witten theory In Seiberg–Witten theory the operator

$$d^* \oplus d^+.$$

likewise appears.

The difference is that, besides a $U(1)$ connection, the Seiberg–Witten equations also involve a spinor. Hence, in addition to

$$d^* \oplus d^+$$

its linearization also involves the Dirac operator

$$D_A : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

Therefore, from the analytic point of view, the two basic elliptic operators that recur throughout 4-dimensional gauge theory are precisely

$$d^* \oplus d^+ \quad \text{and} \quad D_A.$$

The preceding study of the Hodge star, self-dual forms, d^+ and the self-dual complex is precisely a preparation for these gauge theoretic deformation problems.

Connections

2.1 Connections and curvature on principal bundles

The basic geometric objects in gauge theory are principal bundles together with connections on them. Compared with connections on a general vector bundle, the language of principal bundles directly retains the action of the structure group G and of its Lie algebra $\mathfrak{g}$, and is therefore particularly well suited to describing gauge transformations, curvature, and later the anti-self-dual equation.

In this section X is a smooth manifold and G a Lie group, whose Lie algebra we denote by $\mathfrak{g} = T_e G$. All principal bundles are assumed to carry right actions.

2.1.1 Principal bundles

We first recall the definition of a principal bundle.

Definition 2.1.1 (Principal bundle). A **principal bundle** with structure group G (principal G -bundle) is a smooth fiber bundle $\pi : P \rightarrow X$, together with a smooth right action

$$P \times G \longrightarrow P, \quad (p, g) \longmapsto pg,$$

satisfying the following conditions:

1. the right action preserves the fibers, that is, $\pi(pg) = \pi(p)$;
2. for every $x \in X$, G acts freely and transitively on the fiber $P_x := \pi^{-1}(x)$;
3. P is locally equivariantly isomorphic to $U \times G$. More precisely, for any $x \in X$ there exist a neighborhood $U \ni x$ and a diffeomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times G$$

satisfying

$$\Phi(pg) = \Phi(p)g,$$

where G acts on $U \times G$ by $(x, h)g = (x, hg)$.

Freeness in condition 2 means that $pg = p$ can happen only when $g = e$, while transitivity means that for any $p, q \in P_x$ there exists a unique $g \in G$ such that $q = pg$.

Hence every fiber P_x is isomorphic to G , but this isomorphism is not canonical. In fact, once we choose $p \in P_x$, we obtain a diffeomorphism

$$G \longrightarrow P_x, \quad g \longmapsto pg.$$

If we choose a different base point pg_0 , the resulting coordinates change accordingly.

More precisely, therefore, P_x is a G -torsor.

Local sections Local trivializations can also be described by local sections. Let $U \subset X$ be an open set. A local section is a smooth map $s : U \rightarrow P$ satisfying $\pi \circ s = \text{id}_U$.

Once s is chosen, every $p \in P|_U$ can be written uniquely as $p = s(x)g$, where $x = \pi(p)$ and $g \in G$. This gives a local trivialization

$$\begin{aligned} U \times G &\longrightarrow P|_U, \\ (x, g) &\longmapsto s(x)g. \end{aligned}$$

2.1.2 Basic examples of principal bundles

Example 2.1.2 (The trivial principal bundle). For any smooth manifold X and Lie group G , the product $P = X \times G$ is a principal G -bundle.

The projection is $\pi(x, g) = x$, and the right action is $(x, g)h = (x, gh)$.

It has a global section $s(x) = (x, e)$.

The trivial principal bundle is the local model. Although a general principal bundle is locally of the form $U \times G$, a global section need not exist.

In fact, a principal bundle $P \rightarrow X$ is trivial if and only if it admits a global section.

Example 2.1.3 (The frame bundle of a vector bundle). Let $E \rightarrow X$ be a real vector bundle of rank r . For each $x \in X$, set

$$\text{Fr}(E)_x = \{(e_1, \dots, e_r) \mid (e_1, \dots, e_r) \text{ is a basis of } E_x\}.$$

The space $\text{Fr}(E)$ of all frames naturally forms a principal $GL(r, \mathbb{R})$ -bundle.

If we write a frame as a row vector $(e_1, \dots, e_r)$, then the right action of $g \in GL(r, \mathbb{R})$ is defined by

$$(e_1, \dots, e_r)g.$$

Thus a vector bundle can be converted into a principal bundle via its frame bundle.

If E carries a Riemannian metric, we may consider only orthonormal frames, and thereby obtain a principal $O(r)$ -bundle. If E is moreover oriented, we obtain in addition a principal $SO(r)$ -bundle.

Similarly, if E is a Hermitian complex vector bundle of rank r , its unitary frame bundle is a principal $U(r)$ -bundle.

Example 2.1.4 (The Hopf fibration). Consider

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}.$$

$U(1)$ acts on S^3 :

$$(z_1, z_2)\lambda = (z_1\lambda, z_2\lambda), \quad \lambda \in U(1).$$

This action is free, and its quotient space is

$$S^3/U(1) \cong \mathbb{CP}^1 \cong S^2.$$

Hence the Hopf fibration

$$S^1 \longrightarrow S^3 \longrightarrow S^2$$

is a non-trivial principal $U(1)$ -bundle.

This example shows that although a principal bundle can locally be written as $U \times G$, its global topology may be non-trivial.

2.1.3 The vertical tangent space and fundamental vectors

Let $p \in P$, and set $x = \pi(p)$. The fiber P_x is a submanifold of P ; define

$$V_p P := T_p P_x = \ker(d\pi_p).$$

We call $V_p P$ the **vertical tangent space** at p .

It describes the directions of motion along the fibers of the principal bundle.

On the other hand, for any $\xi \in \mathfrak{g}$, the one-parameter subgroup $t \mapsto \exp(t\xi)$ gives the curve $t \mapsto p \exp(t\xi)$.

Define

$$\xi_p^\# := \left. \frac{d}{dt} \right|_{t=0} p \exp(t\xi).$$

We call $\xi_p^\#$ the **fundamental vector** induced by ξ at p .

Since $\pi(p \exp(t\xi)) = \pi(p)$, we have $\xi_p^\# \in V_p P$.

Proposition 2.1.5. For every $p \in P$, the map

$$\begin{aligned} \mathfrak{g} &\longrightarrow V_p P, \\ \xi &\longmapsto \xi_p^\# \end{aligned}$$

is a linear isomorphism.

Proof. Fix $x = \pi(p)$. Since G acts freely and transitively on P_x , the orbit map

$$\Phi_p : G \longrightarrow P_x, \quad g \longmapsto pg$$

is a diffeomorphism.

Hence its differential at the identity, $(d\Phi_p)_e : \mathfrak{g} \rightarrow V_p P$, is a linear isomorphism.

Moreover

$$(d\Phi_p)_e(\xi) = \left. \frac{d}{dt} \right|_{t=0} p \exp(t\xi) = \xi_p^\#.$$

□

Thus every vertical tangent space may be viewed as a copy of $\mathfrak{g}$. However, this isomorphism depends on the choice of the base point p in the fiber.

2.1.4 The adjoint representation and the adjoint bundle

For any $g \in G$, define the conjugation map $C_g(h) = ghg^{-1}$.

Its differential at the identity defines the adjoint representation

$$\text{Ad}_g := (dC_g)_e : \mathfrak{g} \longrightarrow \mathfrak{g}.$$

For a matrix Lie group we have $\text{Ad}_g \xi = g\xi g^{-1}$.

Under the right action the fundamental vectors satisfy

$$(R_g)_* \xi_p^\# = (\text{Ad}_{g^{-1}} \xi)_{pg}^\#.$$

Indeed,

$$\begin{aligned} (R_g)_* \xi_p^\# &= \left. \frac{d}{dt} \right|_{t=0} p \exp(t\xi) g \\ &= \left. \frac{d}{dt} \right|_{t=0} p g \exp(t \operatorname{Ad}_{g^{-1}} \xi). \end{aligned}$$

Hence, when we change the base point along a single fiber, the coordinates in $\mathfrak{g}$ change according to the adjoint representation.

This naturally leads to the associated vector bundle

$$\operatorname{ad} P := P \times_{\operatorname{Ad}} \mathfrak{g}.$$

Concretely,

$$\operatorname{ad} P = (P \times \mathfrak{g}) / \sim,$$

where

$$(p, \xi) \sim (pg, \operatorname{Ad}_{g^{-1}} \xi).$$

The fibers of $\operatorname{ad} P$ are copies of the Lie algebra $\mathfrak{g}$ with no base point chosen.

For $G = U(1)$ the adjoint representation is trivial, so $\operatorname{ad} P \cong X \times i\mathbb{R}$.

For $G = SU(2)$, $\mathfrak{su}(2)$ is a 3-dimensional real vector space, so $\operatorname{ad} P$ is a real vector bundle of rank 3.

2.1.5 Connections on a principal bundle

A connection on a principal bundle can first of all be defined from the point of view of horizontal spaces.

Definition 2.1.6 (Principal connection). A **connection** on a principal G -bundle $P \rightarrow X$ is a smooth distribution

$$p \longmapsto H_p P \subset T_p P,$$

satisfying

$$T_p P = H_p P \oplus V_p P$$

and

$$(R_g)_* H_p P = H_{pg} P.$$

$H_p P$ is called the horizontal subspace at p .

Hence every $v \in T_p P$ decomposes uniquely as $v = v^H + v^V$, where $v^H \in H_p P$ and $v^V \in V_p P$.

The second condition says that the horizontal distribution is compatible with the G -action on the principal bundle.

Since $V_p P = \ker d\pi_p$, the map

$$d\pi_p|_{H_p P} : H_p P \longrightarrow T_{\pi(p)} X$$

is a linear isomorphism.

Therefore, given $v \in T_x X$ and $p \in P_x$, there is a unique horizontal vector $v_p^H \in H_p P$ satisfying $d\pi_p(v_p^H) = v$. We call v_p^H the horizontal lift of v at p .

2.1.6 Horizontal lifts of paths and parallel transport

Let $\gamma : [0, 1] \rightarrow X$ be a smooth path and $p_0 \in P_{\gamma(0)}$.

The connection determines a unique curve $\tilde{\gamma} : [0, 1] \rightarrow P$ satisfying

$$\pi \circ \tilde{\gamma} = \gamma, \quad \tilde{\gamma}(0) = p_0,$$

and $\dot{\tilde{\gamma}}(t) \in H_{\tilde{\gamma}(t)}P$.

We call $\tilde{\gamma}$ the horizontal lift of γ .

This yields the parallel transport

$$\tau_\gamma : P_{\gamma(0)} \longrightarrow P_{\gamma(1)}.$$

Since the horizontal distribution is G -equivariant, parallel transport satisfies $\tau_\gamma(pg) = \tau_\gamma(p)g$.

Thus a principal connection can be understood as a way of comparing points in different fibers.

2.1.7 Connection 1-form

A horizontal distribution can equivalently be encoded by a $\mathfrak{g}$ -valued 1-form.

Definition 2.1.7 (Connection 1-form). The connection 1-form of a connection on a principal bundle is

$$\omega \in \Omega^1(P; \mathfrak{g})$$

satisfying

$$\omega(\xi^\#) = \xi$$

and

$$R_g^* \omega = \text{Ad}_{g^{-1}} \omega.$$

The first condition is called the vertical reproduction property. It says that, in the vertical directions, ω recovers the Lie algebra element generating that direction.

In fact, $\omega_p|_{V_p P} : V_p P \rightarrow \mathfrak{g}$ is a linear isomorphism.

Hence we may define

$$H_p P := \ker \omega_p.$$

Proposition 2.1.8. If ω is a connection 1-form, then

$$T_p P = \ker \omega_p \oplus V_p P$$

and

$$(R_g)_* \ker \omega_p = \ker \omega_{pg}.$$

Proof. Since $\omega_p|_{V_p P}$ is an isomorphism, we have $\ker \omega_p \cap V_p P = \{0\}$. Comparing dimensions gives $T_p P = \ker \omega_p \oplus V_p P$.

If $v \in \ker \omega_p$, then

$$\begin{aligned} \omega_{pg}((R_g)_* v) &= (R_g^* \omega)_p(v) \\ &= \text{Ad}_{g^{-1}} \omega_p(v) = 0. \end{aligned}$$

Hence $(R_g)_* H_p P \subset H_{pg} P$. Since the two have the same dimension, equality must hold. $\square$

Conversely, if a G -equivariant horizontal distribution is given, then every $v \in T_p P$ can be written uniquely as $v = v^H + \xi_p^\#$. Defining $\omega_p(v) = \xi$ yields a connection 1-form.

So a horizontal distribution and a connection 1-form are two equivalent descriptions of one and the same geometric structure.

Remark 2.1.9. The condition $\omega(\xi^\#) = \xi$ alone already guarantees $T_p P = \ker \omega_p \oplus V_p P$.

The role of the condition $R_g^* \omega = \text{Ad}_{g^{-1}} \omega$ is to guarantee that the horizontal spaces satisfy $(R_g)_* H_p P = H_{pg} P$. In other words, it guarantees that the connection genuinely respects the structure of the principal G -bundle.

2.1.8 The local connection form

Take a local section $s : U \rightarrow P$. Define

$$A := s^* \omega \in \Omega^1(U; \mathfrak{g}).$$

A is called the **local connection form** of the connection with respect to the local section s .

The section s gives a local trivialization

$$\Phi_s : U \times G \longrightarrow P|_U, \quad \Phi_s(x, g) = s(x)g.$$

In this trivialization, the connection form satisfies

$$\Phi_s^* \omega = \text{Ad}_{g^{-1}} A + g^{-1} dg.$$

For a matrix Lie group, this can be written as

$$\Phi_s^* \omega = g^{-1} A g + g^{-1} dg.$$

Hence at $(x, g) \in U \times G$, for $(v, \dot{g}) \in T_x U \oplus T_g G$ we have

$$\omega_{(x, g)}(v, \dot{g}) = \text{Ad}_{g^{-1}} A_x(v) + g^{-1} \dot{g}.$$

The local form A is precisely what is usually called the gauge potential in the physics literature.

2.1.9 The local equation for horizontal lifts

Let the path in the base space be $\gamma(t)$, and write its horizontal lift as $\tilde{\gamma}(t) = (\gamma(t), g(t))$.

The horizontality condition is

$$\omega(\dot{\gamma}, \dot{g}) = 0.$$

For a matrix Lie group, this gives

$$g^{-1} A_{\gamma(t)}(\dot{\gamma}(t)) g + g^{-1} \dot{g} = 0.$$

Therefore

$$\dot{g}(t) = -A_{\gamma(t)}(\dot{\gamma}(t))g(t).$$

So the local connection form A is exactly the coefficient in the ordinary differential equation governing parallel transport.

2.1.10 Change of local section

Let $s, s' : U \rightarrow P$ be two local sections. Since they lie in the same fiber at every point, there is a unique smooth map $g : U \rightarrow G$ such that $s' = sg$.

Write

$$A = s^*\omega, \quad A' = (s')^*\omega.$$

Proposition 2.1.10. The two local connection forms satisfy

$$A' = \text{Ad}_{g^{-1}} A + g^{-1}dg.$$

For a matrix Lie group,

$$A' = g^{-1}Ag + g^{-1}dg.$$

This formula shows that A is not a tensor. If A were a tensor, only the homogeneous term $g^{-1}Ag$ would appear in the transformation formula.

The extra term $g^{-1}dg$ is exactly what characterizes a connection.

Example 2.1.11 ($U(1)$). For $G = U(1)$, the adjoint representation is trivial. If we write $g = e^{i\chi}$, then $g^{-1}dg = i d\chi$, hence

$$A' = A + i d\chi.$$

2.1.11 Connections on principal and on vector bundles

Let

$$\rho : G \longrightarrow GL(V)$$

be a finite-dimensional representation.

From the principal bundle P we can construct the associated vector bundle

$$E = P \times_{\rho} V.$$

A principal connection ω automatically induces a connection on the vector bundle E .

If we write $A = s^*\omega$ in a local section s , then the induced connection can be written locally as

$$\nabla = d + \rho_*(A),$$

where $\rho_* : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a Lie algebra representation.

This explains why the language of principal bundles is more fundamental than considering one vector bundle in isolation: the same principal connection simultaneously induces connections on the associated bundles corresponding to all representations of G .

In particular, the difference of two connections on a general vector bundle is an $\text{End}(E)$ -valued 1-form.

Once a reference connection ∇^0 is fixed, every other connection can be written uniquely as $\nabla = \nabla^0 + a$, where $a \in \Omega^1(X; \text{End } E)$.

2.1.12 Connections on duals, tensor products and determinant bundles

The associated-bundle construction is compatible with the usual operations on vector bundles. We record the connection formulas here; their curvature formulas will be used after curvature has been introduced.

Let E and F carry connections ∇^E and ∇^F . The dual connection on E^* is characterized by

$$d(\lambda(s)) = (\nabla^{E^*} \lambda)(s) + \lambda(\nabla^E s),$$

where s is a section of E and λ is a section of E^* . The tensor product connection is given by the Leibniz rule

$$\nabla^{E \otimes F}(s \otimes t) = (\nabla^E s) \otimes t + s \otimes (\nabla^F t).$$

In local frames, if the connection matrices are A_E and A_F , the resulting matrices are

$$A_{E^*} = -A_E^T, \quad A_{E \otimes F} = A_E \otimes I_F + I_E \otimes A_F.$$

Here I_E and I_F denote the identity endomorphisms.

For a rank- r bundle E , the induced connection on $\det E = \Lambda^r E$ acts on a decomposable section by

$$\nabla^{\det E}(s_1 \wedge \cdots \wedge s_r) = \sum_{j=1}^r s_1 \wedge \cdots \wedge \nabla^E s_j \wedge \cdots \wedge s_r.$$

Its local connection form is $\text{tr } A_E$. In particular, if L is a Hermitian line bundle with unitary connection form A , the induced connection on $L^{\otimes m}$ has local form mA . For $m < 0$, this notation uses tensor powers of the dual line bundle.

These formulas also explain how a change of connection propagates. Replacing a connection on a rank- r bundle by $\nabla + aI_E$, where a is a scalar 1-form, changes its determinant connection by ra . For the rank-2 spinor bundles considered later, this is the elementary connection calculation behind the factor $\frac{1}{2}$ relating a change of determinant connection to a change of spinor connection; see Section 3.4.6.

2.1.13 The affine structure of the space of connections

Let ω_0, ω_1 be two connection forms on the principal bundle P , and set $a = \omega_1 - \omega_0$.

For any fundamental vector $\xi^\#$, we have

$$a(\xi^\#) = \omega_1(\xi^\#) - \omega_0(\xi^\#) = 0.$$

Hence a vanishes in the vertical directions.

Moreover,

$$R_g^* a = \text{Ad}_{g^{-1}} a.$$

Thus a is horizontal and Ad -equivariant. Such a form corresponds precisely to an $\text{ad } P$ -valued 1-form on the base space.

Hence the difference of two connections can be regarded as

$$A_1 - A_0 \in \Omega^1(X; \text{ad } P).$$

Once a reference connection A_0 is fixed, all connections form an affine space

$$\mathcal{A}(P) = A_0 + \Omega^1(X; \text{ad } P).$$

Hence its tangent space is naturally $T_A \mathcal{A}(P) \cong \Omega^1(X; \text{ad } P)$.

2.1.14 The geometric origin of curvature

A connection gives a horizontal distribution $H \subset TP$. However, the Lie bracket of two horizontal vector fields is in general no longer horizontal.

Let X, Y be vector fields on the base space, with horizontal lifts denoted by X^H, Y^H respectively.

Although $\omega(X^H) = \omega(Y^H) = 0$, in general

$$\omega([X^H, Y^H]) \neq 0.$$

Hence the vertical part of $[X^H, Y^H]$ precisely measures the extent to which the horizontal distribution fails to be integrable.

This is the geometric origin of curvature.

2.1.15 The curvature form and the structure equation

For $\mathfrak{g}$ -valued 1-forms α, β , define

$$[\alpha \wedge \beta](X, Y) = [\alpha(X), \beta(Y)] - [\alpha(Y), \beta(X)].$$

In particular, $[\alpha \wedge \alpha](X, Y) = 2[\alpha(X), \alpha(Y)]$.

Definition 2.1.12 (Curvature). The **curvature form** of a connection 1-form ω is defined by

$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega] \in \Omega^2(P; \mathfrak{g}).$$

This formula is called the Cartan structure equation.

If X^H, Y^H are horizontal vector fields, then

$$[\omega \wedge \omega](X^H, Y^H) = 0.$$

And

$$\begin{aligned} d\omega(X^H, Y^H) &= X^H(\omega(Y^H)) - Y^H(\omega(X^H)) - \omega([X^H, Y^H]) \\ &= -\omega([X^H, Y^H]). \end{aligned}$$

Hence

$$\Omega(X^H, Y^H) = -\omega([X^H, Y^H]).$$

So the curvature is precisely a measure of the non-integrability of the horizontal distribution.

If $\Omega = 0$, then the Lie bracket of horizontal vector fields is still horizontal. By the Frobenius theorem, the horizontal distribution is locally integrable. In this case the connection is called a **flat connection**.

2.1.16 Curvature is a tensorial form

The curvature form has two important properties.

Proposition 2.1.13. The curvature form Ω satisfies:

1. Ω is horizontal;
- 2.

$$R_g^* \Omega = \text{Ad}_{g^{-1}} \Omega.$$

Hence Ω is a horizontal and Ad -equivariant $\mathfrak{g}$ -valued 2-form.

Such a form corresponds naturally to an $\text{ad } P$ -valued 2-form on the base space.

Hence the curvature of a principal connection is usually written as

$$F_A \in \Omega^2(X; \text{ad } P).$$

Here A denotes the corresponding connection.

2.1.17 Local expression for the curvature

Take a local section $s : U \rightarrow P$ and set $A = s^* \omega$.

The local curvature is defined by $F_A = s^* \Omega$. By the structure equation,

$$F_A = dA + \frac{1}{2}[A \wedge A].$$

For a matrix Lie group, one usually combines matrix multiplication with the wedge product, and writes

$$F_A = dA + A \wedge A.$$

These two formulas differ only in notational convention.

Take local coordinates $(x^1, \dots, x^n)$, and suppose that $A = \sum_i A_i dx^i$; then

$$F_A = \sum_{i < j} (\partial_i A_j - \partial_j A_i + [A_i, A_j]) dx^i \wedge dx^j.$$

Hence if we write $F_A = \sum_{i < j} F_{ij} dx^i \wedge dx^j$, then

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

The last commutator term is one of the basic differences between non-Abelian gauge theory and Abelian gauge theory.

Curvature of the induced connections The operations in Section 2.1.12 also commute with taking curvature. If F_E and F_F denote the curvatures of the connections on E and F , then

$$F_{E^*} = -F_E^T, \quad F_{E \otimes F} = F_E \otimes I_F + I_E \otimes F_F.$$

Indeed, squaring the tensor product covariant derivative cancels the mixed terms by the graded Leibniz rule. For the determinant connection, the local structure equation gives

$$F_{\det E} = d(\text{tr } A_E) = \text{tr}(dA_E + A_E \wedge A_E) = \text{tr } F_E,$$

since the trace of the matrix commutators in $A_E \wedge A_E$ vanishes. In particular, for a Hermitian line bundle L ,

$$F_{L^*} = -F_L, \quad F_{L^{\otimes m}} = mF_L.$$

The same formulas can thus be read either from the representations defining the induced bundles or directly from their local connections.

2.1.18 Basic examples of curvature

Example 2.1.14 ($U(1)$). Since $\mathfrak{u}(1) = i\mathbb{R}$ is an Abelian Lie algebra, the Lie bracket is identically zero.

Hence for a $U(1)$ connection,

$$F_A = dA.$$

If $A = ia$, where a is a real 1-form, then $F_A = i da$.

Example 2.1.15 ($SU(2)$). For

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid X^* = -X, \operatorname{tr} X = 0\},$$

the Lie bracket is the matrix commutator.

Hence

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

In general $[A_i, A_j] \neq 0$, so the curvature is non-linear in A .

2.1.19 Transformation of the curvature under a change of local section

If $s' = sg$, then the local connection forms satisfy

$$A' = g^{-1}Ag + g^{-1}dg.$$

But the transformation formula for the curvature has no derivative term.

Proposition 2.1.16. In the above notation,

$$F_{A'} = g^{-1}F_Ag.$$

Hence the curvature transforms homogeneously under the adjoint representation.

This shows that the local F_A can be glued into a global $\operatorname{ad} P$ -valued 2-form, whereas the local connection forms A themselves cannot be glued together in this way.

Hence the difference between a connection and the curvature can be summarized as: a connection is not a tensor, whereas the curvature is an $\operatorname{ad} P$ -valued tensor.

2.1.20 Covariant exterior derivative

A principal connection induces a vector bundle connection on $\operatorname{ad} P$, and thereby defines the covariant exterior derivative

$$d_A : \Omega^k(X; \operatorname{ad} P) \longrightarrow \Omega^{k+1}(X; \operatorname{ad} P).$$

In a local trivialization,

$$d_A \alpha = d\alpha + [A \wedge \alpha].$$

In particular, for $\xi \in \Omega^0(X; \operatorname{ad} P)$, we have

$$d_A \xi = d\xi + [A, \xi].$$

The ordinary exterior derivative satisfies $d^2 = 0$, whereas d_A in general does not satisfy this property.

In fact,

$$d_A^2 \alpha = [F_A \wedge \alpha].$$

In particular, $d_A^2 \xi = [F_A, \xi]$.

Hence the curvature can also be understood as the obstruction to the covariant exterior derivative having square zero.

When $F_A = 0$, we have $d_A^2 = 0$, and then

$$0 \longrightarrow \Omega^0(\text{ad } P) \xrightarrow{d_A} \Omega^1(\text{ad } P) \xrightarrow{d_A} \Omega^2(\text{ad } P) \xrightarrow{d_A} \dots$$

really becomes a cochain complex.

2.1.21 Bianchi identity

The curvature also satisfies a basic identity.

Proposition 2.1.17 (Bianchi identity). For any connection A , we have

$$d_A F_A = 0.$$

This identity can be deduced from the local structure equation $F_A = dA + A \wedge A$, together with $d^2 = 0$ and the Jacobi identity for the Lie bracket.

The Bianchi identity is very important in Yang–Mills theory later on.

2.1.22 Variation of the curvature under a perturbation of the connection

Fix a connection A , and take $a \in \Omega^1(X; \text{ad } P)$.

Since the space of connections is an affine space, $A + a$ is still a connection.

Using the structure equation directly, we obtain

$$F_{A+a} = F_A + d_A a + a \wedge a$$

which holds under the convention that the wedge product is matrix valued.

If instead one uses the bracket-wedge convention, the last term is written as $\frac{1}{2}[a \wedge a]$.

Hence the curvature map

$$\mathcal{F} : \mathcal{A}(P) \longrightarrow \Omega^2(X; \text{ad } P), \quad A \longmapsto F_A$$

linearizes at A to

$$D\mathcal{F}_A(a) = d_A a.$$

This formula is the basis for the linearization of the ASD equation later on.

2.1.23 The 4-dimensional case and the ASD equation

Now suppose that X is an oriented Riemannian 4-manifold.

The Hodge star gives a decomposition

$$\Omega^2(X; \text{ad } P) = \Omega^+(X; \text{ad } P) \oplus \Omega^-(X; \text{ad } P).$$

Thus the curvature can be written uniquely as

$$F_A = F_A^+ + F_A^-,$$

where

$$F_A^\pm = \frac{1}{2}(F_A \pm *F_A).$$

Definition 2.1.18 (Anti-self-dual connection). If a connection A satisfies

$$F_A^+ = 0,$$

then A is called an **anti-self-dual connection**, or ASD connection for short.

Equivalently, the ASD condition can be written as $*F_A = -F_A$.

Consider a small perturbation $A + a$ of an ASD connection A . By the curvature variation formula,

$$F_{A+a}^+ = F_A^+ + d_A^+ a + (a \wedge a)^+,$$

where

$$d_A^+ := \pi_+ \circ d_A : \Omega^1(X; \text{ad } P) \longrightarrow \Omega^+(X; \text{ad } P).$$

If A itself satisfies $F_A^+ = 0$, then

$$F_{A+a}^+ = d_A^+ a + (a \wedge a)^+.$$

Ignoring the terms of second order in a , the linearization of the ASD equation at A is

$$d_A^+ a = 0.$$

This is precisely the source of the second differential in the ASD deformation complex

$$0 \longrightarrow \Omega^0(X; \text{ad } P) \xrightarrow{d_A} \Omega^1(X; \text{ad } P) \xrightarrow{d_A^+} \Omega^+(X; \text{ad } P) \longrightarrow 0$$

that appears later.

2.2 Chern–Weil theory

We have seen above that a connection A on a principal G -bundle $\pi : P \rightarrow X$ has curvature

$$F_A \in \Omega^2(X; \text{ad } P).$$

The curvature is a differential-geometric object depending on the connection. On the other hand, a principal bundle and its associated vector bundle have Chern classes, Pontryagin classes, and other characteristic classes depending only on the topological type of the bundle.

Chern–Weil theory builds a bridge between these two kinds of objects:

Applying a suitable Lie algebra invariant polynomial to the curvature produces closed differential forms; these forms themselves depend on the connection, but the de Rham cohomology classes they represent are independent of the connection.

Thus Chern–Weil theory takes

$$\text{curvature} \longrightarrow \text{characteristic classes}.$$

This is precisely the basic mechanism in gauge theory that links differential geometry with topology.

2.2.1 Ad-invariant polynomials

Let G be a Lie group with Lie algebra $\mathfrak{g}$.

A symmetric k -multilinear function of degree k

$$p : \mathfrak{g}^k \longrightarrow \mathbb{R}$$

is called **Ad-invariant** if for every $g \in G$ we have

$$p(\text{Ad}_g \xi_1, \dots, \text{Ad}_g \xi_k) = p(\xi_1, \dots, \xi_k).$$

Denote by

$$I^k(\mathfrak{g}) = (S^k \mathfrak{g}^*)^G$$

the space of all Ad-invariant symmetric polynomials of degree k .

For a matrix Lie group, the most basic invariant polynomials come from the trace.

For example,

$$p_k(\xi) = \text{tr}(\xi^k)$$

is invariant under conjugation, because

$$\text{tr}((g\xi g^{-1})^k) = \text{tr}(g\xi^k g^{-1}) = \text{tr}(\xi^k).$$

This is precisely why the trace appears in Chern classes and in the Chern character.

2.2.2 Evaluating an invariant polynomial on the curvature

Let $p \in I^k(\mathfrak{g})$. Since

$$F_A \in \Omega^2(X; \text{ad } P),$$

we can apply p to k copies of the curvature, obtaining an ordinary $2k$ -form

$$p(F_A, \dots, F_A) \in \Omega^{2k}(X).$$

To see this construction clearly, take locally a basis $e_1, \dots, e_N$ of $\mathfrak{g}$, and write

$$F_A = \sum_{\alpha=1}^N F^\alpha e_\alpha, \quad F^\alpha \in \Omega^2(X).$$

Then

$$p(F_A, \dots, F_A) = \sum_{\alpha_1, \dots, \alpha_k} p(e_{\alpha_1}, \dots, e_{\alpha_k}) F^{\alpha_1} \wedge \dots \wedge F^{\alpha_k}.$$

Here each F^α is a 2-form, so permuting the different curvature factors produces no sign.

The key question is: although the formula above is written in a local trivialization, why does it define a global differential form?

When we change the trivialization or the gauge,

$$F_A \longmapsto \text{Ad}_{g^{-1}} F_A.$$

Since p is Ad-invariant,

$$p(\text{Ad}_{g^{-1}} F_A, \dots, \text{Ad}_{g^{-1}} F_A) = p(F_A, \dots, F_A).$$

Hence these local expressions can be glued into a global $2k$ -form on X .

2.2.3 Chern–Weil forms are closed

The first basic property of the Chern–Weil construction is: the differential form obtained from an invariant polynomial is automatically closed.

Proposition 2.2.1. Let $p \in I^k(\mathfrak{g})$. Then

$$dp(F_A, \dots, F_A) = 0.$$

Proof. For an Ad-invariant polynomial, the ordinary exterior derivative and the covariant derivative satisfy

$$dp(\alpha_1, \dots, \alpha_k) = \sum_j (\pm) p(\alpha_1, \dots, d_A \alpha_j, \dots, \alpha_k).$$

In the present situation all the curvature factors are forms of even degree, hence

$$dp(F_A, \dots, F_A) = k p(d_A F_A, F_A, \dots, F_A).$$

By the Bianchi identity 2.1.17,

$$d_A F_A = 0.$$

Therefore

$$dp(F_A, \dots, F_A) = 0.$$

□

Thus $p(F_A, \dots, F_A)$ defines a de Rham cohomology class

$$[p(F_A, \dots, F_A)] \in H_{\text{dR}}^{2k}(X).$$

Here the Bianchi identity is at the heart of the whole construction: it lifts the curvature polynomial from an ordinary differential form to a cohomology class.

2.2.4 Why the cohomology class is independent of the connection

Although the differential form $p(F_A, \dots, F_A)$ itself depends on the connection A , its cohomology class does not depend on A .

This is the most important property of Chern–Weil theory.

Theorem 2.2.2 (Chern–Weil). Let $P \rightarrow X$ be a principal G -bundle, $p \in I^k(\mathfrak{g})$. For any connection A , the form $p(F_A, \dots, F_A)$ is closed, and its de Rham cohomology class

$$[p(F_A, \dots, F_A)] \in H_{\text{dR}}^{2k}(X)$$

is independent of the choice of connection A .

In what follows we explain the mechanism behind connection-independence.

Take two connections A_0, A_1 . Since the space of connections is an affine space, we can write

$$A_1 = A_0 + a, \quad a \in \Omega^1(X; \text{ad } P).$$

Consider the one-parameter family of connections

$$A_t = A_0 + ta, \quad 0 \leq t \leq 1,$$

and write $F_t := F_{A_t}$.

By the curvature variation formula,

$$\frac{d}{dt} F_t = d_{A_t} a.$$

Thus

$$\frac{d}{dt} p(F_t, \dots, F_t) = k p(d_{A_t} a, F_t, \dots, F_t).$$

Using Ad-invariance together with the Bianchi identity $d_{A_t} F_t = 0$, we have

$$p(d_{A_t} a, F_t, \dots, F_t) = d p(a, F_t, \dots, F_t).$$

Therefore

$$\frac{d}{dt} p(F_t, \dots, F_t) = k d p(a, F_t, \dots, F_t).$$

Integrating from $t = 0$ to $t = 1$, we obtain

$$\begin{aligned} p(F_{A_1}, \dots, F_{A_1}) - p(F_{A_0}, \dots, F_{A_0}) \\ = d \left(k \int_0^1 p(a, F_t, \dots, F_t) dt \right). \end{aligned}$$

So the difference of the two characteristic forms is an exact form, and hence

$$[p(F_{A_1})] = [p(F_{A_0})].$$

The $(2k - 1)$ -form in brackets,

$$k \int_0^1 p(a, F_t, \dots, F_t) dt$$

is called the corresponding **transgression form**.

2.2.5 Chern–Weil homomorphism

Hence for every principal G -bundle $P \rightarrow X$ there is a natural map

$$\text{CW}_P : I^k(\mathfrak{g}) \longrightarrow H_{\text{dR}}^{2k}(X),$$

defined by

$$\text{CW}_P(p) = [p(F_A, \dots, F_A)].$$

The Chern–Weil theorem guarantees that the right-hand side is independent of the connection A .

Putting all degrees together, we obtain a graded ring homomorphism

$$\text{CW}_P : I^\bullet(\mathfrak{g}) \longrightarrow H_{\text{dR}}^{2\bullet}(X).$$

So the basic structure of Chern–Weil theory may be summarized as

$$\text{Ad-invariant polynomial on } \mathfrak{g} \longmapsto \text{characteristic cohomology class on } X.$$

2.2.6 Naturality and gauge invariance

The Chern–Weil construction is natural.

Let $f : Y \rightarrow X$ be a smooth map. For the pullback bundle $f^*P \rightarrow Y$, the connection A induces a pullback connection f^*A , and we have

$$F_{f^*A} = f^*F_A.$$

Therefore

$$p(F_{f^*A}) = f^*p(F_A),$$

and hence

$$\text{CW}_{f^*P}(p) = f^* \text{CW}_P(p).$$

On the other hand, a gauge transformation satisfies

$$F_{A^g} = \text{Ad}_{g^{-1}} F_A.$$

So by the Ad-invariance of p ,

$$p(F_{A^g}) = p(F_A).$$

Thus the Chern–Weil form itself is already gauge invariant, and not merely its cohomology class.

2.2.7 Hermitian vector bundles and Chern classes

Now let $E \rightarrow X$ be a rank- r Hermitian vector bundle and A a unitary connection.

In this case

$$F_A \in \Omega^2(X; \mathfrak{u}(E)),$$

so in a unitary frame F_A is a skew-Hermitian matrix-valued 2-form.

Define the total Chern form

$$c(A) := \det \left(I + \frac{i}{2\pi} F_A \right).$$

Expanding it according to differential-form degree,

$$c(A) = 1 + c_1(A) + c_2(A) + \cdots + c_r(A),$$

where $c_j(A)$ is a $2j$ -form.

The Chern–Weil theorem shows that each $c_j(A)$ is closed and that its de Rham class is independent of A .

This cohomology class is precisely the image of the integral Chern class

$$c_j(E) \in H^{2j}(X; \mathbb{Z})$$

under the coefficient change $H^{2j}(X; \mathbb{Z}) \rightarrow H^{2j}(X; \mathbb{R})$ is the image of $c_j(E)$.

In particular,

$$c_1(A) = \frac{i}{2\pi} \text{tr } F_A.$$

And the second Chern form is

$$c_2(A) = \frac{1}{2} \left(\frac{i}{2\pi} \right)^2 [(\operatorname{tr} F_A) \wedge (\operatorname{tr} F_A) - \operatorname{tr}(F_A \wedge F_A)].$$

Therefore

$$[c_1(A)] = c_1(E)_{\mathbb{R}}, \quad [c_2(A)] = c_2(E)_{\mathbb{R}}.$$

Remark 2.2.3. Chern–Weil theory itself first produces a real de Rham cohomology class. “These classes actually come from integral cohomology” is a stronger topological conclusion in characteristic class theory, and cannot be read off from a differential-form computation alone.

2.2.8 The example of a complex line bundle

Let $L \rightarrow X$ be a Hermitian line bundle. Its structure group is $U(1)$, so the curvature is an $i\mathbb{R}$ -valued 2-form.

Since the rank is 1, the first Chern form is

$$c_1(A) = \frac{i}{2\pi} F_A.$$

Therefore

$$c_1(L)_{\mathbb{R}} = \left[\frac{i}{2\pi} F_A \right].$$

If $\Sigma \subset X$ is a closed oriented surface, then

$$\frac{i}{2\pi} \int_{\Sigma} F_A = \langle c_1(L), [\Sigma] \rangle \in \mathbb{Z}.$$

So the integral of the curvature is not an arbitrary real number; it is constrained by an integrality condition coming from the topological type of the bundle.

This may be regarded as the simplest instance of **curvature quantization** in gauge theory.

2.2.9 $SU(r)$ -bundles and the second Chern class

Now let $P \rightarrow X$ be a principal $SU(r)$ -bundle, and take the standard representation

$$SU(r) \longrightarrow U(r).$$

We obtain an associated Hermitian vector bundle

$$E = P \times_{SU(r)} \mathbb{C}^r.$$

Since

$$F_A \in \mathfrak{su}(r),$$

we have

$$\operatorname{tr} F_A = 0.$$

Therefore

$$c_1(E) = 0.$$

The second Chern form simplifies to

$$c_2(A) = \frac{1}{8\pi^2} \operatorname{tr}(F_A \wedge F_A),$$

where we use the convention of these notes: connections and curvatures take values in skew-Hermitian matrices, and the total Chern form is defined as

$$\det \left(I + \frac{i}{2\pi} F_A \right).$$

Thus

$$c_2(E)_{\mathbb{R}} = \left[\frac{1}{8\pi^2} \operatorname{tr}(F_A \wedge F_A) \right].$$

In particular, if X is a closed oriented 4-manifold, then

$$c_2(E)[X] = \frac{1}{8\pi^2} \int_X \operatorname{tr}(F_A \wedge F_A).$$

Although the right-hand side is computed by means of a connection A , the Chern–Weil theorem guarantees that it depends only on the topological type of the bundle E .

2.2.10 Chern character

Another commonly used family of Chern–Weil classes is the Chern character.

Define the differential form

$$\operatorname{ch}(A) = \operatorname{tr} \exp \left(\frac{i}{2\pi} F_A \right).$$

Expanding the exponential function gives

$$\operatorname{ch}(A) = r + \frac{i}{2\pi} \operatorname{tr} F_A + \frac{1}{2!} \operatorname{tr} \left(\frac{i}{2\pi} F_A \right)^2 + \cdots.$$

Its de Rham class is independent of the connection, and is denoted by

$$\operatorname{ch}(E).$$

The low-degree terms satisfy

$$\operatorname{ch}(E) = r + c_1(E) + \frac{1}{2} (c_1(E)^2 - 2c_2(E)) + \cdots.$$

An important advantage of the Chern character is that it has simple algebraic properties with respect to direct sum and tensor product:

$$\operatorname{ch}(E \oplus F) = \operatorname{ch}(E) + \operatorname{ch}(F),$$

and

$$\operatorname{ch}(E \otimes F) = \operatorname{ch}(E) \operatorname{ch}(F).$$

2.2.11 Pontryagin classes

Chern–Weil theory applies equally well to real vector bundles.

Let $E \rightarrow X$ be a real rank- n vector bundle with a metric, and A an orthogonal connection. Then

$$F_A \in \Omega^2(X; \mathfrak{so}(E)).$$

The first Pontryagin form can be written as

$$p_1(A) = -\frac{1}{8\pi^2} \operatorname{tr}(F_A \wedge F_A),$$

where the trace is taken in the defining real representation.

It is closed, and

$$[p_1(A)] = p_1(E)_{\mathbb{R}}.$$

Equivalently, one can define it by complexification:

$$p_1(E) = -c_2(E \otimes \mathbb{C}).$$

In particular, for an oriented Riemannian 4-manifold, applying the Levi–Civita connection to TX yields a differential-form representative of $p_1(TX)$.

Remark 2.2.4. The Hirzebruch signature theorem further gives

$$\langle p_1(TX), [X] \rangle = 3\sigma(X).$$

This equality is not obtained from Chern–Weil theory alone; Chern–Weil is responsible only for expressing $p_1(TX)$ in terms of the curvature, while the signature theorem further relates it to the signature of the intersection form.

2.2.12 Chern–Weil formulas in $SU(2)$ gauge theory

We now return to the most important case in four-dimensional gauge theory.

Let X be a closed oriented Riemannian 4-manifold and $P \rightarrow X$ a principal $SU(2)$ -bundle. Set

$$E = P \times_{SU(2)} \mathbb{C}^2$$

to be the standard associated rank-2 complex vector bundle, and write

$$k := c_2(E)[X] \in \mathbb{Z}.$$

We use on $\mathfrak{su}(2)$ the inner product

$$\langle \xi, \eta \rangle = -\operatorname{tr}(\xi\eta).$$

Then for $F_A \in \Omega^2(X; \operatorname{ad} P)$, we have

$$-\operatorname{tr}(F_A \wedge *F_A) = |F_A|^2 d\operatorname{vol}_g.$$

On the other hand, a Chern–Weil formula gives

$$8\pi^2 k = \int_X \operatorname{tr}(F_A \wedge F_A).$$

Using the Hodge decomposition

$$F_A = F_A^+ + F_A^-,$$

where $*F_A^\pm = \pm F_A^\pm$.

Since self-dual forms are orthogonal to anti-self-dual forms, the cross terms vanish, and

$$\mathrm{tr}(F_A^+ \wedge F_A^+) = -|F_A^+|^2 d\mathrm{vol}_g,$$

while

$$\mathrm{tr}(F_A^- \wedge F_A^-) = |F_A^-|^2 d\mathrm{vol}_g.$$

Hence we obtain the fundamental identity in four-dimensional gauge theory

$$8\pi^2 k = \int_X (|F_A^-|^2 - |F_A^+|^2) d\mathrm{vol}_g.$$

2.2.13 The topological lower bound for the Yang–Mills energy

The Yang–Mills energy is defined as

$$\mathrm{YM}(A) := \int_X |F_A|^2 d\mathrm{vol}_g.$$

By the self-dual decomposition,

$$\mathrm{YM}(A) = \|F_A^+\|_{L^2}^2 + \|F_A^-\|_{L^2}^2.$$

On the other hand,

$$8\pi^2 k = \|F_A^-\|_{L^2}^2 - \|F_A^+\|_{L^2}^2.$$

Eliminating $\|F_A^-\|_{L^2}^2$, we obtain

$$\mathrm{YM}(A) = 8\pi^2 k + 2\|F_A^+\|_{L^2}^2.$$

Thus if $k \geq 0$,

$$\mathrm{YM}(A) \geq 8\pi^2 k,$$

and equality holds if and only if

$$F_A^+ = 0.$$

That is, within the fixed topological class $c_2(E)[X] = k$, the ASD connections are precisely the connections at which the Yang–Mills functional attains its absolute minimum.

In particular, if A is ASD,

$$\mathrm{YM}(A) = \int_X |F_A|^2 d\mathrm{vol}_g = 8\pi^2 k.$$

Hence all ASD connections on a fixed bundle have exactly the same energy.

2.2.14 Why this matters for Uhlenbeck theory

We can now explain where the “fixed total energy” that will be used repeatedly in the Uhlenbeck compactness argument below comes from.

If

$$[A_n] \in \mathcal{M}_k$$

is a sequence of charge- k ASD connections, then Chern–Weil theory automatically gives

$$\int_X |F_{A_n}|^2 d\text{vol}_g = 8\pi^2 k$$

for all n .

Hence this sequence of connections has a uniform gauge-invariant L^2 curvature bound.

When compactness fails, energy can concentrate only in ever smaller regions. After rescaling, a bubble ultimately yields a finite-energy ASD connection on S^4 .

Applying Chern–Weil theory once more, the energy of this bubble must be

$$8\pi^2 m, \quad m \in \mathbb{Z}_{>0}.$$

So the energy quantization appearing in the Uhlenbeck compactification below (Theorem 2.6.5) is not an accidental analytic phenomenon; its integrality ultimately comes from the second Chern class.

2.2.15 Summary of this section

The core chain of Chern–Weil theory is

$$p \in I^k(\mathfrak{g}) \quad \longmapsto \quad p(F_A, \dots, F_A) \quad \longmapsto \quad [p(F_A, \dots, F_A)] \in H_{\text{dR}}^{2k}(X).$$

The first step uses the curvature, the second uses the Bianchi identity, and the Chern–Weil theorem guarantees that the resulting cohomology class is independent of the choice of connection.

For a complex vector bundle, this construction yields the Chern classes; for a real vector bundle, it yields the Pontryagin classes.

In four-dimensional $SU(2)$ gauge theory, the most important special case is

$$c_2(E)[X] = \frac{1}{8\pi^2} \int_X \text{tr}(F_A \wedge F_A),$$

together with the consequence

$$8\pi^2 k = \|F_A^-\|_{L^2}^2 - \|F_A^+\|_{L^2}^2.$$

Thus ASD connections satisfy

$$\text{YM}(A) = 8\pi^2 k.$$

This provides precisely the topological basis for the later study of the instanton moduli space, Uhlenbeck compactness and bubbling.

2.3 The gauge group and its action on the space of connections

We have seen above that a connection on a principal G -bundle $P \rightarrow X$ can be viewed as a G -equivariant horizontal distribution, and equivalently as a connection 1-form $\omega \in \Omega^1(P; \mathfrak{g})$ satisfying suitable conditions.

Next we consider the symmetries of the principal bundle itself. What gauge theory calls gauge symmetry is essentially a principal bundle automorphism that leaves the base space fixed but allows the reference point inside each fiber to be changed according to the structure group G .

Throughout this section we continue to assume that $P \rightarrow X$ is a principal G -bundle and to use the right action convention.

2.3.1 Gauge transformations and the gauge group

Definition 2.3.1 (gauge transformation). A gauge transformation of a principal G -bundle $P \rightarrow X$ is a smooth map

$$\Phi : P \longrightarrow P$$

satisfying

$$\pi \circ \Phi = \pi$$

and

$$\Phi(pg) = \Phi(p)g$$

for any $p \in P$ and $g \in G$.

The first condition says that Φ covers the identity map id_X on the base space, so Φ does not move points of the base space.

The second condition says that Φ is compatible with the right G -action on the principal bundle.

Hence a gauge transformation is a principal bundle automorphism covering id_X .

All gauge transformations form a group under composition.

Definition 2.3.2 (gauge group). Define

$$\mathcal{G}(P) := \text{Aut}_G(P)$$

to be the **gauge group** of the principal bundle P .

Here $\text{Aut}_G(P)$ denotes the principal bundle automorphisms that cover id_X and are compatible with the right G -action.

2.3.2 The equivariant-function description of gauge transformations

A gauge transformation can also be described by a G -valued function defined on the total space P .

Let $\Phi \in \mathcal{G}(P)$. Since $\pi(\Phi(p)) = \pi(p)$, the point $\Phi(p)$ and p lie in the same fiber. Since G acts freely and transitively on each fiber, there is a unique $u(p) \in G$ such that

$$\Phi(p) = p u(p).$$

Hence Φ uniquely determines a smooth map $u : P \rightarrow G$.

Since Φ is compatible with the right action,

$$\Phi(pg) = \Phi(p)g.$$

On the other hand,

$$\Phi(pg) = pg u(pg),$$

while

$$\Phi(p)g = p u(p)g.$$

From the freeness of the right action we obtain

$$g u(pg) = u(p)g.$$

Therefore

$$u(pg) = g^{-1}u(p)g.$$

So the function $u : P \rightarrow G$ corresponding to a gauge transformation must satisfy the conjugate equivariance condition.

Proposition 2.3.3. The gauge group $\mathcal{G}(P)$ is naturally isomorphic to the set of all smooth maps $u : P \rightarrow G$ satisfying

$$u(pg) = g^{-1}u(p)g.$$

Conversely, if $u : P \rightarrow G$ satisfies the above condition, define

$$\Phi_u(p) := p u(p).$$

Then

$$\begin{aligned} \Phi_u(pg) &= pg u(pg) \\ &= pg g^{-1}u(p)g \\ &= p u(p)g \\ &= \Phi_u(p)g, \end{aligned}$$

so Φ_u is indeed a gauge transformation.

2.3.3 The group bundle $\text{Ad } P$

We have already defined the Lie algebra bundle

$$\text{ad } P = P \times_{\text{Ad}} \mathfrak{g}.$$

To describe the gauge group we also need an associated bundle whose fiber is the group G itself.

Definition 2.3.4. Define

$$\mathrm{Ad} P := P \times_{\mathrm{conj}} G,$$

where G acts on itself by conjugation.

More concretely,

$$\mathrm{Ad} P = (P \times G) / \sim,$$

where

$$(p, h) \sim (pg, g^{-1}hg).$$

We need to distinguish two notations:

$$\mathrm{Ad} P \quad \text{and} \quad \mathrm{ad} P.$$

The fiber of the former is the Lie group G , so it is a group bundle; the fiber of the latter is the Lie algebra $\mathfrak{g}$, so it is a vector bundle.

A map $u : P \rightarrow G$ satisfying

$$u(pg) = g^{-1}u(p)g$$

corresponds exactly to a smooth section of $\mathrm{Ad} P$.

Hence there is a natural isomorphism

$$\mathcal{G}(P) \cong \Gamma(\mathrm{Ad} P).$$

Correspondingly, the Lie algebra of the gauge group is

$$\mathrm{Lie} \mathcal{G}(P) \cong \Gamma(\mathrm{ad} P) = \Omega^0(X; \mathrm{ad} P).$$

2.3.4 Basic examples of the gauge group

Example 2.3.5 (The trivial principal bundle). Let

$$P = X \times G.$$

Take the standard section $s(x) = (x, e)$.

A gauge transformation is completely determined by a smooth map $g : X \rightarrow G$, and in the trivialization it acts by

$$\Phi_g(x, h) = (x, g(x)h).$$

Hence

$$\mathcal{G}(X \times G) \cong C^\infty(X, G).$$

Note that a gauge transformation acts on the second factor by *left multiplication* by $g(x)$. This is forced by the requirement that it commute with the right action of the principal bundle:

$$\Phi_g((x, h)k) = (x, g(x)hk) = \Phi_g(x, h)k.$$

Example 2.3.6 ($U(1)$ -principal bundles). Since $U(1)$ is an Abelian group, the conjugation action is trivial. Hence for any principal $U(1)$ -bundle $P \rightarrow X$, even when P itself is non-trivial, we still have

$$\text{Ad } P \cong X \times U(1).$$

Therefore

$$\mathcal{G}(P) \cong C^\infty(X, U(1)).$$

This is a special simplification of Abelian gauge theory.

Example 2.3.7 ($SU(2)$). For a principal $SU(2)$ -bundle $P \rightarrow X$, the gauge group in general cannot simply be written as $C^\infty(X, SU(2))$, unless P is trivial.

In general one must write

$$\mathcal{G}(P) = \Gamma(\text{Ad } P).$$

Its Lie algebra is

$$\Omega^0(X; \text{ad } P).$$

2.3.5 The action of the gauge group on connections

Let $\Phi \in \mathcal{G}(P)$, and let $\omega \in \Omega^1(P; \mathfrak{g})$ be a connection 1-form.

Define

$$\omega^\Phi := \Phi^* \omega.$$

First we need to verify that $\Phi^* \omega$ is still a connection form.

Proposition 2.3.8. If $\Phi \in \mathcal{G}(P)$ and ω is a connection on a principal bundle, then $\Phi^* \omega$ is again a connection on a principal bundle.

Proof. Since Φ is compatible with the right action,

$$\Phi \circ R_g = R_g \circ \Phi.$$

Hence

$$\begin{aligned} R_g^*(\Phi^* \omega) &= \Phi^*(R_g^* \omega) \\ &= \Phi^*(\text{Ad}_{g^{-1}} \omega) \\ &= \text{Ad}_{g^{-1}}(\Phi^* \omega). \end{aligned}$$

On the other hand, since Φ preserves the right action of the principal bundle, it sends the fundamental vector $\xi^\#$ to the fundamental vector $\xi^\#$. Thus

$$(\Phi^* \omega)(\xi^\#) = \omega(\Phi_* \xi^\#) = \omega(\xi^\#) = \xi.$$

Therefore $\Phi^* \omega$ satisfies the two defining conditions of a connection form. □

Consequently the gauge group acts on the space of connections:

$$\mathcal{G}(P) \curvearrowright \mathcal{A}(P).$$

In these notes we fix the pullback convention above.

2.3.6 Local gauge transformation formula

Take a local section $s : U \rightarrow P$, and denote

$$A = s^*\omega.$$

Let the gauge transformation Φ be represented with respect to this local section by $g : U \rightarrow G$, that is,

$$\Phi(s(x)) = s(x)g(x).$$

The connection after the action is $\omega^\Phi = \Phi^*\omega$. Its local connection form is

$$A^g := s^*(\Phi^*\omega).$$

Since

$$\Phi \circ s = sg,$$

we have

$$A^g = (sg)^*\omega.$$

Using the connection transformation formula for a change of local section given above, we obtain

$$A^g = \text{Ad}_{g^{-1}} A + g^{-1}dg.$$

For a matrix Lie group,

$$A^g = g^{-1}Ag + g^{-1}dg.$$

This is the formula for the action of the gauge group on the local connection matrix.

Remark 2.3.9 (Active and passive interpretations). The formula

$$A' = g^{-1}Ag + g^{-1}dg$$

appears in two different situations.

The first is a change of local section, $s' = sg$. In this case the global connection ω has not changed, only the local representation of one and the same connection has changed.

The second is to keep the local section s fixed but to genuinely apply a gauge transformation $\omega \mapsto \Phi^*\omega$ to the connection.

The local formula is the same in the two situations, but the geometric meaning is different.

Remark 2.3.10. The literature also often uses the opposite gauge action convention, for example

$$A \mapsto gAg^{-1} - (dg)g^{-1}.$$

This is equivalent to our convention; it merely amounts to replacing g by g^{-1} , or to adopting the inverse pullback action.

In what follows we always use

$$A^g = g^{-1}Ag + g^{-1}dg.$$

2.3.7 Gauge action for $U(1)$ and $SU(2)$

Example 2.3.11 ($U(1)$). For $U(1)$, since the adjoint action is trivial, the gauge transformation formula reduces to

$$A^g = A + g^{-1}dg.$$

If locally we can write $g = e^{i\chi}$, then

$$g^{-1}dg = i d\chi,$$

and hence

$$A^g = A + i d\chi.$$

Here writing $g = e^{i\chi}$ in general only holds locally; an arbitrary map $g : X \rightarrow U(1)$ need not admit a global real-valued function χ with $g = e^{i\chi}$.

Example 2.3.12 ($SU(2)$). In a local trivialization, $g : U \rightarrow SU(2)$, $A \in \Omega^1(U; \mathfrak{su}(2))$.

The gauge transformation is

$$A^g = g^{-1}Ag + g^{-1}dg.$$

The first term is the adjoint action, and the second term is the derivative term arising because g varies with position in the base space.

2.3.8 How the curvature changes under a gauge transformation

Although the gauge transformation law of the connection form contains $g^{-1}dg$, the curvature has a simpler transformation law.

Proposition 2.3.13. If

$$A^g = g^{-1}Ag + g^{-1}dg,$$

then

$$F_{A^g} = g^{-1}F_Ag.$$

Proof. The most direct approach is in the language of principal bundles. If $\omega^\Phi = \Phi^*\omega$, then

$$\begin{aligned} \Omega_{\omega^\Phi} &= d(\Phi^*\omega) + \frac{1}{2}[\Phi^*\omega \wedge \Phi^*\omega] \\ &= \Phi^* \left(d\omega + \frac{1}{2}[\omega \wedge \omega] \right) \\ &= \Phi^*\Omega_\omega. \end{aligned}$$

In a local trivialization, the equivariance of a tensorial form gives

$$F_{A^g} = \text{Ad}_{g^{-1}} F_A.$$

For a matrix Lie group this is

$$F_{A^g} = g^{-1}F_Ag.$$

□

Thus the connection itself is not gauge invariant, but the curvature transforms according to the adjoint representation.

In the $U(1)$ case the adjoint action is trivial, so

$$F_{A^g} = F_A.$$

2.3.9 Gauge invariance of the ASD equation

Now let X be an oriented Riemannian 4-manifold.

Since a gauge transformation acts only on the $\text{ad } P$ factor, while the Hodge star acts only on the $\Lambda^2 T^* X$ factor, the two commute.

Hence

$$F_{A^g}^+ = g^{-1} F_A^+ g.$$

Therefore

$$F_A^+ = 0$$

if and only if

$$F_{A^g}^+ = 0.$$

Thus the ASD equation is gauge invariant.

This is precisely why the genuine object of study is not a single ASD connection, but rather the gauge-equivalence classes of ASD connections.

2.3.10 The infinitesimal action of the gauge group

We have seen above that

$$\text{Lie } \mathcal{G}(P) = \Omega^0(X; \text{ad } P).$$

Take $\xi \in \Omega^0(X; \text{ad } P)$, and consider the one-parameter gauge transformation

$$g_t = \exp(t\xi).$$

Locally we have

$$A^{g_t} = g_t^{-1} A g_t + g_t^{-1} d g_t.$$

Differentiate at $t = 0$.

First,

$$\left. \frac{d}{dt} \right|_{t=0} g_t^{-1} A g_t = [A, \xi].$$

Second,

$$\left. \frac{d}{dt} \right|_{t=0} g_t^{-1} d g_t = d\xi.$$

Hence

$$\left. \frac{d}{dt} \right|_{t=0} A^{g_t} = d\xi + [A, \xi] = d_A \xi.$$

Thus the tangent space at A of the gauge orbit $\mathcal{G}(P) \cdot A \subset \mathcal{A}(P)$ is

$$T_A(\mathcal{G}(P) \cdot A) = \text{im} \left(d_A : \Omega^0(X; \text{ad } P) \longrightarrow \Omega^1(X; \text{ad } P) \right).$$

Consequently $d_A \xi$ represents an infinitesimal connection deformation produced purely by gauge symmetry.

This is precisely the geometric origin of the first differential

$$d_A : \Omega^0(\text{ad } P) \longrightarrow \Omega^1(\text{ad } P)$$

in the ASD deformation complex below.

2.3.11 Stabilizer

Not every gauge transformation actually changes a given connection.

Definition 2.3.14 (Stabilizer). The stabilizer of a connection A is defined to be

$$\text{Stab}(A) := \{g \in \mathcal{G}(P) \mid A^g = A\}.$$

Thus $\text{Stab}(A)$ is the subgroup of the gauge group consisting of those elements that leave A unchanged.

Its Lie algebra follows immediately from the infinitesimal action.

Proposition 2.3.15. We have

$$\text{Lie Stab}(A) = \ker \left(d_A : \Omega^0(X; \text{ad } P) \rightarrow \Omega^1(X; \text{ad } P) \right).$$

In other words,

$$d_A \xi = 0$$

says exactly that the infinitesimal gauge transformation ξ leaves A unchanged.

Hence the zeroth cohomology of the ASD deformation complex,

$$H_A^0 = \ker d_A,$$

is precisely the infinitesimal stabilizer.

2.3.12 Irreducible and reducible connections

If G has non-trivial center $Z(G)$, then constant central elements always leave every connection unchanged.

In fact, if $z \in Z(G)$ is constant, then

$$A^z = z^{-1} A z + z^{-1} dz = A.$$

Therefore

$$Z(G) \subseteq \text{Stab}(A)$$

holds for every connection A .

Definition 2.3.16. If

$$\text{Stab}(A) = Z(G),$$

then A is called an **irreducible connection**.

If the stabilizer is strictly larger than $Z(G)$, then A is called a **reducible connection**.

For $G = SU(2)$,

$$Z(SU(2)) = \{\pm I\}.$$

So an $SU(2)$ connection being irreducible means that its stabilizer is exactly $\{\pm I\}$.

Central elements act trivially on every connection, so one sometimes considers the effective gauge group

$$\overline{\mathcal{G}}(P) := \mathcal{G}(P)/Z(G).$$

On the irreducible locus, the action of $\overline{\mathcal{G}}(P)$ is locally free.

2.3.13 Gauge orbits and the configuration space

For any connection A , define its gauge orbit to be

$$[A] := \mathcal{G}(P) \cdot A = \{A^g \mid g \in \mathcal{G}(P)\}.$$

From the point of view of gauge theory, connections in the same orbit represent the same geometric object, only expressed in different gauges.

Therefore define

$$\mathcal{B}(P) := \mathcal{A}(P)/\mathcal{G}(P).$$

$\mathcal{B}(P)$ is called the **configuration space of connections modulo gauge**.

A point $[A] \in \mathcal{B}(P)$ is not a concrete connection, but a whole gauge-equivalence class.

Write

$$\mathcal{A}^*(P) = \{A \in \mathcal{A}(P) \mid A \text{ irreducible}\},$$

and define

$$\mathcal{B}^*(P) := \mathcal{A}^*(P)/\mathcal{G}(P).$$

Later, by means of the Coulomb slice theorem, one can show that after a suitable Sobolev completion, $\mathcal{B}^*(P)$ carries the structure of an infinite-dimensional Banach manifold.

Reducible connections, on the other hand, having a larger stabilizer, usually produce singularities in the quotient.

2.3.14 ASD moduli space

Since the ASD equation

$$F_A^+ = 0$$

is invariant under gauge transformations, the solution set

$$\mathcal{A}_{\text{ASD}}(P) := \{A \in \mathcal{A}(P) \mid F_A^+ = 0\}$$

is a gauge-invariant subset.

We therefore define the ASD moduli space

$$\mathcal{M}_{\text{ASD}}(P) := \mathcal{A}_{\text{ASD}}(P)/\mathcal{G}(P).$$

Equivalently,

$$\mathcal{M}_{\text{ASD}}(P) = \{[A] \in \mathcal{B}(P) \mid F_A^+ = 0\}.$$

Thus the relation between the configuration space and the moduli space is

$$\mathcal{M}_{\text{ASD}}(P) \subset \mathcal{B}(P).$$

$\mathcal{B}(P)$ contains all connections modulo gauge, while $\mathcal{M}_{\text{ASD}}(P)$ contains only those gauge-equivalence classes that satisfy the ASD equation.

At an ASD connection A , the first-order deformation of the ASD equation is

$$d_A^+ a = 0$$

while the first-order directions along the gauge orbit are

$$a = d_A \xi.$$

Hence the genuine infinitesimal ASD deformations modulo gauge are

$$H_A^1 = \frac{\ker d_A^+}{\operatorname{im} d_A}.$$

This is precisely the geometric meaning of the ASD deformation complex

$$0 \longrightarrow \Omega^0(X; \operatorname{ad} P) \xrightarrow{d_A} \Omega^1(X; \operatorname{ad} P) \xrightarrow{d_A^+} \Omega^+(X; \operatorname{ad} P) \longrightarrow 0$$

introduced later.

2.3.15 The need for gauge fixing

Above we defined the ASD moduli space (Section 2.3.14)

$$\mathcal{M}_{\text{ASD}}(P) = \{A \in \mathcal{A}(P) \mid F_A^+ = 0\} / \mathcal{G}(P).$$

Because of the action of the gauge group, a gauge-equivalence class $[A]$ contains many different connection representatives. Although these representatives describe the same geometric object, their analytic properties as local connection forms may differ greatly.

In fact, the local action of a gauge transformation is

$$A^g = g^{-1} A g + g^{-1} dg.$$

Hence the size of the connection form itself is not gauge invariant. Even if $A = 0$, after a gauge transformation it becomes $g^{-1} dg$, and this term may have very large derivatives.

By contrast, the curvature transforms as $F_{A^g} = g^{-1} F_A g$ and hence, when the structure group is a compact Lie group, the natural norm of the curvature is gauge invariant.

This gives rise to a basic analytic problem in gauge theory: given a gauge-equivalence class, how does one choose a representative with good analytic properties?

On the other hand, the linearization of the ASD equation is $d_A^+ a = 0$, while the infinitesimal action of the gauge group is given by $d_A \xi$. Thus the linearization naturally contains gauge directions. Unless these redundant directions are removed first, the ASD equation itself is not an equation to which elliptic theory can be applied directly.

A standard procedure is to choose a Coulomb-type gauge condition, separating the gauge directions from the genuine geometric deformations. In local analysis and in compactness problems, one needs further to establish a quantitative gauge fixing result: under suitable control of the curvature, choose a gauge so that the connection form itself satisfies appropriate differential conditions and Sobolev estimates.

This is precisely the problem solved by Uhlenbeck gauge fixing theory. The next section introduces this result and its role in the analysis of the ASD moduli space.

2.4 Abelian gauge theory on Hermitian line bundles

The general discussion of gauge orbits has a particularly explicit form for $U(1)$. We now work this out on a line bundle which need not be trivial, and on a manifold which need not be simply connected. Besides providing a model for gauge fixing, these results will supply the connection estimates and the reducible equation in Seiberg–Witten theory.

Throughout this section, X is a closed, connected, oriented Riemannian 4-manifold, L is a Hermitian line bundle on X , and A_0 is a fixed smooth unitary connection on L . We retain the convention

$$A^g = A + g^{-1}dg, \quad c_1(L)_{\mathbb{R}} = \left[\frac{i}{2\pi} F_A \right].$$

The Chern–Weil normalization is the one fixed in Section 2.2.8. All connection differences and curvatures are imaginary-valued forms. We identify their norms with the norms of the corresponding real forms by multiplication by $-i$.

2.4.1 A reference connection on a possibly non-trivial line bundle

Although a connection on a non-trivial line bundle is represented by different 1-forms in different local frames, the difference of two connections is a globally defined 1-form. Since the adjoint representation of $U(1)$ is trivial, we have

$$\mathcal{A}(L) = A_0 + i\Omega^1(X).$$

Thus every unitary connection can be written uniquely as $A = A_0 + a$, with $a \in i\Omega^1(X)$. The Abelian curvature formula is

$$F_A = F_{A_0} + da, \quad F_A^+ = F_{A_0}^+ + d^+a.$$

There is no quadratic term: the wedge square of a scalar 1-form vanishes. These are global identities, even when neither A nor A_0 can be represented by a single connection form on X .

Likewise, the gauge group is $\mathcal{G}(L) = C^\infty(X, U(1))$ regardless of the topological type of L . Its action on the difference form is $a \mapsto a + g^{-1}dg$.

2.4.2 Large gauge transformations and integral periods

Locally, $g = e^{i\chi}$ and $g^{-1}dg = i d\chi$. Globally, $g^{-1}dg$ is still closed, but it need not be exact. For every closed oriented loop γ ,

$$\frac{1}{2\pi i} \int_{\gamma} g^{-1}dg = \deg(g \circ \gamma) \in \mathbb{Z}.$$

Here the degree is the winding number of the map from the loop to $U(1)$.

Conversely, suppose that $\beta \in i\Omega^1(X)$ is closed and has periods in $2\pi i\mathbb{Z}$. Fix $x_0 \in X$ and define

$$g(x) = \exp \left(\int_{x_0}^x \beta \right).$$

Changing the path changes the integral by an element of $2\pi i\mathbb{Z}$, so g is well-defined. Local primitives of β show that g is smooth and $g^{-1}dg = \beta$.

It follows that a gauge transformation admits a global logarithm $g = e^{i\chi}$ exactly when $[g^{-1}dg] = 0$ in de Rham cohomology. Such transformations form the identity component $\mathcal{G}_0(L)$: the path $e^{it\chi}$ joins 1 to g , while the integral periods are constant under any homotopy. Consequently

$$\mathcal{G}(L)/\mathcal{G}_0(L) \cong H^1(X; \mathbb{Z}).$$

We use here the natural injection of $H^1(X; \mathbb{Z})$ into $H^1(X; \mathbb{R})$; $H^1(X; \mathbb{Z})$ has no torsion. A transformation outside the identity component is called a *large gauge transformation*.

Let $\mathcal{H}^1(X; \mathbb{R})$ be the harmonic real 1-forms, and let

$$\mathcal{H}_{\mathbb{Z}}^1 = \{h \in \mathcal{H}^1(X; \mathbb{R}) : [h] \in H^1(X; \mathbb{Z})\}.$$

Then $2\pi i \mathcal{H}_{\mathbb{Z}}^1$ is a full lattice in $i\mathcal{H}^1(X; \mathbb{R})$. Every element of this lattice is $g^{-1}dg$ for some gauge transformation. Thus large gauge transformations can translate the harmonic part of a connection difference along this lattice.

2.4.3 Global Coulomb gauge and the remaining harmonic freedom

Proposition 2.4.1 (Global Abelian gauge fixing). Fix a closed fundamental parallelepiped K for $2\pi i \mathcal{H}_{\mathbb{Z}}^1$ in $i\mathcal{H}^1(X; \mathbb{R})$. For every smooth unitary connection A on L , there is $g \in \mathcal{G}(L)$ such that, writing $A^g = A_0 + a'$,

$$d^*a' = 0, \quad \text{Harm}(a') \in K.$$

Here Harm denotes the L^2 orthogonal projection onto harmonic forms.

Proof. Write $A = A_0 + a$. Solve for an imaginary-valued function $f \in i\Omega^0(X)$:

$$d^*df = -d^*a, \quad \int_X f \, d\text{vol} = 0.$$

The right-hand side is orthogonal to constants, since $\langle d^*a, 1 \rangle = \langle a, d1 \rangle = 0$. The Poisson equation therefore has a unique mean-zero solution. The gauge transformation $g_0 = e^f$ gives $a_1 = a + df$, with $d^*a_1 = 0$.

Next choose a lattice element $\ell \in 2\pi i \mathcal{H}_{\mathbb{Z}}^1$ such that $\text{Harm}(a_1) + \ell \in K$. By Section 2.4.2, there is a gauge transformation g_1 with $g_1^{-1}dg_1 = \ell$. Since ℓ is harmonic, adding it preserves the Coulomb condition. The product $g = g_0g_1$ has the required properties. $\square$

The first step removes the exact part of a ; the second controls its harmonic part. The Coulomb condition alone leaves the latter untouched. If $b_1(X) = 0$, the harmonic space is zero and the second step is unnecessary. In general, the residual harmonic parameters form the torus

$$\frac{i\mathcal{H}^1(X; \mathbb{R})}{2\pi i \mathcal{H}_{\mathbb{Z}}^1}.$$

Constant gauge transformations still fix every connection. Also, points on the boundary of K may represent the same orbit. The proposition selects bounded representatives, rather than a unique global slice.

2.4.4 An estimate using only the self-dual curvature

To turn this choice of gauge into an estimate, consider

$$D = d^* \oplus d^+ : i\Omega^1(X) \longrightarrow i\Omega^0(X) \oplus i\Omega^+(X).$$

The underlying complex was studied in Section 1.4.5. Its operator symbol is computed in Section A.2.10; the removal of the kernel from an elliptic estimate is explained in Section A.7.

First, $\ker D = i\mathcal{H}^1(X; \mathbb{R})$. Indeed, if $Da = 0$, write $a = ib$ with b real. Then db is anti-self-dual, so Stokes' theorem gives

$$-\|db\|_{L^2}^2 = \int_X db \wedge db = \int_X d(b \wedge db) = 0.$$

Thus $da = 0$, and together with $d^*a = 0$ this says that a is harmonic.

Proposition 2.4.2 (Abelian connection estimate). For $1 < p < \infty$, there is a constant C_p depending on X , its metric and p such that every smooth $a \in i\Omega^1(X)$ satisfies

$$\|a\|_{W^{1,p}} \leq C_p \left(\|d^*a\|_{L^p} + \|d^+a\|_{L^p} + \|\text{Harm}(a)\|_{L^2} \right).$$

Consequently, after the gauge transformation in Proposition 2.4.1,

$$\|A^g - A_0\|_{W^{1,p}} \leq C_{p,A_0} \left(\|F_A^+\|_{L^p} + 1 \right).$$

The constant in the second estimate may also depend on A_0 and the chosen fundamental parallelepiped, but not on A .

Proof. For $\text{Harm}(b) = 0$, we claim that

$$\|b\|_{W^{1,p}} \leq C_p \|Db\|_{L^p}.$$

For completeness, if this failed there would be smooth b_n with $\text{Harm}(b_n) = 0$, $\|b_n\|_{W^{1,p}} = 1$ and $Db_n \rightarrow 0$ in L^p . Rellich compactness (Section A.4.5) gives a subsequence converging in L^p . The ordinary elliptic estimate applied to differences then gives convergence in $W^{1,p}$ to a non-zero element of $\ker D$. Its harmonic projection is zero, a contradiction. The projection is continuous in L^p , since it is given by integration against finitely many smooth harmonic forms.

Apply this estimate to $b = a - \text{Harm}(a)$. All norms on the finite-dimensional harmonic space are equivalent, which proves the first inequality. In the gauge of Proposition 2.4.1, $d^*a' = 0$, the harmonic part is bounded, and

$$d^+a' = F_{A^g}^+ - F_{A_0}^+ = F_A^+ - F_{A_0}^+.$$

This proves the second inequality. □

This estimate requires no small-curvature hypothesis. In Seiberg–Witten theory, a bound on the spinor will give a bound on F_A^+ through the curvature equation. The proposition will then control the connection in $W^{1,p}$, including for $p > 4$.

2.4.5 Prescribing the self-dual curvature

There is also a precise existence criterion for the Abelian equation $F_A^+ = \eta$, where $\eta \in i\Omega^+(X)$ is prescribed and is not assumed closed. Let $c_1(L)_g^+$ denote the self-dual part of the harmonic representative of $c_1(L)_\mathbb{R}$, using the metric-dependent splitting in Section 1.4.4.

Proposition 2.4.3. There is a smooth unitary connection A on L satisfying $F_A^+ = \eta$ if and only if

$$\text{Harm}(\eta) = -2\pi i c_1(L)_g^+.$$

Here the harmonic projection of a self-dual form takes values in $i\mathcal{H}_g^+(X)$.

Proof. The Chern–Weil convention gives $\text{Harm}(F_A) = -2\pi i c_1(L)_g$, where $c_1(L)_g$ is the full harmonic representative. Harmonic projection commutes with the Hodge star, so the displayed condition is necessary.

For sufficiency, write $A = A_0 + a$. It is enough to solve

$$d^+a = \eta - F_{A_0}^+.$$

The image of d^+ is the orthogonal complement of $i\mathcal{H}_g^+(X)$. To see this, for a self-dual 2-form β the adjoint of d^+ is $d^*\beta$, and $d^*\beta = -*d\beta$. Its kernel therefore consists precisely of self-dual harmonic forms. Hodge theory for the elliptic self-dual complex gives closed range and identifies the image with the orthogonal complement of this kernel; see Section A.9.6. Equivalently, on self-dual 2-forms d^+d^* is elliptic and has kernel $i\mathcal{H}_g^+(X)$; solve

$$d^+d^*\beta = \eta - F_{A_0}^+, \quad a = d^*\beta.$$

The assumed harmonic condition makes the right-hand side orthogonal to the kernel. The Fredholm alternative (Section A.8.8) and elliptic regularity give a smooth solution. $\square$

If the equation is solvable and A_* is one solution, any other solution is $A_* + a$ with $d^+a = 0$. The Stokes argument above, which does not require $d^*a = 0$ to prove $da = 0$, shows that a is closed. Modulo ordinary line-bundle gauge transformations, the solution space is therefore an affine torus modeled on

$$\frac{i\mathcal{H}^1(X; \mathbb{R})}{2\pi i\mathcal{H}_\mathbb{Z}^1}.$$

Choosing A_* identifies this affine torus with the displayed group; there is no preferred origin without such a choice.

In particular, L admits an ASD unitary connection exactly when $c_1(L)_g^+ = 0$. This condition concerns the real Chern class. It does not detect its torsion part. For example, a torsion line bundle admits a flat unitary connection: choose a connection with harmonic curvature by subtracting a primitive of the exact part of its curvature; the harmonic representative of its real Chern class is zero.

2.4.6 The Abelian energy identity

The sign in the energy identity can be fixed without ambiguity. Write $F_A = if_A$ with f_A a real 2-form. Then

$$\int_X F_A \wedge F_A = \|F_A^-\|_{L^2}^2 - \|F_A^+\|_{L^2}^2 = -4\pi^2 \langle c_1(L)^2, [X] \rangle.$$

The last equality follows by squaring the Chern–Weil representative $\frac{i}{2\pi}F_A$. Hence

$$\|F_A\|_{L^2}^2 = 2\|F_A^+\|_{L^2}^2 - 4\pi^2\langle c_1(L)^2, [X] \rangle.$$

For a fixed line bundle, a bound on the self-dual curvature thus gives an L^2 bound on the full curvature. The preceding connection estimate is stronger: it uses the self-dual curvature directly to control a representative of the connection modulo gauge.

2.4.7 Passing to the Seiberg–Witten convention

The preceding results concern ordinary unitary connections and the right action $A^g = A + g^{-1}dg$. Later, A will be the connection on the determinant line of a fixed Spin^c structure. The convention in that setting is the left action

$$u \cdot (A, \psi) = (A - 2u^{-1}du, u\psi).$$

The spinor transforms by u , while the determinant line transforms by u^2 . The determinant character is constructed in Section 3.2.9; the action and Dirac covariance are derived in Sections 4.1.10 and 4.1.12.

For the connection component, the two formulas agree upon setting $g = u^{-2}$. However, an arbitrary map $g : X \rightarrow U(1)$ need not admit such a square root, so this substitution does not identify the two global orbit relations. With the Seiberg–Witten action, the harmonic translation lattice is

$$4\pi i\mathcal{H}_{\mathbb{Z}}^1$$

instead of $2\pi i\mathcal{H}_{\mathbb{Z}}^1$. It is still a full lattice. The proofs of global Coulomb gauge and the connection estimate apply with this new lattice: for $u = e^f$, solve $d^*df = \frac{1}{2}d^*a$ to impose $d^*(a - 2df) = 0$, and then use large gauge transformations to bound the harmonic part.

There is a further distinction concerning stabilizers. Every constant $U(1)$ transformation fixes a line-bundle connection, whereas it multiplies a spinor by that constant. On connected X , a transformation fixing (A, ψ) must be constant, and if $\psi \not\equiv 0$ it must be 1. Thus reducibility of a Seiberg–Witten configuration concerns the pair (A, ψ) , rather than just the underlying Abelian connection.

These observations give the preparation needed later: the equation $F_A^+ = \eta$ determines the reducible wall in Section 4.4.7, and the global Abelian estimate supplies the connection bound in Section 4.5.8. We now return to general structure groups, for which gauge fixing is a nonlinear problem.

2.5 The Coulomb gauge and Uhlenbeck gauge fixing

We have seen above that the ASD equation has gauge symmetry. At an ASD connection A , the first-order linearization of the equation is $d_A^+a = 0$, while the infinitesimal action of the gauge group is $a = d_A\xi$.

Hence, unless the gauge directions are removed, the ASD equation itself is not an elliptic equation. In order to use elliptic estimates, one must first choose a suitable gauge representative.

This section introduces the most basic Coulomb gauge, as well as Uhlenbeck’s local gauge fixing theorem for connections of small curvature.

2.5.1 Elliptic estimates and the need for gauge fixing

We first recall the basic estimate for an elliptic operator. Let

$$D : W^{k,p}(E) \longrightarrow W^{k-m,p}(F)$$

be an elliptic differential operator of order m . Under suitable compactness or boundary conditions, there is an elliptic estimate of the form

$$\|s\|_{W^{k,p}} \leq C \left(\|Ds\|_{W^{k-m,p}} + \|s\|_{L^p} \right)$$

Therefore, if one can write a geometric equation as an elliptic equation, then control of the right-hand side of the equation yields Sobolev control of the solution itself, and one can then use the Sobolev embedding, Rellich compactness and elliptic bootstrapping to obtain higher regularity and compactness.

However, the ASD equation $F_A^+ = 0$ itself has gauge symmetry. For example, even if a connection is written as $A = 0$ in some gauge, after a gauge transformation it also becomes $A^g = g^{-1}dg$, and the latter may have very large derivatives.

So the Sobolev norm of the connection form itself is not gauge invariant.

By contrast, $F_{A^g} = g^{-1}F_Ag$. For a compact structure group, one can take an Ad-invariant inner product on $\mathfrak{g}$, so that $|F_{A^g}| = |F_A|$.

Hence the basic analytic problem in gauge theory is:

How does one use a gauge-invariant curvature bound to select, in every gauge orbit, a representative with good Sobolev control?

The Coulomb gauge is precisely the basic tool for solving this problem.

2.5.2 The Coulomb gauge in the $U(1)$ case

The Abelian construction in Proposition 2.4.1 provides the model. Writing $A = A_0 + a$ and taking $g = e^f$, with f imaginary-valued, the Coulomb condition is the linear equation

$$d^*(a + df) = 0, \quad \text{or equivalently} \quad d^*df = -d^*a.$$

The right-hand side is orthogonal to constants, so a mean-zero solution exists. On a general closed manifold, one must also control the harmonic component using the integral lattice. This is the extra global step established in Section 2.4.2.

For a non-Abelian structure group, the transformed connection contains both $g^{-1}Ag$ and $g^{-1}dg$. The equation for g is therefore nonlinear. A reference connection gives the appropriate local formulation of the Coulomb condition.

2.5.3 Coulomb gauge relative to a reference connection

For a general non-Abelian structure group, the natural Coulomb condition should be defined relative to a fixed reference connection.

Fix a connection A_0 , and write nearby connections as

$$B = A_0 + a, \quad a \in \Omega^1(X; \text{ad } P).$$

We say that B is in Coulomb gauge relative to A_0 if

$$d_{A_0}^* a = 0.$$

That is,

$$d_{A_0}^*(B - A_0) = 0.$$

At the infinitesimal level, the tangent space to the gauge orbit through A_0 is $\text{im } d_{A_0}$. And by the definition of the formal adjoint,

$$\ker d_{A_0}^* = (\text{im } d_{A_0})^\perp.$$

Hence the geometric meaning of the Coulomb condition is: choose the connection representative in the direction orthogonal to the gauge orbit.

2.5.4 The local Coulomb slice and the implicit function theorem

Near a fixed connection, the Coulomb gauge can be constructed by the implicit function theorem.

Fix a smooth connection A_0 . To avoid the technical problems caused by the critical Sobolev exponent, take $p > 2$, and consider

$$A_0 + a, \quad a \in W^{1,p}\Omega^1(X; \text{ad } P).$$

Near the identity gauge transformation, write $g = e^\xi$, where $\xi \in W^{2,p}\Omega^0(X; \text{ad } P)$. Define the map

$$\mathcal{F}(\xi, a) = d_{A_0}^* \left((A_0 + a)^{e^\xi} - A_0 \right).$$

Clearly $\mathcal{F}(0, 0) = 0$.

According to the gauge action convention adopted in these notes,

$$\left. \frac{d}{dt} \right|_{t=0} A_0^{e^{t\xi}} = d_{A_0} \xi.$$

Hence the linearization of $\mathcal{F}$ in the ξ direction at $(0, 0)$ is

$$D_\xi \mathcal{F}_{(0,0)}(\xi) = d_{A_0}^* d_{A_0} \xi.$$

The operator $d_{A_0}^* d_{A_0}$ is an elliptic operator, and

$$\ker(d_{A_0}^* d_{A_0}) = \ker d_{A_0}.$$

Hence if A_0 is irreducible, then at the level of the Lie algebra we usually have $\ker d_{A_0} = 0$, so that $d_{A_0}^* d_{A_0}$ is invertible.

More generally, if A_0 has a stabilizer, we can restrict the gauge parameter to $(\ker d_{A_0})^\perp$, or else use the based gauge group.

By the Banach implicit function theorem, when $\|a\|_{W^{1,p}}$ is sufficiently small, there exists a small gauge transformation $g = e^\xi$ such that

$$d_{A_0}^* \left((A_0 + a)^g - A_0 \right) = 0.$$

Therefore, near A_0 , every gauge orbit can be represented locally by the Coulomb slice

$$\{A_0 + a \mid d_{A_0}^* a = 0\}$$

to represent each gauge orbit locally.

Remark 2.5.1. The above conclusion is the basic analytic mechanism behind the local slice theorem. Its hypothesis is that the connection itself is already sufficiently close to some fixed reference connection A_0 .

What makes Uhlenbeck gauge fixing important is this: it does not presuppose that the connection is close to some fixed connection, but starts only from a small-energy condition on the curvature.

2.5.5 Uhlenbeck gauge fixing theorem

In what follows we consider the unit ball $B^4 \subset \mathbb{R}^4$. Since B^4 is contractible, every principal G -bundle over B^4 is trivial.

Hence, after choosing a trivialization, every connection can be written as

$$A = d + a, \quad a \in \Omega^1(B^4; \mathfrak{g}).$$

Theorem 2.5.2 (Uhlenbeck gauge fixing). Let G be a compact Lie group. There exist constants $\varepsilon_0 > 0$ and $C > 0$ such that for every connection A defined on the trivial principal G -bundle $B^4 \times G \rightarrow B^4$, if

$$\|F_A\|_{L^2(B^4)} < \varepsilon_0,$$

then there exists a gauge transformation u such that

$$A^u = d + a$$

satisfies

$$d^*a = 0$$

together with the boundary condition

$$a(\nu) = 0 \quad \text{on } \partial B^4,$$

where ν is the outward unit normal vector to ∂B^4 .

Moreover, we have the estimate

$$\|a\|_{W^{1,2}(B^4)} \leq C \|F_A\|_{L^2(B^4)}.$$

The boundary condition $a(\nu) = 0$ is also often written as $*a|_{\partial B^4} = 0$.

The key point of this theorem is not merely that a Coulomb gauge exists, but that it provides the quantitative estimate

$$\|a\|_{W^{1,2}} \leq C \|F_A\|_{L^2}.$$

The left-hand side is the Sobolev norm of the connection form itself, whereas the right-hand side is determined entirely by the gauge-invariant curvature.

Therefore the Uhlenbeck theorem converts curvature control into connection control.

2.5.6 Why L^2 is natural in dimension four

In dimension four, the L^2 -norm of the curvature is scale invariant.

In fact, when we rescale the coordinates of $\mathbb{R}^4$ by $x \mapsto \lambda x$, the curvature scales by the inverse square of length, while the volume element scales by the fourth power. Hence

$$\int |F_A|^2 d\text{vol}$$

stays unchanged.

Thus $\|F_A\|_{L^2}$ is the critical quantity in four-dimensional Yang–Mills theory.

At the same time, the four-dimensional Sobolev embedding gives

$$W^{1,2}(B^4) \hookrightarrow L^4(B^4).$$

So the Uhlenbeck estimate further gives

$$\|a\|_{L^4} \leq C\|F_A\|_{L^2}.$$

This is exactly what is needed to control the nonlinear term $a \wedge a$ in the curvature, since

$$\|a \wedge a\|_{L^2} \leq C\|a\|_{L^4}^2.$$

2.5.7 The ASD equation in the Coulomb gauge

Now suppose that $A = d + a$ is an ASD connection on B^4 that has already been brought by Uhlenbeck gauge fixing into the form satisfying $d^*a = 0$.

The curvature is

$$F_A = da + a \wedge a.$$

The ASD equation $F_A^+ = 0$ therefore becomes

$$d^+a + (a \wedge a)^+ = 0.$$

So a satisfies the system of equations

$$\begin{cases} d^*a = 0, \\ d^+a = -(a \wedge a)^+. \end{cases}$$

In other words,

$$(d^* \oplus d^+)a = (0, -(a \wedge a)^+).$$

The operator

$$d^* \oplus d^+ : \Omega^1(B^4; \mathfrak{g}) \longrightarrow \Omega^0(B^4; \mathfrak{g}) \oplus \Omega^+(B^4; \mathfrak{g})$$

is elliptic.

Hence, once the Coulomb gauge is fixed, the ASD equation becomes an elliptic non-linear equation.

2.5.8 Interior regularity

Combining Uhlenbeck gauge fixing with the ellipticity of the ASD equation, one obtains interior estimates of higher order.

Theorem 2.5.3 (Interior estimate for ASD connections). There exists $\varepsilon_1 > 0$ for which the following holds.

Let $A = d + a$ be an ASD connection on B^4 satisfying

$$d^*a = 0, \quad \|a\|_{L^4(B^4)} < \varepsilon_1.$$

Then for every $D \Subset B^4$ and every integer $k \geq 1$, there exists a constant $C_{k,D}$ such that

$$\|a\|_{W^{k,2}(D)} \leq C_{k,D} \|F_A\|_{L^2(B^4)}.$$

Here $D \Subset B^4$ means that $\overline{D}$ is a compact subset of B^4 .

The basic mechanism of the proof is elliptic bootstrapping. From

$$(d^* \oplus d^+)a = (0, -(a \wedge a)^+)$$

and the elliptic estimate, one can first control higher Sobolev norms of a .

Since $\|a\|_{L^4}$ is sufficiently small, the nonlinear term $a \wedge a$ can be absorbed into the left-hand side in the estimates. Applying elliptic regularity and Sobolev multiplication repeatedly then yields interior estimates of arbitrarily high order.

In particular, an ASD connection of Sobolev class is in fact smooth in the Coulomb gauge.

2.5.9 Local compactness

The two preceding theorems immediately give local compactness for ASD connections.

Corollary 2.5.4 (Small-energy local compactness). There exists $\varepsilon > 0$ for which the following holds.

Let $\{A_n\}$ be a sequence of ASD connections on the trivial principal G -bundle $B^4 \times G \rightarrow B^4$, satisfying the uniform small-energy condition

$$\|F_{A_n}\|_{L^2(B^4)} < \varepsilon.$$

Then there exist gauge transformations u_n and a subsequence such that for every $D \Subset B^4$, $A_n^{u_n}$ converges on D to a smooth ASD connection A_∞ .

More precisely, after passing to a diagonal subsequence, we obtain

$$A_n^{u_n} \longrightarrow A_\infty \quad \text{in } C_{\text{loc}}^\infty(B^4).$$

Proof. By the Uhlenbeck gauge fixing theorem 2.5.2, for each n we can choose u_n so that

$$A_n^{u_n} = d + a_n, \quad d^*a_n = 0,$$

and

$$\|a_n\|_{W^{1,2}(B^4)} \leq C \|F_{A_n}\|_{L^2(B^4)} \leq C\varepsilon.$$

By the four-dimensional Sobolev embedding,

$$\|a_n\|_{L^4(B^4)} \leq C_S \|a_n\|_{W^{1,2}(B^4)} \leq C_S C\varepsilon.$$

Take ε small enough that $C_S C \varepsilon < \varepsilon_1$, where ε_1 is the constant in Theorem 2.5.3 involved.

Hence for every $D \Subset B^4$ and every $k \geq 1$ we have the uniform estimate

$$\|a_n\|_{W^{k,2}(D)} \leq C_{k,D} \varepsilon.$$

By the Rellich compactness theorem and the Sobolev embedding, we can pass to subsequences level by level. Using the diagonal argument as well, we obtain a subsequence that converges in C^∞ on every $D \Subset B^4$ to some smooth connection A_∞ .

Since the ASD equation is preserved under smooth convergence, A_∞ still satisfies

$$F_{A_\infty}^+ = 0.$$

□

2.5.10 Further directions

What Uhlenbeck gauge fixing gives is essentially local: as soon as the curvature energy in a ball is sufficiently small, one can find a Coulomb gauge, and the curvature then controls all interior derivatives of the connection.

Hence when we study a sequence of ASD connections of bounded total energy, the only places where compactness can genuinely fail are those at which the curvature energy cannot be kept small.

These points correspond to curvature concentration. As we shall see later, in dimension four this phenomenon produces bubbling, and ultimately leads to Uhlenbeck compactness and to the compactification of the ASD moduli space.

2.6 Uhlenbeck compactness and bubbling

We saw in the previous section that on a 4-dimensional ball, if the L^2 -norm of the curvature is sufficiently small, Uhlenbeck gauge fixing (Theorem 2.5.2) can put the connection into Coulomb gauge and use the curvature to control the $W^{1,2}$ -norm of the connection form.

If the connection moreover satisfies the ASD equation, then elliptic bootstrapping yields Sobolev estimates of arbitrarily high order on interior regions, and hence smooth subsequential convergence.

Therefore, for a sequence of ASD connections, the only places where compactness can genuinely fail are those at which the curvature energy concentrates.

We discuss this phenomenon below.

2.6.1 The energy of an ASD connection

Let X be a closed, oriented Riemannian 4-manifold, and let $P \rightarrow X$ be a principal $SU(2)$ -bundle.

Write

$$k = c_2(P)[X].$$

In what follows we use the standard Chern–Weil normalization, so that for an ASD connection A ,

$$\int_X |F_A|^2 d\text{vol} = 8\pi^2 k.$$

Thus on a fixed bundle P , all ASD connections have the same Yang–Mills energy.

This point is very important. If $\{A_n\}$ is a sequence of ASD connections, then although their curvature may be redistributed over different regions, the total energy always satisfies

$$\int_X |F_{A_n}|^2 d\text{vol} = 8\pi^2 k.$$

So a sequence of ASD connections automatically has a uniform global L^2 curvature bound.

2.6.2 Why bounded global energy still does not directly give compactness

The local compactness of the previous section (Corollary 2.5.4) requires that on some ball B we have

$$\|F_A\|_{L^2(B)} < \varepsilon_0,$$

where ε_0 is the small-energy threshold in the Uhlenbeck gauge fixing theorem.

But the global estimate

$$\int_X |F_A|^2 \leq C$$

does not guarantee small energy on balls of every fixed scale.

There may exist a sequence of points $x_n \rightarrow x$ such that more and more curvature energy concentrates in smaller and smaller $B_r(x)$.

In other words, even if the total energy stays constant, it may still happen that

$$|F_{A_n}|^2 d\text{vol}$$

concentrates towards certain points.

This phenomenon is called **curvature concentration**.

In ASD gauge theory, curvature concentration ultimately produces the so-called **bubbling phenomenon**.

2.6.3 Concentration points

Let $\{A_n\}$ be a sequence of ASD connections on a fixed principal $SU(2)$ -bundle $P \rightarrow X$.

Write

$$\mu_n := |F_{A_n}|^2 d\text{vol}.$$

Since

$$\mu_n(X) = 8\pi^2 k$$

is uniformly bounded, after passing to a subsequence, μ_n converges in the weak topology of Radon measures to some finite measure μ .

Take the small-energy constant $\varepsilon_0 > 0$ of the Uhlenbeck gauge fixing theorem. Define the concentration set

$$\Sigma := \{x \in X \mid \mu(\{x\}) \geq \varepsilon_0\}.$$

Since $\mu(X) = 8\pi^2 k$, the set Σ must be finite. In fact,

$$\#\Sigma \leq \frac{8\pi^2 k}{\varepsilon_0}.$$

Hence we may write

$$\Sigma = \{x_1, \dots, x_N\}.$$

These points are precisely the possible locations of bubbling.

2.6.4 Convergence away from the concentration points

Let $K \Subset X \setminus \Sigma$ be a compact subset.

Since K contains no concentration point, we can cover K by finitely many small geodesic balls $B_1, \dots, B_m$, such that for all sufficiently large n , on each ball we have

$$\|F_{A_n}\|_{L^2(B_j)} < \varepsilon_0.$$

Hence on each B_j we may apply the Uhlenbeck gauge fixing theorem. There exist local gauge transformations $u_{n,j}$ such that

$$A_n^{u_{n,j}} = d + a_{n,j}, \quad d^* a_{n,j} = 0,$$

and

$$\|a_{n,j}\|_{W^{1,2}(B_j)} \leq C \|F_{A_n}\|_{L^2(B_j)}.$$

Since A_n satisfies the ASD equation, these Coulomb-gauge representatives also satisfy

$$d^+ a_{n,j} + (a_{n,j} \wedge a_{n,j})^+ = 0.$$

By elliptic regularity and bootstrapping, on any smaller inner ball $B'_j \Subset B_j$, for every $r \geq 1$ there is a uniform estimate

$$\|a_{n,j}\|_{W^{r,2}(B'_j)} \leq C_{r,j}.$$

Hence, after passing to a subsequence, we obtain on B'_j C^∞ convergence.

Note that the Coulomb gauges obtained on different balls are not automatically consistent with one another. To obtain convergence on all of K , we must also use a gauge patching argument to glue these local gauges together appropriately.

After this standard step, we obtain gauge transformations defined on $X \setminus \Sigma$, such that some subsequence converges to a smooth connection A_∞ .

Since the ASD equation is preserved under smooth convergence, we have

$$F_{A_\infty}^+ = 0.$$

Thus A_∞ is an ASD connection on $X \setminus \Sigma$.

2.6.5 Uhlenbeck compactness theorem

The analysis above can be organized into the following basic theorem.

Theorem 2.6.1 (Uhlenbeck compactness, ASD version). Let X be a closed, oriented Riemannian 4-manifold, $P \rightarrow X$ a principal $SU(2)$ -bundle, and

$$c_2(P)[X] = k.$$

Let $\{A_n\}$ be a sequence of smooth ASD connections on P .

Then, after passing to a subsequence, there exist a finite set of points

$$\Sigma = \{x_1, \dots, x_N\} \subset X,$$

a smooth ASD connection defined on $X \setminus \Sigma$, A_∞ , and suitable bundle isomorphisms and gauge transformations such that

$$A_n \longrightarrow A_\infty \quad \text{in } C_{\text{loc}}^\infty(X \setminus \Sigma)$$

modulo gauge.

More precisely, for any $K \Subset X \setminus \Sigma$, one can choose bundle identifications and gauges on K so that the corresponding connection forms converge in $C^\infty(K)$.

We must emphasize one point here: the convergence holds only away from finitely many concentration points.

Hence the original ASD moduli space $\mathcal{M}_k$ is in general not compact.

2.6.6 Removable singularity theorem

Although the limiting connection A_∞ is initially defined only on $X \setminus \Sigma$, it nevertheless has finite energy near each puncture.

Here we may use the Uhlenbeck removable singularity theorem.

Theorem 2.6.2 (Removable singularity). Let A be a smooth ASD connection on the punctured ball $B^4 \setminus \{0\}$, and

$$\int_{B^4 \setminus \{0\}} |F_A|^2 < \infty.$$

Then there exists a gauge transformation such that A extends smoothly across the origin, yielding a smooth ASD connection on B^4 .

Thus A_∞ can be extended at each $x_i \in \Sigma$.

But note: the global bundle $P_\infty \rightarrow X$ obtained after extension need not have the same topological type as the original P .

Its second Chern number may decrease.

Write

$$c_2(P_\infty)[X] = k - \ell.$$

Here $\ell \geq 0$ is precisely the total instanton charge lost in the bubbling process.

2.6.7 The energy defect and bubbling

Although on $X \setminus \Sigma$ A_n converges smoothly to A_∞ , the total energy does not in general converge entirely to the energy of A_∞ .

The excess energy is concentrated in Σ .

Hence the limit of the curvature measures has the form

$$|F_{A_n}|^2 d\text{vol} \rightharpoonup |F_{A_\infty}|^2 d\text{vol} + \sum_{i=1}^N \theta_i \delta_{x_i},$$

where δ_{x_i} is the Dirac measure concentrated at x_i , and $\theta_i > 0$ is the curvature energy lost at that point.

These Dirac masses are the traces of bubbling on the original manifold.

To understand θ_i , choose near x_i a point at which the curvature is maximal together with a suitable scale $\lambda_n \rightarrow 0$, and then rescale the connection by the scale λ_n^{-1} .

Since the four-dimensional Yang–Mills energy is scale invariant, the rescaled connection still retains non-zero finite energy.

The limit gives a non-trivial finite-energy ASD connection on $\mathbb{R}^4$.

By adding a point at infinity and applying the removable singularity theorem (Theorem 2.6.2), this connection can be viewed as an ASD instanton on S^4 .

This is a **bubble**.

2.6.8 Energy quantization

A finite-energy $SU(2)$ instanton on S^4 has integral second Chern number.

Hence the energy lost at each concentration point is not an arbitrary real number, but an integer multiple of $8\pi^2$.

There exist positive integers

$$m_1, \dots, m_N \in \mathbb{Z}_{>0}$$

such that

$$\theta_i = 8\pi^2 m_i.$$

Therefore

$$|F_{A_n}|^2 d\text{vol} \rightharpoonup |F_{A_\infty}|^2 d\text{vol} + 8\pi^2 \sum_{i=1}^N m_i \delta_{x_i}.$$

If we write

$$\ell := \sum_{i=1}^N m_i,$$

then conservation of energy gives

$$k = (k - \ell) + \ell.$$

That is,

$$c_2(P_\infty)[X] = k - \ell.$$

Thus bubbling does not really make the instanton charge disappear; it merely transfers ℓ of its units from the smooth limiting connection to several bubble points.

Remark 2.6.3. A concentration point x_i may have $m_i > 1$.

This does not necessarily mean that at a single scale there is only one charge- m_i instanton. In a finer blow-up analysis several distinct scales may occur, forming what is called a bubble tree.

The Uhlenbeck compactification records only the total multiplicity m_i , and not these finer scale parameters.

2.6.9 Ideal instantons

To include bubbling limits in the moduli space as well, define the symmetric product

$$\mathrm{Sym}^\ell(X) := X^\ell / \mathfrak{S}_\ell.$$

Its elements can be written as formal sums

$$x = m_1 x_1 + \cdots + m_N x_N, \quad \sum_{i=1}^N m_i = \ell.$$

Here m_i denotes the multiplicity of the point x_i .

Definition 2.6.4 (Ideal instanton). An **ideal instanton** of charge k is a pair

$$([A_\infty], m_1 x_1 + \cdots + m_N x_N),$$

where A_∞ is an ASD connection of charge $k - \ell$, and

$$m_1 x_1 + \cdots + m_N x_N \in \mathrm{Sym}^\ell(X).$$

Here $[A_\infty]$ records the remaining smooth instanton, while the point in the symmetric product records the locations and multiplicities of the bubbles.

Hence the limit of a bubbling sequence is no longer just an ordinary connection, but

$$\left([A_\infty], \sum_i m_i x_i \right).$$

2.6.10 Uhlenbeck convergence

Let A_n be charge- k ASD connections, and let

$$\left([A_\infty], \sum_i m_i x_i \right)$$

be an ideal instanton.

We say that $[A_n]$ Uhlenbeck-converges to this ideal instanton if the following two conditions hold.

First, away from all bubble points, after suitable bundle identifications and gauge transformations, we have

$$A_n \longrightarrow A_\infty \quad \text{in } C_{\mathrm{loc}}^\infty(X \setminus \{x_1, \dots, x_N\}).$$

Second, the curvature measures satisfy

$$|F_{A_n}|^2 d \mathrm{vol} \rightharpoonup |F_{A_\infty}|^2 d \mathrm{vol} + 8\pi^2 \sum_i m_i \delta_{x_i}.$$

The first condition records the convergence of the connection itself away from the bubble points, and the second records the energy lost at these points.

Neither can be omitted.

2.6.11 Uhlenbeck compactification

Let $\mathcal{M}_k$ denote the charge- k ASD moduli space.

The Uhlenbeck compactness theorem shows that if one allows ideal instantons to appear in the limit, then the limit points missing from the ordinary moduli space because of bubbling can be filled in.

In the present $SU(2)$ case, its set-theoretic stratification can be written as

$$\overline{\mathcal{M}}_k^U = \coprod_{\ell=0}^k \mathcal{M}_{k-\ell} \times \text{Sym}^\ell(X).$$

The part with $\ell = 0$ is the original $\mathcal{M}_k$.

After endowing it with the appropriate Uhlenbeck topology, $\overline{\mathcal{M}}_k^U$ is compact.

Theorem 2.6.5 (Uhlenbeck compactification). Let X be a closed oriented Riemannian 4-manifold, and consider the $SU(2)$ ASD moduli space $\mathcal{M}_k$ of fixed charge k .

Then every sequence

$$[A_n] \in \mathcal{M}_k$$

has a subsequence that Uhlenbeck converges to some ideal instanton

$$\left([A_\infty], \sum_i m_i x_i \right) \in \mathcal{M}_{k-\ell} \times \text{Sym}^\ell(X).$$

Consequently, $\overline{\mathcal{M}}_k^U$ is compact with respect to the Uhlenbeck topology.

It must be emphasized that $\overline{\mathcal{M}}_k^U$ is in general not a smooth manifold with boundary.

Different ℓ correspond to different strata, and the symmetric product itself carries an additional stratified structure along the diagonal where several points coincide.

Hence the Uhlenbeck compactification is more naturally regarded as a compact stratified space.

2.6.12 The simplest bubbling example

The most basic example comes from the charge-1 instanton on S^4 .

Under stereographic projection, it can be viewed as the BPST instanton on $\mathbb{R}^4$. This solution carries a scale parameter $\lambda > 0$.

Fix a bubble center x_0 , and let

$$\lambda_n \longrightarrow 0.$$

As λ_n becomes smaller and smaller, the curvature concentrates more and more near x_0 , but the total energy always remains

$$8\pi^2.$$

On $S^4 \setminus \{x_0\}$, after a suitable gauge transformation, this sequence of instantons converges to a flat connection.

However, on the whole of S^4 there is no smooth convergence, because all of the $8\pi^2$ of curvature energy has concentrated at x_0 .

Hence its Uhlenbeck limit is

$$[A_{\text{flat}}] + x_0,$$

or, more precisely, the ideal instanton

$$([A_{\text{flat}}], x_0) \in \mathcal{M}_0 \times S^4.$$

This example shows clearly that: the non-compactness of the original charge-1 moduli space comes from the scale $\lambda \rightarrow 0$, and the Uhlenbeck compactification fills in this limit by adding a single bubble point.

2.6.13 The expected dimension of the lower strata

In the regular irreducible case, the expected dimension of the $SU(2)$ charge- k ASD moduli space is

$$d_k = 8k - 3(1 - b_1(X) + b_2^+(X)).$$

If ℓ units of instanton charge are lost, then

$$d_{k-\ell} = d_k - 8\ell.$$

On the other hand, the top dimension of $\text{Sym}^\ell(X)$ is 4ℓ .

Therefore the expected dimension of the ideal stratum

$$\mathcal{M}_{k-\ell} \times \text{Sym}^\ell(X)$$

is

$$d_{k-\ell} + 4\ell = d_k - 4\ell.$$

Hence the first bubbling stratum $\ell = 1$ has expected codimension 4.

More generally, the ℓ -bubble stratum has expected codimension 4ℓ .

The dimension count here applies only to regular irreducible strata, and away from the diagonal singularities of the symmetric product.

2.6.14 The logical structure of Uhlenbeck compactness

The core mechanism of Uhlenbeck compactness can be summarized as

$$\text{fixed ASD energy} \implies \text{only finitely many concentration points},$$

while away from these points,

$$\begin{aligned} \text{small energy} &\implies \text{Uhlenbeck Coulomb gauge} \implies W^{1,2} \text{ control} \\ &\implies \text{elliptic bootstrapping} \implies C_{\text{loc}}^\infty \text{ convergence.} \end{aligned}$$

At the concentration points, rescaling yields non-trivial instantons on S^4 , and hence energy quantization

$$8\pi^2 m_i.$$

Finally, the removable singularity theorem is used to extend the limiting connection to all of X , while the lost instanton charge is recorded in

$$\sum_i m_i x_i \in \mathrm{Sym}^\ell(X)$$

itself.

This is precisely the basic structure of the Uhlenbeck compactification.

Spin^c geometry and the Dirac operator

Seiberg–Witten theory is built on the Spin^c geometry of four-manifolds. In order to define the spinor bundle and the Dirac operator later on, we first need to understand where they come from at the level of linear algebra.

The basic algebraic object is the Clifford algebra. Given a Euclidean vector space (V, g) , the Clifford algebra encodes the metric g into an associative algebra. A spinor, then, comes from a complex representation of this Clifford algebra.

In dimension four the spinor representation splits further into a positive and a negative part. The bundles S^+ and S^- below are precisely the manifold version of this fact.

This section discusses only the linear algebra of a single Euclidean vector space.

3.1 Clifford algebras and the four-dimensional spinor representation

3.1.1 Clifford algebras

Let (V, g) be a finite-dimensional real Euclidean vector space, and write $|v|^2 = g(v, v)$.

We begin with the tensor algebra

$$T(V) = \mathbb{R} \oplus V \oplus (V \otimes V) \oplus (V^{\otimes 3}) \oplus \cdots.$$

The tensor algebra uses only the linear structure of V . In order to encode the metric g into the multiplication, we need to impose the relation $v^2 = -|v|^2$.

Definition 3.1.1 (Clifford algebra). The **Clifford algebra** of a Euclidean vector space (V, g) is defined to be

$$\text{Cl}(V, g) := T(V) / \langle v \otimes v + |v|^2 \mid v \in V \rangle.$$

When no confusion can arise we write $\text{Cl}(V)$ for short.

By definition, $v^2 = -|v|^2$ holds in $\text{Cl}(V)$. Applying the same relation to $v + w$ yields the Clifford relation.

Proposition 3.1.2 (Clifford relation). For all $v, w \in V$ we have

$$vw + wv = -2g(v, w).$$

Proof. From $(v + w)^2 = -|v + w|^2$, expanding the left-hand side and substituting $v^2 = -|v|^2$ and $w^2 = -|w|^2$ gives

$$vw + wv = -|v + w|^2 + |v|^2 + |w|^2 = -2g(v, w).$$

□

In particular, if v and w are orthogonal, then $vw = -wv$. If $e_1, \dots, e_n$ is an orthonormal basis, then $e_i^2 = -1$, and $e_i e_j = -e_j e_i$ whenever $i \neq j$.

Thus the Clifford algebra may be viewed as the algebra obtained from the tensor algebra by forcing distinct orthogonal directions to anticommute while requiring unit vectors to square to -1 .

3.1.2 The $\mathbb{Z}/2$ -grading

The tensor algebra is naturally $\mathbb{Z}$ -graded. After passing to the Clifford quotient the full integer grading is no longer preserved, but the parity of the degree still is.

Hence there is a natural decomposition

$$\text{Cl}(V) = \text{Cl}^0(V) \oplus \text{Cl}^1(V).$$

Here $\text{Cl}^0(V)$ is spanned by products of an even number of vectors and is called the even Clifford algebra; $\text{Cl}^1(V)$ is spanned by products of an odd number of vectors.

In particular, $V \subset \text{Cl}^1(V)$. Later on, this fact will imply that Clifford multiplication by a vector interchanges positive and negative spinors.

3.1.3 Complex Clifford algebras

Seiberg–Witten theory uses complex spinors, so the Clifford algebra has to be complexified.

Definition 3.1.3 (Complex Clifford algebra). Define

$$\text{Cl}^{\mathbb{C}}(V) := \text{Cl}(V) \otimes_{\mathbb{R}} \mathbb{C}.$$

If $\dim_{\mathbb{R}} V = n$, then $\dim_{\mathbb{R}} \text{Cl}(V) = 2^n$.

In dimension four, the standard classification of complex Clifford algebras gives

$$\text{Cl}_4^{\mathbb{C}} \cong M_4(\mathbb{C}), \quad \text{Cl}_4^{0,\mathbb{C}} \cong M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).$$

The dimension of the four-dimensional spinor representation below is determined precisely by these two algebra isomorphisms.

3.1.4 Clifford modules

Definition 3.1.4 (Complex Clifford module). A complex vector space Δ is called a **Clifford module** for $\text{Cl}^{\mathbb{C}}(V)$ if there is given a complex algebra homomor-

phism

$$c : \text{Cl}^{\mathbb{C}}(V) \longrightarrow \text{End}_{\mathbb{C}}(\Delta).$$

Thus every $a \in \text{Cl}^{\mathbb{C}}(V)$ gives a linear transformation $c(a)$ of Δ .

In particular, V is naturally contained in the Clifford algebra, so every $v \in V$ induces $c(v) : \Delta \rightarrow \Delta$. This action is called **Clifford multiplication**.

Since c is an algebra homomorphism, the Clifford relation on the representation space becomes

$$c(v)c(w) + c(w)c(v) = -2g(v, w)I.$$

In particular, $c(v)^2 = -|v|^2 I$.

So Clifford multiplication is not a map specified separately: it is simply the action obtained by restricting the Clifford algebra representation to the generators V .

3.1.5 The four-dimensional spinor space

We now take $V = \mathbb{R}^4$.

Since $\text{Cl}_4^{\mathbb{C}} \cong M_4(\mathbb{C})$, it has a 4-dimensional irreducible complex representation.

Definition 3.1.5 (Four-dimensional complex spinor space). Write

$$\Delta_4 \cong \mathbb{C}^4$$

for the standard irreducible complex module over $\text{Cl}_4^{\mathbb{C}}$, and call it the four-dimensional **complex spinor space**.

Here Δ_4 is still just a fixed vector space. It is not a vector bundle over a manifold. The spinor bundle below will be constructed by the associated bundle construction with Δ_4 as fiber.

3.1.6 The chirality decomposition

Take an oriented orthonormal basis e_1, e_2, e_3, e_4 of $\mathbb{R}^4$, and define the volume element

$$\omega = e_1 e_2 e_3 e_4.$$

Using the Clifford relation one checks that $\omega^2 = 1$. Hence the eigenvalues of $c(\omega)$ can only be $+1$ and -1 .

Definition 3.1.6 (Half-spinor spaces). Define

$$\Delta^+ := \ker(c(\omega) - I), \quad \Delta^- := \ker(c(\omega) + I).$$

These are called the positive and negative half-spinor spaces, respectively.

Thus

$$\Delta_4 = \Delta^+ \oplus \Delta^-,$$

and in dimension four $\dim_{\mathbb{C}} \Delta^{\pm} = 2$.

Hence one may identify $\Delta^{\pm} \cong \mathbb{C}^2$ non-canonically.

The naming of the signs depends on the orientation and chirality conventions. If the conventions are changed, Δ^+ and Δ^- may be interchanged.

3.1.7 Clifford multiplication and chirality

For every $v \in \mathbb{R}^4$ we have $v\omega = -\omega v$. Hence

$$c(v)c(\omega) = -c(\omega)c(v).$$

Let $\psi \in \Delta^+$. Since $c(\omega)\psi = \psi$,

$$c(\omega)c(v)\psi = -c(v)c(\omega)\psi = -c(v)\psi.$$

Hence $c(v)\psi \in \Delta^-$. Similarly, $c(v)$ maps Δ^- into Δ^+ .

In other words,

$$c(v) : \Delta^{\pm} \longrightarrow \Delta^{\mp}.$$

This is the linear-algebraic origin of the map $c : T^*X \otimes S^+ \rightarrow S^-$ on manifolds later on.

3.1.8 A matrix model in dimension four

To write Clifford multiplication more concretely, take the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $v = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$, define

$$Q(v) = x_4 I_2 + i(x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3).$$

The multiplication relations for the Pauli matrices give $Q(v)^* Q(v) = |v|^2 I_2$.

With respect to the decomposition $\Delta_4 = \Delta^+ \oplus \Delta^-$, define

$$c(v) = \begin{pmatrix} 0 & -Q(v)^* \\ Q(v) & 0 \end{pmatrix}.$$

A direct computation then gives $c(v)^2 = -|v|^2 I_4$, so this matrix representation does satisfy the Clifford relation.

Restricted to Δ^+ , Clifford multiplication is

$$Q(v) : \Delta^+ \longrightarrow \Delta^-.$$

Restricted to Δ^- it is

$$-Q(v)^* : \Delta^- \longrightarrow \Delta^+.$$

Thus in dimension four one can understand Clifford multiplication concretely as turning a vector $v \in \mathbb{R}^4$ into a complex-linear map from Δ^+ to Δ^- .

3.1.9 Summary of this section

The role of the Clifford algebra is to encode the Euclidean metric into a non-commutative algebra. Its basic relation is

$$vw + wv = -2g(v, w).$$

The four-dimensional complex Clifford algebra satisfies $\text{Cl}_4^{\mathbb{C}} \cong M_4(\mathbb{C})$, hence it has a 4-dimensional irreducible complex module Δ_4 .

Via the chirality operator, Δ_4 decomposes into two 2-dimensional half-spinor spaces Δ^+ and Δ^- .

Clifford multiplication is the action of the Clifford algebra on the spinor space, and Clifford multiplication by vectors interchanges the two half-spinor spaces.

Next we shall introduce $\text{Spin}(4)$, show that

$$\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-,$$

and explain how the two $SU(2)$ factors act on Δ^+ and Δ^- respectively.

3.2 $\text{Spin}(4)$ and $\text{Spin}^c(4)$

In the previous section we started from the Clifford algebra and constructed the four-dimensional complex spinor space

$$\Delta_4 = \Delta^+ \oplus \Delta^-, \quad \Delta^{\pm} \cong \mathbb{C}^2.$$

Next we need to introduce the Spin group. Its role is to lift the standard action of $SO(4)$ on $\mathbb{R}^4$ to the spinor spaces.

In dimension four there is a special isomorphism $\text{Spin}(4) \cong SU(2) \times SU(2)$. This fact makes four-dimensional spin geometry very concrete, and it is also a low-dimensional phenomenon of particular importance in the Seiberg–Witten theory that follows.

3.2.1 Spin group

Let (V, g) be an n -dimensional Euclidean vector space. We have already defined the Clifford algebra $\text{Cl}(V)$ and its even part $\text{Cl}^0(V)$.

The Spin group can be defined directly inside the Clifford algebra.

Definition 3.2.1 (Spin group). Define

$$\text{Spin}(V) := \{v_1 \cdots v_{2k} \in \text{Cl}^0(V) \mid |v_1| = \cdots = |v_{2k}| = 1\}.$$

That is, $\text{Spin}(V)$ is generated by Clifford products of an even number of unit vectors. When $V = \mathbb{R}^n$ one usually writes $\text{Spin}(n)$.

Although the definition takes place inside the Clifford algebra, the Spin group acts naturally on the original vector space V . For $a \in \text{Spin}(V)$ define

$$\lambda(a)(v) = av a^{-1}.$$

Since a is an even Clifford element, this conjugation action preserves $V \subset \text{Cl}(V)$, and it also preserves the Euclidean inner product.

The standard theory of Clifford algebras gives a group homomorphism

$$\lambda : \text{Spin}(n) \longrightarrow SO(n).$$

Theorem 3.2.2. The map

$$\lambda : \text{Spin}(n) \longrightarrow SO(n)$$

is a double cover, and there is a short exact sequence

$$1 \longrightarrow \{\pm 1\} \longrightarrow \text{Spin}(n) \xrightarrow{\lambda} SO(n) \longrightarrow 1.$$

Hence every oriented orthogonal transformation has exactly two lifts in the Spin group, and the two differ by the central element -1 .

One point requires care here: $\text{Spin}(n)$ is not another matrix representation of $SO(n)$, but rather its non-trivial double cover.

3.2.2 How the Spin group acts on spinors

Since

$$\text{Spin}(4) \subset \text{Cl}_4^0,$$

and Δ_4 is a module over the Clifford algebra, restricting the Clifford representation to the Spin group gives

$$\rho : \text{Spin}(4) \longrightarrow GL(\Delta_4).$$

Since the Spin group lies in the even part of the Clifford algebra, its action preserves the chirality decomposition. Hence we obtain two representations

$$\rho^\pm : \text{Spin}(4) \longrightarrow GL(\Delta^\pm).$$

After choosing a Hermitian metric compatible with the Clifford structure, these representations in fact take values in $U(2)$.

ρ^+ and ρ^- are called **half-spin representations**.

Thus there are three distinct but related representations:

$$\text{Spin}(4) \rightarrow SO(4), \quad \text{Spin}(4) \rightarrow U(\Delta^+), \quad \text{Spin}(4) \rightarrow U(\Delta^-).$$

The first representation describes how the Spin group acts on vectors, and the last two describe how it acts on spinors.

When we define the spinor bundle later on, it is precisely the last two representations that are used as the representations in the associated bundle construction.

3.2.3 Quaternions and $SU(2)$

The four-dimensional Spin group can be described concretely in terms of quaternions.

Write the quaternion algebra as

$$\mathbb{H} = \{x_4 + x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k} \mid x_i \in \mathbb{R}\}.$$

As a real vector space there is a natural identification

$$\mathbb{H} \cong \mathbb{R}^4.$$

Quaternions come with a conjugation

$$\bar{q} = x_4 - x_1\mathbf{i} - x_2\mathbf{j} - x_3\mathbf{k},$$

and define the norm

$$|q|^2 = q\bar{q}.$$

The unit quaternions form a group

$$Sp(1) = \{q \in \mathbb{H} \mid |q| = 1\}.$$

This group is isomorphic to $SU(2)$.

Concretely, send $q = x_4 + x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}$ to the matrix

$$Q(q) = x_4 I_2 + i(x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3),$$

where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices of the previous section.

We have seen above that

$$Q(q)^* Q(q) = |q|^2 I_2.$$

When $|q| = 1$, $Q(q)$ is a unitary matrix. On the other hand,

$$\det Q(q) = |q|^2 = 1.$$

Hence $Q(q) \in SU(2)$.

Conversely, every $SU(2)$ matrix can be written uniquely in this form. Thus we obtain a group isomorphism

$$Sp(1) \cong SU(2).$$

3.2.4 $Spin(4) \cong SU(2)_+ \times SU(2)_-$

The four-dimensional Spin group has a special decomposition

$$Spin(4) \cong Sp(1) \times Sp(1) \cong SU(2)_+ \times SU(2)_-.$$

Here the subscripts $+$ and $-$ serve to distinguish two different $SU(2)$ factors.

Under this isomorphism, the half-spin representations take a very simple form.

If we write

$$g = (A_+, A_-) \in SU(2)_+ \times SU(2)_-,$$

then

$$\rho^+(A_+, A_-) = A_+, \quad \rho^-(A_+, A_-) = A_-.$$

Thus Δ^+ sees only the first $SU(2)$ factor, while Δ^- sees only the second factor.

This explains the origin of the notation $SU(2)_+$ and $SU(2)_-$.

In particular,

$$\Delta^+ \cong \mathbb{C}^2$$

is the standard two-dimensional representation of $SU(2)_+$, while

$$\Delta^- \cong \mathbb{C}^2$$

is the standard two-dimensional representation of $SU(2)_-$.

3.2.5 Vector representation

We now explain how the vector representation of Spin(4) on $\mathbb{R}^4$ is related to the two half-spin representations.

In the previous section, every $v \in \mathbb{R}^4$, via Clifford multiplication, gave a complex-linear map

$$c(v) : \Delta^+ \longrightarrow \Delta^-.$$

Thus we may regard $\mathbb{R}^4$ as a 4-dimensional real subspace of $\text{Hom}_{\mathbb{C}}(\Delta^+, \Delta^-)$.

Under the matrix convention adopted above, v corresponds to the matrix $Q(v)$.

If

$$g = (A_+, A_-) \in SU(2)_+ \times SU(2)_-,$$

then the vector representation can be written as

$$Q(v) \longmapsto A_- Q(v) A_+^{-1}.$$

That is,

$$Q(\lambda(g)v) = A_- Q(v) A_+^{-1}.$$

This formula uses both $SU(2)$ factors at the same time, in sharp contrast with the half-spin representations, which use only one factor.

If

$$(A_+, A_-) = (-I, -I),$$

then

$$(-I)Q(v)(-I)^{-1} = Q(v).$$

So $(-I, -I)$ acts trivially in the vector representation.

Hence

$$\ker \lambda = \{(I, I), (-I, -I)\},$$

and once again we obtain

$$\text{Spin}(4)/\{\pm(1, 1)\} \cong SO(4).$$

On the other hand, $(-I, -I)$ acts as $-I$ on both $\Delta^{\pm}$. So the half-spin representations do not descend to representations of $SO(4)$.

This is exactly why, on a general oriented Riemannian manifold, the $SO(4)$ frame bundle alone is still not enough to construct the spinor bundle directly.

3.2.6 Equivariance of Clifford multiplication

Clifford multiplication is compatible with the action of the Spin group.

Concretely, for any $g \in \text{Spin}(4)$, $v \in \mathbb{R}^4$ and $\psi \in \Delta^+$, we have

$$c(\lambda(g)v)\rho^+(g)\psi = \rho^-(g)c(v)\psi.$$

In the matrix model of $\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-$, this formula is precisely

$$Q(\lambda(g)v)A_+ = A_- Q(v).$$

It follows from the relation obtained above

$$Q(\lambda(g)v) = A_- Q(v) A_+^{-1}.$$

This equivariance property is very important.

Once we construct the associated bundles on the manifold later on, it is precisely because Clifford multiplication satisfies this equivariance that the map at a single point

$$\mathbb{R}^4 \otimes \Delta^+ \longrightarrow \Delta^-$$

can be glued into a global bundle map

$$T^*X \otimes S^+ \longrightarrow S^-.$$

3.2.7 The definition of $\text{Spin}^c(4)$

A spin structure requires lifting the $SO(4)$ frame bundle to $\text{Spin}(4)$. But a general oriented 4-manifold need not admit a spin structure.

Seiberg–Witten theory uses the slightly larger structure group $\text{Spin}^c(4)$.

Definition 3.2.3 ($\text{Spin}^c(4)$). Define

$$\text{Spin}^c(4) := \frac{\text{Spin}(4) \times U(1)}{\{(1, 1), (-1, -1)\}}.$$

Here the $\mathbb{Z}/2$ in the quotient is the diagonal embedding

$$\{\pm 1\} \longrightarrow \text{Spin}(4) \times U(1), \quad \varepsilon \longmapsto (\varepsilon, \varepsilon).$$

Thus

$$(g, z) \sim (-g, -z).$$

The corresponding equivalence class is denoted

$$[g, z].$$

Note that the map from $\text{Spin}^c(4)$ to $SO(4)$ is no longer a double cover.

Define

$$\lambda^c : \text{Spin}^c(4) \longrightarrow SO(4), \quad [g, z] \longmapsto \lambda(g).$$

This map is well-defined, since g and $-g$ have the same image in $SO(4)$.

Its kernel is isomorphic to $U(1)$, so there is a short exact sequence

$$1 \longrightarrow U(1) \longrightarrow \text{Spin}^c(4) \xrightarrow{\lambda^c} SO(4) \longrightarrow 1.$$

Thus, compared with the Spin case, above each orthogonal frame there are no longer just two lifts, but rather a $U(1)$ family.

3.2.8 Spin^c half-spin representations

The Spin group already has representations $\rho^\pm$ on $\Delta^\pm$.

In order to obtain representations of $\text{Spin}^c(4)$, we let the extra $U(1)$ factor act on spinors by scalar multiplication.

Define

$$\rho_c^\pm : \text{Spin}^c(4) \longrightarrow U(\Delta^\pm)$$

by

$$\rho_c^\pm([g, z]) = z \rho^\pm(g).$$

First we need to check that this definition is compatible with the quotient.

Indeed, in the spinor representation we have

$$\rho^\pm(-g) = -\rho^\pm(g).$$

Hence

$$(-z)\rho^\pm(-g) = z\rho^\pm(g).$$

Thus (g, z) and $(-g, -z)$ give exactly the same linear transformation, so that $\rho_c^\pm$ is well-defined.

Using $\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-$, if we write an element as

$$[A_+, A_-, z],$$

then

$$\rho_c^+([A_+, A_-, z]) = zA_+,$$

while

$$\rho_c^-([A_+, A_-, z]) = zA_-.$$

These two representations will be used later to define the positive and negative spinor bundles.

3.2.9 Determinant character

$\text{Spin}^c(4)$ has another very important one-dimensional representation.

At first glance one might try to define $[g, z] \mapsto z$, but this map is not well-defined, because

$$[g, z] = [-g, -z]$$

while z and $-z$ are in general distinct.

After squaring, this problem disappears.

Definition 3.2.4 (Determinant character). Define

$$\delta : \text{Spin}^c(4) \longrightarrow U(1), \quad \delta([g, z]) = z^2.$$

Since $(-z)^2 = z^2$, this definition is well-defined.

This character agrees exactly with the determinant of the half-spin representations.

Indeed, in dimension four both $\Delta^\pm$ are 2-dimensional complex vector spaces. If $g = (A_+, A_-)$, then $A_\pm \in SU(2)$, so $\det A_\pm = 1$.

Hence

$$\det \rho_c^\pm([g, z]) = \det(zA_\pm) = z^2 = \delta([g, z]).$$

In other words,

$$\det \circ \rho_c^+ = \det \circ \rho_c^- = \delta.$$

The associated complex line bundle constructed from δ below is exactly the determinant line bundle of the Spin^c structure.

3.2.10 Clifford equivariance in the Spin^c case

Adding the extra $U(1)$ factor does not destroy the equivariance of Clifford multiplication.

Let $u = [g, z] \in \text{Spin}^c(4)$. Since $\lambda^c(u) = \lambda(g)$, while the $U(1)$ factor does not act on the vector representation, we have

$$c(\lambda^c(u)v)\rho_c^+(u) = \rho_c^-(u)c(v).$$

Indeed, the left-hand side is

$$c(\lambda(g)v)z\rho^+(g),$$

which by Spin equivariance may be written as

$$z\rho^-(g)c(v),$$

and this is precisely

$$\rho_c^-(u)c(v).$$

Thus Clifford multiplication at a single point,

$$\mathbb{R}^4 \otimes \Delta^+ \longrightarrow \Delta^-$$

is in fact a $\text{Spin}^c(4)$ -equivariant map.

This property will be used directly in the next section to construct Clifford multiplication on manifolds.

3.2.11 Summary of this section

The Spin group is a Lie group defined inside the even part of the Clifford algebra, and through the double cover

$$\text{Spin}(4) \longrightarrow SO(4)$$

it controls the lifting of oriented orthogonal transformations.

In dimension four there is a special isomorphism

$$\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-.$$

The two factors give respectively the half-spin representations on Δ^+ and on Δ^- , while the vector representation uses both factors at once:

$$Q(v) \longmapsto A_- Q(v) A_+^{-1}.$$

In order to handle a general oriented 4-manifold, we enlarge the Spin group to

$$\text{Spin}^c(4) = \frac{\text{Spin}(4) \times U(1)}{\mathbb{Z}/2}.$$

It has two basic families of representations: the half-spin representations

$$\rho_c^\pm([g, z]) = z\rho^\pm(g),$$

and the determinant character

$$\delta([g, z]) = z^2.$$

The next step is to turn these fixed representation spaces into genuine vector bundles on a manifold.

Concretely, starting from the oriented orthonormal frame bundle of X we shall define a Spin^c structure, and then use the associated bundle construction to obtain

$$S^+, \quad S^-, \quad L = \det S^+,$$

and to globalize Clifford multiplication at a single point to

$$T^*X \otimes S^+ \longrightarrow S^-.$$

3.3 Spin^c structures on 4-manifolds

The objects discussed in the previous section, $\Delta^\pm$, $\rho_c^\pm$ and the determinant character δ , are all merely fixed vector spaces and fixed Lie group representations. In order to obtain genuine spinor fields on a manifold X , we need to globalize these representations by the associated bundle construction.

Let (X, g) be an oriented Riemannian 4-manifold. The Riemannian metric and the orientation give the oriented orthonormal frame bundle

$$P_{SO}(X) \longrightarrow X,$$

which is a principal $SO(4)$ -bundle.

The role of a Spin^c structure is to lift this $SO(4)$ -bundle along $\text{Spin}^c(4) \rightarrow SO(4)$ to a principal $\text{Spin}^c(4)$ -bundle.

3.3.1 The definition of a Spin^c structure

Recall the group homomorphism from the previous section

$$\lambda^c : \text{Spin}^c(4) \longrightarrow SO(4).$$

Definition 3.3.1 (Spin^c structure). A Spin^c structure on X is a principal $\text{Spin}^c(4)$ -bundle

$$P_{\mathfrak{s}} \longrightarrow X,$$

together with a smooth map covering id_X

$$\Lambda_{\mathfrak{s}} : P_{\mathfrak{s}} \longrightarrow P_{SO}(X),$$

satisfying the equivariance condition

$$\Lambda_{\mathfrak{s}}(pu) = \Lambda_{\mathfrak{s}}(p)\lambda^c(u)$$

for all $p \in P_{\mathfrak{s}}$ and $u \in \text{Spin}^c(4)$.

Thus $\Lambda_{\mathfrak{s}}$ is not an ordinary principal bundle morphism, since the two principal bundles have different structure groups. It is an equivariant map with respect to the group homomorphism λ^c .

Equivalently, one may first construct, by extension of structure group, the principal $SO(4)$ -bundle

$$P_{\mathfrak{s}} \times_{\lambda^c} SO(4),$$

and then the condition for a Spin^c structure is precisely the requirement that

$$P_{\mathfrak{s}} \times_{\lambda^c} SO(4) \cong P_{SO}(X).$$

Hence a Spin^c structure may be understood as follows: the given oriented orthonormal frame bundle can be recovered from some principal $\text{Spin}^c(4)$ -bundle by a change of structure group.

3.3.2 Isomorphisms of Spin^c structures

Two Spin^c structures should not be regarded as the same merely because their underlying principal bundles are isomorphic; the isomorphism must also be compatible with their lifts to the frame bundle.

Definition 3.3.2. Let $(P_{\mathfrak{s}}, \Lambda_{\mathfrak{s}})$ and $(P_{\mathfrak{t}}, \Lambda_{\mathfrak{t}})$ be two Spin^c structures. If there is a principal $\text{Spin}^c(4)$ -bundle isomorphism

$$F : P_{\mathfrak{s}} \longrightarrow P_{\mathfrak{t}}$$

satisfying

$$\Lambda_{\mathfrak{t}} \circ F = \Lambda_{\mathfrak{s}},$$

then the two Spin^c structures are said to be isomorphic.

Write the set of all isomorphism classes as

$$\text{Spin}^c(X).$$

Note here that $\text{Spin}^c(X)$ is at first only a set, and carries no natural group structure.

We shall see later that when it is non-empty, $H^2(X; \mathbb{Z})$ acts naturally on $\text{Spin}^c(X)$, and that this action is free and transitive. Hence $\text{Spin}^c(X)$ is an $H^2(X; \mathbb{Z})$ -torsor.

3.3.3 Spinor bundles

Fix a Spin^c structure $\mathfrak{s}$.

The previous section already produced two half-spin representations

$$\rho_c^{\pm} : \text{Spin}^c(4) \longrightarrow U(\Delta^{\pm}).$$

Hence we may use the associated bundle construction to define two complex rank-2 vector bundles.

Definition 3.3.3 (Spinor bundles). Define

$$S_{\mathfrak{s}}^{\pm} := P_{\mathfrak{s}} \times_{\rho_c^{\pm}} \Delta^{\pm}.$$

These are called the positive and negative spinor bundles associated with $\mathfrak{s}$.

Since $\Delta^\pm \cong \mathbb{C}^2$, both bundles are complex rank-2 Hermitian vector bundles.

Once $\mathfrak{s}$ is fixed one usually writes S^+ and S^- for short.

Their smooth sections

$$\psi \in \Gamma(S^\pm)$$

are called spinor fields.

We must distinguish once more: $\Delta^\pm$ are fixed representation spaces, whereas $S^\pm$ are vector bundles defined over all of X .

3.3.4 Determinant line bundle

The previous section defined the determinant character $\delta : \text{Spin}^c(4) \rightarrow U(1)$, with the formula $\delta([g, z]) = z^2$ (Definition 3.2.4).

Hence we can likewise construct an associated complex line bundle.

Definition 3.3.4 (Determinant line bundle). Define

$$L_{\mathfrak{s}} := P_{\mathfrak{s}} \times_{\delta} \mathbb{C}.$$

It is called the **determinant line bundle** of $\mathfrak{s}$.

Once $\mathfrak{s}$ is fixed one usually writes L for short.

The name comes from the relation between $L_{\mathfrak{s}}$ and the determinant of the spinor bundles.

Proposition 3.3.5. There is a natural isomorphism

$$L_{\mathfrak{s}} \cong \det S_{\mathfrak{s}}^+ \cong \det S_{\mathfrak{s}}^-.$$

Proof. For the positive half-spin representation, if we write $u = [A_+, A_-, z]$, then $\rho_c^+(u) = zA_+$.

Since $A_+ \in SU(2)$, we have $\det A_+ = 1$. Hence

$$\det \rho_c^+(u) = z^2 = \delta(u).$$

Therefore the representations $\det \circ \rho_c^+$ and δ are exactly the same, and the corresponding associated line bundles are naturally isomorphic as well.

The proof for S^- is exactly the same. $\square$

This is also the reason why, later on, the $U(1)$ connection in the Seiberg–Witten equations is taken on $\det S^+$.

3.3.5 The transition function description

The above construction can also be understood in terms of transition functions.

Take a good cover $\{U_\alpha\}$, and choose local sections of $P_{\mathfrak{s}}$. Let their transition functions be

$$\tilde{g}_{\alpha\beta} : U_\alpha \cap U_\beta \longrightarrow \text{Spin}^c(4).$$

Writing

$$\tilde{g}_{\alpha\beta} = [g_{\alpha\beta}, z_{\alpha\beta}]$$

is only a local notation; since $\text{Spin}^c(4)$ is itself a quotient, $g_{\alpha\beta}$ and $z_{\alpha\beta}$ cannot necessarily be chosen separately and canonically over the whole overlap.

The transition functions of $S^\pm$ are $\rho_c^\pm(\tilde{g}_{\alpha\beta})$, while the transition functions of the determinant line are

$$\delta(\tilde{g}_{\alpha\beta}).$$

If one chooses local representatives $[g_{\alpha\beta}, z_{\alpha\beta}]$, then the determinant transition function is written $z_{\alpha\beta}^2$.

Hence the “square” does not appear by accident. It comes from the

$$(g, z) \sim (-g, -z)$$

in the definition of $\text{Spin}^c(4)$.

The individual z cannot descend to the quotient, but z^2 can.

3.3.6 Clifford multiplication on manifolds

At a single point we already have the $\text{Spin}^c(4)$ -equivariant map

$$\mathbb{R}^4 \otimes \Delta^+ \longrightarrow \Delta^-.$$

We now globalize it by means of $P_\mathfrak{s}$.

Since

$$P_\mathfrak{s} \times_{\lambda^c} \mathbb{R}^4 \cong TX,$$

and using the Riemannian metric to identify TX with T^*X , we obtain a natural bundle map

$$c : T^*X \otimes S^+ \longrightarrow S^-.$$

There likewise exists

$$c : T^*X \otimes S^- \longrightarrow S^+.$$

Here c is still called Clifford multiplication.

Its fiberwise Clifford relation is

$$c(\alpha)c(\beta) + c(\beta)c(\alpha) = -2g(\alpha, \beta)I.$$

In particular, for every $\alpha \in T_x^*X$, we have $c(\alpha)^2 = -|\alpha|^2 I$.

Thus Clifford multiplication is a pointwise algebraic structure determined jointly by the Riemannian metric and the Spin^c structure.

It does not by itself use any connection.

3.3.7 $c_1(L_\mathfrak{s})$ and $w_2(TX)$

The first Chern class of the determinant line bundle and the second Stiefel–Whitney class of the tangent bundle satisfy a basic relation.

Proposition 3.3.6. For any Spin^c structure $\mathfrak{s}$, we have

$$c_1(L_\mathfrak{s}) \equiv w_2(TX) \pmod{2}.$$

Hence $c_1(L_{\mathfrak{s}})$ is an integral lift of $w_2(TX)$.

This relation can be seen to originate in the transition functions.

Let $g_{\alpha\beta}$ be the transition functions of $P_{SO}(X)$. On each double overlap one can make a local choice of a lift $\widehat{g}_{\alpha\beta}$ in $\text{Spin}(4)$.

Since a genuine spin structure does not exist in general, these lifts need not satisfy the cocycle condition on triple overlaps. Their discrepancy is

$$\widehat{g}_{\alpha\beta}\widehat{g}_{\beta\gamma}\widehat{g}_{\gamma\alpha} = \varepsilon_{\alpha\beta\gamma}, \quad \varepsilon_{\alpha\beta\gamma} \in \{\pm 1\}.$$

This $\mathbb{Z}/2$ -valued Čech 2-cocycle is precisely a representative of $w_2(TX)$.

A Spin^c lift, on the other hand, can be written as $[\widehat{g}_{\alpha\beta}, z_{\alpha\beta}]$. The genuine cocycle condition requires that $z_{\alpha\beta}z_{\beta\gamma}z_{\gamma\alpha}$ compensate exactly for the above signs $\varepsilon_{\alpha\beta\gamma}$.

On the other hand, the transition functions of the determinant line are $z_{\alpha\beta}^2$. The first Chern class obtained from them, reduced mod 2, recovers precisely the class represented by $\varepsilon_{\alpha\beta\gamma}$.

This is the transition-function explanation of $c_1(L_{\mathfrak{s}}) \equiv w_2(TX) \pmod{2}$.

3.3.8 Existence of Spin and Spin^c

A spin structure and a Spin^c structure have different obstructions.

An oriented manifold admits a spin structure if and only if

$$w_2(TX) = 0.$$

The existence of a Spin^c structure, by contrast, only requires that $w_2(TX)$ lift to an integral cohomology class.

Equivalently, if

$$\beta : H^2(X; \mathbb{Z}_2) \longrightarrow H^3(X; \mathbb{Z})$$

is the Bockstein homomorphism, then the obstruction is

$$W_3(TX) := \beta(w_2(TX)).$$

Hence

$$X \text{ admits a } \text{Spin}^c \text{ structure} \iff W_3(TX) = 0.$$

For an oriented smooth 4-manifold there is the following standard result.

Theorem 3.3.7. Every oriented smooth 4-manifold admits a Spin^c structure.

Thus in four-dimensional Seiberg–Witten theory a Spin^c structure can always be chosen; what really has to be recorded is which $\mathfrak{s}$ has been chosen.

3.3.9 Twisting by a line bundle

Different Spin^c structures can be related to one another by complex line bundles.

Let $E \rightarrow X$ be a complex line bundle, with transition functions

$$h_{\alpha\beta} : U_{\alpha} \cap U_{\beta} \longrightarrow U(1).$$

There is a central embedding

$$\iota : U(1) \longrightarrow \text{Spin}^c(4), \quad z \longmapsto [1, z].$$

If the transition functions of $\mathfrak{s}$ are $\tilde{g}_{\alpha\beta}$, define new transition functions

$$\tilde{g}'_{\alpha\beta} = \tilde{g}_{\alpha\beta} \iota(h_{\alpha\beta}).$$

Since $\iota(U(1))$ lies in the center of $\text{Spin}^c(4)$, and $h_{\alpha\beta}$ satisfies the cocycle condition of a line bundle, $\tilde{g}'_{\alpha\beta}$ still satisfies the cocycle condition.

On the other hand,

$$\lambda^c \circ \iota = 1.$$

Hence the image of $\tilde{g}'_{\alpha\beta}$ in $SO(4)$ is exactly the same as that of $\tilde{g}_{\alpha\beta}$, so it still lifts the same oriented frame bundle.

This yields a new Spin^c structure, denoted

$$\mathfrak{s} \otimes E.$$

3.3.10 The effect of twisting on the determinant line

Since the determinant character satisfies

$$\delta(\iota(z)) = z^2,$$

after twisting the determinant transition functions acquire an extra factor $h_{\alpha\beta}^2$.

Hence there is a natural isomorphism

$$L_{\mathfrak{s} \otimes E} \cong L_{\mathfrak{s}} \otimes E^{\otimes 2}.$$

Taking first Chern classes gives

$$c_1(L_{\mathfrak{s} \otimes E}) = c_1(L_{\mathfrak{s}}) + 2c_1(E).$$

This explains why all determinant classes take the same value $w_2(TX)$ after reduction mod 2.

3.3.11 The $H^2(X; \mathbb{Z})$ -torsor

Isomorphism classes of complex line bundles are classified by the first Chern class:

$$\text{Pic}_{\text{top}}(X) \cong H^2(X; \mathbb{Z}).$$

Hence the twisting construction gives an action of $H^2(X; \mathbb{Z})$ on $\text{Spin}^c(X)$.

The standard classification theorem shows that this action is free and transitive.

Theorem 3.3.8. If $\text{Spin}^c(X)$ is non-empty, then it is an $H^2(X; \mathbb{Z})$ -torsor.

By a torsor we mean that for any two Spin^c structures $\mathfrak{s}$ and $\mathfrak{t}$, there exists a unique line bundle E such that

$$\mathfrak{t} \cong \mathfrak{s} \otimes E.$$

Hence once a reference structure $\mathfrak{s}_0$ is chosen arbitrarily, one obtains a non-canonical bijection

$$H^2(X; \mathbb{Z}) \longrightarrow \text{Spin}^c(X), \quad a \longmapsto \mathfrak{s}_0 \otimes E_a,$$

where $c_1(E_a) = a$.

But this identification depends on the choice of $\mathfrak{s}_0$. So $\text{Spin}^c(X)$ itself has no natural “zero element”, and this is exactly what it means for it to be a torsor rather than a natural Abelian group.

3.3.12 Does the determinant class determine the Spin^c structure?

By the twisting formula, if $\mathfrak{t} = \mathfrak{s} \otimes E$, then

$$c_1(L_{\mathfrak{t}}) - c_1(L_{\mathfrak{s}}) = 2c_1(E).$$

Hence if two Spin^c structures have the same determinant first Chern class, then

$$2c_1(E) = 0.$$

So the determinant class determines the Spin^c structure uniquely only when $H^2(X; \mathbb{Z})$ has no 2-torsion.

In particular, for a simply connected closed oriented 4-manifold, $H^2(X; \mathbb{Z})$ is torsion-free. Hence in this case one can use

$$c_1(L_{\mathfrak{s}})$$

to label Spin^c structures.

In general, however, one must distinguish a Spin^c structure $\mathfrak{s}$ from its determinant class $c_1(L_{\mathfrak{s}})$.

3.3.13 Basic examples

Example 3.3.9 (S^4). Since $H^2(S^4; \mathbb{Z}) = 0$, while S^4 admits a Spin^c structure, the torsor property immediately shows that S^4 has only one isomorphism class of Spin^c structures.

The first Chern class of its determinant line bundle is 0.

Example 3.3.10 ($\mathbb{CP}^2$). Let

$$H \in H^2(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}$$

be the positive generator.

We have

$$w_2(T\mathbb{CP}^2) \equiv H \pmod{2}.$$

Hence any Spin^c determinant class must be an integral lift of w_2 , that is, of the form

$$(2k+1)H, \quad k \in \mathbb{Z}.$$

Since $H^2(\mathbb{CP}^2; \mathbb{Z})$ contains no 2-torsion, these odd classes correspond one-to-one to Spin^c structures.

Example 3.3.11 (K3). The K3 surface is a spin manifold, hence

$$w_2(TK3) = 0.$$

So the Spin^c determinant classes are all even classes.

In particular, the spin structure itself induces a Spin^c structure, whose determinant line bundle is trivial, so that

$$c_1(L) = 0.$$

3.3.14 Summary of this section

A Spin^c structure is first of all a lift of the oriented orthonormal frame bundle along

$$\text{Spin}^c(4) \rightarrow SO(4).$$

Once $\mathfrak{s}$ is fixed, the half-spin representations give the spinor bundles $S^\pm$, while the determinant character gives the complex line bundle $L_{\mathfrak{s}}$.

These bundles satisfy

$$L_{\mathfrak{s}} \cong \det S^+ \cong \det S^-.$$

Clifford multiplication at a single point globalizes by equivariance to

$$c : T^*X \otimes S^\pm \longrightarrow S^\mp.$$

The determinant class satisfies

$$c_1(L_{\mathfrak{s}}) \equiv w_2(TX) \pmod{2}.$$

Different Spin^c structures are related by twisting with complex line bundles, and satisfy

$$L_{\mathfrak{s} \otimes E} \cong L_{\mathfrak{s}} \otimes E^{\otimes 2}.$$

Hence $\text{Spin}^c(X)$ is an $H^2(X; \mathbb{Z})$ -torsor.

At this point the global algebro-geometric objects needed for Seiberg–Witten theory have all been constructed:

$$S^+, \quad S^-, \quad L, \quad c : T^*X \otimes S^\pm \rightarrow S^\mp.$$

The next step is to introduce a connection on these bundles. The Levi–Civita connection together with a unitary connection on the determinant line bundle L will jointly induce a spinor connection on $S^\pm$, and only then can the Spin^c Dirac operator be defined.

3.4 Spin^c connections and the Dirac operator

In the previous section we fixed a Spin^c structure $\mathfrak{s}$, and thereby obtained the spinor bundles $S^\pm$, the determinant line bundle L , and Clifford multiplication

$$c : T^*X \otimes S^\pm \longrightarrow S^\mp.$$

These objects are still merely pointwise algebraic structures. In order to define the Dirac operator, we need a connection that is able to differentiate a spinor field.

This connection is determined jointly by two pieces of geometric data: partly by the Levi–Civita connection of the Riemannian metric, and partly by a unitary connection on the determinant line bundle L .

3.4.1 Why one has to choose a $U(1)$ connection

The Riemannian metric g gives the Levi–Civita connection on the tangent bundle TX ,

$$\nabla^{LC}.$$

Equivalently, it gives a principal $SO(4)$ -connection on the oriented orthonormal frame bundle $P_{SO}(X)$.

If X carries a genuine spin structure, the Levi–Civita connection can be lifted to the spin bundle, and thus directly induces a connection on the spinor bundle.

For a Spin^c structure, however, there is an extra $U(1)$ direction in the structure group. The Levi–Civita connection determines only the $SO(4)$ part, and does not determine this extra $U(1)$ part.

Hence one also has to choose a unitary connection on the determinant line bundle

$$L = \det S^+$$

In what follows we denote this connection by A throughout.

Thus A in Seiberg–Witten theory is not a connection on the tangent bundle, but rather a unitary connection on the determinant line bundle L .

3.4.2 The Lie algebra of $\mathfrak{spin}^c(4)$

In order to understand how the two connections combine, we first look at the level of Lie algebras.

Since

$$\text{Spin}^c(4) = \frac{\text{Spin}(4) \times U(1)}{\mathbb{Z}/2},$$

and taking the quotient by a discrete subgroup does not change the Lie algebra, we have

$$\mathfrak{spin}^c(4) \cong \mathfrak{spin}(4) \oplus i\mathbb{R}.$$

At the level of Lie algebras the group homomorphism $\lambda^c : \text{Spin}^c(4) \rightarrow SO(4)$ gives

$$d\lambda^c : \mathfrak{spin}^c(4) \longrightarrow \mathfrak{so}(4),$$

which is an isomorphism on the $\mathfrak{spin}(4)$ part and vanishes on the $i\mathbb{R}$ part.

On the other hand, the differential of the determinant character $\delta([g, z]) = z^2$ is

$$d\delta : \mathfrak{spin}^c(4) \longrightarrow i\mathbb{R}.$$

If the central direction is written $it \in i\mathbb{R}$, then

$$d\delta(it) = 2it.$$

The coefficient 2 appearing here is precisely the origin of the $\frac{1}{2}A$ in the spinor connection below.

3.4.3 Spin^c connections

Fix the Spin^c structure (P_s, Λ_s) of Definition 3.3.1. Write ω^{LC} for the principal Levi–Civita connection form on $P_{SO}(X)$.

Let $P_{U(1)}(L)$ be the bundle of unit vectors in the determinant line $L = P_s \times_\delta \mathbb{C}$. There is a canonical map

$$\Delta_s : P_s \longrightarrow P_{U(1)}(L), \quad \Delta_s(p) = [p, 1],$$

satisfying $\Delta_s(pu) = \Delta_s(p)\delta(u)$. For a unitary connection A on L , denote its principal connection form on $P_{U(1)}(L)$ by $\omega_A^{\det}$. This distinguishes the global principal form from the local connection form, which we continue to denote by A .

Definition 3.4.1 (**Spin^c connection**). A Spin^c **connection** compatible with the Riemannian metric is a principal connection form

$$\omega \in \Omega^1(P_s; \mathfrak{spin}^c(4))$$

whose induced $SO(4)$ connection is Levi–Civita:

$$d\lambda^c \circ \omega = \Lambda_s^* \omega^{LC}.$$

Given a unitary connection A on L , the corresponding connection ω_A is further characterized by

$$\begin{aligned} d\lambda^c \circ \omega_A &= \Lambda_s^* \omega^{LC}, \\ d\delta \circ \omega_A &= \Delta_s^* \omega_A^{\det}. \end{aligned}$$

Both equalities are identities of Lie-algebra-valued 1-forms on P_s . The pullbacks are essential: the original Levi–Civita and determinant connection forms live on different principal bundles.

To see that these conditions determine a connection uniquely, use the Lie algebra splitting of Section 3.4.2. Set

$$j = d\lambda^c|_{\mathfrak{spin}(4)} : \mathfrak{spin}(4) \longrightarrow \mathfrak{so}(4).$$

Then j is an isomorphism, and so is

$$\begin{aligned} J = (d\lambda^c, d\delta) : \mathfrak{spin}^c(4) &\longrightarrow \mathfrak{so}(4) \oplus i\mathbb{R}, \\ (\xi, it) &\longmapsto (j(\xi), 2it). \end{aligned}$$

The required form is therefore explicitly

$$\omega_A = J^{-1}(\Lambda_s^* \omega^{LC}, \Delta_s^* \omega_A^{\det}).$$

It is indeed a principal connection: equivariance of Λ_s and Δ_s gives

$$R_u^* \omega_A = \text{Ad}_{u^{-1}} \omega_A, \quad \omega_A(\zeta^\#) = \zeta \quad (\zeta \in \mathfrak{spin}^c(4)),$$

which are the axioms in Definition 2.1.7. Here $\zeta^\#$ denotes the fundamental vector field generated by ζ . Conversely, applying J to any connection satisfying the two projection identities gives the same pair of pulled-back forms, so the connection is unique.

In a local section $s : U \rightarrow P_s$, use the induced orthonormal frame $\Lambda_s \circ s$ and unit frame $\Delta_s \circ s$ of L . Set

$$\omega_U^{LC} = (\Lambda_s \circ s)^* \omega^{LC}, \quad A_U = (\Delta_s \circ s)^* \omega_A^{\det}.$$

The same definition then takes the local form

$$s^* \omega_A = \left(j^{-1}(\omega_U^{LC}), \frac{1}{2} A_U \right) \in \Omega^1(U; \mathfrak{spin}(4) \oplus i\mathbb{R}).$$

Thus, with the metric and Spin^c structure fixed, compatible Spin^c connections correspond bijectively to unitary connections on L . Applying the half-spin representations to this principal connection gives the spinor covariant derivatives in the next section.

3.4.4 The induced connection on the spinor bundles

Since

$$S^\pm = P_s \times_{\rho_c^\pm} \Delta^\pm,$$

the principal Spin^c connection induces Hermitian connections on $S^\pm$ via the representations $\rho_c^\pm$, respectively.

We still denote them by

$$\nabla_A : \Gamma(S^\pm) \longrightarrow \Omega^1(X; S^\pm).$$

Thus for a spinor field $\psi \in \Gamma(S^\pm)$, $\nabla_A \psi$ is an $S^\pm$ -valued 1-form.

In other words, at every point $x \in X$,

$$(\nabla_A \psi)_x : T_x X \longrightarrow S_x^\pm$$

is a linear map.

Given a vector field V , define the directional covariant derivative by

$$\nabla_{A,V} \psi := (\nabla_A \psi)(V).$$

Thus $\nabla_A \psi$ records at once the covariant derivatives of ψ in all tangent directions.

3.4.5 Local expression

Take an open set $U \subset X$, and choose an oriented orthonormal frame $e_1, \dots, e_4$ on U .

Write its dual coframe as $e^1, \dots, e^4$.

The Levi-Civita connection can be written as

$$\nabla^{LC} e_i = \sum_j \omega_{ij} e_j,$$

where $\omega_{ij} \in \Omega^1(U)$ satisfies $\omega_{ij} = -\omega_{ji}$.

Next choose a local unitary frame of the determinant line bundle L . With respect to this trivialization, the unitary connection A is represented by an $i\mathbb{R}$ -valued 1-form.

Proposition 3.4.2. With these local choices, the spinor connection is written

$$\nabla_A = d + \frac{1}{4} \sum_{i,j} \omega_{ij} c(e^i) c(e^j) + \frac{1}{2} A.$$

The first term d is the ordinary derivative, the second term is the part produced by the Levi–Civita connection after passing through the spin representation, and the final $\frac{1}{2}A$ comes from the central $U(1)$ factor of Spin^c .

Proof. We derive it from the local form of the Spin^c connection in Section 3.4.3, and then verify the coefficients independently by requiring Clifford multiplication to be parallel.

First we explain what a local expression means.

A local section

$$s : U \longrightarrow P_s$$

trivializes the Spin^c frame bundle over U . It induces the orthonormal frame $e_1, \dots, e_4$ used above, together with a unit frame of the determinant line L .

By Section 3.4.3, in this trivialization the principal connection form becomes

$$s^* \omega_A = \left(j^{-1}(\omega_U^{LC}), \frac{1}{2} A_U \right) \in \Omega^1(U; \mathfrak{spin}(4) \oplus i\mathbb{R}).$$

The spinor bundles are associated to P_s by the half-spin representations,

$$S^\pm = P_s \times_{\rho_c^\pm} \Delta^\pm.$$

Hence, in the same trivialization, the connection form of ∇_A is the image of $s^* \omega_A$ under the differential

$$(\rho_c^\pm)_* : \mathfrak{spin}^c(4) \longrightarrow \mathfrak{u}(\Delta^\pm).$$

So the local expression of ∇_A is exactly

$$\nabla_A = d + (\rho_c^\pm)_*(s^* \omega_A).$$

It remains to evaluate the two summands separately.

First, the central summand. Recall from Section 3.2.8 that

$$\rho_c^\pm([g, z]) = z \rho^\pm(g).$$

Putting $g = 1$ gives the action of the central $U(1)$,

$$\rho_c^\pm([1, z]) = z I_2.$$

Differentiating this at $z = e^{it}$ gives

$$(\rho_c^\pm)_*(it) = it I_2.$$

So the $i\mathbb{R}$ -component $\frac{1}{2} A_U$ of $s^* \omega_A$ contributes

$$\frac{1}{2} A_U I_2,$$

which is precisely the final term $\frac{1}{2} A$, acting on spinors by scalar multiplication.

No further factor 2 appears at this stage. The factor 2 has already been absorbed in the definition of J , through $d\delta(it) = 2it$ of Section 3.4.2. Its effect is exactly that the spinor representation sees $\frac{1}{2}A$ rather than A , as discussed in Section 3.4.6.

Next, the $\mathfrak{spin}(4)$ summand. We have to evaluate

$$(\rho^\pm)_* : \mathfrak{spin}(4) \longrightarrow \mathfrak{u}(\Delta^\pm).$$

Since $\text{Spin}(4) \subset \text{Cl}_4^0$, and $\rho^\pm$ is the restriction of the Clifford algebra representation, $\rho^\pm(g)$ is left Clifford multiplication by g . Hence the differential is simply left Clifford multiplication as well,

$$(\rho^\pm)_*(\xi) = c(\xi), \quad \xi \in \mathfrak{spin}(4) \subset \text{Cl}_4^0.$$

Here, as everywhere in this section, the frame and coframe indices are identified through the metric, so that $c(e_i e_j) = c(e^i) c(e^j)$.

It therefore remains to identify the image of ω_U^{LC} under the inverse isomorphism

$$j^{-1} : \mathfrak{so}(4) \longrightarrow \mathfrak{spin}(4).$$

Write $G_{ij} \in \mathfrak{so}(4)$ for the infinitesimal rotation carrying e_i towards e_j , that is,

$$G_{ij} e_k = \delta_{ik} e_j - \delta_{jk} e_i.$$

Then the Levi-Civita form is the $\mathfrak{so}(4)$ -valued 1-form

$$\omega_U^{LC} = \sum_{i < j} \omega_{ij} G_{ij}.$$

Indeed, both sides act on e_k by

$$\sum_{j \neq k} \omega_{kj} e_j.$$

Recall from Section 3.2.5 that the vector representation is characterized by

$$Q(\lambda(g)v) = \rho^-(g)Q(v)\rho^+(g)^{-1}.$$

Differentiate this identity at $g = 1$ in the direction $\xi \in \mathfrak{spin}(4) \subset \text{Cl}_4^0$, that is, put $g = \exp(t\xi)$ and differentiate at $t = 0$. The left-hand side gives $c(d\lambda(\xi)v)$, while the right-hand side gives the commutator with $c(\xi)$. Hence

$$c(d\lambda(\xi)v) = [c(\xi), c(v)].$$

Now take $\xi = \frac{1}{2}e_i e_j$ with $i \neq j$. Using the Clifford relation $c(e^i)c(e^k) + c(e^k)c(e^i) = -2\delta^{ik}$ of Proposition 3.1.2, a direct computation gives

$$\left[\frac{1}{2}c(e^i)c(e^j), c(e^k) \right] = \delta_{ik}c(e^j) - \delta_{jk}c(e^i).$$

The right-hand side is Clifford multiplication by the vector $\delta_{ik}e_j - \delta_{jk}e_i$, which is exactly $G_{ij}e_k$. Therefore

$$d\lambda\left(\frac{1}{2}e_i e_j\right) = G_{ij},$$

and hence

$$j^{-1}(G_{ij}) = \frac{1}{2}e_i e_j.$$

Applying $(\rho^\pm)_*$ to $j^{-1}(\omega_U^{LC})$ now gives

$$(\rho^\pm)_* \left(j^{-1}(\omega_U^{LC}) \right) = \sum_{i < j} \omega_{ij} (\rho^\pm)_* \left(\frac{1}{2}e_i e_j \right) = \frac{1}{2} \sum_{i < j} \omega_{ij} c(e^i) c(e^j).$$

Since $\omega_{ij} = -\omega_{ji}$, the sum over $i < j$ is half of the sum over all ordered pairs,

$$\sum_{i < j} \omega_{ij} c(e^i) c(e^j) = \frac{1}{2} \sum_{i, j} \omega_{ij} c(e^i) c(e^j),$$

so that

$$(\rho^\pm)_* \left(j^{-1}(\omega_U^{LC}) \right) = \frac{1}{4} \sum_{i, j} \omega_{ij} c(e^i) c(e^j).$$

The coefficient $\frac{1}{4}$ therefore has two independent origins,

$$\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

One factor $\frac{1}{2}$ comes from $j^{-1}(G_{ij}) = \frac{1}{2}e_i e_j$, that is, from the double cover $\text{Spin}(4) \rightarrow \text{SO}(4)$; the other comes from the antisymmetry $\omega_{ij} = -\omega_{ji}$, which replaces a sum over the six pairs $i < j$ by half of a sum over all sixteen ordered pairs.

Adding the central part computed above yields the local expression

$$\nabla_A = d + \frac{1}{4} \sum_{i, j} \omega_{ij} c(e^i) c(e^j) + \frac{1}{2}A.$$

The same Levi-Civita part can be obtained without mentioning $\mathfrak{so}(4)$ at all, by requiring Clifford multiplication to be parallel with respect to ∇_A , that is,

$$\nabla_{A,X} (c(\alpha)\psi) = c(\nabla_X^{LC} \alpha)\psi + c(\alpha)\nabla_{A,X} \psi$$

for every 1-form α and every spinor field ψ . This gives an independent check of the coefficient $\frac{1}{4}$.

Write

$$M \in \Omega^1(U; \mathfrak{u}(2))$$

for the matrix part of ∇_A in a trivialization, so that $\nabla_A = d + M$. Since the frame is orthonormal, the matrices $c(e^i)$ are constant, and the identity above becomes

$$[M(X), c(\alpha)] = -c(X(\alpha) - \nabla_X^{LC} \alpha).$$

For $\alpha = \sum_j \alpha_j e^j$ we have

$$X(\alpha) - \nabla_X^{LC} \alpha = \sum_{k, j} \omega_{kj}(X) \alpha_j e^k.$$

On the other hand, taking

$$M = \frac{1}{4} \sum_{i, j} \omega_{ij} c(e^i) c(e^j)$$

and using

$$[c(e^i)c(e^j), c(e^k)] = 2\delta_{ik}c(e^j) - 2\delta_{jk}c(e^i)$$

gives

$$[M(X), c(e^j)] = -\sum_k \omega_{kj}(X)c(e^k),$$

hence

$$[M(X), c(\alpha)] = -\sum_{i,j} \omega_{ij}(X)\alpha_j c(e^i),$$

which is exactly the required right-hand side. So this choice of M does produce a connection compatible with Clifford multiplication.

Moreover, the compatibility condition determines M only up to an element of the centralizer of Clifford multiplication, because any $Z \in \mathfrak{u}(2)$ satisfying $[Z, c(\alpha)] = 0$ for all α may be added to M without changing the identity. On $\Delta^\pm$ the even Clifford algebra

$$\text{Cl}_4^{0,\mathbb{C}} \cong M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$$

acts irreducibly, so this centralizer is $\mathbb{C}I$; unitarity then forces it to be $i\mathbb{R}I$. Hence the most general unitary connection compatible with Clifford multiplication is

$$d + \frac{1}{4} \sum_{i,j} \omega_{ij} c(e^i) c(e^j) + a$$

with $a \in i\Omega^1(U)$. The determinant condition in Definition 3.4.1 then fixes $a = \frac{1}{2}A$.

Thus the two terms of the local expression have different natures: the first is forced by compatibility with Clifford multiplication, and the second is forced by the prescribed determinant connection.

Finally, two consistency checks.

First, the Levi–Civita term is traceless. Indeed $\omega_{ij} = -\omega_{ji}$ makes the diagonal terms vanish, and for $i \neq j$ the product $c(e^i)c(e^j)$ is conjugate to its negative, hence traceless. Consequently the induced connection on

$$\det S^+ = \det S^- = L$$

has connection form exactly A , in agreement with Section 3.4.6.

Second, differentiating the local expression gives the curvature of the spinor connection,

$$R^{S^A}(X, Y) = \frac{1}{4} \sum_{i,j} R_{XYij} c(e^i) c(e^j) + \frac{1}{2} F_A(X, Y) I,$$

in agreement with Section 3.5.7. Likewise, replacing A by $A + a$ shifts ∇_A by $\frac{1}{2}a$, as recorded in Section 3.4.7. $\square$

3.4.6 Why it is $\frac{1}{2}A$

This coefficient easily causes confusion among the notations of different references, so we explain it separately.

In the spinor representation, a central element $z \in U(1)$ acts on $S^\pm$ by scalar multiplication by z .

But on the determinant line $L = \det S^+$, the same element acts by

$$z^2$$

Hence, infinitesimally, if the local connection form of the determinant connection is A , then the central component of the Spin^c principal connection can only be $\frac{1}{2}A$.

Thus what the spinor representation sees is

$$\frac{1}{2}A.$$

This can also be verified directly from the determinant connection formula in Section 2.1.12.

Since S^+ is a complex rank-2 bundle, if the scalar part of the connection is $\frac{1}{2}A I_2$, then its trace is

$$\text{tr} \left(\frac{1}{2}A I_2 \right) = A.$$

So the induced connection on $\det S^+$ is exactly the original determinant connection A .

Remark 3.4.3. Some references do not denote the connection on the determinant line by A , but instead first choose a formal “square-root $U(1)$ connection” and denote it by A .

Under that convention, the formula for the spinor connection contains no $\frac{1}{2}$.

Hence when reading different references, one must first make sure whether A denotes the determinant connection or the central Spin^c connection.

3.4.7 Changing the determinant connection

As explained in Section 2.4.1, unitary connections on a fixed line bundle form an affine space.

If A is a unitary connection, and

$$a \in i\Omega^1(X),$$

then $A + a$ is again a unitary connection.

By the local formula of the previous section, the corresponding spinor connections satisfy

$$\nabla_{A+a} = \nabla_A + \frac{1}{2}a.$$

Here a acts on spinors by scalar multiplication.

Thus for $\psi \in \Gamma(S^\pm)$ we have

$$\nabla_{A+a}\psi = \nabla_A\psi + \frac{1}{2}a \otimes \psi.$$

This simple variation formula will be used repeatedly later on when linearizing the Seiberg–Witten equations.

3.4.8 Local expansion of $\nabla_A \psi$

Take a local frame $e_1, \dots, e_4$ together with its dual coframe $e^1, \dots, e^4$.

Since

$$\nabla_A \psi \in \Omega^1(X; S^\pm),$$

it can be expanded, just as an ordinary 1-form, as

$$\nabla_A \psi = \sum_{j=1}^4 e^j \otimes \nabla_{A, e_j} \psi.$$

Here

$$\nabla_{A, e_j} \psi = (\nabla_A \psi)(e_j).$$

That is, $\nabla_{A, e_j} \psi$ is exactly the coefficient obtained by applying the $S^\pm$ -valued 1-form $\nabla_A \psi$ to the vector e_j .

This is completely parallel to the differential of an ordinary function f ,

$$df = \sum_j e_j(f) e^j$$

The only difference is that $e_j(f)$ is a scalar, whereas $\nabla_{A, e_j} \psi$ lies in the spinor fiber $S^\pm$.

3.4.9 Dirac operator

We now have two natural maps: first the spinor connection

$$\nabla_A : \Gamma(S^+) \longrightarrow \Gamma(T^*X \otimes S^+),$$

and second Clifford multiplication

$$c : T^*X \otimes S^+ \longrightarrow S^-.$$

Composing them gives the Dirac operator.

Definition 3.4.4 (Spin^c Dirac operator). Define

$$D_A^+ := c \circ \nabla_A : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

One can likewise define

$$D_A^- : \Gamma(S^-) \longrightarrow \Gamma(S^+).$$

Usually, in the Seiberg–Witten equations, one writes D_A^+ simply as D_A when no confusion can arise.

3.4.10 The local formula for the Dirac operator

We work in the local frame $e_1, \dots, e_4$ of Section 3.4.8, with dual coframe $e^1, \dots, e^4$.

Proposition 3.4.5. With these local choices, the Dirac operator is given by

$$D_A^+ \psi = \sum_{j=1}^4 c(e^j) \nabla_{A, e_j} \psi, \quad \psi \in \Gamma(S^+).$$

Unfolding the coefficients by means of Proposition 3.4.2, this reads explicitly

$$D_A^+ \psi = \sum_j c(e^j) e_j(\psi) + \frac{1}{4} \sum_{j,k,l} \omega_{kl}(e_j) c(e^j) c(e^k) c(e^l) \psi + \frac{1}{2} \sum_j A(e_j) c(e^j) \psi.$$

Proof. By Definition 3.4.4, the Dirac operator is the composition

$$D_A^+ = c \circ \nabla_A,$$

so the only thing to determine is what Clifford multiplication does to the spinor-valued 1-form $\nabla_A \psi$.

The expansion of $\nabla_A \psi$ in the frame reads

$$\nabla_A \psi = \sum_{j=1}^4 e^j \otimes \nabla_{A, e_j} \psi \in \Gamma(T^*X \otimes S^\pm),$$

where $\nabla_{A, e_j} \psi = (\nabla_A \psi)(e_j)$ is the coefficient of e^j , as explained in Section 3.4.8.

The point is that Clifford multiplication is C^∞ -linear. It comes from a bundle map

$$c : T^*X \otimes S^\pm \longrightarrow S^\mp,$$

and not from a differential operator, so it commutes with sums and with multiplication by functions. Applying it term by term, and using $c(\alpha \otimes \psi) = c(\alpha)\psi$, we obtain

$$D_A^+ \psi = c(\nabla_A \psi) = c \left(\sum_{j=1}^4 e^j \otimes \nabla_{A, e_j} \psi \right) = \sum_{j=1}^4 c(e^j \otimes \nabla_{A, e_j} \psi) = \sum_{j=1}^4 c(e^j) \nabla_{A, e_j} \psi.$$

This is the point at which the two ingredients behave differently. The covariant derivative is not C^∞ -linear, since

$$\nabla_{A, e_j}(f\psi) = f \nabla_{A, e_j} \psi + e_j(f) \psi,$$

and it is a differential operator rather than a bundle map: the functions cannot be moved across it. Clifford multiplication, by contrast, is a bundle map, and that is exactly what allows the sum above to be taken term by term. The order of the composition therefore matters: one differentiates first, and contracts with c afterwards.

It remains to verify the second formula stated in Proposition 3.4.5, for which we unfold the coefficients. Inserting the local expression of Proposition 3.4.2 into the coefficients of $\nabla_A \psi$ gives

$$\nabla_{A, e_j} \psi = e_j(\psi) + \frac{1}{4} \sum_{k,l} \omega_{kl}(e_j) c(e^k) c(e^l) \psi + \frac{1}{2} A(e_j) \psi,$$

which, after multiplication by $c(e^j)$ and summation over j , is precisely the second display of Proposition 3.4.5.

In that display, writing

$$c(A) := \sum_{j=1}^4 A(e_j)c(e^j)$$

for Clifford multiplication by the 1-form A , the last term is simply $\frac{1}{2}c(A)\psi$. The three terms are the exact analogues of the three terms of Proposition 3.4.2: the first is the derivative relative to the frame, the second is the correction produced by the Levi-Civita connection, and the third is the contribution of the determinant connection A .

Finally, some consistency checks.

First, the formula does not depend on the chosen frame. If

$$e'_j = \sum_k R_{jk}e_k$$

is another oriented orthonormal frame, with dual coframe $e'^j = \sum_k R_{jk}e^k$, then

$$\sum_{j=1}^4 e'^j \otimes (\nabla_A \psi)(e'_j) = \nabla_A \psi,$$

simply because a 1-form is decomposed this way in any frame. Since c is C^∞ -linear, applying it gives the same spinor field for every choice of frame.

Second, the formula is compatible with the grading. Each $c(e^j)$ is an odd element of the Clifford algebra, hence interchanges S^+ and S^- . The formula therefore exhibits

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-)$$

as a sum of four maps that each change the chirality, and the same sum applied to $\Gamma(S^-)$ is the operator $D_A^- : \Gamma(S^-) \rightarrow \Gamma(S^+)$.

Third, the dependence on the determinant connection is the expected one. Only the third term of the unfolded expression involves A , so replacing A by $A + a$ gives

$$D_{A+a}^+ \psi = D_A^+ \psi + \frac{1}{2}c(a)\psi,$$

which is the variation recorded in Section 3.4.11.

The leading term is also visible directly. Replacing ψ by $f\psi$ with $f \in C^\infty(X)$, and using $df = \sum_j e_j(f)e^j$, we get

$$D_A^+(f\psi) = fD_A^+\psi + \sum_j c(e^j)e_j(f)\psi = fD_A^+\psi + c(df)\psi,$$

so that

$$\sigma_{D_A}(df)\psi = D_A^+(f\psi) - fD_A^+\psi = c(df)\psi.$$

Thus the principal symbol of D_A is Clifford multiplication, as used in Section 3.4.13. $\square$

Thus the Dirac operator may be understood as follows: first differentiate the spinor covariantly along the four orthonormal directions separately, and then use Clifford multiplication to contract the T^*X direction with the spinor.

One has to be careful that e_j and e^j are objects of different types: $e_j \in TX$, whereas $e^j \in T^*X$.

In an orthonormal frame, the musical isomorphism given by the Riemannian metric satisfies $e_j^\flat = e^j$, so some references write the two with the same symbol in the Clifford notation.

3.4.11 How the Dirac operator changes when A changes

From

$$\nabla_{A+a} = \nabla_A + \frac{1}{2}a$$

we immediately obtain

$$D_{A+a}\psi = D_A\psi + \frac{1}{2}c(a)\psi.$$

Here

$$c(a)\psi$$

means that the 1-form a is first regarded, via Clifford multiplication, as a bundle map $S^+ \rightarrow S^-$, and then applied to ψ .

So the dependence of the Dirac operator on the determinant connection A is affine-linear.

This formula gives the connection term in the linearization of the Seiberg–Witten equations in Section 4.2.2.

3.4.12 The full Dirac operator

Combining the two half-spinor bundles, write

$$S = S^+ \oplus S^-.$$

Clifford multiplication interchanges chirality, so with respect to this decomposition the full Dirac operator is written

$$D_A = \begin{pmatrix} 0 & D_A^- \\ D_A^+ & 0 \end{pmatrix}.$$

Hence

$$D_A : \Gamma(S) \longrightarrow \Gamma(S).$$

Only on this full spinor bundle is D_A^2 a literal square of an operator.

If we discuss only $D_A^+ : \Gamma(S^+) \rightarrow \Gamma(S^-)$, then the corresponding second-order operator should be written

$$D_A^- D_A^+$$

rather than vaguely as $(D_A^+)^2$.

This distinction is very important in the Weitzenböck formula below.

3.4.13 The principal symbol of the Dirac operator

The Dirac operator is a first-order differential operator. Its principal symbol is determined entirely by Clifford multiplication.

Adopting the symbol convention

$$\sigma_D(df)\psi = D(f\psi) - fD\psi,$$

we have

$$\sigma_{D_A}(\xi) = c(\xi), \quad \xi \in T_x^*X.$$

If one uses the symbol convention with i that appears in the PDE literature, then one writes $ic(\xi)$. These two convention have no effect on ellipticity.

By the Clifford relation,

$$c(\xi)^2 = -|\xi|^2 I.$$

Hence for $\xi \neq 0$, $c(\xi)$ is invertible, with inverse

$$-\frac{1}{|\xi|^2} c(\xi).$$

Thus we obtain:

Proposition 3.4.6. The Spin^c Dirac operator

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-)$$

is a first-order elliptic differential operator.

Here the source of ellipticity is not some computation in local coordinates, but precisely the Clifford relation $c(\xi)^2 = -|\xi|^2 I$ from the beginning.

3.4.14 Formal adjoint

Spinor bundles carry a natural Hermitian metric, and the Riemannian metric gives a volume form. Hence one can define an L^2 inner product on spinor fields.

Clifford multiplication by a real 1-form satisfies

$$c(\alpha)^* = -c(\alpha).$$

On the other hand, ∇_A is a Hermitian connection. Using integration by parts we obtain:

Proposition 3.4.7. If X has no boundary, then

$$(D_A^+)^* = D_A^-.$$

Hence the full Dirac operator

$$D_A : \Gamma(S^+ \oplus S^-) \longrightarrow \Gamma(S^+ \oplus S^-)$$

is formally self-adjoint.

In particular, the natural second-order operator acting on positive spinors is

$$(D_A^+)^* D_A^+ = D_A^- D_A^+.$$

The Weitzenböck formula in the next section (Theorem 3.5.3) is precisely meant to study this second-order operator.

3.4.15 Summary of this section

Once a Spin^c structure is fixed, the Levi-Civita connection determines only the $SO(4)$ part of the Spin^c connection. In order to make up the central $U(1)$ direction, one must also choose a unitary connection A on the determinant line $L = \det S^+$.

Together the two induce the spinor connection

$$\nabla_A : \Gamma(S^\pm) \longrightarrow \Omega^1(X; S^\pm).$$

Since the determinant character is $z \mapsto z^2$, under the determinant-connection convention $\frac{1}{2}A$ appears in the local spinor connection.

Composing the spinor connection with Clifford multiplication gives the Dirac operator

$$D_A^+ = c \circ \nabla_A.$$

Locally,

$$D_A^+ \psi = \sum_j c(e^j) \nabla_{A, e_j} \psi.$$

Its principal symbol is Clifford multiplication, so the Clifford relation immediately guarantees that D_A^+ is elliptic.

Moreover, if the determinant connection is changed, $A \mapsto A + a$, then

$$D_{A+a} \psi = D_A \psi + \frac{1}{2} c(a) \psi.$$

The next step is to define ∇_A^* and the connection Laplacian $\nabla_A^* \nabla_A$, and then to prove that the square of the Dirac operator satisfies the Weitzenböck formula. This formula will connect the Dirac operator, the Riemannian scalar curvature and the $U(1)$ curvature with one another.

3.5 The connection Laplacian and the Weitzenböck formula

The previous section defined the Spin^c Dirac operator (Definition 3.4.4)

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

The Dirac operator is a first-order elliptic operator. One basic question is: how is it related to the usual second-order Laplace-type operator?

The Weitzenböck formula gives the answer to this question. It decomposes the square of the Dirac operator into a connection Laplacian and several curvature terms.

Before entering the formula, we first need to define the formal adjoint of a connection and the corresponding connection Laplacian.

3.5.1 The L^2 inner product on spinor fields

The spinor bundles $S^\pm$ carry the Hermitian metric induced by the Spin^c representation. On the other hand, the Riemannian metric g gives the volume form $d \text{vol}_g$.

Hence for $\psi, \phi \in \Gamma(S^\pm)$ one can define

$$(\psi, \phi)_{L^2} = \int_X \langle \psi, \phi \rangle d \text{vol}_g.$$

Similarly, $T^*X \otimes S^\pm$ carries the tensor product metric coming from g and the spinor Hermitian metric at the same time.

If $\eta, \theta \in \Omega^1(X; S^\pm)$, and one takes a local orthonormal frame $e_1, \dots, e_4$, then the pointwise inner product is

$$\langle \eta, \theta \rangle = \sum_{j=1}^4 \langle \eta(e_j), \theta(e_j) \rangle.$$

This likewise defines an L^2 inner product on $\Omega^1(X; S^\pm)$.

3.5.2 Formal adjoint

The spinor connection is a first-order differential operator

$$\nabla_A : \Gamma(S^\pm) \longrightarrow \Omega^1(X; S^\pm).$$

Since X is a closed manifold, there are no boundary terms to handle.

Definition 3.5.1 (Formal adjoint). The **formal adjoint** of ∇_A is the operator

$$\nabla_A^* : \Omega^1(X; S^\pm) \longrightarrow \Gamma(S^\pm),$$

defined by the condition

$$(\nabla_A \psi, \eta)_{L^2} = (\psi, \nabla_A^* \eta)_{L^2}$$

for all smooth ψ and η .

Here the symbol $*$ denotes the L^2 formal adjoint, and has nothing to do with the Hodge star.

This definition is completely parallel to the adjoint in finite-dimensional linear algebra. The only difference is that here the inner product is obtained by integrating over the whole manifold.

3.5.3 The local formula for ∇_A^*

In order to write a local formula for ∇_A^* , one has to note that $\eta \in \Omega^1(X; S)$ is itself an S -valued 1-form.

Hence when taking the covariant derivative of η , one not only has to act on the spinor component, but also has to use the Levi-Civita connection on T^*X at the same time.

More precisely, one uses the tensor product connection on $T^*X \otimes S$. For vector fields X, Y , we have

$$(\nabla_X \eta)(Y) = \nabla_{A,X}(\eta(Y)) - \eta(\nabla_X^{LC} Y).$$

Taking a local orthonormal frame $e_1, \dots, e_4$, the local formula for the formal adjoint is

$$\nabla_A^* \eta = - \sum_{j=1}^4 (\nabla_{e_j} \eta)(e_j).$$

After expanding, this is

$$\nabla_A^* \eta = - \sum_{j=1}^4 \left[\nabla_{A,e_j}(\eta(e_j)) - \eta(\nabla_{e_j}^{LC} e_j) \right].$$

This formula may be regarded as a vector-bundle-valued divergence with a minus sign in front.

This minus sign has the same origin as

$$\left(\frac{d}{dx}\right)^* = -\frac{d}{dx}$$

in the ordinary integration by parts.

3.5.4 Connection Laplacian

We can now compose ∇_A with its formal adjoint.

Definition 3.5.2 (Connection Laplacian). Define

$$\nabla_A^* \nabla_A : \Gamma(S^\pm) \longrightarrow \Gamma(S^\pm)$$

to be the **connection Laplacian** on the spinor bundle.

It is also often called **rough Laplacian** or **Bochner Laplacian**.

By the local formula above, we have

$$\nabla_A^* \nabla_A \psi = - \sum_{j=1}^4 \left(\nabla_{A, e_j} \nabla_{A, e_j} \psi - \nabla_{A, \nabla_{e_j}^{LC} e_j} \psi \right).$$

The second term cannot simply be omitted. Its role is to ensure that the whole expression is independent of the choice of local frame.

At a fixed point $p \in X$, however, one can always choose a local orthonormal frame such that

$$\nabla_{e_i}^{LC} e_j(p) = 0.$$

Such a frame is called normal at p . Thus at that point alone, the connection Laplacian reduces to

$$(\nabla_A^* \nabla_A \psi)(p) = - \sum_j \nabla_{A, e_j} \nabla_{A, e_j} \psi(p).$$

Hence the connection Laplacian can be understood as the covariant version of the ordinary Euclidean Laplacian $-\sum_j \partial_j^2$.

3.5.5 Clifford multiplication by 2-forms

In the Weitzenböck formula the action of the curvature 2-form on spinors will appear, so we need to extend Clifford multiplication from 1-forms to 2-forms.

For $\alpha, \beta \in T_x^* X$, define

$$c(\alpha \wedge \beta) = \frac{1}{2} (c(\alpha)c(\beta) - c(\beta)c(\alpha)).$$

Then extend to all of $\Lambda^2 T_x^* X$ by linearity.

If α, β are orthogonal, then the Clifford relation gives $c(\alpha)c(\beta) = -c(\beta)c(\alpha)$, so in this case

$$c(\alpha \wedge \beta) = c(\alpha)c(\beta).$$

Since 2-forms correspond to the even part of the Clifford algebra, their Clifford multiplication preserves chirality. Hence

$$c(\omega) : S^\pm \longrightarrow S^\pm, \quad \omega \in \Lambda^2 T^*X.$$

3.5.6 The self-dual decomposition in dimension four

In dimension four, the Hodge star on 2-forms satisfies $*^2 = 1$, so there is a decomposition

$$\Lambda^2 = \Lambda_+^2 \oplus \Lambda_-^2.$$

On the other hand, we have seen above that

$$\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-.$$

Under the orientation and chirality conventions adopted in these notes, self-dual 2-forms correspond to $\mathfrak{su}(2)_+$, while anti-self-dual 2-forms correspond to $\mathfrak{su}(2)_-$.

Hence Λ_+^2 acts non-trivially only on S^+ , while Λ_-^2 acts non-trivially only on S^- .

More concretely, if $\omega = \omega^+ + \omega^-$, then on S^+

$$c(\omega) = c(\omega^+),$$

while on S^-

$$c(\omega) = c(\omega^-).$$

Thus in the Weitzenböck formula for positive spinors, the determinant curvature will in the end produce only F_A^+ .

3.5.7 The curvature of the spinor connection

For an arbitrary vector bundle connection ∇ , its curvature is defined by

$$R^\nabla(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}.$$

The spinor connection ∇_A contains both a Levi-Civita part and a determinant $U(1)$ part, so its curvature likewise splits into two parts.

Take a local oriented orthonormal frame $e_1, \dots, e_4$. Denote the components of the Riemann curvature tensor by

$$R_{XYij} = \langle R^{LC}(X, Y)e_i, e_j \rangle.$$

Then the curvature of the spinor connection is

$$R^{SA}(X, Y) = \frac{1}{4} \sum_{i,j} R_{XYij} c(e^i)c(e^j) + \frac{1}{2} F_A(X, Y)I.$$

The first term comes from the Levi-Civita curvature, the second term from the curvature of the determinant connection.

The coefficient $\frac{1}{2}$ has the same origin as the $\frac{1}{2}A$ appearing in the spinor connection in the previous section.

3.5.8 The basic computation of the square of the Dirac operator

In what follows we study the full Dirac operator

$$D_A = \sum_i c(e^i) \nabla_{A, e_i}.$$

Fix a point $p \in X$, and choose a normal orthonormal frame at p . Since the spinor connection is compatible with Clifford multiplication, computing at p produces no extra first-order terms from the frame derivatives.

Thus

$$D_A^2 \psi = \sum_{i,j} c(e^i) c(e^j) \nabla_{A, e_i} \nabla_{A, e_j} \psi.$$

First consider the part with $i = j$. Since $c(e^i)^2 = -1$, it gives

$$- \sum_i \nabla_{A, e_i} \nabla_{A, e_i} \psi,$$

which is exactly the value at p of $\nabla_A^* \nabla_A \psi$.

Next consider the terms with $i \neq j$. Pairing (i, j) with (j, i) , using $c(e^i)c(e^j) = -c(e^j)c(e^i)$, we obtain

$$\sum_{i < j} c(e^i) c(e^j) \left(\nabla_{A, e_i} \nabla_{A, e_j} - \nabla_{A, e_j} \nabla_{A, e_i} \right) \psi.$$

Since a normal frame at p satisfies $[e_i, e_j](p) = 0$, the expression in parentheses is exactly the spinor connection curvature:

$$R^{SA}(e_i, e_j) \psi.$$

Hence

$$D_A^2 = \nabla_A^* \nabla_A + \sum_{i < j} c(e^i) c(e^j) R^{SA}(e_i, e_j).$$

This already reveals the basic mechanism of the Weitzenböck formula:

the second-order principal term of the Dirac operator produces the connection Laplacian, while the covariant derivatives in different directions do not commute, and their commutator is precisely the curvature.

3.5.9 The Riemann curvature term

Substituting the Levi-Civita part of the spinor curvature into the formula above, one obtains a contraction formed by the Riemann curvature tensor together with four Clifford matrices.

Using the symmetries of the Riemann curvature tensor

$$R_{ijkl} = -R_{jikl}, \quad R_{ijkl} = -R_{ijlk}, \quad R_{ijkl} = R_{klij},$$

together with the first Bianchi identity, all non-scalar parts cancel one another in the Clifford contraction.

In the end only the scalar curvature remains,

$$s = \sum_{i,j} R_{ijij}.$$

The standard Lichnerowicz computation gives the action of the Riemann curvature part on a spinor as

$$\frac{s}{4}I.$$

The $\frac{1}{4}$ here is not a normalization inserted by hand, but is produced jointly by the coefficient of the curvature in the spin representation and the Clifford contraction.

3.5.10 The $U(1)$ curvature term

The other part of the spinor curvature is

$$\frac{1}{2}F_A(X, Y)I.$$

Substituting this into the curvature contraction of the Dirac square gives

$$\frac{1}{2}c(F_A).$$

Hence on the full spinor bundle we have

$$D_A^2 = \nabla_A^* \nabla_A + \frac{s}{4} + \frac{1}{2}c(F_A).$$

When restricted to positive spinors, the action of the anti-self-dual part F_A^- vanishes, so we obtain

$$D_A^- D_A^+ = \nabla_A^* \nabla_A + \frac{s}{4} + \frac{1}{2}c(F_A^+).$$

Similarly, on negative spinors we have

$$D_A^+ D_A^- = \nabla_A^* \nabla_A + \frac{s}{4} + \frac{1}{2}c(F_A^-).$$

3.5.11 Weitzenböck formula

We collect the computations above into the following theorem.

Theorem 3.5.3 (Spin^c Weitzenböck formula). Let A be a unitary connection on the determinant line bundle $L = \det S^+$. Then for any $\psi \in \Gamma(S^+)$ we have

$$D_A^- D_A^+ \psi = \nabla_A^* \nabla_A \psi + \frac{s}{4} \psi + \frac{1}{2}c(F_A^+) \psi.$$

Since $(D_A^+)^* = D_A^-$, this can also be written as

$$(D_A^+)^* D_A^+ = \nabla_A^* \nabla_A + \frac{s}{4} + \frac{1}{2}c(F_A^+).$$

Remark 3.5.4. The coefficients above depend on the conventions adopted in these notes: A denotes a connection on the determinant line bundle, and the Clifford relation is $c(\xi)^2 = -|\xi|^2$.

If a different curvature, Clifford multiplication or Spin^c connection normalization is used, then the sign and coefficient in front of the F_A term may change.

But the structure of the formula is always the same: the square of the Dirac operator equals the connection Laplacian plus curvature terms.

3.5.12 Untwisted Lichnerowicz formula

If X is itself a spin manifold, and one considers the untwisted spin Dirac operator induced by the Levi–Civita connection, then there is no additional $U(1)$ curvature term.

In this case the Weitzenböck formula reduces to

$$D^2 = \nabla^* \nabla + \frac{s}{4}.$$

This formula is usually called the **Lichnerowicz formula**.

Hence the Spin^c Weitzenböck formula may be viewed as the Lichnerowicz formula plus a determinant curvature term.

3.5.13 Vanishing under positive scalar curvature

A basic application of the Weitzenböck formula is the study of harmonic spinors.

Let

$$D\psi = 0$$

and let X be a closed spin manifold. By the Lichnerowicz formula,

$$0 = D^2\psi = \nabla^* \nabla \psi + \frac{s}{4}\psi.$$

Taking the L^2 inner product with ψ , we obtain

$$0 = \|\nabla \psi\|_{L^2}^2 + \frac{1}{4} \int_X s |\psi|^2 d\text{vol}.$$

If the scalar curvature satisfies $s > 0$ everywhere, then the two terms on the right are both non-negative, while the second term is strictly positive when $\psi \not\equiv 0$.

Hence the only possibility is

$$\psi \equiv 0.$$

So positive scalar curvature can force the kernel of the Dirac operator to vanish.

Later, in Seiberg–Witten theory, exactly the same idea will be combined with the nonlinear curvature equation to give a pointwise bound for the Seiberg–Witten spinor as well as compactness of the moduli space.

3.5.14 A check in flat space

Consider flat $\mathbb{R}^4$, and take the trivial spin structure and the trivial connection.

In this case

$$s = 0, \quad F_A = 0,$$

and ∇_A is just the ordinary derivative.

So the Weitzenböck formula reduces to

$$D^2 = - \sum_{j=1}^4 \partial_j^2.$$

This agrees with the result obtained in the previous section by computing the square of the Dirac operator directly via Clifford matrices.

Thus the curvature terms are precisely the correction produced by curved geometry relative to the flat Dirac operator.

3.5.15 Summary of this section

The connection Laplacian $\nabla_A^* \nabla_A$ is the covariant version of the ordinary Laplacian on the spinor bundle.

When the Dirac operator is squared, the second-order principal term gives the connection Laplacian, while the cross terms are controlled by the commutator of the covariant derivatives. This commutator is precisely the curvature of the spinor connection.

The curvature of the spinor connection in turn splits into the Levi–Civita curvature and the determinant $U(1)$ curvature. After Clifford contraction the former leaves only the scalar curvature $s/4$, while the latter produces $c(F_A)/2$.

Hence on positive spinors the Weitzenböck formula reads

$$D_A^- D_A^+ = \nabla_A^* \nabla_A + \frac{s}{4} + \frac{1}{2} c(F_A^+).$$

At this point the linear Spin^c geometry needed for Seiberg–Witten theory has essentially been prepared.

The next step is to introduce a new pointwise quadratic map

$$q : S^+ \longrightarrow i\Lambda_+^2,$$

which turns a positive spinor into a self-dual 2-form.

Using this quadratic map, we can couple the Dirac equation with the determinant curvature equation, and thereby obtain the Seiberg–Witten equations.

The Seiberg–Witten equations and invariants

4.1 The Seiberg–Witten equations

Previously we obtained, for a fixed Spin^c structure $\mathfrak{s}$, the corresponding spinor bundles $S^\pm$, determinant line bundle L , Clifford multiplication and Dirac operator

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

The unknowns in the Seiberg–Witten equations consist of two parts: a unitary connection A on the determinant line bundle L , and a positive spinor $\psi \in \Gamma(S^+)$.

The Dirac equation

$$D_A \psi = 0$$

supplies one of the two equations.

In order to obtain the second equation, we still need to turn the positive spinor ψ into a self-dual 2-form. This process is carried out by a natural quadratic map.

4.1.1 Rank-one endomorphisms

Fix a point $x \in X$, and take $\psi \in S_x^+$.

By the Hermitian inner product, ψ defines a rank-one endomorphism

$$\psi\psi^* : S_x^+ \longrightarrow S_x^+,$$

acting by

$$(\psi\psi^*)(\phi) = \psi\langle\psi, \phi\rangle.$$

Here ψ^* may be understood as the covector $\phi \mapsto \langle\psi, \phi\rangle$.

This endomorphism is Hermitian, and

$$\text{tr}(\psi\psi^*) = |\psi|^2.$$

Since S_x^+ is a complex rank-2 space, its trace-free part is

$$(\psi\psi^*)_0 = \psi\psi^* - \frac{1}{2}|\psi|^2 I.$$

Hence $(\psi\psi^*)_0$ lies in $\text{Herm}_0(S_x^+)$, the real vector space of trace-free Hermitian endomorphisms.

4.1.2 Self-dual 2-forms and trace-free endomorphisms

In four-dimensional Spin^c geometry there is a natural real-linear isomorphism

$$\rho : i\Lambda_+^2 \longrightarrow \text{Herm}_0(S^+).$$

It comes essentially from Clifford multiplication of self-dual 2-forms on positive spinors.

The factor i here serves to convert Clifford multiplication from skew-Hermitian endomorphisms into Hermitian endomorphisms.

Different references do not use entirely consistent conventions at this point. Some identify Λ_+^2 directly with the skew-Hermitian trace-free endomorphisms, while others use $i\Lambda_+^2$ and the Hermitian trace-free endomorphisms.

We adopt the latter.

4.1.3 Quadratic map

Definition 4.1.1 (Quadratic map). Define

$$q : S^+ \longrightarrow i\Lambda_+^2$$

such that

$$\rho(q(\psi)) = (\psi\psi^*)_0.$$

In other words, $q(\psi)$ is the unique imaginary-valued self-dual 2-form whose Clifford action equals $(\psi\psi^*)_0$.

Since

$$(z\psi)(z\psi)^* = |z|^2\psi\psi^*,$$

we have

$$q(z\psi) = |z|^2q(\psi).$$

So q is quadratic in the spinor variable.

In particular, if $u \in U(1)$, then $|u| = 1$, and therefore

$$q(u\psi) = q(\psi).$$

This property will later guarantee that the second Seiberg–Witten equation is gauge invariant.

4.1.4 A basic identity for the quadratic map

By definition,

$$\rho(q(\psi)) = \psi\psi^* - \frac{1}{2}|\psi|^2I.$$

Letting this endomorphism act on ψ , we obtain

$$\begin{aligned} \rho(q(\psi))\psi &= \psi\langle\psi, \psi\rangle - \frac{1}{2}|\psi|^2\psi \\ &= \frac{1}{2}|\psi|^2\psi. \end{aligned}$$

Hence

$$\rho(q(\psi))\psi = \frac{1}{2}|\psi|^2\psi.$$

Later, combined with the Weitzenböck formula, this identity will produce the most important quartic term $|\psi|^4$ in Seiberg–Witten theory.

4.1.5 Configuration space

Fix a Spin^c structure $\mathfrak{s}$, and write its determinant line bundle as L .

All unitary connections on L form an affine space $\mathcal{A}(L)$.

Hence the configuration space of the Seiberg–Witten equations is defined to be

$$\mathcal{C}_{\mathfrak{s}} = \mathcal{A}(L) \times \Gamma(S^+).$$

A point

$$(A, \psi) \in \mathcal{C}_{\mathfrak{s}}$$

consists of a determinant connection and a positive spinor field.

After fixing a reference connection A_0 , any connection can be written as

$$A = A_0 + a, \quad a \in i\Omega^1(X).$$

Hence $\mathcal{A}(L)$ is an affine space modeled on $i\Omega^1(X)$.

Correspondingly, the tangent space of the configuration space at (A, ψ) is naturally

$$T_{(A, \psi)}\mathcal{C}_{\mathfrak{s}} \cong i\Omega^1(X) \oplus \Gamma(S^+).$$

4.1.6 Curvature equation

For a determinant connection A , its curvature is

$$F_A \in i\Omega^2(X).$$

The four-dimensional Hodge decomposition gives

$$F_A = F_A^+ + F_A^-,$$

where $F_A^+ \in i\Omega_+^2(X)$.

By the quadratic map above (Definition 4.1.1), $q(\psi)$ likewise belongs to $i\Omega_+^2(X)$.

Hence the two have exactly the same tensor type and can be compared directly.

This produces the second equation of Seiberg–Witten theory.

4.1.7 Seiberg–Witten equations

Definition 4.1.2 (Seiberg–Witten equations). For a configuration $(A, \psi) \in \mathcal{C}_{\mathfrak{s}}$, the Seiberg–Witten equations are defined to be

$$\begin{cases} D_A\psi = 0, \\ F_A^+ = q(\psi). \end{cases}$$

The first equation is the Dirac equation. It is a first-order elliptic equation for the spinor, but its coefficients depend on the connection A .

The second equation is the curvature equation. Its left-hand side is first-order in the connection A , while its right-hand side is quadratic in the spinor ψ .

Hence the Seiberg–Witten equations form a coupled nonlinear first-order system.

4.1.8 Perturbed Seiberg–Witten equations

In order to study the smoothness of the moduli space, one usually needs to add a perturbation to the second equation.

Take

$$\eta \in i\Omega_+^2(X).$$

Consider the equations

$$\begin{cases} D_A\psi = 0, \\ F_A^+ = q(\psi) + \eta. \end{cases}$$

These are called the perturbed Seiberg–Witten equations.

Some references use $F_A^+ - q(\psi) = i\eta$ or another sign convention. These differences do not change the basic structure of the theory.

Later, when we discuss transversality, η will be used as an adjustable parameter.

4.1.9 Seiberg–Witten map

Putting the two equations together, we can define a nonlinear map

$$\mathcal{F}_{s,\eta} : \mathcal{C}_s \longrightarrow \Gamma(S^-) \oplus i\Omega_+^2(X)$$

by

$$\mathcal{F}_{s,\eta}(A, \psi) = (D_A\psi, F_A^+ - q(\psi) - \eta).$$

Then the perturbed Seiberg–Witten solutions are precisely

$$\mathcal{F}_{s,\eta}^{-1}(0).$$

Hence, from the point of view of nonlinear analysis, Seiberg–Witten theory studies the zero set of a nonlinear map on an infinite-dimensional configuration space.

But this zero set is not yet the final moduli space, because the equations have a gauge symmetry.

4.1.10 Gauge group

For a Spin^c structure, the gauge group can be written as

$$\mathcal{G} = C^\infty(X, U(1)).$$

This is because the central $U(1)$ produces natural bundle automorphisms of the spinor bundles and of the determinant line bundle.

We continue to use the determinant-connection convention.

For $u \in \mathcal{G}$, define its action on the configuration space to be

$$u \cdot (A, \psi) = (A - 2u^{-1}du, u\psi).$$

Here the coefficient 2 in front of the connection part comes from the determinant character $z \mapsto z^2$.

The spinor itself transforms by u , while the determinant line transforms by u^2 , so the determinant connection must produce twice the gauge term.

Remark 4.1.3. There are several action conventions in the gauge theory literature. For example, some authors write

$$(A, \psi) \mapsto (A + 2u^{-1}du, u^{-1}\psi).$$

These conventions are equivalent to one another, but the infinitesimal action below and the linearization formulas must always stay consistent with the convention chosen.

4.1.11 Gauge invariance of the curvature

Since $U(1)$ is an Abelian group, for the determinant connection we have

$$F_{A-2u^{-1}du} = F_A - 2d(u^{-1}du).$$

The Maurer–Cartan equation, in the Abelian case, gives

$$d(u^{-1}du) = 0.$$

Hence

$$F_{A-2u^{-1}du} = F_A.$$

In particular,

$$F_{A-2u^{-1}du}^+ = F_A^+.$$

Hence the left-hand side of the curvature equation is invariant under a gauge transformation.

On the other hand, we have seen above that $q(u\psi) = q(\psi)$.

Hence the second Seiberg–Witten equation is gauge invariant.

4.1.12 Gauge covariance of the Dirac equation

The Dirac operator itself is not gauge invariant, but rather gauge covariant.

More precisely, we have

$$D_{A-2u^{-1}du}(u\psi) = uD_A\psi.$$

Hence if $D_A\psi = 0$, then

$$D_{A-2u^{-1}du}(u\psi) = 0.$$

So the first Seiberg–Witten equation is also preserved by the gauge group.

Combining the two equations, if (A, ψ) is a solution, then for any $u \in \mathcal{G}$, $u \cdot (A, \psi)$ is still a solution.

4.1.13 Solution space and moduli space

Define the perturbed solution space

$$\mathcal{S}_{s,\eta} = \{(A, \psi) \in \mathcal{C}_s \mid \mathcal{F}_{s,\eta}(A, \psi) = 0\}.$$

Since the Seiberg–Witten equations are gauge invariant, $\mathcal{S}_{s,\eta}$ is a $\mathcal{G}$ -invariant subset.

Hence the object that really has geometric meaning is not an individual solution, but its gauge-equivalence class.

Definition 4.1.4 (Seiberg–Witten moduli space). Define

$$\mathcal{M}_{s,\eta} := \mathcal{S}_{s,\eta} / \mathcal{G}.$$

Hence a point $[(A, \psi)] \in \mathcal{M}_{s,\eta}$ represents an entire gauge orbit of solutions of the Seiberg–Witten equations.

This has exactly the same basic structure as

$$\mathcal{M}_{\text{ASD}} = \{\text{ASD connections}\} / \mathcal{G}$$

in the ASD theory above.

4.1.14 Reducible and irreducible solutions

The action of the gauge group on the configuration space is not always free.

Definition 4.1.5. A Seiberg–Witten solution (A, ψ) for which

$$\psi \equiv 0,$$

is called a **reducible solution**.

If $\psi \not\equiv 0$, it is called an **irreducible solution**.

If $\psi = 0$, every constant gauge transformation $u \in U(1)$ leaves $(A, 0)$ unchanged.

Hence a reducible solution has a non-trivial stabilizer.

Conversely, let (A, ψ) be irreducible, and assume that X is connected. If u fixes (A, ψ) , then

$$A - 2u^{-1}du = A,$$

so $du = 0$, that is, u is constant.

On the other hand, $u\psi = \psi$. Since ψ is not identically zero, there is a point at which $\psi(x) \neq 0$, and hence $u = 1$.

Hence on the irreducible locus the action of the gauge group is free.

This distinction will be very important later: near regular irreducible solutions the moduli space can have a smooth manifold structure, whereas reducibles are often the source of singular behavior.

4.1.15 Unperturbed reducible equation

If (A, ψ) is a reducible solution of the unperturbed equation, then $\psi = 0$, so the second equation degenerates into

$$F_A^+ = 0.$$

Hence a reducible Seiberg–Witten solution is exactly an anti-self-dual $U(1)$ connection on the determinant line bundle.

Since the $U(1)$ theory is Abelian, this condition can ultimately be described by Hodge theory and a harmonic representative of $c_1(L)$.

After adding the perturbation η , the reducible equation becomes

$$F_A^+ = \eta.$$

The so-called reducible wall discussed later is precisely the study of when this equation has a solution.

4.1.16 Summary of this section

A configuration of Seiberg–Witten theory consists of a determinant connection A and a positive spinor ψ .

The quadratic map

$$q : S^+ \longrightarrow i\Lambda_+^2$$

is defined by the trace-free rank-one endomorphism $(\psi\psi^*)_0$.

Hence the Seiberg–Witten equations read

$$D_A\psi = 0, \quad F_A^+ = q(\psi),$$

or, in the perturbed case,

$$F_A^+ = q(\psi) + \eta.$$

The gauge group $\mathcal{G} = C^\infty(X, U(1))$ acts on configurations, and preserves the solution set of the equations.

So the object that is really studied is

$$\mathcal{M}_{s,\eta} = \frac{\{\text{Seiberg–Witten solutions}\}}{\mathcal{G}}.$$

At this point the Seiberg–Witten moduli space has been defined as a set (Definition 4.1.4).

The next step is to study its local structure. Concretely, one has to analyze at the linear level both the linearization of the Seiberg–Witten equations and the infinitesimal action of the gauge group.

This will produce the Seiberg–Witten deformation complex, and explain why the equations are elliptic after gauge fixing.

4.2 Seiberg–Witten deformation complex

Above we have already defined the perturbed Seiberg–Witten equations

$$D_A\psi = 0, \quad F_A^+ = q(\psi) + \eta,$$

and the corresponding moduli space $\mathcal{M}_{\mathfrak{s},\eta}$.

Next we study the local structure of this quotient space near a solution.

Here there are two problems that have to be handled simultaneously. First, the Seiberg–Witten equations have to be linearized; second, the infinitesimal directions produced by the gauge group have to be removed.

Combining the two gives the Seiberg–Witten deformation complex.

4.2.1 Infinitesimal variations of a configuration

Fix a solution (A, ψ) .

Since $\mathcal{A}(L)$ is an affine space modeled on $i\Omega^1(X)$, an infinitesimal variation of the connection can be written as

$$A \longmapsto A + ta, \quad a \in i\Omega^1(X).$$

An infinitesimal variation of the spinor is written

$$\psi \longmapsto \psi + t\phi, \quad \phi \in \Gamma(S^+).$$

Hence the tangent space of the configuration space at (A, ψ) is

$$T_{(A,\psi)}\mathcal{C}_{\mathfrak{s}} = i\Omega^1(X) \oplus \Gamma(S^+).$$

Below we compute the linearizations of the two Seiberg–Witten equations separately.

4.2.2 Linearization of the Dirac equation

Section 3.4.11 gives the formula for a change of the determinant connection

$$D_{A+a}\psi = D_A\psi + \frac{1}{2}c(a)\psi.$$

Hence, for $A_t = A + ta$ and $\psi_t = \psi + t\phi$, we have

$$D_{A_t}\psi_t = D_A\psi + tD_A\phi + \frac{t}{2}c(a)\psi + O(t^2).$$

So the linearization of the first equation at (A, ψ) is

$$(a, \phi) \longmapsto D_A\phi + \frac{1}{2}c(a)\psi.$$

Here the part involving ϕ is a first-order Dirac operator, whereas $c(a)\psi$ is merely a zeroth-order term in a .

4.2.3 Linearization of the curvature equation

Since the structure group of the determinant line bundle is $U(1)$, this is an Abelian gauge theory.

Hence if $A_t = A + ta$, then

$$F_{A_t} = F_A + t da.$$

Taking the self-dual part gives

$$F_{A_t}^+ = F_A^+ + t d^+ a,$$

where

$$d^+ := \pi_+ \circ d : i\Omega^1(X) \longrightarrow i\Omega_+^2(X).$$

On the other hand, the differential of the quadratic map q at ψ is written

$$dq_\psi : \Gamma(S^+) \longrightarrow i\Omega_+^2(X).$$

It is a real-linear map.

From $\rho(q(\psi)) = (\psi\psi^*)_0$ one obtains directly

$$\rho(dq_\psi(\phi)) = (\phi\psi^* + \psi\phi^*)_0.$$

Hence the linearization of the second equation is

$$(a, \phi) \longmapsto d^+ a - dq_\psi(\phi).$$

The perturbation η is fixed, so it makes no contribution to the linearization.

4.2.4 Linearization of the Seiberg–Witten map

Putting the two parts together, the differential of the Seiberg–Witten map at the solution (A, ψ) is

$$d_1 : i\Omega^1(X) \oplus \Gamma(S^+) \longrightarrow i\Omega_+^2(X) \oplus \Gamma(S^-),$$

where

$$d_1(a, \phi) = \left(d^+ a - dq_\psi(\phi), D_A \phi + \frac{1}{2} c(a) \psi \right).$$

Here interchanging the order of the two target factors is only a matter of notation. These notes adopt $i\Omega_+^2 \oplus \Gamma(S^-)$ as the order.

If we ignore the gauge symmetry for the moment, a pair satisfying $d_1(a, \phi) = 0$ describes a first-order deformation of the solution.

But some of these deformations come only from gauge transformations, and do not correspond to genuinely different points of the moduli space.

4.2.5 The infinitesimal action of the gauge group

The gauge group is $\mathcal{G} = C^\infty(X, U(1))$, with Lie algebra

$$\text{Lie } \mathcal{G} = i\Omega^0(X).$$

Take $f \in i\Omega^0(X)$, and consider the one-parameter gauge transformations

$$u_t = e^{tf}.$$

The gauge action adopted in these notes is

$$u \cdot (A, \psi) = (A - 2u^{-1}du, u\psi).$$

Differentiate at $t = 0$. Since

$$\left. \frac{d}{dt} \right|_{t=0} u_t^{-1} du_t = df,$$

the connection part gives $-2df$.

The spinor part satisfies

$$\left. \frac{d}{dt} \right|_{t=0} u_t \psi = f\psi.$$

Hence the infinitesimal gauge action is

$$\begin{aligned} d_0 : i\Omega^0(X) &\longrightarrow i\Omega^1(X) \oplus \Gamma(S^+), \\ d_0(f) &= (-2df, f\psi). \end{aligned}$$

Thus $\text{im } d_0$ is precisely the tangent space at this point of the gauge orbit through (A, ψ) .

4.2.6 Why $d_1 d_0 = 0$

Since the Seiberg–Witten equations are invariant under gauge transformations, a purely gauge deformation cannot change the first-order value of the equations.

Hence at a solution (A, ψ) one must have

$$d_1 \circ d_0 = 0.$$

This can also be verified directly.

For the first equation, the connection Leibniz rule gives

$$D_A(f\psi) = c(df)\psi + fD_A\psi.$$

On the other hand, the connection variation $-2df$ produces

$$\frac{1}{2}c(-2df)\psi = -c(df)\psi.$$

Since (A, ψ) is a solution, $D_A\psi = 0$, the two terms cancel exactly.

For the second equation, we have $d^+df = 0$. On the other hand, since $q(e^{tf}\psi) = q(\psi)$, differentiating with respect to t gives

$$dq_\psi(f\psi) = 0.$$

So the second component vanishes as well.

Hence indeed $d_1 d_0 = 0$.

4.2.7 Deformation complex

Thus we obtain a cochain complex

$$0 \longrightarrow i\Omega^0(X) \xrightarrow{d_0} i\Omega^1(X) \oplus \Gamma(S^+) \xrightarrow{d_1} i\Omega_+^2(X) \oplus \Gamma(S^-) \longrightarrow 0.$$

This is called the **Seiberg–Witten deformation complex** at (A, ψ) . Define its cohomology groups

$$H_{(A, \psi)}^0 = \ker d_0,$$

$$H_{(A, \psi)}^1 = \frac{\ker d_1}{\operatorname{im} d_0},$$

and

$$H_{(A, \psi)}^2 = \operatorname{coker} d_1.$$

These three spaces each have a direct geometric meaning.

4.2.8 H^0 and the stabilizer

$H_{(A, \psi)}^0 = \ker d_0$ describes the infinitesimal stabilizer.

In fact, $f \in \ker d_0$ if and only if

$$df = 0, \quad f\psi = 0.$$

If X is connected and (A, ψ) is irreducible, then ψ is non-zero at some point. From $df = 0$ it follows that f is constant, and then $f\psi = 0$ gives $f = 0$.

Hence an irreducible solution satisfies

$$H_{(A, \psi)}^0 = 0.$$

If $(A, 0)$ is reducible, then every constant imaginary-valued function lies in the kernel. Therefore

$$H_{(A, 0)}^0 \cong i\mathbb{R}.$$

This is exactly the constant $U(1)$ stabilizer of a reducible solution.

4.2.9 H^1 and the infinitesimal moduli space

$\ker d_1$ describes all deformations that preserve the Seiberg–Witten equations to first order.

Among them, however, $\operatorname{im} d_0$ comes only from infinitesimal gauge transformations.

Hence the genuine infinitesimal deformations modulo gauge are

$$H_{(A, \psi)}^1 = \frac{\ker d_1}{\operatorname{im} d_0}.$$

At an irreducible regular solution, the moduli space is smooth and its tangent space is exactly $H_{(A, \psi)}^1$.

Hence the first cohomology of the deformation complex is the true tangent space of the moduli problem.

4.2.10 H^2 and the obstruction

$H^2_{(A,\psi)} = \text{coker } d_1$ is called the obstruction space.
If

$$H^2_{(A,\psi)} = 0,$$

then d_1 is surjective on a suitable gauge slice.

In this case the Banach implicit function theorem allows one to identify the zero set of the nonlinear Seiberg–Witten equations locally with a finite-dimensional smooth manifold.

Hence a solution satisfying $H^2_{(A,\psi)} = 0$ is called a regular solution.

If H^2 is non-zero, then linearization cannot guarantee that the nonlinear equation has the expected smooth local structure. This is exactly why perturbation and transversality theory are needed later on.

4.2.11 Why gauge fixing is needed

Although d_1 is the linearization of the Seiberg–Witten equations, d_1 alone is not an elliptic operator that is suitable for direct use.

The reason is that

$$d_1 d_0 = 0.$$

Hence all infinitesimal gauge directions in d_0 automatically lie in $\ker d_1$.

This is exactly the same problem as the one produced by $d_A^+ d_A = 0$ in ASD theory above.

In order to obtain an elliptic problem, one has to choose a slice transverse to the gauge orbit.

The most natural choice is to take the L^2 orthogonal complement of the gauge orbit.

4.2.12 Coulomb-type slice

Fix a configuration $C = (A, \psi)$. All adjoints below use real L^2 pairings; the Hermitian product $\langle \psi, \phi \rangle$ is linear in the second variable, as fixed earlier in this chapter. For $f = ih$, with h real, the infinitesimal action is $d_0 f = (-2df, f\psi)$. Integration by parts and the identity

$$\text{Re}\langle ih\psi, \phi \rangle = h \text{Im}\langle \psi, \phi \rangle$$

give

$$(d_0 f, (a, \phi))_{L^2} = (f, -2d^*a + i \text{Im}\langle \psi, \phi \rangle)_{L^2}.$$

Thus the formal adjoint and the slice condition are

$$d_0^*(a, \phi) = -2d^*a + i \text{Im}\langle \psi, \phi \rangle, \quad d_0^*(a, \phi) = 0.$$

The affine slice through C is a small neighborhood of C in $C + \ker d_0^*$. Its tangent vectors are orthogonal to the full gauge orbit of the pair, including its spinor component. The gauge-fixed linearization is therefore $d_0^* \oplus d_1$.

This condition differs from $d^*a = 0$, used later to estimate the connection globally. The latter leaves constant phase rotations of the spinor unrestricted. The coupled condition above is adapted to local coordinates on the configuration quotient; the connection-only condition is sufficient for the compactness estimates of Section 4.5.7.

4.2.13 $d_0^*d_0$ and the local slice

The preceding formula gives the full operator

$$d_0^*d_0f = 4d^*df + |\psi|^2f, \quad (d_0^*d_0f, f)_{L^2} = 4\|df\|_{L^2}^2 + \|f\psi\|_{L^2}^2.$$

On connected X , its kernel is zero when $\psi \neq 0$: $df = 0$ makes f constant, and $f\psi = 0$ then forces $f = 0$. The self-adjoint elliptic operator $d_0^*d_0$ is consequently an isomorphism $W^{k+1,2} \rightarrow W^{k-1,2}$ at an irreducible configuration.

Here is how it produces a local gauge choice. Work with $k \geq 3$ in the completions of Section 4.2.16. For $C' = (A + a, \psi + \phi)$ close to C , define

$$\Phi(f; a, \phi) = d_0^*(a - 2df, e^f(\psi + \phi) - \psi), \quad f \in W^{k+1,2}(i\mathbb{R}).$$

The adjoint in this expression is fixed at C . This is a smooth map to $W^{k-1,2}(i\mathbb{R})$, and

$$\Phi(0; 0, 0) = 0, \quad D_f\Phi(0; 0, 0) = d_0^*d_0.$$

The Banach implicit function theorem gives a unique small $f = f(a, \phi)$ such that $e^f \cdot C'$ lies in $C + \ker d_0^*$. This proves existence and uniqueness for gauge transformations near the identity. To obtain uniqueness under the full gauge group, one also uses properness of the action, verified in Section 4.2.19: after shrinking the slice, any transformation identifying two of its points must be close to the stabilizer. At an irreducible configuration that stabilizer is trivial.

At a reducible configuration, the kernel consists of constant imaginary functions. Restricting to their orthogonal complement removes this kernel for the local gauge-fixing argument, but leaves a residual S^1 action. This is the stabilizer appearing in the local model of Section 4.2.21.

4.2.14 The principal part of the gauge-fixed operator

In order to decide ellipticity, it suffices to keep the terms of highest order.

In d_1 , both $dq_\psi(\phi)$ and $c(a)\psi$ are zeroth-order terms.

Hence the principal part of d_1 is

$$(a, \phi) \mapsto (d^+a, D_A\phi).$$

Similarly, the principal part of d_0^* is just d^*a , up to a non-zero constant factor.

Hence the principal part of the gauge-fixed operator $d_0^* \oplus d_1$ is equivalent to

$$(d^* \oplus d^+) \oplus D_A.$$

In other words, at the level of the highest linear order the Seiberg–Witten deformation problem splits into exactly two familiar elliptic operators: one is

$$d^* \oplus d^+ : i\Omega^1 \longrightarrow i\Omega^0 \oplus i\Omega_+^2,$$

and the other is

$$D_A : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

4.2.15 Symbol complex

Ellipticity can also be verified directly from the symbol sequence.

Fix $x \in X$ and a non-zero covector $\xi \in T_x^*X$.

Ignoring the irrelevant constants and the i in the symbol convention, the symbol sequence of the deformation complex is

$$0 \longrightarrow i\mathbb{R} \longrightarrow iT_x^*X \oplus S_x^+ \longrightarrow i\Lambda_{+,x}^2 \oplus S_x^- \longrightarrow 0,$$

where the first symbol sends f to $(\xi f, 0)$, and the second symbol is

$$(a, \phi) \longmapsto (\pi_+(\xi \wedge a), c(\xi)\phi).$$

In what follows we verify that this sequence is exact for $\xi \neq 0$.

First, the first symbol is clearly injective.

Next, if $c(\xi)\phi = 0$, then, since the Clifford relation gives $c(\xi)^2 = -|\xi|^2 I$, $c(\xi)$ is invertible when $\xi \neq 0$, so $\phi = 0$.

What remains is the condition

$$\pi_+(\xi \wedge a) = 0.$$

Take an orthonormal coframe such that $e^1 = \xi/|\xi|$, and write

$$a = a_1 e^1 + a_2 e^2 + a_3 e^3 + a_4 e^4.$$

The three self-dual forms $\pi_+(e^1 \wedge e^2)$, $\pi_+(e^1 \wedge e^3)$, $\pi_+(e^1 \wedge e^4)$ are linearly independent, so the condition $\pi_+(\xi \wedge a) = 0$ forces $a_2 = a_3 = a_4 = 0$.

Hence a must be parallel to ξ , and therefore lies exactly in the image of the first symbol.

Finally, $c(\xi) : S_x^+ \rightarrow S_x^-$ is an isomorphism, while

$$a \longmapsto \pi_+(\xi \wedge a)$$

restricted to $\xi^\perp$ is an isomorphism from a 3-dimensional real vector space to Λ_+^2 . So the last symbol is surjective.

Hence the symbol sequence is exact.

Proposition 4.2.1. The Seiberg–Witten deformation complex is an elliptic complex. Equivalently, the gauge-fixed operator

$$d_0^* \oplus d_1$$

is an elliptic Fredholm operator.

4.2.16 Sobolev completion

In order to apply the Banach implicit function theorem rigorously, one has to replace the smooth spaces above by suitable Sobolev completions.

For example, for a fixed integer $k \geq 2$, one can consider

$$\mathcal{C}_k = (A_0 + W^{k,2}(i\Lambda^1)) \times W^{k,2}(S^+),$$

together with the Sobolev gauge group

$$\mathcal{G}_{k+1} = W^{k+1,2}(X, U(1)).$$

Since in dimension four $k+1 > 4/2$, the Sobolev multiplication theorem guarantees that $\mathcal{G}_{k+1}$ has a Banach Lie group structure, and that its action on $\mathcal{C}_k$ is smooth.

The Seiberg–Witten map extends to

$$\mathcal{F} : \mathcal{C}_k \longrightarrow W^{k-1,2}(S^-) \oplus W^{k-1,2}(i\Lambda_+^2).$$

Only after these functional-analytic constructions are complete can the local slice theorem and the Banach implicit function theorem be applied rigorously.

On the other hand, since the Seiberg–Witten equations are elliptic, a Sobolev solution is, after elliptic regularity and in a suitable gauge, in fact smooth.

Hence the smooth moduli space obtained at the end does not depend on which sufficiently high Sobolev completion was chosen to begin with.

4.2.17 Why the moduli space is finite-dimensional

On the Sobolev spaces just introduced, the elliptic operator $d_0^* \oplus d_1$ is Fredholm. Its kernel and cokernel are finite-dimensional, and hence so are the deformation groups H^0 , H^1 and H^2 . This is the finite-dimensional linear data from which the local nonlinear model is constructed below.

4.2.18 Local moduli space structure

Near an irreducible solution (A, ψ) , the local slice theorem allows us to choose, for every sufficiently close gauge orbit, a unique small representative satisfying

$$d_0^*(a, \phi) = 0.$$

Thus the quotient problem locally becomes a nonlinear equation on the slice.

If in addition we assume the solution to be regular, that is, $H_{(A,\psi)}^2 = 0$, then the linearization is surjective on the slice.

The Banach implicit function theorem therefore gives: $\mathcal{M}_{\mathfrak{s},\eta}$ is a smooth finite-dimensional manifold near $[(A, \psi)]$, whose tangent space is

$$T_{[(A,\psi)]}\mathcal{M}_{\mathfrak{s},\eta} \cong H_{(A,\psi)}^1.$$

If the solution is reducible, then, since the stabilizer is non-trivial, the quotient generally cannot carry such a smooth manifold structure directly.

Therefore, when we construct the Seiberg–Witten invariant later on, an important task is to use generic perturbation to avoid reducible solutions and to ensure regularity.

4.2.19 Topology of the configuration quotient

For the analytic quotient, fix $k \geq 3$ and give $\mathcal{B}_k = \mathcal{C}_k / \mathcal{G}_{k+1}$ the quotient topology. The smooth moduli space will instead be given the quotient of the C^∞ topology on smooth

solutions. Elliptic regularity connects these descriptions; the compactness proof below establishes smooth convergence directly.

The Sobolev gauge action is proper. To see the relevant compactness criterion, suppose $C_n \rightarrow C$ and $u_n \cdot C_n \rightarrow C'$ in $\mathcal{C}_k$. Write their connection components as A_n and B_n . The transformation law gives

$$du_n = \frac{1}{2}(A_n - B_n)u_n, \quad |u_n| = 1.$$

Since $k > 2$, $W^{k,2}$ embeds in C^0 and is a multiplication algebra. The displayed equation first bounds du_n uniformly; differentiating it and using Sobolev multiplication bounds u_n in $W^{k+1,2}$. Rellich compactness gives a subsequence converging strongly in $W^{k,2}$ to a circle-valued map u . The right-hand sides then converge in $W^{k,2}$, so the derivative equation improves the convergence to $W^{k+1,2}$. Taking the limit in the action yields $C' = u \cdot C$. This is the sequential criterion for properness in these metrizable spaces.

In particular the orbit relation is closed. The quotient map is open, because the saturation of an open set is a union of its gauge translates. These facts imply that $\mathcal{B}_k$ is Hausdorff. Together with the local slices above, they give local quotient charts; on the irreducible locus the charts are Hilbert manifold charts. For the compactness theorem we use the more direct compact-set argument in Section 4.5.11, so no metrizability assertion about an arbitrary quotient is needed.

4.2.20 Local representatives and global choices

Choosing one representative from each orbit as a set-theoretic matter does not make that choice continuous. A local slice gives more: on a neighborhood V of an irreducible class, the quotient map $\pi : \mathcal{C}_k^* \rightarrow \mathcal{B}_k^*$ has a smooth section

$$\sigma : V \longrightarrow \mathcal{C}_k^*, \quad \pi \circ \sigma = \text{id}_V.$$

The construction depends on a reference configuration and the chosen slice. A global continuous section would have to make compatible choices over the whole quotient; local existence does not imply this. Even a global section, when it exists, need not be canonical, since its construction may require additional choices.

The same distinctions apply after restricting to the solution space. One must not infer that every particular moduli space has a global obstruction to choosing representatives: for a finite discrete moduli space, for example, a choice at each point is continuous, but is not thereby distinguished geometrically. The relation between global sections and the universal line bundle is explained in Remark 4.7.3.

4.2.21 The local obstruction map

The regularity condition $H_{(A,\psi)}^2 = 0$ has a nonlinear meaning. To see it, fix an irreducible solution $C = (A, \psi)$ and use the local slice constructed above. Write the equation on this slice as

$$F_C(\xi) = 0, \quad F_C(0) = 0,$$

where ξ is a small displacement from C and $D = DF_C(0)$ is the linearization restricted to the slice. Its kernel is H_C^1 and its cokernel is H_C^2 . These are real vector spaces.

Choose closed complements in the Sobolev domain and target:

$$T_C \mathcal{S}_C = H_C^1 \oplus K', \quad \mathcal{E}_C = \operatorname{im} D \oplus H_C^2.$$

The restriction of D to K' is an isomorphism onto its image. Consequently, the implicit function theorem solves the image component of $F_C(h + v) = 0$ uniquely as $v = v(h)$ for small $h \in H_C^1$. The remaining equation is

$$\kappa(h) = 0, \quad \kappa(h) := \pi_{H_C^2} F_C(h + v(h)).$$

Thus a neighborhood of $[C]$ in the moduli space is modeled on

$$\kappa^{-1}(0) \subset N \subset H_C^1, \quad \kappa : N \longrightarrow H_C^2.$$

Here N is a neighborhood of the origin. Differentiating the image component gives $Dv(0) = 0$, and projection to the cokernel kills the linear part of F_C . Hence

$$\kappa(0) = 0, \quad D\kappa(0) = 0.$$

This is the Kuranishi construction of Section A.11, applied to the SW slice.

If $H_C^2 = 0$, there is no remaining equation, and the moduli space is locally an open set in H_C^1 . If $H_C^2 \neq 0$, the higher-order terms of κ decide which infinitesimal deformations extend to solutions. A nonzero obstruction space does not by itself imply that the zero set is singular, but the regular-level-set argument no longer applies. In particular, H_C^1 is the infinitesimal deformation space; at an obstructed point it must not automatically be identified with the tangent space of a smooth zero set.

At a reducible solution, the stabilizer is the constant S^1 . One can choose the complements and the construction equivariantly, and the local model becomes

$$\kappa^{-1}(0)/S^1.$$

This residual action explains why taking a slice alone does not remove the singular behavior at reducibles.

4.2.22 Expected dimension as Fredholm index

The gauge-fixed deformation operator

$$d_0^* \oplus d_1$$

is a Fredholm operator.

Define its Fredholm index by

$$\operatorname{ind}(d_0^* \oplus d_1) = \dim \ker(d_0^* \oplus d_1) - \dim \operatorname{coker}(d_0^* \oplus d_1).$$

By the Hodge theory of the elliptic complex, this index can be written as

$$\dim H^1 - \dim H^0 - \dim H^2.$$

It is called the **expected dimension** or the **virtual dimension** of the Seiberg–Witten moduli problem.

At an irreducible regular solution, $H^0 = H^2 = 0$, so the expected dimension is exactly the actual local dimension $\dim H^1$.

More importantly, the Fredholm index stays unchanged when (A, ψ) is varied continuously. Hence the expected dimension depends only on the Spin^c structure and on the topological data of the underlying manifold, and not on the particular solution.

The next step is to compute this Fredholm index by means of the Atiyah–Singer index theorem, which finally gives

$$d(\mathfrak{s}) = \frac{c_1(L_{\mathfrak{s}})^2 - (2\chi(X) + 3\sigma(X))}{4}.$$

This formula turns the local dimension of an infinite-dimensional nonlinear gauge problem into characteristic numbers of the 4-manifold.

4.3 The expected dimension of the Seiberg–Witten moduli space

We have seen in the previous section that, near a solution (A, ψ) , the Seiberg–Witten deformation problem is controlled, after gauge fixing, by an elliptic Fredholm operator.

Its principal part splits into

$$d^* \oplus d^+ : i\Omega^1(X) \longrightarrow i\Omega^0(X) \oplus i\Omega_+^2(X)$$

and

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

All terms involving ψ are zeroth-order terms, so they do not change the Fredholm index.

Hence the computation of the expected dimension can be split into two mutually independent problems: first compute the real index of $d^* \oplus d^+$, and then compute the complex index of the Spin^c Dirac operator.

4.3.1 The index of $d^* \oplus d^+$

Consider the operator

$$P = d^* \oplus d^+ : \Omega^1(X) \longrightarrow \Omega^0(X) \oplus \Omega_+^2(X).$$

Here we temporarily omit the $i\mathbb{R}$ coefficients, since multiplying by i does not change the underlying real vector spaces or their index.

This operator comes from the self-dual complex of Section 1.4.5, whose elliptic symbol is computed in Section A.2.10:

$$0 \longrightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d^+} \Omega_+^2(X) \longrightarrow 0.$$

We first describe the cohomology of this complex.

In degree 0,

$$H^0 = \ker d.$$

Since X is connected, $\ker d$ consists of the constant functions, so

$$\dim H^0 = 1.$$

In degree 1,

$$H^1 = \frac{\ker d^+}{\operatorname{im} d}.$$

This space is naturally isomorphic to the ordinary de Rham cohomology $H^1(X; \mathbb{R})$.

To see this, let $a \in \Omega^1(X)$ satisfy $d^+a = 0$. Then da is an anti-self-dual 2-form. On the other hand, da is an exact form, so

$$\int_X da \wedge da = 0.$$

But since $*da = -da$, we have

$$\int_X da \wedge da = - \int_X |da|^2 d\operatorname{vol}.$$

Hence $da = 0$. Therefore $\ker d^+ = \ker d$, and thus

$$H^1 \cong H_{\text{dR}}^1(X).$$

Hence $\dim H^1 = b_1(X)$.

In degree 2, the cohomology of this elliptic complex is naturally isomorphic to the space of self-dual harmonic 2-forms $\mathcal{H}^+(X; g)$. By the Hodge theorem,

$$\dim \mathcal{H}^+(X; g) = b_2^+(X).$$

Hence the Euler characteristic of the elliptic complex is

$$1 - b_1(X) + b_2^+(X).$$

But the domain of the operator $P = d^* \oplus d^+$ lies in the middle term of the complex, so its Fredholm index differs from the above Euler characteristic by a minus sign.

Thus

$$\operatorname{ind}_{\mathbb{R}}(d^* \oplus d^+) = b_1(X) - 1 - b_2^+(X).$$

It can also be written as

$$\operatorname{ind}_{\mathbb{R}}(d^* \oplus d^+) = -(1 - b_1(X) + b_2^+(X)).$$

4.3.2 The index of the Spin^c Dirac operator

In what follows we compute the index of

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

Although the explicit expression for D_A^+ depends on the metric g and on the connection A , as an elliptic operator its Fredholm index depends only on the topological type of the Spin^c structure.

The Atiyah–Singer index theorem, applied to the Spin^c Dirac operator, gives

$$\operatorname{ind}_{\mathbb{C}}(D_A^+) = \left\langle e^{c_1(L)/2} \hat{A}(TX), [X] \right\rangle.$$

Here $L = L_{\mathfrak{s}}$ is the determinant line bundle, while $\hat{A}(TX)$ is the $\hat{A}$ -class of the tangent bundle.

Since X is a 4-manifold, only the degree-4 part has to be kept.

Write

$$c = c_1(L).$$

The exponential factor expands as

$$e^{c/2} = 1 + \frac{1}{2}c + \frac{1}{8}c^2 + \cdots,$$

while

$$\hat{A}(TX) = 1 - \frac{1}{24}p_1(TX) + \cdots.$$

So the degree-4 part of their product is

$$\frac{1}{8}c^2 - \frac{1}{24}p_1(TX).$$

Hence

$$\text{ind}_{\mathbb{C}}(D_A^+) = \frac{1}{8}\langle c_1(L)^2, [X] \rangle - \frac{1}{24}\langle p_1(TX), [X] \rangle.$$

By the Hirzebruch signature theorem,

$$\langle p_1(TX), [X] \rangle = 3\sigma(X).$$

So we obtain

$$\text{ind}_{\mathbb{C}}(D_A^+) = \frac{c_1(L)^2 - \sigma(X)}{8}.$$

From now on we write

$$c_1(L)^2 := \langle c_1(L) \smile c_1(L), [X] \rangle.$$

Note that this is the complex Fredholm index. In the Seiberg–Witten deformation problem what we compute is the real dimension, so the contribution of the Dirac operator to the real index is

$$2 \text{ind}_{\mathbb{C}}(D_A^+) = \frac{c_1(L)^2 - \sigma(X)}{4}.$$

4.3.3 The total index

At the principal-symbol level, the gauge-fixed Seiberg–Witten operator is $(d^* \oplus d^+) \oplus D_A^+$.

Hence its real Fredholm index is

$$\begin{aligned} d(\mathfrak{s}) &= \text{ind}_{\mathbb{R}}(d^* \oplus d^+) + 2 \text{ind}_{\mathbb{C}}(D_A^+) \\ &= b_1(X) - 1 - b_2^+(X) + \frac{c_1(L)^2 - \sigma(X)}{4}. \end{aligned}$$

This is already a commonly used expression for the expected dimension.

However, one usually wants to write it in a form that involves only the Euler characteristic and the signature.

4.3.4 Simplifying with χ and σ

For a closed oriented 4-manifold, Poincaré duality gives $b_3 = b_1$ and $b_4 = b_0 = 1$.

Hence the Euler characteristic is

$$\chi(X) = 2 - 2b_1 + b_2^+ + b_2^-.$$

On the other hand, the signature of the intersection form is

$$\sigma(X) = b_2^+ - b_2^-.$$

From this we obtain

$$\frac{\chi(X) + \sigma(X)}{2} = 1 - b_1(X) + b_2^+(X).$$

Therefore

$$b_1 - 1 - b_2^+ = -\frac{\chi + \sigma}{2}.$$

Substituting into the total index, we have

$$\begin{aligned} d(\mathfrak{s}) &= \frac{c_1(L)^2 - \sigma}{4} - \frac{\chi + \sigma}{2} \\ &= \frac{c_1(L)^2 - 2\chi - 3\sigma}{4}. \end{aligned}$$

We thus obtain the expected dimension formula.

Theorem 4.3.1 (Seiberg–Witten expected dimension). Let X be a closed, connected, oriented smooth 4-manifold, let $\mathfrak{s}$ be a Spin^c structure on it, and let $L = L_{\mathfrak{s}}$ be the determinant line bundle. Then the expected dimension of the Seiberg–Witten moduli problem is

$$d(\mathfrak{s}) = \frac{c_1(L)^2 - (2\chi(X) + 3\sigma(X))}{4}.$$

4.3.5 Why this number must be an integer

Judging from the final formula itself, the right-hand side contains a division by 4, so its integrality is not formally obvious.

But $d(\mathfrak{s})$ was originally defined as the index of a Fredholm operator, and hence must belong to $\mathbb{Z}$.

One can also see directly from the congruences among characteristic classes that this integrality is no accident.

Since

$$c_1(L) \equiv w_2(TX) \pmod{2},$$

$c_1(L)$ is a characteristic element of the intersection form.

For a closed oriented 4-manifold, characteristic elements satisfy

$$c_1(L)^2 \equiv \sigma(X) \pmod{8}.$$

Therefore

$$\frac{c_1(L)^2 - \sigma(X)}{8}$$

is indeed an integer, which is also consistent with the Spin^c Dirac index formula.

4.3.6 Expected dimension and actual dimension

The index calculation gives

$$d(\mathfrak{s}) = \dim H^1 - \dim H^0 - \dim H^2.$$

At an irreducible regular solution, $H^0 = H^2 = 0$, so the local manifold of Section 4.2.18 has dimension $d(\mathfrak{s})$. Without regularity, this number remains an index; the actual zero set is governed by the obstruction map of Section 4.2.21. The perturbation theorem in the next section supplies regularity on the irreducible locus.

4.3.7 Negative expected dimension

When $d(\mathfrak{s}) < 0$, an irreducible solution cannot be regular: otherwise the local model would have negative dimension. Section 4.4.13 will turn this observation into an emptiness statement after perturbation and wall avoidance.

4.3.8 The zero-dimensional case

When $d(\mathfrak{s}) = 0$, regularity and compactness will reduce the moduli space to finitely many points. Orientations then allow a signed count, defined in Section 4.7.1. The index formula therefore identifies the case in which no additional cohomology insertions are needed.

4.3.9 Example: K3

Consider the K3 surface.

Its topological invariants are

$$\chi(K3) = 24, \quad \sigma(K3) = -16.$$

K3 is a spin manifold, so the Spin^c structure induced by the spin structure has trivial determinant line, that is, $c_1(L) = 0$.

Hence

$$d(\mathfrak{s}) = \frac{0 - 2 \cdot 24 - 3(-16)}{4} = 0.$$

So this Spin^c structure corresponds to a zero-dimensional Seiberg–Witten moduli problem.

This is exactly why the basic Seiberg–Witten invariant of K3 can be defined by a signed count.

4.3.10 Example: $\mathbb{CP}^2$

Let $H \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the positive generator, satisfying $H^2 = 1$.

We have seen above that the Spin^c determinant classes are all of the form

$$c_1(L) = (2k + 1)H.$$

On the other hand,

$$\chi(\mathbb{CP}^2) = 3, \quad \sigma(\mathbb{CP}^2) = 1.$$

Therefore

$$d(\mathfrak{s}_k) = \frac{(2k+1)^2 - 9}{4} = k^2 + k - 2.$$

For example, when $c_1(L) = 3H$, we have $d(\mathfrak{s}) = 0$.

However, $b_2^+(\mathbb{CP}^2) = 1$, so even though the expected dimension is zero, the Seiberg–Witten invariant still involves chamber dependence.

This shows that the expected dimension only settles the question of the dimension of the moduli space, and cannot replace the transversality, compactness and wall-crossing analysis that come later.

4.3.11 Summary of this section

The principal part of the Seiberg–Witten deformation operator splits into $d^* \oplus d^+$ and the Spin^c Dirac operator D_A^+ .

Hodge theory gives

$$\text{ind}_{\mathbb{R}}(d^* \oplus d^+) = b_1 - 1 - b_2^+,$$

while the Atiyah–Singer index theorem gives

$$\text{ind}_{\mathbb{C}}(D_A^+) = \frac{c_1(L)^2 - \sigma(X)}{8}.$$

Since the Dirac operator is complex linear, its real index is twice its complex index.

We finally obtain

$$d(\mathfrak{s}) = \frac{c_1(L)^2 - (2\chi + 3\sigma)}{4}.$$

This formula is the first core bridge leading from analysis to topology in Seiberg–Witten theory: the virtual dimension of an infinite-dimensional nonlinear gauge equation is in the end determined only by $c_1(L)$, the Euler characteristic and the signature of the intersection form.

The next problem to be solved is: how to guarantee that the actual moduli space really becomes a smooth manifold of the expected dimension.

This requires introducing generic perturbations, studying reducible solutions, and using the Sard–Smale transversality theorem.

4.4 Perturbation, transversality and the reducible wall

The previous section computed the expected dimension of the Seiberg–Witten moduli problem (Theorem 4.3.1). But the expected dimension is only a Fredholm index, and it cannot guarantee that the actual moduli space really is a smooth manifold of the corresponding dimension.

There are two sources of trouble.

First, the linearization of the Seiberg–Witten map may fail to be surjective, so that a solution is not regular.

Second, reducible solutions may occur in the moduli space. In this case the action of the gauge group has a non-trivial stabilizer, and the local structure of the quotient is no longer an ordinary smooth manifold.

Hence, before we can really define the Seiberg–Witten invariant, we have to handle both problems at once by means of perturbation.

4.4.1 Why perturbation is needed

The unperturbed Seiberg–Witten equations are

$$D_A\psi = 0, \quad F_A^+ = q(\psi).$$

Even if a solution is irreducible, the linearization

$$d_1 : i\Omega^1 \oplus \Gamma(S^+) \longrightarrow i\Omega_+^2 \oplus \Gamma(S^-)$$

need not be surjective.

If $\text{coker } d_1 \neq 0$, then the obstruction space H^2 of the deformation complex above is non-zero, and in this case one cannot directly obtain from the implicit function theorem a smooth moduli space of the expected dimension.

The standard remedy is to add to the curvature equation a self-dual 2-form perturbation

$$\eta \in i\Omega_+^2(X),$$

and then to consider

$$D_A\psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

By varying η , one can make the nonlinear equation transverse for a generic parameter.

4.4.2 Universal Seiberg–Witten map

In order to study all perturbations at once, we also take η as a variable.

Write the perturbation space as

$$\mathcal{P} = i\Omega_+^2(X),$$

For the Banach-manifold argument fix an integer $k \geq 3$ and use $\mathcal{P}_{k-1} = W^{k-1,2}(i\Lambda^+)$, configurations of class $W^{k,2}$, and gauge transformations of class $W^{k+1,2}$. The target has class $W^{k-1,2}$. Sobolev multiplication makes the universal section smooth between these completions. All smooth perturbations belong to this parameter space; the passage back to their C^∞ topology will be justified below.

Define the universal Seiberg–Witten map

$$\mathfrak{F} : \mathcal{C}_s \times \mathcal{P} \longrightarrow \Gamma(S^-) \oplus i\Omega_+^2(X)$$

by

$$\mathfrak{F}(A, \psi, \eta) = (D_A\psi, F_A^+ - q(\psi) - \eta).$$

Once η is fixed, $\mathfrak{F}(\cdot, \eta) = 0$ is exactly the corresponding perturbed Seiberg–Witten equations.

At a solution (A, ψ, η) , differentiating in the perturbation direction $\nu \in i\Omega_+^2(X)$ makes the second component acquire an extra term $-\nu$.

Hence the universal linearization has more adjustable directions than when η is fixed.

This is precisely the basic analytic mechanism by which perturbation removes the obstruction.

4.4.3 Surjectivity of the universal linearization

The perturbation direction fills the curvature component of the target. We must still explain why no obstruction remains in the spinor component. At a smooth solution the full differential is

$$\mathcal{L}(a, \phi, \nu) = \left(D_A \phi + \frac{1}{2} c(a) \psi, d^+ a - dq_\psi(\phi) - \nu \right).$$

Here $a \in i\Omega^1$, $\phi \in \Gamma(S^+)$ and $\nu \in i\Omega_+^2$ are independent variations. All adjoints and orthogonality statements in this argument use real L^2 pairings.

Proposition 4.4.1. At an irreducible solution, the universal linearization is surjective, also after restricting configuration variations to a local gauge slice.

Proof. For fixed perturbation, the linearization on the slice is Fredholm. Its image is therefore closed and has finite codimension. Adding the perturbation directions still gives a closed image: its image in the finite-dimensional quotient of the target is a closed subspace. It suffices to show that its annihilator is zero. Elliptic regularity for the adjoint of the gauge-fixed operator makes an annihilating Sobolev dual vector smooth, so it may be represented by an L^2 pair

$$(\chi, \beta) \in \Gamma(S^-) \oplus i\Omega_+^2.$$

Varying ν arbitrarily gives $\beta = 0$. Varying ϕ then gives

$$D_A^* \chi = 0.$$

Finally, variations of a give

$$\operatorname{Re} \int_X \langle \chi, c(a) \psi \rangle d\operatorname{vol} = 0 \quad \text{for every } a \in i\Omega^1(X).$$

Using variations supported in a coordinate ball makes this a pointwise orthogonality condition.

At a point x with $\psi(x) \neq 0$, consider the real-linear map

$$iT_x^* X \longrightarrow S_x^-, \quad a \longmapsto c(a) \psi(x).$$

For $a = i\alpha$ with α real, the Clifford relations imply

$$|c(i\alpha) \psi(x)|^2 = |\alpha|^2 |\psi(x)|^2.$$

The map is injective, and both real dimensions are four, so it is an isomorphism. The pointwise orthogonality therefore forces $\chi(x) = 0$. Since the solution is irreducible, ψ is nonzero on some nonempty open set. The unique continuation theorem for the Dirac-type operator D_A^* implies that a solution of $D_A^* \chi = 0$ vanishing on such an open set vanishes on connected X . Thus $\chi = 0$.

At a zero of the equations, infinitesimal gauge directions are killed by the linearization. Decomposing a configuration variation into a slice direction and a gauge direction therefore does not change its image. This proves the assertion on the slice as well. $\square$

Unique continuation is the analytic input in this proof; it does not follow merely from ellipticity. The additional perturbation term and the nonvanishing of the spinor are both needed before it can be applied. The Sobolev universal section is C^∞ , so its differentiability exceeds the Fredholm index required by Sard–Smale. Elliptic regularity makes solutions for smooth data smooth modulo gauge. It does not, by itself, transfer a residual subset of a Sobolev parameter space to the space of smooth parameters.

4.4.4 Generic regularity

After the Sobolev completion, the universal moduli problem can be placed in the framework of Banach manifolds and Banach bundles.

For irreducible solutions, Proposition 4.4.1 shows that the universal section is transverse to the zero section at its zeros.

The projection from the universal zero set to $\mathcal{P}_{k-1}$ is Fredholm of index $d(\mathfrak{s})$, and its regular values are exactly the parameters for which all irreducible solutions are regular; see Section A.12.5. Sard–Smale gives a residual set in this Sobolev parameter space. For smooth parameters the corresponding statement is:

Theorem 4.4.2 (Generic transversality). Fix a Riemannian metric g and a Spin^c structure $\mathfrak{s}$. For a residual set of perturbations

$$\eta \in i\Omega_+^2(X),$$

all irreducible Seiberg–Witten solutions are regular.

Proof. We give the extra argument needed for the C^∞ topology. It uses compactness from Sections 4.5.11 and 4.5.12; those estimates do not use transversality, so there is no circularity. They also apply to smoothly convergent data and to compact finite-dimensional families, with the same uniform bounds and subsequence argument.

For $n \geq 1$, let $\mathcal{O}_n \subset i\Omega_+^2(X)$ consist of perturbations for which every solution with $\|\psi\|_{L^2} \geq 1/n$ is regular. This set is open. Otherwise, nonregular solutions for a sequence $\eta_j \rightarrow \eta \in \mathcal{O}_n$ would converge modulo gauge to a solution with the same lower bound. On a slice the linearizations converge in operator norm, and surjectivity is open for Fredholm operators, a contradiction.

To prove density, fix a smooth η_0 . The solutions for η_0 with $\|\psi\|_{L^2} \geq 1/(2n)$ form a compact subset of the irreducible quotient. At each such solution universal surjectivity allows finitely many smooth perturbing forms to span the cokernel of the fixed-parameter linearization. Indeed, smooth forms are dense in $\mathcal{P}_{k-1}$, and spanning a finite-dimensional quotient is an open condition. Surjectivity persists on a neighborhood of the solution. A finite subcover therefore supplies one finite-dimensional space $V \subset i\Omega_+^2(X)$ which makes the family $\eta_0 + v$, $v \in V$, transverse near this compact set.

For sufficiently small v , every solution with $\|\psi\|_{L^2} \geq 1/n$ lies in these neighborhoods. Failure would, by compactness, give a limiting solution for η_0 outside them. The regular-value theorem now makes this local universal zero set a finite-dimensional manifold. Sard’s theorem for its projection to V supplies arbitrarily small v for which all the relevant solutions are regular. Smallness in V implies smallness in any prescribed C^∞ neighborhood. Thus $\mathcal{O}_n$ is dense.

The smooth parameter space is a Fréchet Baire space. Consequently $\bigcap_{n \geq 1} \mathcal{O}_n$ is residual and dense. Every irreducible solution has positive L^2 spinor norm, so this intersection is precisely the set of perturbations in the statement. $\square$

Here residual means a set containing a countable intersection of open dense sets. A residual set in a Baire space is dense, so such perturbations are usually called generic for short.

Hence, as long as there are also no reducible solutions, the generic perturbed moduli space is a smooth manifold of the expected dimension $d(\mathfrak{s})$.

4.4.5 Regularity and the actual dimension

Combining Theorem 4.4.2 with the local model of Section 4.2.18 and the index calculation above, the irreducible moduli space for generic perturbation is smooth of dimension $d(\mathfrak{s})$. To apply this statement to the whole moduli space, we must also exclude reducible solutions.

4.4.6 Solvability of the reducible equation

At a reducible solution $\psi = 0$, so the equations reduce to

$$F_A^+ = \eta.$$

Fix the metric g , and let $c_1(L)_g^+$ be the self-dual projection of the harmonic representative of $c_1(L)_{\mathbb{R}}$. Write Harm for harmonic projection. Our Chern–Weil convention $[\frac{i}{2\pi}F_A] = c_1(L)_{\mathbb{R}}$ gives

$$\text{Harm}(F_A^+) = -2\pi i c_1(L)_g^+.$$

Thus reducibility requires

$$\text{Harm}(\eta) = -2\pi i c_1(L)_g^+.$$

This condition is also sufficient by Proposition 2.4.3: writing $A = A_0 + a$, one solves $d^+a = \eta - F_{A_0}^+$ precisely when the right-hand side has zero harmonic projection. No further condition comes from the Dirac equation, since the spinor is zero.

4.4.7 Reducible wall

The harmonic projection

$$\text{Harm} : i\Omega_+^2(X) \longrightarrow i\mathcal{H}_g^+(X)$$

is surjective, and its target has dimension $b_2^+(X)$. The inverse image of the value specified above is therefore an affine subspace of codimension $b_2^+(X)$ in the perturbation space. We call it the **reducible wall**. Its codimension controls whether a single perturbation, or a path of perturbations, can avoid reducibles.

4.4.8 Why $b_2^+ > 0$ avoids reducibles

If

$$b_2^+(X) > 0,$$

the reducible wall has positive codimension in the perturbation space.

Hence a generic single perturbation does not lie on the wall.

Therefore when $b_2^+ > 0$, for generic η , the moduli space contains no reducible solutions.

Combining this with the generic regularity of the previous section (Theorem 4.4.2) yields:

Proposition 4.4.3. If $b_2^+(X) > 0$, then for a generic perturbation η , the Seiberg–Witten moduli space $\mathcal{M}_{\mathfrak{s},\eta}$ contains only irreducible regular solutions, and is therefore a smooth manifold of the expected dimension $d(\mathfrak{s})$.

Compactness has not been discussed yet, so for the moment we only get smoothness.

4.4.9 Why defining the invariant usually requires $b_2^+ > 1$

Merely to choose a generic perturbation avoiding reducibles, $b_2^+ > 0$ already suffices.

But to prove that the Seiberg–Witten invariant is independent of the choices of perturbation and metric, one also has to connect two sets of auxiliary data.

Let

$$(g_0, \eta_0), \quad (g_1, \eta_1)$$

be two sets of generic data, and consider the one-parameter family

$$(g_t, \eta_t), \quad t \in [0, 1].$$

In order to construct the parameterized moduli space connecting the two moduli spaces, we want the whole path to avoid the reducible wall.

A one-dimensional path generically avoids a subset of codimension at least 2.

Hence we require

$$b_2^+(X) > 1.$$

This is the reason why the Seiberg–Witten invariant, in its simplest chamber-independent form, is usually assumed to satisfy $b_2^+(X) > 1$, and this is the reason.

One has to distinguish two different levels:

$$b_2^+ > 0$$

is enough for a generic single perturbation to have no reducibles;

whereas

$$b_2^+ > 1$$

guarantees that a generic one-parameter family can also avoid the reducible wall.

The two conditions must not be conflated.

4.4.10 $b_2^+ = 1$ and the chamber structure

When

$$b_2^+(X) = 1,$$

the reducible wall is an affine subspace of codimension 1.

Hence the perturbation space is divided by the wall into different connected components, called **chambers**.

Within a single chamber, one can choose a generic path avoiding the wall, so the corresponding Seiberg–Witten invariant stays unchanged.

But a path connecting different chambers usually has to cross the wall.

When the path crosses the wall, reducible solutions appear, and the signed count may change.

This phenomenon is called **wall crossing**.

Hence in the case $b_2^+ = 1$, the Seiberg–Witten invariant generally depends not only on the Spin^c structure but also on the choice of chamber.

$\mathbb{CP}^2$ is precisely the basic example of this situation.

4.4.11 Identifying the bundles along a path of metrics

When the metric changes, the notation D_{A,g_t} involves changing spinor bundles as well as changing coefficients of an operator. Fix the underlying topological Spin^c structure and let g_t , $0 \leq t \leq 1$, be a smooth path of metrics. Relative to g_0 , write

$$g_t(v, w) = g_0(B_t v, w),$$

where B_t is the positive g_0 -self-adjoint bundle automorphism of TX . The map $B_t^{-1/2}$ sends a g_0 -orthonormal frame to a g_t -orthonormal frame. Transporting the fixed lift of the frame bundle along these identifications gives spinor bundles over $X \times [0, 1]$. Choose smooth identifications with the bundles at $t = 0$. The determinant line can likewise be identified with the fixed L .

The spaces Λ_{+,g_t}^2 also form a smooth bundle over the interval and can be identified with a fixed rank-three bundle. After these choices, D_{A,g_t} , self-dual projection, and q_{g_t} are smooth families of operators on fixed Sobolev spaces. They are not constant operators: their coefficients still depend on t . A perturbation η_t means a smooth section of the varying bundle $i\Lambda_{+,g_t}^2$, expressed in these identifications.

In local trivializations the gauge-equivariant equations descend to a section of a Banach bundle over the irreducible quotient. Explicitly, for a fixed metric its fiber model is

$$\mathcal{E}_k = \left(\mathcal{C}_k^* \times (W^{k-1,2}(S^-) \oplus W^{k-1,2}(i\Lambda_+^2)) \right) / \mathcal{G}_{k+1},$$

where u acts by u on the negative spinor and trivially on the form. The local expression for this section is the SW map. This formulation makes the parameterized zero set meaningful without choosing a global gauge slice. Different smooth bundle identifications give isomorphic presentations of the same zero-set problem.

4.4.12 Parameterized moduli space

Fix regular endpoint data with no reducible solutions. Choose a smooth path (g_t, η_t) , $t \in [0, 1]$, joining them and constant on small endpoint intervals. We first explain how to choose its interior so that the parameterized equation is transverse.

Define the parameterized solution space to consist of all triples (t, A, ψ) satisfying

$$D_{A,g_t} \psi = 0, \quad F_{A,g_t}^+ = q_{g_t}(\psi) + \eta_t$$

for some $t \in [0, 1]$.

Dividing by the gauge group, we obtain the parameterized moduli space

$$\mathcal{W} = \bigcup_{t \in [0,1]} \{t\} \times \mathcal{M}_{\mathfrak{s},g_t,\eta_t}.$$

Since one extra parameter t has been added, its expected dimension is

$$d(\mathfrak{s}) + 1.$$

Proposition 4.4.4 (Relative path transversality). Assume $b_2^+(X) > 1$. For any two regular endpoint data with no reducibles, there is a smooth joining path, constant near its endpoints, which avoids reducibles and for which the parameterized moduli space is a smooth manifold of dimension $d(\mathfrak{s}) + 1$, with boundary its two endpoint moduli spaces.

Proof. Choose a smooth joining metric path, constant near the endpoints, and use the bundle identifications of Section 4.4.11. The reducible condition is the vanishing of the harmonic projection of the curvature equation. Along the interval the self-dual harmonic spaces form a rank- b_2^+ vector bundle. Perturbing the harmonic part of η_t by smooth sections supported in the interior makes this section transverse to zero, with its endpoint values fixed. Since the interval has dimension one and $b_2^+ > 1$, a transverse section has no zeros. This constructs a wall-avoiding path. Its positive minimum distance from zero in the harmonic bundle shows that wall avoidance persists under sufficiently small perturbations.

It remains to make the parameterized equation transverse. At an interior solution, variations of the form $\beta(t)\nu$, where ν is a smooth imaginary self-dual form and β is supported near that time with value one there, supply the same perturbation directions as in Proposition 4.4.1. Near the endpoints no such variations are needed: the path is constant and its endpoint linearizations on slices are already surjective.

The compactness estimates of Section 4.5.12 apply before transversality is imposed. The solution space over the chosen path is therefore compact, and all its solutions are irreducible. As in the proof of Theorem 4.4.2, finitely many smooth perturbation directions, now supported away from the endpoint intervals, make the universal path family transverse near every solution. Compactness ensures that all solutions for sufficiently small family parameters remain in these neighborhoods. The projection from this finite-dimensional universal zero set to its parameter space has regular values arbitrarily close to zero by Sard's theorem. Choose one small enough to preserve wall avoidance. The resulting path has the required properties.

Adding the variable t raises the Fredholm index by one. On each constant endpoint interval the zero set is the product of that interval with the regular endpoint moduli space. These are the boundary collars. Thus no additional boundary component appears. Orientation and compactness of the resulting cobordism are treated below. $\square$

Only the parameterized equation is required to be transverse. An individual fiber at an interior time may fail to be regular; such a time is a critical value of the projection $\mathcal{W} \rightarrow [0, 1]$.

Hence, once compactness also holds, $\mathcal{W}$ will become a compact oriented cobordism joining the two moduli spaces at the endpoints.

This is precisely the core mechanism by which one later proves that the Seiberg–Witten invariant is independent of the metric and of the perturbation.

4.4.13 Generic emptiness in the negative-dimensional case

Suppose $d(\mathfrak{s}) < 0$ and choose generic data for which the moduli space contains no reducibles. Theorem 4.4.2 would make any nonempty moduli space a smooth manifold of dimension $d(\mathfrak{s})$, which is impossible. Hence

$$\mathcal{M}_{\mathfrak{s},\eta} = \emptyset.$$

For $b_2^+(X) > 0$, generic perturbations can satisfy both regularity and wall avoidance, as explained above. The conclusion concerns these generic data; special data can still admit nonregular solutions.

4.4.14 Summary of this section

The expected dimension formula does not by itself guarantee that the moduli space is smooth.

By adding a perturbation $\eta \in i\Omega_+^2(X)$ and using the Sard–Smale theorem, we can make the generic irreducible solutions regular.

On the other hand, reducible solutions satisfy

$$F_A^+ = \eta.$$

Hodge theory shows that this equation is solvable if and only if the self-dual harmonic part of η equals the fixed value determined by $c_1(L)$ and the metric.

Hence the perturbations that allow reducibles form an affine wall of codimension $b_2^+(X)$.

When $b_2^+ > 0$, a single generic perturbation can avoid the wall; when $b_2^+ > 1$, a generic one-parameter family can avoid the wall as well.

Therefore in the standard case $b_2^+ > 1$, the generic perturbed moduli space is a smooth manifold of the expected dimension, and different generic auxiliary data can be related to one another through the parameterized moduli space.

One key property is still missing at this point: even if the moduli space is already smooth, it may still fail to be compact.

The next section must therefore prove the compactness of the Seiberg–Witten moduli space. There the Weitzenböck formula will once again play a central role.

4.5 Compactness of the Seiberg–Witten moduli space

Above we used perturbations and transversality to discuss the smoothness of the Seiberg–Witten moduli space.

In order to finally construct a topological invariant from the moduli space, we need another basic property: compactness.

Unlike non-Abelian ASD theory, the gauge group of Seiberg–Witten theory is $U(1)$. This Abelian nature makes the compactness of the connection much simpler than in instanton theory.

At the heart of it is the Weitzenböck formula. It first gives a pointwise bound for the spinor, and the curvature equation then gives control of F_A^+ . Combining this with Abelian gauge fixing and elliptic estimates, we finally obtain the convergence of solutions modulo gauge.

The convergence sought here is convergence modulo gauge. For example, fix a configuration (A, ψ) and a nonconstant smooth real function f . The maps $u_n = e^{inf}$ give

$$u_n \cdot (A, \psi) = (A - 2in df, e^{inf} \psi).$$

All these configurations represent the same quotient point, although $\|A - 2in df - A\|_{L^2} = 2n\|df\|_{L^2}$ diverges. If the original configuration is a solution, so is every term. The estimates below must therefore be applied after choosing suitable representatives.

4.5.1 The Weitzenböck formula and the Seiberg–Witten equations

Let (A, ψ) satisfy the perturbed Seiberg–Witten equations

$$D_A \psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

Theorem 3.5.3 gives the Spin^c Weitzenböck formula on positive spinors

$$D_A^- D_A^+ \psi = \nabla_A^* \nabla_A \psi + \frac{s}{4} \psi + \frac{1}{2} c(F_A^+) \psi.$$

Since $D_A \psi = 0$, the left-hand side is zero. Substituting the curvature equation then gives

$$0 = \nabla_A^* \nabla_A \psi + \frac{s}{4} \psi + \frac{1}{2} c(q(\psi)) \psi + \frac{1}{2} c(\eta) \psi.$$

According to the normalization adopted earlier in this text, q satisfies

$$c(q(\psi)) = (\psi \psi^*)_0.$$

And we computed above that

$$(\psi \psi^*)_0 \psi = \frac{1}{2} |\psi|^2 \psi.$$

Hence we obtain

$$0 = \nabla_A^* \nabla_A \psi + \frac{s}{4} \psi + \frac{1}{4} |\psi|^2 \psi + \frac{1}{2} c(\eta) \psi.$$

This is the basic second-order equation obtained by combining the Seiberg–Witten equations with the Weitzenböck formula.

Remark 4.5.1. Here the constant in front of $\frac{1}{4} |\psi|^2 \psi$ depends on the normalization of q and of Clifford multiplication.

The constant may differ from one reference to another, but what matters is that a positive cubic term $|\psi|^2 \psi$ always appears.

4.5.2 Bochner identity

In order to obtain a pointwise estimate from the above second-order equation, we need to study the function $|\psi|^2$.

In this text we use the non-negative Laplacian on functions

$$\Delta = d^* d.$$

For any spinor field ψ , the standard Bochner computation gives

$$\frac{1}{2} \Delta |\psi|^2 = \langle \nabla_A^* \nabla_A \psi, \psi \rangle - |\nabla_A \psi|^2.$$

Substituting the Seiberg–Witten Weitzenböck equation from above, we obtain

$$\begin{aligned} \frac{1}{2} \Delta |\psi|^2 = & - |\nabla_A \psi|^2 - \frac{s}{4} |\psi|^2 - \frac{1}{4} |\psi|^4 \\ & - \frac{1}{2} \langle c(\eta) \psi, \psi \rangle. \end{aligned}$$

The last term may have no fixed sign, but it can be estimated using the pointwise norm of Clifford multiplication.

There exists a constant $C > 0$ depending only on the normalization of this text, such that

$$|\langle c(\eta)\psi, \psi \rangle| \leq C|\eta| |\psi|^2.$$

Hence the scalar curvature and the perturbation can be used to control $|\psi|$.

4.5.3 The maximum principle and the L^∞ bound

Let $x_0 \in X$ be a point where the function $|\psi|^2$ attains its maximum.

Since this text adopts the sign convention $\Delta = d^*d$, at the maximum point we have

$$\Delta|\psi|^2(x_0) \geq 0.$$

If $\psi(x_0) = 0$, then $\psi \equiv 0$, and no further discussion is needed.

Suppose $\psi(x_0) \neq 0$. Applying the Bochner identity of the previous section (Section 4.5.2) at x_0 , and discarding the non-positive term $-|\nabla_A \psi|^2$, we obtain an estimate of the form

$$|\psi(x_0)|^2 \leq C_1 s_-(x_0) + C_2 |\eta(x_0)|$$

where

$$s_- := \max\{-s, 0\}.$$

Hence there exists a constant C depending only on the fixed metric and on the normalization, such that all solutions satisfy

$$\|\psi\|_{L^\infty}^2 \leq C (\|s_-\|_{L^\infty} + \|\eta\|_{L^\infty}).$$

The key point is that the right-hand side is completely determined by the fixed background data g and η , and is independent of the particular solution (A, ψ) .

Thus a family of Seiberg–Witten solutions has a uniform spinor L^∞ bound.

4.5.4 The unperturbed equations and positive scalar curvature

As a special case, let the perturbation be zero, that is, $\eta = 0$.

If the scalar curvature satisfies

$$s > 0$$

everywhere, then taking the L^2 inner product of the Weitzenböck equation of the previous section with ψ gives

$$0 = \int_X \left(|\nabla_A \psi|^2 + \frac{s}{4} |\psi|^2 + \frac{1}{4} |\psi|^4 \right) d\text{vol}.$$

All the terms are non-negative, so the only possibility is

$$\psi \equiv 0.$$

Therefore positive scalar curvature forces all unperturbed Seiberg–Witten solutions to be reducible.

This is also the basic analytic source of the fact that Seiberg–Witten invariants can give an obstruction to positive scalar curvature.

4.5.5 Control of F_A^+

From the curvature equation

$$F_A^+ = q(\psi) + \eta$$

and the quadratic nature of q , there is a constant C such that

$$|q(\psi)| \leq C|\psi|^2.$$

Hence the L^∞ bound on the spinor immediately gives a uniform L^∞ bound on F_A^+ . In particular, for every $1 \leq p < \infty$ there is a uniform estimate

$$\|F_A^+\|_{L^p} \leq C_p.$$

Here the constant depends only on g , η , the Spin^c structure and the normalization chosen.

However, in order to control the connection, we also have to consider the full curvature F_A , and not only its self-dual part.

4.5.6 The L^2 bound on the full curvature

Since A is a unitary connection on a fixed line bundle L , the de Rham cohomology class of its curvature is fixed:

$$\left[\frac{i}{2\pi} F_A \right] = c_1(L)_{\mathbb{R}}.$$

Hence the integral

$$\int_X F_A \wedge F_A$$

depends only on $c_1(L)$, and not on the particular connection A .

On the other hand, the four-dimensional Hodge decomposition gives

$$F_A = F_A^+ + F_A^-.$$

With the imaginary-valued curvature convention fixed in Section 2.4.6, the identity is

$$\|F_A\|_{L^2}^2 = 2\|F_A^+\|_{L^2}^2 - 4\pi^2 \langle c_1(L)^2, [X] \rangle.$$

Therefore, once $\|F_A^+\|_{L^2}$ is uniformly controlled, the fixed Chern–Weil topological term immediately gives uniform control of $\|F_A^-\|_{L^2}$.

Consequently there is a constant C such that all solutions satisfy

$$\|F_A\|_{L^2} \leq C.$$

Unlike in non-Abelian Yang–Mills theory, here the curvature equation in fact gives control stronger than a mere energy bound.

4.5.7 A reference connection and global gauge choice

Fix a smooth unitary connection A_0 on L and write $A = A_0 + a$. The Abelian curvature identity gives

$$F_A = F_{A_0} + da, \quad d^+a = q(\psi) + \eta - F_{A_0}^+.$$

To estimate a , apply Proposition 2.4.1 with the determinant convention of Section 2.4.7. Since the gauge action is $a \mapsto a - 2u^{-1}du$, a representative can be chosen with

$$d^*a = 0, \quad \text{Harm}(a) \in K,$$

where K is a fixed compact fundamental domain for the lattice $4\pi i\mathcal{H}_{\mathbb{Z}}^1$ in $i\mathcal{H}^1(X)$. The factor $4\pi i$ comes from the coefficient 2 in the action and the $2\pi i\mathbb{Z}$ periods of $u^{-1}du$. Controlling the harmonic component is necessary when $b_1(X) > 0$; the Coulomb condition alone leaves it unrestricted.

Constant gauge transformations preserve the connection and rotate the spinor. Thus these conditions select bounded representatives, not a unique representative of each configuration orbit. This remains true when $b_1(X) = 0$, even though no harmonic reduction is then needed. For $b_1(X) > 0$, the boundary identifications of the compact fundamental domain give another source of nonuniqueness; see Section 2.4.3. Compactness requires neither a canonical choice nor continuous dependence of this choice on the orbit.

4.5.8 Elliptic estimate for the connection

In the above gauge, a satisfies

$$d^*a = 0, \quad d^+a = q(\psi) + \eta - F_{A_0}^+.$$

The operator

$$d^* \oplus d^+ : \Omega^1(X) \longrightarrow \Omega^0(X) \oplus \Omega_+^2(X)$$

is elliptic.

By Proposition 2.4.2, for every $1 < p < \infty$ there is the elliptic estimate

$$\|a\|_{W^{1,p}} \leq C_p \left(\|d^*a\|_{L^p} + \|d^+a\|_{L^p} + \|\text{Harm}(a)\| \right).$$

The first term vanishes, the second is controlled by $q(\psi) + \eta - F_{A_0}^+$, and the third is likewise uniformly bounded, since we have already restricted to a compact fundamental domain.

Therefore

$$\|a\|_{W^{1,p}} \leq C_p$$

holds for all solutions.

In particular, we may take $p > 4$. The four-dimensional Sobolev embedding gives

$$W^{1,p} \hookrightarrow C^{0,\alpha}$$

with $\alpha = 1 - \frac{4}{p} > 0$.

So after gauge fixing, the connections already enjoy uniform $C^{0,\alpha}$ control.

4.5.9 First-order control of the spinor

The Dirac equation is

$$D_A \psi = 0.$$

Substituting $A = A_0 + a$ and using

$$D_A = D_{A_0} + \frac{1}{2}c(a),$$

we obtain

$$D_{A_0} \psi = -\frac{1}{2}c(a)\psi.$$

Since a already has an L^∞ bound and ψ has the preceding L^∞ bound, the right-hand side is uniformly bounded in every finite L^p .

Applying the elliptic estimate to the fixed elliptic operator D_{A_0} , we obtain

$$\|\psi\|_{W^{1,p}} \leq C_p.$$

Hence after gauge fixing, both the connection and the spinor acquire uniform first-order Sobolev control.

4.5.10 Elliptic bootstrapping

We can now use the two Seiberg–Witten equations repeatedly, following the bootstrapping scheme of Section A.6.8.

The Dirac equation gives

$$D_{A_0} \psi = -\frac{1}{2}c(a)\psi,$$

while the curvature equation together with the Coulomb gauge gives

$$(d^* \oplus d^+)a = (0, q(\psi) + \eta - F_{A_0}^+).$$

The right-hand sides are zeroth-order nonlinear expressions in a and ψ .

Since the first step already gives $W^{1,p}$ control, the Sobolev multiplication theorem allows us to control the derivatives of these nonlinear terms further.

Applying elliptic regularity repeatedly to the two elliptic equations, we obtain, for every integer $k \geq 1$, uniform estimates

$$\|a\|_{W^{k,p}} + \|\psi\|_{W^{k,p}} \leq C_{k,p}.$$

Sobolev embedding then gives C^r bounds of every order.

This is the mechanism of standard elliptic bootstrapping in Seiberg–Witten compactness.

4.5.11 The Seiberg–Witten compactness theorem

Combining the estimates above, we obtain the following basic theorem.

Theorem 4.5.2 (Seiberg–Witten compactness). Let X be a closed, oriented Riemannian 4-manifold, let $\mathfrak{s}$ be a fixed Spin^c structure, and let $\eta \in i\Omega_+^2(X)$ be a fixed smooth perturbation.

Let

$$(A_n, \psi_n)$$

be a sequence of perturbed Seiberg–Witten solutions.

Then there exist gauge transformations $u_n \in \mathcal{G}$ and a subsequence such that

$$u_n \cdot (A_n, \psi_n) \longrightarrow (A_\infty, \psi_\infty)$$

converges in the C^∞ topology, where (A_∞, ψ_∞) is again a perturbed Seiberg–Witten solution.

Proof. Fix $p > 4$. For each n , choose one smooth gauge transformation u_n as in Section 4.5.7 and write

$$(A'_n, \psi'_n) := u_n \cdot (A_n, \psi_n) = (A_0 + a_n, u_n \psi_n).$$

Gauge covariance preserves the equations and $|\psi'_n| = |\psi_n|$. The estimates above, in these same gauges, give

$$\|a_n\|_{W^{m,p}} + \|\psi'_n\|_{W^{m,p}} \leq C_{m,p} \quad \text{for every } m \geq 1.$$

For each finite r , choose $m > r + 4/p$. The compact Sobolev embedding into C^r gives a convergent subsequence. Successively extracting subsequences for $r = 0, 1, 2, \dots$ and taking a diagonal subsequence produces a single limit (a_∞, ψ_∞) in every C^r norm. In particular the limit is smooth.

Passing to the limit in

$$D_{A_0} \psi'_n = -\frac{1}{2}c(a_n)\psi'_n, \quad F_{A_0}^+ + d^+ a_n = q(\psi'_n) + \eta$$

is justified by C^1 convergence and continuity of the products. Thus $(A_0 + a_\infty, \psi_\infty)$ solves the same equations. $\square$

There is also a direct topological justification of compactness, without assuming that sequential compactness suffices in an arbitrary quotient. Let $\mathcal{K}$ be the set of smooth solutions satisfying $d^*(A - A_0) = 0$ and $\text{Harm}(A - A_0) \in K$, with K the closed compact fundamental domain chosen above. Every orbit meets $\mathcal{K}$. The proof gives sequential compactness of $\mathcal{K}$; the equations and gauge conditions are preserved under smooth limits. The smooth configuration space is metrizable by its countable family of C^r seminorms, so $\mathcal{K}$ is compact. Its continuous image in the quotient is the whole moduli space, which is therefore compact.

Here compactness does not require the perturbation to be generic. Genericity serves to ensure that the moduli space is smooth, while compactness is a separate analytic property.

4.5.12 Uniform compactness along a path

The invariance argument uses a compact parameter interval, so the estimates must hold uniformly as the metric and perturbation vary. Suppose (g_t, η_t) is smooth on $[0, 1]$ and

the determinant line L is fixed. Use the identifications of Section 4.4.11. All metrics are uniformly equivalent, and their coefficients and those of η_t have bounded derivatives of every fixed order. The scalar-curvature lower bound and the perturbation bound in the maximum-principle argument can therefore be chosen independently of t . This gives a uniform bound on $|\psi|$ and hence on F_A^+ .

For the connection estimate, impose $d_{g_t}^*(A - A_0) = 0$. Choose an integral basis of $H^1(X; \mathbb{Z})/\text{tors}$ and its g_t -harmonic representatives. These representatives vary smoothly. Large gauge transformations put the harmonic coefficients of $A - A_0$ in a bounded fundamental parallelepiped for the lattice $4\pi i \mathcal{H}_{\mathbb{Z}, g_t}^1$. The harmonic part is thus uniformly bounded as t varies.

On its orthogonal complement, the estimate for $d_{g_t}^* \oplus d_{g_t}^+$ is uniform. One way to verify this is by contradiction: parameters with diverging constants have a convergent subsequence, while normalized almost-kernel forms converge, by the elliptic estimate and compact embedding, to a nonzero harmonic form orthogonal to the limiting harmonic space. This is impossible. The remaining estimates and the bootstrap now give

$$\|A - A_0\|_{W^{m,p}} + \|\psi\|_{W^{m,p}} \leq C_{m,p}, \quad p > 4,$$

with constants independent of t , after gauge fixing.

Given a sequence $(t_j, [A_j, \psi_j])$ of solutions, first pass to $t_j \rightarrow t_\infty$. The uniform estimates and a diagonal subsequence then give smooth convergence modulo gauge to a solution for $(g_{t_\infty}, \eta_{t_\infty})$. Hence the parameterized moduli space is compact. No regularity or wall-avoidance assumption is needed for this compactness statement; those assumptions are used to make the compact space a smooth cobordism.

4.5.13 Why there is no instanton bubbling here

This is an important difference from ASD theory.

In non-Abelian $SU(2)$ gauge theory, even if the total Yang–Mills energy is bounded, the curvature can still concentrate on ever smaller scales, thereby producing bubbling.

Hence the ASD moduli space itself is in general not compact, and one has to add ideal instantons to obtain the Uhlenbeck compactification.

In Seiberg–Witten theory the situation is different.

First, the gauge group is Abelian, so the curvature has no quadratic nonlinearity such as $A \wedge A$.

More importantly, the curvature equation

$$F_A^+ = q(\psi) + \eta$$

together with the Weitzenböck estimate controls the self-dual curvature directly.

Hodge theory and the fixed $c_1(L)$ then control the full curvature, while the Coulomb gauge turns this curvature control into global Sobolev control of the connection.

Hence the phenomenon of non-Abelian instanton theory, in which rescaling produces a new S^4 instanton bubble, does not occur.

So the Seiberg–Witten moduli space needs no extra Uhlenbeck ideal-point compactification.

4.5.14 Summary of this section

The estimates proceed from the Weitzenböck formula to a uniform L^∞ bound on ψ , then to control of F_A^+ and of $A - A_0$ in global Coulomb gauge. The Dirac equation and elliptic bootstrapping give bounds of all orders and smooth subsequential convergence modulo gauge. Section 4.5.12 makes these bounds uniform along a compact smooth path of data.

These estimates give compactness without assuming regularity. Transversality and wall avoidance are separate inputs: together with compactness, they give a compact smooth moduli space of dimension $d(\mathfrak{s})$. To extract a signed count or a cohomology pairing, we still need the orientation constructed next.

4.6 Orientation of the Seiberg–Witten moduli space

The previous section has already given: when $b_2^+(X) > 0$ and the perturbation is generic, if there are no reducible solutions, then $\mathcal{M}_{\mathfrak{s},\eta}$ is a compact smooth manifold of dimension $d(\mathfrak{s})$.

In order to define an integer-valued invariant from this manifold, we still need to choose an orientation on it.

In particular, when $d(\mathfrak{s}) = 0$, every point of an oriented 0-manifold carries a sign $+1$ or -1 . The Seiberg–Witten invariant is precisely the sum of these signs.

The orientation of the moduli space comes from the determinant line of the deformation operator.

4.6.1 The determinant line of a Fredholm operator

Let

$$D : E \longrightarrow F$$

be a Fredholm operator between two real Banach spaces.

Since $\ker D$ and $\operatorname{coker} D$ are both finite-dimensional, we can define the one-dimensional real vector space

$$\det D = \det(\ker D) \otimes \det(\operatorname{coker} D)^*.$$

Here, if V is a finite-dimensional real vector space, we write $\det V = \Lambda^{\dim V} V$.

Definition 4.6.1 (Determinant line). The above one-dimensional real vector space $\det D$ is called the **determinant line** of the Fredholm operator D .

Choosing an orientation on $\det D$ is called choosing an orientation for the Fredholm problem.

If D is surjective, then $\operatorname{coker} D = 0$, so

$$\det D \cong \det(\ker D).$$

Thus, at a regular point, an orientation of the Fredholm determinant line induces precisely an orientation of the tangent space of the solution space.

4.6.2 The Seiberg–Witten deformation operator

At an irreducible solution (A, ψ) , consider the gauge-fixed deformation operator

$$\mathcal{D}_{(A, \psi)} = d_0^* \oplus d_1.$$

It is a real Fredholm operator

$$i\Omega^1 \oplus \Gamma(S^+) \longrightarrow i\Omega^0 \oplus i\Omega_+^2 \oplus \Gamma(S^-).$$

Hence there is a corresponding determinant line

$$\det \mathcal{D}_{(A, \psi)}.$$

When the solution is regular, $\mathcal{D}_{(A, \psi)}$ is surjective, while irreducibility in turn eliminates the infinitesimal stabilizer. Thus $\ker \mathcal{D}_{(A, \psi)}$ is naturally identified with

$$T_{[(A, \psi)]} \mathcal{M}_{\bar{s}, \eta}.$$

So if $\det \mathcal{D}_{(A, \psi)}$ can be oriented consistently, then the whole moduli space can be oriented.

4.6.3 Zeroth-order terms do not affect the orientation problem

We have seen above that $\mathcal{D}_{(A, \psi)}$ differs from its principal part

$$(d^* \oplus d^+) \oplus D_A^+$$

only by zeroth-order terms.

By continuously scaling these zeroth-order terms, we can deform $\mathcal{D}_{(A, \psi)}$ continuously into $(d^* \oplus d^+) \oplus D_A^+$.

Fredholm determinant lines form a line bundle over such a continuous Fredholm family, so the orientation problem reduces to these two simpler operators.

Hence we need to understand separately the real determinant line of

$$d^* \oplus d^+$$

and the determinant line of the Spin^c Dirac operator D_A^+ .

4.6.4 The complex orientation of the Dirac operator

The Spin^c Dirac operator

$$D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-)$$

is a complex-linear operator.

Hence $\ker D_A^+$ and $\text{coker } D_A^+$ are both complex vector spaces.

Every complex vector space carries a natural underlying real orientation: if $v_1, \dots, v_k$ is a complex basis, then

$$v_1, iv_1, \dots, v_k, iv_k$$

gives a canonical real orientation.

Hence the determinant line of the Dirac operator carries a natural orientation, called the **complex orientation**.

In other words, the Spin^c Dirac part requires no additional orientation choice.

The part that genuinely requires an artificial choice of orientation comes from $d^* \oplus d^+$.

4.6.5 The determinant line of $d^* \oplus d^+$

Consider

$$P = d^* \oplus d^+ : \Omega^1(X) \longrightarrow \Omega^0(X) \oplus \Omega_+^2(X).$$

The kernel and cokernel were identified in Section 4.3.1 using harmonic forms. We now use those identifications to orient the determinant line.

More precisely, the determinant line reduces naturally to the top exterior powers of $H^1(X; \mathbb{R})$ and of the self-dual harmonic space

$$\mathcal{H}_g^+(X) \subset H^2(X; \mathbb{R}).$$

Since X is connected, $H^0(X; \mathbb{R}) \cong \mathbb{R}$ carries the canonical orientation given by the constant function 1.

Hence the remaining non-canonical orientation data are precisely

$$H^1(X; \mathbb{R}) \oplus \mathcal{H}_g^+(X).$$

Note that $\mathcal{H}_g^+(X)$ depends on the metric as a concrete subspace of $H^2(X; \mathbb{R})$, but its orientation bundle can be tracked naturally and continuously when the metric is changed.

4.6.6 Homology orientation

Definition 4.6.2 (Homology orientation). A **homology orientation** is an orientation of the real vector space

$$H^1(X; \mathbb{R}) \oplus \mathcal{H}^+(X; \mathbb{R}).$$

Here $\mathcal{H}^+(X; \mathbb{R})$ denotes the maximal positive subspace of the intersection form; once a metric is given, one can concretely take self-dual harmonic representatives.

Once a homology orientation is chosen, the determinant line of $d^* \oplus d^+$ acquires an orientation.

Combining this with the canonical complex orientation of the Dirac operator, we obtain the orientation of the entire Seiberg–Witten deformation operator.

Hence:

Proposition 4.6.3. After fixing a homology orientation of X , every Spin^c Seiberg–Witten moduli space acquires a natural orientation on its irreducible regular locus.

Here “natural” means: once the homology orientation is fixed, no further choices have to be made separately for different Spin^c structures or for different solutions.

4.6.7 The simplification when $b_1 = 0$

If

$$b_1(X) = 0,$$

then $H^1(X; \mathbb{R}) = 0$.

So the homology orientation degenerates to choosing an orientation of

$$\mathcal{H}^+(X; \mathbb{R}).$$

In particular, many simply connected 4-manifolds belong to this case.

Hence in the most common case $b_1 = 0$, one only has to choose an orientation for the positive part of the intersection form.

4.6.8 Signs in a zero-dimensional moduli space

Now assume that

$$d(\mathfrak{s}) = 0,$$

and that η is generic, so that the moduli space contains only irreducible regular solutions.

Then $\mathcal{M}_{\mathfrak{s},\eta}$ is a compact oriented 0-manifold, hence a set of finitely many points.

To each point

$$p = [(A, \psi)] \in \mathcal{M}_{\mathfrak{s},\eta},$$

the orientation assigns a sign

$$\varepsilon(p) \in \{+1, -1\}.$$

Intuitively, $\varepsilon(p) = +1$ means that the determinant orientation of the deformation operator is compatible with the global orientation chosen in advance; if the two orientations are opposite, then $\varepsilon(p) = -1$.

Hence the oriented count is defined by

$$\#\mathcal{M}_{\mathfrak{s},\eta} = \sum_{p \in \mathcal{M}_{\mathfrak{s},\eta}} \varepsilon(p).$$

This is the basic numerical source of the zero-dimensional Seiberg–Witten invariant.

4.6.9 Changing the homology orientation

If the homology orientation is reversed, then the orientations of all Seiberg–Witten moduli spaces are reversed at the same time.

Hence in the zero-dimensional case, the sign of every point changes simultaneously, so that

$$\#\mathcal{M}_{\mathfrak{s},\eta}$$

changes sign as a whole.

Therefore the integer-valued Seiberg–Witten invariant strictly speaking depends on the homology orientation of X .

In many concrete references, if a natural orientation choice exists, it is usually fixed by default and not written out repeatedly.

4.6.10 Orientation of the parameterized moduli space

Let (g_t, η_t) , $t \in [0, 1]$, be a generic path avoiding reducibles, with regular endpoints. For $b_2^+(X) > 1$ such paths can be chosen as in Section 4.4.9. Fix a homology orientation of X and the standard orientation of $[0, 1]$. Write $\mathcal{M}_i = \mathcal{M}_{\mathfrak{s}, g_i, \eta_i}$ for the endpoint moduli spaces, $i = 0, 1$.

The orientation of a parameterized moduli space must remain defined at parameters for which the individual fiber is not regular. The determinant line provides this extension. For the linearization L_t on a slice, put

$$\det L_t = \Lambda^{\max} \ker L_t \otimes (\Lambda^{\max} \operatorname{coker} L_t)^*.$$

Although kernel dimensions may jump, these lines assemble into a continuous line bundle. Locally, choose a finite-dimensional target space V such that $L_t \oplus \iota_V$ is surjective. The formula

$$\det L_t \cong \Lambda^{\max} \ker(L_t \oplus \iota_V) \otimes (\Lambda^{\max} V)^*$$

defines the bundle across the jumps and agrees on overlaps.

Scaling the zeroth-order terms, as above, is a homotopy of these line bundles, not just an equality of numerical indices. The complex orientation of the Dirac factor and the fixed homology orientation orient the determinant bundle throughout the path. The spaces $\mathcal{H}_{g_t}^1 \oplus \mathcal{H}_{g_t}^+$ carry the continuously transported homology orientation. Replacing a chosen operator homotopy by a homotopy of homotopies does not change this transport. Gauge transformations act complex linearly on the Dirac factor and trivially on the real cohomology factors. They therefore preserve the resulting orientation, which descends to the quotient.

At a parameterized regular solution, orient the total kernel using

$$\det L_{\text{par}} \cong \det T_t[0, 1] \otimes \det L_t,$$

with the parameter factor first. Take paths constant near their regular endpoints, so the total moduli space has collars there. The outward-normal-first convention then gives, in every dimension,

$$\partial \mathcal{W} = \mathcal{M}_1 \sqcup (-\mathcal{M}_0).$$

At $t = 1$ the outward normal is $+\partial_t$; at $t = 0$ it is $-\partial_t$. This accounts for the sign used in the cobordism proof.

4.6.11 Summary of this section

The complex Dirac factor has a canonical orientation. The remaining choice is a homology orientation of

$$H^1(X; \mathbb{R}) \oplus \mathcal{H}^+(X; \mathbb{R}).$$

Together they orient the deformation determinant line, the regular moduli spaces and the parameterized cobordisms. Reversing the homology orientation reverses the resulting moduli-space orientation. The next section uses this choice to define the invariant and the endpoint boundary signs to prove its independence of auxiliary data.

4.7 Seiberg–Witten invariant

Above we have completed the main analytic preparation needed to define the Seiberg–Witten invariant.

Fix a closed, connected, oriented smooth 4-manifold X , a homology orientation, and a Spin^c structure $\mathfrak{s}$.

When

$$b_2^+(X) > 1$$

one can choose a generic metric and perturbation so that the corresponding Seiberg–Witten moduli space has no reducible solutions and all solutions are regular.

By the compactness theorem, in this case $\mathcal{M}_{\mathfrak{s}}$ is a compact oriented smooth manifold, whose dimension is

$$d(\mathfrak{s}) = \frac{c_1(L_{\mathfrak{s}})^2 - (2\chi(X) + 3\sigma(X))}{4}.$$

The Seiberg–Witten invariant is precisely the topological information extracted from this oriented moduli space.

4.7.1 The zero-dimensional case

For $d(\mathfrak{s}) = 0$, generic data avoiding reducibles give a compact oriented zero-manifold. Use the orientation signs $\varepsilon(p)$ constructed in the preceding section to define its signed count.

Definition 4.7.1 (Seiberg–Witten invariant, zero-dimensional case). Assume that

$$b_2^+(X) > 1, \quad d(\mathfrak{s}) = 0.$$

Fix a homology orientation of X . Define

$$SW_X(\mathfrak{s}) := \sum_{p \in \mathcal{M}_{\mathfrak{s},g,\eta}} \varepsilon(p).$$

On the face of it this integer depends on the chosen metric g and perturbation η .

In what follows we explain that when $b_2^+ > 1$ it is in fact independent of these auxiliary choices.

4.7.2 Metric and perturbation independence

Take two sets of generic data

$$(g_0, \eta_0), \quad (g_1, \eta_1).$$

Since $b_2^+(X) > 1$, one can choose a generic path

$$(g_t, \eta_t), \quad t \in [0, 1],$$

such that the whole family avoids the reducible wall and satisfies parametric transversality.

Consider the parameterized moduli space

$$\mathcal{W} = \{(t, [A, \psi]) \mid (A, \psi) \text{ solves the equations for } (g_t, \eta_t)\}.$$

Since the original moduli space has expected dimension 0, after the parameter t has been added, the expected dimension of $\mathcal{W}$ is 1.

Genericity guarantees that $\mathcal{W}$ is a smooth oriented 1-manifold.

The compactness argument above can be carried out uniformly over the compact parameter interval $[0, 1]$, so $\mathcal{W}$ is compact.

Hence $\mathcal{W}$ is a compact oriented 1-manifold with boundary.

Its boundary is

$$\partial\mathcal{W} = \mathcal{M}_{\mathfrak{s},g_1,\eta_1} \sqcup (-\mathcal{M}_{\mathfrak{s},g_0,\eta_0}).$$

But the signed total of the boundary points of any compact oriented 1-manifold is zero.

Hence

$$\#\mathcal{M}_{\mathfrak{s},g_1,\eta_1} - \#\mathcal{M}_{\mathfrak{s},g_0,\eta_0} = 0.$$

Therefore we obtain:

Theorem 4.7.2 (Metric and perturbation independence). Assume that

$$b_2^+(X) > 1, \quad d(\mathfrak{s}) = 0.$$

Once a homology orientation is fixed, the integer

$$SW_X(\mathfrak{s})$$

is independent of the choice of the generic metric and of the generic perturbation.

Hence $SW_X(\mathfrak{s})$ is an invariant of the oriented smooth 4-manifold together with its Spin^c structure.

4.7.3 Why this is a smooth invariant

The definition of the Seiberg–Witten invariant uses a Riemannian metric, the Spin^c Dirac operator, connections, and a nonlinear elliptic PDE.

But the integer finally obtained is independent of all these auxiliary analytic choices. Therefore, if

$$f : X \longrightarrow Y$$

is an orientation-preserving diffeomorphism, then f matches up Spin^c structures, homology orientations and Seiberg–Witten moduli problems in a natural way.

Correspondingly,

$$SW_X(\mathfrak{s}) = SW_Y(f_*\mathfrak{s})$$

holds under a compatible orientation convention.

Hence the Seiberg–Witten invariant is an invariant in the smooth category.

It is not something automatically determined by the homotopy type or by the intersection form alone.

This is precisely one of the fundamental reasons why it is able to distinguish homeomorphic but non-diffeomorphic 4-manifolds.

4.7.4 Negative expected dimension

For $d(\mathfrak{s}) < 0$, the generic moduli space is empty under the wall-avoidance assumptions of Section 4.4.13. Accordingly, we set

$$SW_X(\mathfrak{s}) = 0.$$

4.7.5 The problem in positive expected dimension

If

$$d(\mathfrak{s}) > 0,$$

then the generic moduli space is a compact oriented manifold of positive dimension.

In this case one cannot simply “count how many points the moduli space contains” and obtain an integer.

In order to extract a number from a positive-dimensional moduli space, one has to choose cohomology classes of the appropriate degree on it, and pair them with the fundamental class.

These cohomology classes come from the universal line bundle on the configuration space.

4.7.6 Irreducible configuration space

Write the irreducible configuration space as

$$\mathcal{C}_s^* = \{(A, \psi) \in \mathcal{C}_s \mid \psi \neq 0\}.$$

Since X is connected, the gauge group $\mathcal{G}$ acts freely on $\mathcal{C}_s^*$.

Define

$$\mathcal{B}_s^* = \mathcal{C}_s^* / \mathcal{G}.$$

To make the bundle assertion precise, use the Sobolev completions with $k \geq 3$. The slice construction and properness in Sections 4.2.13 and 4.2.19 give a local section σ over each sufficiently small open $V \subset \mathcal{B}_k^*$. The map

$$V \times \mathcal{G}_{k+1} \longrightarrow \pi^{-1}(V), \quad (b, u) \longmapsto u \cdot \sigma(b)$$

is a smooth local trivialization. Thus $\mathcal{C}_k^* \rightarrow \mathcal{B}_k^*$ is a principal gauge-group bundle. Freeness supplies uniqueness of u ; the slice theorem supplies the local section and smooth inverse. For the right-action convention of Section 2.1, one may set $C \cdot u = u^{-1} \cdot C$. We suppress the Sobolev indices in the bundle constructions below.

The generic Seiberg–Witten moduli space is naturally contained in $\mathcal{B}_s^*$.

Hence the cohomology classes on $\mathcal{B}_s^*$ can be restricted to the moduli space.

4.7.7 Based gauge transformations and the residual circle

Fix $x_0 \in X$. Evaluation at this point gives an exact sequence

$$1 \longrightarrow \mathcal{G}_{x_0} \longrightarrow \mathcal{G} \xrightarrow{\text{ev}_{x_0}} S^1 \longrightarrow 1, \quad \mathcal{G}_{x_0} = \{u : u(x_0) = 1\}.$$

It is split by the constant gauge transformations. The based group acts freely even on reducible configurations: a transformation fixing a connection satisfies $du = 0$, and its value at x_0 then forces it to be the identity.

On irreducible configurations the quotient

$$\tilde{\mathcal{B}}_{x_0}^* = \mathcal{C}^* / \mathcal{G}_{x_0} \longrightarrow \mathcal{B}^*$$

is a principal circle bundle, with residual action induced by constants. Indeed, if a constant transformation fixes a based class, its difference from a based transformation fixes the original irreducible configuration; freeness of the full action then makes that constant equal to 1. At a reducible class the residual circle is instead a stabilizer.

Use the explicit weight-one quotient convention

$$\ell_{x_0} = (\tilde{\mathcal{B}}_{x_0}^* \times \mathbb{C})/S^1, \quad (b, z) \sim (vb, vz), \quad v \in S^1.$$

Writing the equivalence relation fixes its sign; using the inverse character would replace the line by its dual.

For fixed regular data, let $\mathcal{M}^*$ be the irreducible moduli space and write $\mathcal{Z}^* \subset \mathcal{C}^*$ for its solution set before quotienting. The **framed moduli space** is

$$\widehat{\mathcal{M}}_{x_0}^* = \mathcal{Z}^*/\mathcal{G}_{x_0}.$$

The map $\widehat{\mathcal{M}}_{x_0}^* \rightarrow \mathcal{M}^*$ is the pullback of $\tilde{\mathcal{B}}_{x_0}^* \rightarrow \mathcal{B}^*$ under the inclusion $\mathcal{M}^* \hookrightarrow \mathcal{B}^*$. Thus it is a principal circle bundle, and its associated line with the same weight-one convention is $\ell_{x_0}|_{\mathcal{M}^*}$. Its first Chern class is the restriction of the class U defined below. For a nonempty regular moduli space, the framed space has dimension $d(\mathfrak{s}) + 1$. This additional dimension records the phase at x_0 , which the full gauge quotient removes.

4.7.8 Universal line bundle

Allowing the evaluation point to vary gives a line bundle over $X \times \mathcal{B}^*$:

$$\mathbb{L} = (X \times \mathcal{C}^* \times \mathbb{C})/\mathcal{G},$$

where the equivalence relation is

$$(x, A, \psi, z) \sim (x, A - 2u^{-1}du, u\psi, u(x)z).$$

The initial line over $X \times \mathcal{C}^*$ is trivial; its gauge action has weight one. Restricting the quotient to $\{x_0\} \times \mathcal{B}^*$ gives ℓ_{x_0} . Hence

$$U := c_1(\ell_{x_0}) = c_1(\mathbb{L})|_{\{x_0\} \times \mathcal{B}^*}.$$

Because X is connected, moving x_0 along a path gives homotopic inclusions into $X \times \mathcal{B}^*$ and leaves U unchanged.

The universal *determinant* line is a different bundle. Denote it by $\mathbb{L}_{\det}$: before quotienting, its fiber is L_x , and u acts on this fiber by $u(x)^2$. With p_X the projection to X , our quotient conventions give

$$\mathbb{L}_{\det} \cong p_X^* L \otimes \mathbb{L}^{\otimes 2}, \quad c_1(\mathbb{L}_{\det}) = p_X^* c_1(L) + 2c_1(\mathbb{L}).$$

In particular its restriction at a point has first Chern class $2U$. Its slant product with a 1-cycle is likewise twice the weight-one class. It cannot be substituted for $\mathbb{L}$ in the definition of the standard insertions: using $(2U)^k$ would multiply the degree- $2k$ evaluation by 2^k . The notation $\mathbb{L}$ below always means the weight-one bundle.

Remark 4.7.3 (Global representatives and the class U). A global continuous section $\sigma : \mathcal{B}^* \rightarrow \mathcal{C}^*$ would trivialize the principal gauge-group bundle by $(b, u) \mapsto u \cdot \sigma(b)$. Passing to based gauge classes would then give a section of the residual circle bundle, and hence a nowhere-zero section of its associated line ℓ_{x_0} . Consequently,

$$\text{a global continuous choice of representatives} \implies U = 0.$$

Thus nonvanishing of U is an obstruction to such a choice. The converse is not asserted: trivializing this one associated line need not trivialize the entire gauge-group bundle. On a particular moduli space $\mathcal{M} \subset \mathcal{B}^*$, the corresponding obstruction is $U|_{\mathcal{M}}$, which can vanish even when the ambient class does not. This explains why the distinction between local and global choices in Section 4.2.20 matters without claiming an obstruction on every moduli space.

4.7.9 The μ -map

For

$$\alpha \in H_i(X; \mathbb{Z}),$$

define

$$\mu(\alpha) = c_1(\mathbb{L})/\alpha.$$

Here “/” denotes the slant product.

Thus

$$\mu(\alpha) \in H^{2-i}(\mathcal{B}_s^*; \mathbb{Z}).$$

For a point $x \in X$, the slant product recovers the class already defined:

$$\mu([x]) = U \in H^2(\mathcal{B}_s^*; \mathbb{Z}).$$

On the other hand, if

$$\gamma \in H_1(X; \mathbb{Z}),$$

then

$$\mu(\gamma) \in H^1(\mathcal{B}_s^*; \mathbb{Z}).$$

These classes are precisely the insertions needed to define the general Seiberg–Witten invariant.

4.7.10 The Seiberg–Witten algebra

In order to record these insertions in a uniform way, define the graded algebra

$$\mathbb{A}(X) = \Lambda^*(H_1(X; \mathbb{Z})/\text{tors}) \otimes \mathbb{Z}[U].$$

Prescribe the grading by

$$\deg(\gamma) = 1 \quad (\gamma \in H_1(X)/\text{tors}), \quad \deg U = 2.$$

The μ -map takes this algebra naturally to $H^*(\mathcal{B}_s^*; \mathbb{Z})$.

If

$$z = \gamma_1 \wedge \cdots \wedge \gamma_r U^k,$$

then its total degree is

$$\deg z = r + 2k,$$

and it corresponds to the cohomology class

$$\mu(z) = \mu(\gamma_1) \smile \cdots \smile \mu(\gamma_r) \smile U^k.$$

4.7.11 The general Seiberg–Witten invariant

Assume that $b_2^+(X) > 1$, and choose generic data so that $\mathcal{M}_s$ becomes a compact oriented smooth manifold.

If the degree of

$$z \in \mathbb{A}(X)$$

satisfies

$$\deg z = d(\mathfrak{s}),$$

then $\mu(z)$ restricted to $\mathcal{M}_s$ is a top-degree cohomology class.

Hence it can be paired with the fundamental class.

Definition 4.7.4 (General Seiberg–Witten invariant). Define

$$SW_{X,s}(z) := \langle \mu(z), [\mathcal{M}_s] \rangle, \quad \deg z = d(\mathfrak{s}).$$

If the degree is not equal to $d(\mathfrak{s})$, the corresponding evaluation is usually declared to be zero.

Hence in the general case the Seiberg–Witten invariant is more accurately viewed as a graded functional

$$SW_{X,s} : \mathbb{A}(X) \longrightarrow \mathbb{Z},$$

which can be non-zero only in degree $d(\mathfrak{s})$.

4.7.12 Simplification when $b_1 = 0$

If

$$b_1(X) = 0,$$

then there are no degree-1 generators.

Thus

$$\mathbb{A}(X) = \mathbb{Z}[U].$$

Hence only even degrees can be obtained from powers of U .

If

$$d(\mathfrak{s}) = 2k \geq 0,$$

then the definition simplifies to

$$SW_X(\mathfrak{s}) = \langle U^k, [\mathcal{M}_s] \rangle.$$

In particular, when $d(\mathfrak{s}) = 0$, we have $k = 0$, so that

$$U^0 = 1,$$

and the above definition degenerates precisely to the oriented point count of the moduli space.

Hence the zero-dimensional definition is a special case of the general definition.

If $b_1(X) = 0$ while $d(\mathfrak{s})$ is odd, then $\mathbb{Z}[U]$ contains no element of the corresponding odd degree, so the ordinary invariant is zero.

4.7.13 Euler classes and zero sets

The geometric interpretation of the insertions uses the following identity. Let M be a closed oriented n -manifold, and let $E \rightarrow M$ be an oriented real vector bundle of even rank r . A transverse section s has zero set Z of dimension $n - r$. Orient Z by the normal-bundle identification $\nu Z \cong E|_Z$, with the normal factor first. If $i : Z \hookrightarrow M$ and $\alpha \in H^{n-r}(M; \mathbb{Z})$, then

$$\langle i^* \alpha, [Z] \rangle = \langle \alpha \smile e(E), [M] \rangle.$$

To verify this identity, take the Thom class of E . Pulling it back by s gives the class dual to Z , because the derivative of s identifies the normal directions with the oriented fibers of E . The section s is homotopic to the zero section, whose pullback of the Thom class is $e(E)$. Hence $i_*[Z] = e(E) \cap [M]$, and the identity follows by the cup-cap pairing.

For a complex line, the Euler class of the underlying oriented real plane bundle is its first Chern class. We apply this fact to the universal line restricted to the moduli space in the next subsection.

There is also an Euler-class interpretation of the equations as a zero-section problem, but it requires care. The SW section lives in an infinite-dimensional Banach bundle over $\mathcal{B}^*$, which has no ordinary finite-dimensional fundamental class to use in the displayed identity. Thus an expression pairing an ordinary Euler class with $[\mathcal{B}^*]$ is only formal unless an appropriate construction has first been supplied. The local Kuranishi maps of Section 4.2.21 give finite-dimensional zero-set models near solutions; they do not automatically form one global vector bundle over a closed finite-dimensional base. The definition used here remains the pairing on the compact oriented moduli space, or equivalently its cut-down count.

4.7.14 Cut-down moduli spaces

The pairing defining $SW_{X,\mathfrak{s}}$ has a geometric realization. Let $\mathcal{M} = \mathcal{M}_{\mathfrak{s},g,\eta}$ be compact, regular and irreducible, with its chosen orientation. Restrict ℓ_{x_0} to $\mathcal{M}$ and choose a transverse smooth section s_1 . By Section 4.7.13, its zero set has real codimension two. The complex orientation of the normal line gives an orientation satisfying

$$[s_1^{-1}(0)] = U \cap [\mathcal{M}].$$

Thus inserting one copy of U means imposing one complex equation. For k copies, take a transverse section of $\ell_{x_0}^{\oplus k}|_{\mathcal{M}}$. Its zero set represents $U^k \cap [\mathcal{M}]$, since $e(\ell_{x_0}^{\oplus k}|_{\mathcal{M}}) = U^k|_{\mathcal{M}}$.

A degree-one class $\mu(\gamma)$ can be represented by a smooth map $h_\gamma : \mathcal{M} \rightarrow S^1$, since S^1 classifies integral cohomology in degree one. The inverse image of a regular value is a cooriented hypersurface representing its Poincaré dual. Choose all sections and level sets mutually transverse. For

$$z = \gamma_1 \wedge \cdots \wedge \gamma_r U^k, \quad r + 2k = d(\mathfrak{s}),$$

their common zero and level set is a compact oriented zero-manifold, and its signed cardinality is $SW_{X,\mathfrak{s}}(z)$. Order its normal factors as the displayed cup product: interchanging two degree-one factors changes the sign, in agreement with the exterior algebra. Changing these auxiliary representatives gives an oriented cut-down cobordism and leaves the count unchanged.

For example, when $d(\mathfrak{s}) = 2$, the evaluation $\langle U, [\mathcal{M}] \rangle$ counts the zeros of a generic section of $\ell_{x_0}|_{\mathcal{M}}$. When $d(\mathfrak{s}) = 0$, no cut-down condition is imposed and the original signed count is recovered.

4.7.15 Invariance in the general dimension

Assume $b_2^+(X) > 1$ and fix a homology orientation. Join two generic sets of data by a path supplied by Proposition 4.4.4, avoiding reducibles and constant near its regular endpoints. The resulting parameterized moduli space $\mathcal{W}$ is a compact oriented cobordism with

$$\partial\mathcal{W} = \mathcal{M}_1 \sqcup (-\mathcal{M}_0).$$

Here compactness is supplied by Section 4.5.12 and the boundary sign by Section 4.6.10.

The weight-one quotient construction defines $\mathbb{L}$ over X times the parameterized irreducible configuration space. Its endpoint restrictions are the universal bundles used to define the invariants. For an insertion z of degree $d = d(\mathfrak{s})$, the slant and cup products therefore give a class $\tilde{\mu}(z) \in H^d(\mathcal{W}; \mathbb{Z})$ restricting to the two endpoint insertions.

Let $j : \partial\mathcal{W} \hookrightarrow \mathcal{W}$. The relative fundamental class and the exact sequence of the pair give

$$j_*[\partial\mathcal{W}] = 0 \quad \text{in } H_d(\mathcal{W}; \mathbb{Z}).$$

Pairing with $\tilde{\mu}(z)$ yields

$$0 = \langle \tilde{\mu}(z), j_*[\partial\mathcal{W}] \rangle = \langle \mu(z), [\mathcal{M}_1] \rangle - \langle \mu(z), [\mathcal{M}_0] \rangle.$$

This proves independence of the metric and generic perturbation over the integers. For $d = 0$ and $z = 1$, it recovers the signed-count proof of Section 4.7.2.

4.7.16 Basic classes

In many four-dimensional topological problems it is more customary to use the determinant class

$$K = c_1(L_{\mathfrak{s}})$$

to label the possibly non-zero Seiberg–Witten invariants.

Definition 4.7.5 (Seiberg–Witten basic class). If the Seiberg–Witten invariant of some Spin^c structure $\mathfrak{s}$ is non-zero, then

$$c_1(L_{\mathfrak{s}})$$

is called a **Seiberg–Witten basic class**.

Note that if $H^2(X; \mathbb{Z})$ contains 2-torsion, then different Spin^c structures may have the same determinant class.

Hence, strictly speaking, the Seiberg–Witten invariant is first of all defined on Spin^c structures.

Only when the determinant class determines the Spin^c structure uniquely can one write $SW_X(K)$ directly without ambiguity.

For example, for a simply connected closed 4-manifold, $H^2(X; \mathbb{Z})$ is torsion-free, so this problem does not arise.

4.7.17 The basic example: K3

Consider the K3 surface.

Section 4.3.9 shows that the Spin^c structure induced by the spin structure satisfies

$$c_1(L) = 0$$

and

$$d(\mathfrak{s}) = 0.$$

K3 satisfies

$$b_1 = 0, \quad b_2^+ = 3 > 1.$$

Hence its Seiberg–Witten invariant belongs to the chamber-independent standard case. A classical computation gives

$$SW_{K3}(\mathfrak{s}_0) = \pm 1,$$

where the sign depends on the homology orientation convention.

Hence the zero class is a Seiberg–Witten basic class of K3.

This example will later be connected with the symplectic structure of K3 and with Taubes’ theorem.

4.7.18 Chamber dependence when $b_2^+ = 1$

For $b_2^+(X) = 1$, the wall geometry of Section 4.4.10 restricts the preceding cobordism argument to data in the same chamber. The definition thus requires a chamber as additional data; the notation for the invariant must record that choice. Data in different chambers need not give the same value.

4.7.19 Summary of this section

When $b_2^+(X) > 1$ and $d(\mathfrak{s}) = 0$, a generic Seiberg–Witten moduli space is a compact oriented 0-manifold.

Its signed count of points

$$SW_X(\mathfrak{s}) = \sum_{p \in \mathcal{M}_{\mathfrak{s}}} \varepsilon(p)$$

is independent of the metric and the perturbation, hence gives a smooth invariant.

When $d(\mathfrak{s}) > 0$, one has to pair the μ -classes coming from the universal line bundle with the fundamental class of the moduli space.

The general definition reads

$$SW_{X,\mathfrak{s}}(z) = \langle \mu(z), [\mathcal{M}_{\mathfrak{s}}] \rangle, \quad \deg z = d(\mathfrak{s}).$$

Thus the essence of the Seiberg–Witten invariant can be summarized as follows:

first construct a compact oriented moduli space by means of nonlinear elliptic gauge equations, and then extract integral-valued topological information from its orientation and cohomology.

The next step is to begin studying the geometric consequences of the Seiberg–Witten invariant.

The most direct is first to use the Weitzenböck formula to study positive scalar curvature; afterwards one can discuss the blow-up formula, the adjunction inequality, and Taubes’ theorem on symplectic 4-manifolds.

Applications of Seiberg–Witten theory

5.1 Positive scalar curvature and the vanishing of the Seiberg–Witten invariant

The first direct application of the Seiberg–Witten invariant is to provide an obstruction to metrics of positive scalar curvature.

The basic mechanism has already appeared in the Weitzenböck formula above: when the scalar curvature is everywhere positive, the Dirac equation and the curvature equation force the spinor to vanish.

If in addition one chooses a perturbation avoiding the reducible wall, then the whole Seiberg–Witten moduli space must be empty.

5.1.1 Vanishing in the unperturbed equations

Let (X, g) be a closed oriented Riemannian 4-manifold, and assume that

$$s_g > 0$$

holds everywhere.

Fix a Spin^c structure $\mathfrak{s}$, and consider the unperturbed Seiberg–Witten equations

$$D_A \psi = 0, \quad F_A^+ = q(\psi).$$

We have obtained above

$$0 = \nabla_A^* \nabla_A \psi + \frac{s_g}{4} \psi + \frac{1}{4} |\psi|^2 \psi.$$

Taking the L^2 inner product with ψ gives

$$0 = \int_X \left(|\nabla_A \psi|^2 + \frac{s_g}{4} |\psi|^2 + \frac{1}{4} |\psi|^4 \right) d\text{vol}_g.$$

Since $s_g > 0$, the three terms in the integrand are all non-negative, so the only possibility is

$$\psi \equiv 0.$$

Therefore:

Proposition 5.1.1. If the scalar curvature of g is positive everywhere, then every unperturbed Seiberg–Witten solution is reducible.

Note that this conclusion alone does not yet imply that the Seiberg–Witten invariant vanishes.

The reason is that reducible solutions may still exist for the unperturbed equations, while defining the invariant usually requires choosing a generic perturbation.

5.1.2 Vanishing under a small perturbation

Now consider the perturbed equations

$$D_A\psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

The corresponding Weitzenböck equation is

$$0 = \nabla_A^* \nabla_A \psi + \frac{s_g}{4} \psi + \frac{1}{4} |\psi|^2 \psi + \frac{1}{2} c(\eta) \psi.$$

Taking the L^2 inner product with ψ gives

$$\begin{aligned} 0 &= \|\nabla_A \psi\|_{L^2}^2 + \frac{1}{4} \int_X s_g |\psi|^2 d\text{vol}_g \\ &\quad + \frac{1}{4} \int_X |\psi|^4 d\text{vol}_g + \frac{1}{2} \int_X \langle c(\eta) \psi, \psi \rangle d\text{vol}_g. \end{aligned}$$

There exists a constant $C > 0$ depending only on the Clifford multiplication normalization, such that

$$|\langle c(\eta) \psi, \psi \rangle| \leq C |\eta| |\psi|^2.$$

Set

$$s_0 := \min_X s_g > 0.$$

If η is small enough in the L^∞ norm, so that

$$C \|\eta\|_{L^\infty} < \frac{s_0}{2},$$

then the perturbation term cannot cancel the positive scalar-curvature term.

Hence the integral identity above again forces

$$\psi \equiv 0.$$

So positive scalar curvature excludes not only unperturbed irreducible solutions, but also all irreducible solutions under sufficiently small perturbations.

5.1.3 Avoiding the reducible wall as well

If $\psi = 0$, the perturbed equations reduce to

$$F_A^+ = \eta.$$

We have seen above that the perturbations admitting a solution of this equation form the reducible wall in $i\Omega_+^2(X)$, whose codimension is $b_2^+(X)$.

When

$$b_2^+(X) > 0$$

this wall has positive codimension.

Hence one can choose a generic perturbation η in an arbitrarily small neighborhood of 0 satisfying simultaneously:

1. $\|\eta\|_{L^\infty}$ is small enough, so that the Weitzenböck argument forces $\psi = 0$;
2. η does not lie on the reducible wall, so that no reducible solution of $F_A^+ = \eta$ exists.

Thus for such a perturbation, the Seiberg–Witten moduli space is completely empty:

$$\mathcal{M}_{s,g,\eta} = \emptyset.$$

5.1.4 The positive scalar curvature vanishing theorem

Combining the analysis above with the independence of the Seiberg–Witten invariant from auxiliary choices gives the following basic conclusion.

Theorem 5.1.2 (Positive scalar curvature vanishing). Let X be a closed, connected, oriented smooth 4-manifold, and suppose that

$$b_2^+(X) > 1.$$

If X admits a metric of positive scalar curvature, then for every Spin^c structure $\mathfrak{s}$, the corresponding Seiberg–Witten invariant is zero.

Proof. Take a metric g satisfying $s_g > 0$.

For any fixed Spin^c structure $\mathfrak{s}$, choose a sufficiently small generic perturbation η , and require that it avoid the reducible wall.

By the Weitzenböck argument above, all solutions must satisfy $\psi = 0$. But since η does not lie on the reducible wall, the reducible equation has no solution either.

Hence

$$\mathcal{M}_{\mathfrak{s},g,\eta} = \emptyset.$$

Therefore the Seiberg–Witten invariant computed from this generic pair (g, η) is zero.

When $b_2^+(X) > 1$, the Seiberg–Witten invariant is independent of the choice of metric and of generic perturbation, and thus

$$SW_X(\mathfrak{s}) = 0.$$

□

5.1.5 The obstruction given by a non-zero Seiberg–Witten invariant

The contrapositive of the theorem above is used more often in four-dimensional topology.

Corollary 5.1.3. Let X be a closed oriented smooth 4-manifold, with $b_2^+(X) > 1$.

If there exists a Spin^c structure $\mathfrak{s}$ such that

$$SW_X(\mathfrak{s}) \neq 0,$$

then X admits no metric of positive scalar curvature.

Hence the Seiberg–Witten invariant can not only distinguish smooth structures, but also constrain which Riemannian metrics a smooth 4-manifold can carry.

The logical direction here matters:

a non-zero Seiberg–Witten invariant implies that no metric of positive scalar curvature exists;

but the vanishing of all Seiberg–Witten invariants does not conversely guarantee that a metric of positive scalar curvature exists.

So this is an obstruction theorem, not a characterization of metrics of positive scalar curvature.

5.1.6 Example: the K3 surface

The K3 surface satisfies

$$b_2^+(K3) = 3 > 1.$$

We mentioned above that its canonical Spin^c structure satisfies

$$SW_{K3}(\mathfrak{s}_0) = \pm 1.$$

Hence

$$SW_{K3}(\mathfrak{s}_0) \neq 0.$$

The positive scalar curvature vanishing theorem (Theorem 5.1.2) now gives immediately:

Corollary 5.1.4. The K3 surface admits no metric of positive scalar curvature.

This conclusion can also be obtained from the $\hat{A}$ -genus and the Lichnerowicz theorem, since K3 is a spin manifold.

But the advantage of Seiberg–Witten theory is that it does not require the underlying manifold itself to be spin, only the existence of a non-zero Seiberg–Witten basic class.

5.1.7 The caveat when $b_2^+ = 1$

If

$$b_2^+(X) = 1,$$

then one cannot use the chamber-independent vanishing theorem above directly.

A metric of positive scalar curvature still allows one to use the Weitzenböck argument to find a nearby chamber, in which the corresponding Seiberg–Witten invariant vanishes.

But the invariant in the other chamber may be different.

Hence when $b_2^+ = 1$, the correct conclusion has to be stated in terms of the chamber structure together with the wall-crossing formula.

For example, $\mathbb{CP}^2$ carries a metric of positive scalar curvature and at the same time satisfies $b_2^+ = 1$. So one cannot apply the conclusion of the case $b_2^+ > 1$ to $\mathbb{CP}^2$ without modification.

5.1.8 The mechanism behind this application

This proof exhibits a basic pattern that recurs throughout Seiberg–Witten theory.

First, the Weitzenböck formula turns the first-order Dirac equation into a second-order equation involving the curvature.

Second, the Seiberg–Witten curvature equation replaces F_A^+ by a quadratic expression in the spinor.

Once the two are combined, there appears a quartic term of definite positive sign,

$$|\psi|^4.$$

Positive scalar curvature in turn supplies an additional positive zeroth-order term.

Hence the scalar curvature condition in differential geometry ultimately forces the gauge-theoretic spinor to vanish.

The deformation invariance of the invariant then promotes the analytic vanishing for one particular metric to a topological obstruction for the whole smooth manifold.

This is exactly the typical way in which Seiberg–Witten theory connects Riemannian geometry, nonlinear PDE and smooth 4-manifold topology.

5.1.9 Summary of this section

If g has positive scalar curvature, then for a sufficiently small perturbation the Weitzenböck formula forces the Seiberg–Witten spinor to satisfy $\psi \equiv 0$.

If the perturbation is in addition chosen to avoid the reducible wall, then the moduli space is completely empty.

Hence when $b_2^+(X) > 1$, the existence of a metric of positive scalar curvature implies that all Seiberg–Witten invariants vanish.

Equivalently, any non-zero Seiberg–Witten invariant is an obstruction to positive scalar curvature.

The next step is to study how the Seiberg–Witten invariant changes under the blow-up operation

$$X \longmapsto X \# \overline{\mathbb{CP}}^2$$

This will give one of the most important structural formulas in Seiberg–Witten theory: the blow-up formula.

5.2 Blow-up and the Seiberg–Witten invariant

One of the most basic smooth operations in four-manifold theory is blow-up.

From the point of view of smooth topology, performing a blow-up on an oriented smooth 4-manifold X amounts to taking the connected sum

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2.$$

Here $\overline{\mathbb{CP}}^2$ denotes the manifold obtained by reversing the orientation of $\mathbb{CP}^2$.

Under blow-up the Seiberg–Witten invariant has a very explicit transformation law. This formula is not only used to compute new Seiberg–Witten invariants, but is also closely related to symplectic minimality, exceptional spheres and the smooth classification of four-manifolds.

5.2.1 The topology of a blow-up

Let

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2.$$

Since the connected sum gives a direct sum decomposition on second cohomology, we have

$$H^2(\widetilde{X}; \mathbb{Z}) \cong H^2(X; \mathbb{Z}) \oplus H^2(\overline{\mathbb{CP}}^2; \mathbb{Z}).$$

Take

$$e \in H^2(\widetilde{X}; \mathbb{Z})$$

to be the standard generator of the $\overline{\mathbb{CP}}^2$ summand, and adopt the convention

$$e^2 = -1.$$

Then the intersection form satisfies

$$Q_{\widetilde{X}} \cong Q_X \oplus \langle -1 \rangle.$$

If instead one uses homology classes, let

$$E \in H_2(\widetilde{X}; \mathbb{Z})$$

be the corresponding exceptional sphere class; then

$$E \cdot E = -1, \quad e = \text{PD}(E).$$

In the standard blow-up construction, E can be represented by a smoothly embedded 2-sphere. This sphere is called the **exceptional sphere**.

5.2.2 Euler characteristic and signature

The connected sum satisfies

$$\chi(X \# Y) = \chi(X) + \chi(Y) - 2.$$

Since

$$\chi(\overline{\mathbb{CP}^2}) = 3,$$

we have

$$\chi(\widetilde{X}) = \chi(X) + 1.$$

On the other hand, the signature is additive under connected sum, and

$$\sigma(\overline{\mathbb{CP}^2}) = -1.$$

Hence

$$\sigma(\widetilde{X}) = \sigma(X) - 1.$$

So the combination $2\chi + 3\sigma$ satisfies

$$2\chi(\widetilde{X}) + 3\sigma(\widetilde{X}) = 2\chi(X) + 3\sigma(X) - 1.$$

Moreover, since what is added is a negative direction, we have

$$b_2^+(\widetilde{X}) = b_2^+(X), \quad b_2^-(\widetilde{X}) = b_2^-(X) + 1.$$

Hence blow-up does not change b_2^+ .

In particular, if $b_2^+(X) > 1$, then $b_2^+(\widetilde{X}) > 1$, so that before and after the blow-up one stays within chamber-independent Seiberg–Witten theory.

5.2.3 Spin^c structures on the blow-up

Now fix a Spin^c structure $\mathfrak{s}$ on X , and write

$$K = c_1(L_{\mathfrak{s}}) \in H^2(X; \mathbb{Z}).$$

By the natural inclusion, in what follows we shall still regard K as an element of $H^2(\widetilde{X}; \mathbb{Z})$.

Since

$$w_2(\overline{\mathbb{CP}}^2) \equiv e \pmod{2},$$

the Spin^c determinant classes on $\overline{\mathbb{CP}}^2$ must be odd multiples of e .

Hence on the blow-up, the family of Spin^c structures corresponding to $\mathfrak{s}$ has determinant classes

$$K + (2m + 1)e, \quad m \in \mathbb{Z}.$$

Here $K \cdot e = 0$, since the two classes come from different summands of the connected sum.

Therefore

$$(K + (2m + 1)e)^2 = K^2 - (2m + 1)^2.$$

This makes the expected dimension after the blow-up directly computable.

5.2.4 Change of the expected dimension

Let $\widetilde{\mathfrak{s}}_m$ be the Spin^c structure on $\widetilde{X}$ satisfying

$$c_1(L_{\widetilde{\mathfrak{s}}_m}) = K + (2m + 1)e.$$

By the expected dimension formula,

$$d(\widetilde{\mathfrak{s}}_m) = \frac{(K + (2m + 1)e)^2 - (2\chi(\widetilde{X}) + 3\sigma(\widetilde{X}))}{4}.$$

Substituting the two formulas of the previous section gives

$$\begin{aligned} d(\widetilde{\mathfrak{s}}_m) &= \frac{K^2 - (2m + 1)^2 - (2\chi(X) + 3\sigma(X) - 1)}{4} \\ &= d(\mathfrak{s}) - \frac{(2m + 1)^2 - 1}{4}. \end{aligned}$$

On the other hand,

$$\frac{(2m + 1)^2 - 1}{4} = m(m + 1).$$

Hence

$$d(\widetilde{\mathfrak{s}}_m) = d(\mathfrak{s}) - m(m + 1).$$

In particular, when $m = 0$ or $m = -1$, one obtains respectively the determinant classes

$$K + e, \quad K - e,$$

and

$$d(\widetilde{\mathfrak{s}}_0) = d(\widetilde{\mathfrak{s}}_{-1}) = d(\mathfrak{s}).$$

This explains why $K + e$ and $K - e$ appear in a particularly prominent way in the blow-up formula.

They are precisely the two most natural Spin^c structures that leave the expected dimension unchanged after the blow-up.

5.2.5 The blow-up formula

In what follows we state the fundamental blow-up theorem of Seiberg–Witten theory.

Its complete proof requires studying the Seiberg–Witten equations on the connected-sum neck and using gluing theory. We shall not carry out all the analytic details here.

Theorem 5.2.1 (Seiberg–Witten blow-up formula). Let X be a closed, connected, oriented smooth 4-manifold, with

$$b_2^+(X) > 1.$$

Let $\mathfrak{s}$ be a Spin^c structure on X , and write

$$K = c_1(L_{\mathfrak{s}}).$$

On

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2$$

take two Spin^c structures $\widetilde{\mathfrak{s}}_+$ and $\widetilde{\mathfrak{s}}_-$, satisfying

$$c_1(L_{\widetilde{\mathfrak{s}}_{\pm}}) = K \pm e.$$

Under compatible homology orientations, we have

$$SW_{\widetilde{X}, \widetilde{\mathfrak{s}}_{\pm}} = SW_{X, \mathfrak{s}}.$$

If $d(\mathfrak{s}) = 0$, this formula is an equality between ordinary integer-valued invariants.

If $d(\mathfrak{s}) > 0$, the above identity should be read as saying that the corresponding Seiberg–Witten functionals on $\mathbb{A}(X)$ are equal under the natural identification.

5.2.6 Treating the homology orientation under a blow-up

A blow-up adds only one negative direction.

Hence

$$H^1(\widetilde{X}; \mathbb{R}) \cong H^1(X; \mathbb{R}),$$

while the positive subspace can likewise be identified naturally with

$$\mathcal{H}^+(\widetilde{X}; \mathbb{R}) \cong \mathcal{H}^+(X; \mathbb{R}).$$

So a homology orientation chosen on X induces a homology orientation on $\widetilde{X}$ naturally.

The comparison of signs in the blow-up formula is carried out under this compatible orientation choice.

5.2.7 Why the two classes $K \pm e$ appear

The index calculation shows that $K \pm e$ do not appear by accident.

A general candidate determinant class is

$$K + (2m + 1)e.$$

Its expected dimension is

$$d(\mathfrak{s}) - m(m + 1).$$

Only $m = 0, -1$ leave the dimension completely unchanged.

So from the point of view of deformation theory, $K + e$ and $K - e$ are exactly the two Spin^c structures that can preserve the virtual dimension of the original moduli problem.

From the point of view of gluing theory, these two choices correspond to the two most basic Spin^c solutions on the $\overline{\mathbb{CP}}^2$ side.

Gluing them to the solutions on the X side along the long neck, one finally obtains the blow-up solutions corresponding to the original moduli space.

5.2.8 The geometric picture of the gluing proof

Although the complete gluing proof involves a fair amount of nonlinear elliptic analysis, its geometric mechanism can be summarized as follows.

When the connected sum is written as

$$X \# \overline{\mathbb{CP}}^2$$

one removes a small 4-ball from each of the two manifolds, and then glues along the boundary S^3 .

Afterwards one stretches the neck, turning the connected-sum metric into a metric carrying a very long cylinder

$$[-T, T] \times S^3$$

segment.

When $T \rightarrow \infty$, a Seiberg–Witten solution on the blow-up should approximately decompose into:

1. the Seiberg–Witten solution on X with one point removed;
2. the corresponding solution on $\overline{\mathbb{CP}}^2$ with one point removed;
3. matching asymptotic behavior of the two on the cylinder end.

For the determinant classes $K \pm e$, the moduli problem on the $\overline{\mathbb{CP}}^2$ side has basic solutions exactly suited to gluing.

The gluing theorem shows that when the neck is long enough, these limiting configurations correspond one-to-one to genuine solutions on $\widehat{X}$.

Moreover, the orientation comparison is compatible with the original moduli space.

Hence one finally obtains

$$SW_{\widehat{X}}(K \pm e) = SW_X(K).$$

The genuinely difficult part here is to establish a uniform right inverse, control the nonlinear error, and use the contraction mapping principle or the implicit function theorem to correct the approximate solution into an exact solution.

5.2.9 Seiberg–Witten simple type

The blow-up formula is especially simple in the case of the so-called Seiberg–Witten simple type.

Definition 5.2.2 (Seiberg–Witten simple type). Let $b_2^+(X) > 1$. If every Spin^c structure $\mathfrak{s}$ satisfying

$$SW_X(\mathfrak{s}) \neq 0$$

also satisfies

$$d(\mathfrak{s}) = 0,$$

then X is said to be of **Seiberg–Witten simple type**.

In other words, all non-zero Seiberg–Witten invariants occur on zero-dimensional moduli spaces.

In this case, the basic classes satisfy

$$K^2 = 2\chi(X) + 3\sigma(X).$$

Since this is exactly the equivalent expression of $d(\mathfrak{s}) = 0$.

5.2.10 Blow-up in the simple type case

If X is of Seiberg–Witten simple type, then its blow-up

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2$$

is likewise of Seiberg–Witten simple type.

Moreover, the basic classes after the blow-up are obtained exactly by adding or subtracting the exceptional class to or from the original basic classes.

More concretely, if K is a basic class of X , then

$$K + e \quad \text{and} \quad K - e$$

are both basic classes of $\widetilde{X}$, and satisfy

$$SW_{\widetilde{X}}(K + e) = SW_{\widetilde{X}}(K - e) = SW_X(K).$$

Conversely, in the simple type case all basic classes of $\widetilde{X}$ are of this form.

Hence if the basic-class set of X is written $\mathcal{B}(X)$, then

$$\mathcal{B}(\widetilde{X}) = \{K + e, K - e \mid K \in \mathcal{B}(X)\}.$$

This is the most commonly used form of the blow-up formula.

5.2.11 Repeated blow-up

If one blows up k times in succession, one obtains

$$X \# k \overline{\mathbb{CP}}^2.$$

Write the exceptional classes as

$$e_1, \dots, e_k,$$

satisfying

$$e_i^2 = -1, \quad e_i \cdot e_j = 0 \quad (i \neq j).$$

If X is of Seiberg–Witten simple type, then every basic class K produces

$$K + \varepsilon_1 e_1 + \dots + \varepsilon_k e_k, \quad \varepsilon_i \in \{\pm 1\},$$

altogether 2^k basic classes.

The Seiberg–Witten invariants of these classes are all equal to the original $SW_X(K)$, under a compatible orientation convention.

Hence each blow-up formally doubles the number of simple-type basic classes.

5.2.12 Example: the K3 blow-up

The standard basic class of the K3 surface is

$$K = 0,$$

and

$$SW_{K3}(0) = \pm 1.$$

Consider

$$\widetilde{X} = K3 \# \overline{\mathbb{CP}}^2.$$

By the blow-up formula 5.2.1, $\widetilde{X}$ has two basic classes

$$e \quad \text{and} \quad -e,$$

and

$$SW_{\widetilde{X}}(e) = SW_{\widetilde{X}}(-e) = \pm 1.$$

Blowing up once more,

$$K3 \# 2 \overline{\mathbb{CP}}^2$$

has four corresponding basic classes

$$\pm e_1 \pm e_2.$$

This example shows very concretely how blowing up affects the basic-class set.

5.2.13 Blow-up and minimality

A smooth 4-manifold containing a smoothly embedded sphere

$$E \subset X$$

satisfying

$$E \cdot E = -1$$

is said to contain an exceptional sphere.

If no such sphere exists, then the manifold is called minimal in the corresponding category.

Blowing up evidently produces an exceptional sphere, so

$$X \# \overline{\mathbb{CP}}^2$$

is not minimal.

The blow-up formula shows that an exceptional class is directly reflected in the Seiberg–Witten basic classes: the basic classes occur in pairs

$$K + e, \quad K - e.$$

Hence the Seiberg–Witten invariants can detect the traces left by the blow-up operation in the smooth topology.

In a symplectic 4-manifold, this phenomenon is more strongly related to symplectic exceptional spheres and to symplectic minimality.

5.2.14 An important dimension consistency check

Assume that X is of simple type, and that K is a basic class.

Then

$$K^2 = 2\chi(X) + 3\sigma(X).$$

After blowing up,

$$(K \pm e)^2 = K^2 - 1.$$

On the other hand,

$$2\chi(\widetilde{X}) + 3\sigma(\widetilde{X}) = 2\chi(X) + 3\sigma(X) - 1.$$

Hence

$$(K \pm e)^2 = 2\chi(\widetilde{X}) + 3\sigma(\widetilde{X}).$$

Therefore the classes $K \pm e$ after blowing up still have expected dimension 0.

This is completely consistent with the blow-up formula.

5.2.15 Summary of this section

A single blow-up is

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2.$$

It adds to the intersection form an exceptional direction e of self-intersection -1 .

If $\mathfrak{s}$ is a Spin^c structure on X with determinant class K , then determinant classes

$$K + (2m + 1)e$$

arise naturally on the blow-up.

Their expected dimension is

$$d(\widetilde{\mathfrak{s}}_m) = d(\mathfrak{s}) - m(m + 1).$$

In particular, $K + e$ and $K - e$ keep the expected dimension unchanged.
The Seiberg–Witten blow-up theorem gives

$$SW_{\widehat{X}}(K + e) = SW_{\widehat{X}}(K - e) = SW_X(K).$$

The complete proof relies on neck-stretching and gluing theory.

If X is of Seiberg–Witten simple type, then after blowing up all basic classes are exactly of the form

$$K \pm e.$$

The next step is naturally to enter another core application of Seiberg–Witten theory: the **adjunction inequality**.

It relates a basic class K to the genus and the self-intersection of an embedded surface

$$\Sigma \subset X,$$

typically in the form

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |K \cdot [\Sigma]|.$$

This inequality is one of the main tools by which the Seiberg–Witten invariants constrain the topology of embedded surfaces in four-manifolds.

5.3 Adjunction inequality

The Seiberg–Witten invariant can constrain not only Riemannian metrics, but also the topological type of embedded surfaces in a smooth 4-manifold.

Its core conclusion is the **adjunction inequality**.

It relates a Seiberg–Witten basic class to the genus and the self-intersection of an embedded surface.

5.3.1 Basic setup

Let X be a closed, connected, oriented smooth 4-manifold, and assume that

$$b_2^+(X) > 1.$$

Let

$$\Sigma \subset X$$

be a smoothly embedded closed connected oriented surface.

Write its homology class as $[\Sigma] \in H_2(X; \mathbb{Z})$, its genus as $g(\Sigma)$, and its self-intersection as

$$[\Sigma]^2 = [\Sigma] \cdot [\Sigma].$$

On the other hand, if $\mathfrak{s}$ is a Spin^c structure with non-zero Seiberg–Witten invariant, write

$$K = c_1(L_{\mathfrak{s}}).$$

Such a K is called a Seiberg–Witten basic class.

The goal of the adjunction inequality is to control

$$K \cdot [\Sigma].$$

5.3.2 Adjunction inequality

We now give the most commonly used version.

Theorem 5.3.1 (Seiberg–Witten adjunction inequality). Let X be a closed oriented smooth 4-manifold satisfying $b_2^+(X) > 1$.

Let K be a Seiberg–Witten basic class, that is, there exists a Spin^c structure $\mathfrak{s}$ satisfying

$$K = c_1(L_{\mathfrak{s}}), \quad SW_X(\mathfrak{s}) \neq 0.$$

If $\Sigma \subset X$ is a closed connected smoothly embedded surface satisfying

$$g(\Sigma) > 0, \quad [\Sigma]^2 \geq 0,$$

and $[\Sigma]$ is not torsion, then

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |K \cdot [\Sigma]|.$$

Hence a non-zero Seiberg–Witten basic class gives a direct constraint on the lowest genus that can occur in a homology class.

Equivalently,

$$g(\Sigma) \geq 1 + \frac{[\Sigma]^2 + |K \cdot [\Sigma]|}{2}.$$

It should be emphasized that this conclusion concerns **smoothly embedded surfaces**, and not merely abstract homology classes.

5.3.3 Comparison with the adjunction formula in algebraic geometry

The name “adjunction inequality” comes from the adjunction formula for complex surfaces.

Let X be a complex surface, and let $C \subset X$ be a smooth complex curve.

Write the canonical class of the complex surface as

$$K_X = c_1(K_X) = -c_1(TX, J).$$

The classical adjunction formula gives

$$2g(C) - 2 = C^2 + K_X \cdot C.$$

This is an equality.

By contrast, in Seiberg–Witten theory, for an arbitrary smooth embedded surface, in general one can only obtain

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |K \cdot [\Sigma]|.$$

Hence the Seiberg–Witten adjunction inequality (Theorem 5.3.1) can be understood as a generalization of the complex adjunction formula to the smooth category.

In symplectic geometry, Taubes’ theorem will show that the symplectic canonical class is a basic class. For symplectic surfaces, the inequality above becomes an equality again in suitable situations.

5.3.4 Why the absolute value appears

We first consider the weaker one-sided form

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + K \cdot [\Sigma].$$

Spin^c structures admit a conjugation operation. If the determinant class of $\mathfrak{s}$ is K , then its conjugate Spin^c structure $\bar{\mathfrak{s}}$ satisfies

$$c_1(L_{\bar{\mathfrak{s}}}) = -K.$$

The Seiberg–Witten invariant under conjugation may change its overall sign, but nonvanishing is preserved. Hence if K is a basic class, then $-K$ is also a basic class.

Applying the one-sided adjunction inequality to K and to $-K$ respectively, we obtain

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + K \cdot [\Sigma]$$

and

$$2g(\Sigma) - 2 \geq [\Sigma]^2 - K \cdot [\Sigma].$$

Combining the two gives exactly

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |K \cdot [\Sigma]|.$$

So the absolute value comes from the symmetry between K and $-K$ in Seiberg–Witten theory.

5.3.5 The analytic mechanism of the proof

The full proof of the adjunction inequality is much more complicated than the positive scalar curvature argument above.

It cannot be obtained immediately by simply substituting some formula into the Weitzenböck identity.

The central idea of the standard proof is to study a tubular neighborhood of Σ and to perform neck stretching along its boundary.

If

$$[\Sigma]^2 = 0,$$

then a tubular neighborhood of Σ may be regarded as

$$\Sigma \times D^2.$$

After stretching the metric near its boundary, a long cylinder appears in the manifold. The behavior of Seiberg–Witten solutions on this region can be related to lower-dimensional gauge equations on the surface Σ .

After restricting the Spin^c structure to Σ , $K \cdot [\Sigma]$ controls the degree of the corresponding line bundle.

On the other hand, the vortex-type equations on a surface can have solutions only when the degree of the line bundle and the genus satisfy a certain relation.

This degree restriction ultimately turns into the estimate

$$K \cdot [\Sigma] \leq 2g(\Sigma) - 2$$

in the basic case of zero self-intersection.

For $[\Sigma]^2 > 0$, one can combine the blow-up operation with the corresponding gluing argument to reduce the problem to the case of zero self-intersection, and finally obtain

$$[\Sigma]^2 + K \cdot [\Sigma] \leq 2g(\Sigma) - 2.$$

Applying the conjugate basic class $-K$ again then gives the absolute-value form.

Remark 5.3.2. The above is only a proof mechanism, not a complete proof of the adjunction inequality.

A complete proof requires neck stretching, cylindrical-end Seiberg–Witten theory, as well as the corresponding compactness and gluing analysis. This section uses the adjunction inequality as a standard theorem of Seiberg–Witten theory.

5.3.6 The case of zero self-intersection

If

$$[\Sigma]^2 = 0,$$

the adjunction inequality simplifies to

$$|K \cdot [\Sigma]| \leq 2g(\Sigma) - 2.$$

In particular, if Σ is an embedded torus, then $g(\Sigma) = 1$, so

$$K \cdot [\Sigma] = 0.$$

Hence every Seiberg–Witten basic class must be orthogonal to a torus of self-intersection zero.

This simple conclusion is very important when studying elliptic surfaces, fiber sums and knot surgery.

5.3.7 Minimal genus

Given a homology class

$$\alpha \in H_2(X; \mathbb{Z}),$$

one may ask:

among all connected smoothly embedded oriented surfaces representing α , what is the smallest possible genus?

Define the minimal genus function

$$g_X(\alpha) = \min \{g(\Sigma) \mid [\Sigma] = \alpha\},$$

provided α can be represented by an embedded oriented surface.

If $\alpha^2 \geq 0$ and the hypotheses of the adjunction inequality hold, then for every basic class K we have

$$2g_X(\alpha) - 2 \geq \alpha^2 + |K \cdot \alpha|.$$

Hence

$$g_X(\alpha) \geq 1 + \frac{\alpha^2 + |K \cdot \alpha|}{2}.$$

If several basic classes exist, one may take the maximum over all K to obtain a stronger lower bound:

$$g_X(\alpha) \geq 1 + \frac{1}{2} \left(\alpha^2 + \max_{K \in \mathcal{B}(X)} |K \cdot \alpha| \right).$$

Thus Seiberg–Witten basic classes can give direct estimates for the minimal genus of smooth embedded surfaces.

5.3.8 Example: K3

The Seiberg–Witten basic class of K3 is

$$K = 0.$$

Hence for $\alpha \in H_2(K3; \mathbb{Z})$ satisfying $\alpha^2 \geq 0$ and represented by a positive-genus embedded surface, the adjunction inequality gives

$$2g(\Sigma) - 2 \geq \alpha^2.$$

So

$$g(\Sigma) \geq 1 + \frac{\alpha^2}{2}.$$

For example, if

$$\alpha^2 = 2,$$

then every smooth embedded surface satisfying the above hypotheses and representing α must satisfy

$$g(\Sigma) \geq 2.$$

If $\alpha^2 = 0$, then one only obtains

$$g(\Sigma) \geq 1.$$

This is compatible with the existence of self-intersection zero torus fibers in the K3 elliptic fibration.

5.3.9 Why the exceptional sphere is not a contradiction

After the blow-up above there appears the exceptional sphere

$$E \subset X \# \overline{\mathbb{CP}}^2, \quad E^2 = -1.$$

Its genus is 0.

This does not contradict the adjunction inequality stated in this section, because the standard version adopted here requires

$$[\Sigma]^2 \geq 0$$

and $g(\Sigma) > 0$.

The exceptional sphere violates both of these hypotheses at once.

For surfaces of negative self-intersection, there exist finer generalized adjunction inequalities, but their hypotheses and form require additional discussion.

Hence the formula above cannot be applied unconditionally to an arbitrary embedded surface.

5.3.10 Relation to the blow-up formula

There is a close relation between the blow-up formula and the adjunction inequality.

If

$$\Sigma^2 = n > 0,$$

choose n distinct points on Σ and blow up there.

Write the corresponding exceptional classes as

$$e_1, \dots, e_n.$$

The class represented by the proper transform $\tilde{\Sigma}$ of Σ is

$$[\tilde{\Sigma}] = [\Sigma] - e_1 - \dots - e_n.$$

Then

$$[\tilde{\Sigma}]^2 = [\Sigma]^2 - n = 0.$$

On the other hand, the blow-up formula turns the original basic class K into

$$K \pm e_1 \pm \dots \pm e_n.$$

By choosing the signs appropriately, and applying the inequality of the zero self-intersection case to $\tilde{\Sigma}$, one recovers the original inequality for $\Sigma^2 > 0$:

$$2g(\Sigma) - 2 \geq \Sigma^2 + |K \cdot [\Sigma]|.$$

Thus the blow-up technique is an important tool for extending the square-zero adjunction inequality to the case of nonnegative self-intersection.

5.3.11 Why this is a smooth-topological obstruction

The minimal genus is not merely a homological quantity.

For one and the same topological 4-manifold with different smooth structures, how low a genus a smooth surface representing a given homology class can have may change.

The adjunction inequality controls the minimal genus through the smooth invariant SW_X , and hence can detect smooth structure.

This is also one of the basic strategies frequently used in Seiberg–Witten theory to distinguish exotic smooth structures:

first compute or control the basic classes, and then use the adjunction inequality to compare the smooth embedding behavior of certain homology classes.

5.3.12 Summary of this section

Let $b_2^+(X) > 1$, let K be a Seiberg–Witten basic class, and let $\Sigma \subset X$ be a positive-genus, non-torsion, self-intersection nonnegative connected embedded oriented surface.

The adjunction inequality gives

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + |K \cdot [\Sigma]|.$$

It may be viewed as the gauge-theoretic analogue of the complex adjunction formula in the smooth category.

This inequality links information from three different levels:

- the genus of the embedded surface;
- the self-intersection given by the intersection form;
- the pairing of the Seiberg–Witten basic class with the surface.

Hence Seiberg–Witten invariants can give lower bounds for the minimal genus, and can further detect smooth structure.

The next step leads naturally to the central link between symplectic 4-manifolds and Seiberg–Witten theory: Taubes'

$$SW = Gr$$

theorem.

It will prove that the symplectic canonical class is itself a Seiberg–Witten basic class, and will relate gauge-theoretic solutions to pseudoholomorphic curves.

5.4 Symplectic 4-manifolds and Taubes' theorem

The deepest link between Seiberg–Witten theory and symplectic geometry comes from the work of Taubes.

Let

$$(X, \omega)$$

be a closed symplectic 4-manifold.

Choose an almost complex structure J compatible with ω . Then TX becomes a complex rank-2 vector bundle, and one can define the symplectic canonical bundle

$$K_X = \Lambda^{2,0} T^* X.$$

Write its first Chern class as

$$K_\omega := c_1(K_X) = -c_1(TX, J).$$

Although the compatible almost complex structure J is not unique, all compatible J form a contractible space, so K_ω depends only on the symplectic structure ω .

The basic conclusion of Taubes' theorem is this: a symplectic structure forces certain Seiberg–Witten invariants to be non-zero, and, going further, the Seiberg–Witten invariants can actually be computed by means of pseudoholomorphic curves.

5.4.1 The canonical Spin^c structure induced by an almost complex structure

An almost complex structure on any oriented Riemannian 4-manifold induces a Spin^c structure naturally.

For J compatible with ω , write this Spin^c structure as

$$\mathfrak{S}_\omega.$$

Its spinor bundles can be written concretely as

$$S^+ \cong \Lambda^{0,0} \oplus \Lambda^{0,2}, \quad S^- \cong \Lambda^{0,1}.$$

Here $\Lambda^{0,q}$ denotes the $(0, q)$ -forms with respect to the almost complex structure J . Hence

$$\det S^+ \cong \Lambda^{0,2}.$$

As a complex line bundle, $\Lambda^{0,2}$ has the same first Chern class as K_X^{-1} , so the determinant line of the canonical Spin^c structure satisfies

$$L_{\mathfrak{s}_\omega} \cong K_X^{-1}.$$

Therefore

$$c_1(L_{\mathfrak{s}_\omega}) = -K_\omega = c_1(TX, J).$$

Here there is a sign convention that is easy to confuse.

These notes have always defined the basic class of a Spin^c structure to be $c_1(L_s)$ (Definition 4.7.5). Under this convention, the basic class corresponding to the canonical Spin^c structure is

$$-K_\omega.$$

Since conjugation negates the basic class, in the end both K_ω and $-K_\omega$ appear as basic classes.

5.4.2 Characteristic numbers of an almost complex 4-manifold

For a closed almost complex 4-manifold, the first Chern class satisfies the basic identity

$$c_1(TX, J)^2 = 2\chi(X) + 3\sigma(X).$$

Since $K_\omega = -c_1(TX, J)$, we likewise have

$$K_\omega^2 = 2\chi(X) + 3\sigma(X).$$

Hence the expected dimension of the canonical Spin^c structure is

$$d(\mathfrak{s}_\omega) = 0.$$

So when $b_2^+ > 1$ its Seiberg–Witten invariant can be defined directly as the signed count of the zero-dimensional moduli space.

5.4.3 Taubes nonvanishing theorem

The first basic conclusion of Taubes is that the symplectic canonical Spin^c structure has non-zero Seiberg–Witten invariant.

Theorem 5.4.1 (Taubes nonvanishing theorem). Let (X, ω) be a closed symplectic 4-manifold, and assume that

$$b_2^+(X) > 1.$$

Let $\mathfrak{s}_\omega$ be the canonical Spin^c structure induced by a compatible almost complex structure. Then

$$SW_X(\mathfrak{s}_\omega) = \pm 1.$$

According to the basic-class convention of these notes, this says that

$$-K_\omega$$

is a Seiberg–Witten basic class.

Using the conjugation symmetry again, we see that

$$K_\omega$$

is likewise a basic class.

Hence:

Corollary 5.4.2. If (X, ω) is a closed symplectic 4-manifold and $b_2^+(X) > 1$, then

$$\pm K_\omega$$

are both Seiberg–Witten basic classes.

This is one of the most basic bridges between symplectic topology and Seiberg–Witten theory.

5.4.4 The positive scalar curvature obstruction obtained immediately

Combining this with the positive scalar curvature vanishing theorem of the previous section, we obtain immediately:

Corollary 5.4.3. Let (X, ω) be a closed symplectic 4-manifold, and suppose that

$$b_2^+(X) > 1.$$

Then X admits no positive scalar curvature metric.

The proof is very direct.

Taubes' theorem gives a non-zero Seiberg–Witten invariant, while positive scalar curvature forces all Seiberg–Witten invariants to vanish.

Hence the two cannot occur simultaneously.

5.4.5 Twisting the canonical Spin^c structure

In order to state the more general $SW = Gr$ theorem, consider a twisting of the canonical Spin^c structure.

Let

$$\mathcal{E} \longrightarrow X$$

be a complex line bundle, and write

$$e = c_1(\mathcal{E}) \in H^2(X; \mathbb{Z}).$$

Define

$$\mathfrak{s}_e = \mathfrak{s}_\omega \otimes \mathcal{E}.$$

We showed above that Spin^c twisting turns the determinant line into

$$L_{\mathfrak{s}_e} \cong L_{\mathfrak{s}_\omega} \otimes \mathcal{E}^2.$$

Hence

$$c_1(L_{\mathfrak{s}_e}) = -K_\omega + 2e.$$

The corresponding expected dimension is

$$\begin{aligned} d(\mathfrak{s}_e) &= \frac{(-K_\omega + 2e)^2 - K_\omega^2}{4} \\ &= e^2 - K_\omega \cdot e. \end{aligned}$$

This expression is exactly the natural index of the class $PD(e)$ in pseudoholomorphic curve theory.

This is no coincidence, but is precisely the first numerical sign that the $SW = Gr$ correspondence can hold.

5.4.6 The almost complex decomposition of the spinor

Under the canonical Spin^c structure, a positive spinor can be written as

$$\psi = (\alpha, \beta),$$

where

$$\alpha \in \Omega^{0,0}(X), \quad \beta \in \Omega^{0,2}(X).$$

If we twist by $\mathcal{E}$ as well, then

$$S_{\mathfrak{s}_e}^+ \cong \mathcal{E} \oplus (\Lambda^{0,2} \otimes \mathcal{E}).$$

Hence in this case

$$\alpha \in \Gamma(\mathcal{E}), \quad \beta \in \Gamma(\Lambda^{0,2} \otimes \mathcal{E}).$$

So the first component of the positive spinor is in fact a section of the line bundle $\mathcal{E}$. Its zero set naturally represents

$$PD(c_1(\mathcal{E})).$$

The key phenomenon in Taubes theory is this: in a suitable large-parameter limit, the Seiberg–Witten equations force the zero set of α to converge to pseudoholomorphic curves.

5.4.7 The large-parameter perturbation of Taubes

Taubes does not study the Seiberg–Witten equations under an arbitrary perturbation directly.

He chooses a family of perturbations closely tied to the symplectic form ω , and introduces a large parameter

$$r \longrightarrow \infty.$$

Under a suitable convention, a term of large size proportional to

$$ir\omega$$

appears in the curvature equation.

The signs and constants in the explicit formula depend on the normalization of Clifford multiplication and of the quadratic map, but its structure has the following form:

$$D_A \psi = 0, \quad F_A^+ = q(\psi) - ir \omega + \text{lower-order terms}.$$

At the same time one performs a suitable r -dependent rescaling of the spinor.

This special perturbation makes the symplectic form enter the Seiberg–Witten equations directly.

As $r \rightarrow \infty$, the equations originally force the two components of the spinor to exhibit very different asymptotic behavior.

5.4.8 α and β in the large- r limit

Let

$$\psi = (\alpha, \beta).$$

The a priori estimates of Taubes show that, when r is large, β is usually very small, while on most of the region α is close to a non-zero section of fixed norm.

Hence on most of X α is nowhere vanishing.

But since

$$c_1(\mathcal{E})$$

may be non-zero, α cannot be nowhere vanishing on all of X .

So topology forces α to have a zero set.

As $r \rightarrow \infty$, the curvature and the energy concentrate more and more near this zero set.

The core analytic result of Taubes shows that, in a suitable sense, these zero sets converge to a collection of J -holomorphic curves.

Hence a gauge-theoretic object ultimately produces a symplectic-geometric object.

5.4.9 Pseudo-holomorphic curves

Let J be an ω -compatible almost complex structure.

A smooth map

$$u : (\Sigma, j) \longrightarrow (X, J)$$

is called J -holomorphic if it satisfies the Cauchy–Riemann equation

$$du \circ j = J \circ du.$$

If u is an embedding, its image is called an embedded J -holomorphic curve.

For a non-constant J -holomorphic curve C , the symplectic form is positive on C , hence

$$\int_C \omega > 0.$$

So pseudoholomorphic curves are naturally compatible with symplectic geometry.

Taubes theory shows that Seiberg–Witten solutions produce such curves in the large-parameter limit.

5.4.10 From Seiberg–Witten solutions to curves

Let $\mathfrak{s}_e = \mathfrak{s}_\omega \otimes \mathcal{E}$, and assume that the corresponding Seiberg–Witten invariant is non-zero. For Taubes’ large-parameter equations, take a sequence of solutions

$$(A_r, \psi_r), \quad r \rightarrow \infty.$$

Write

$$\psi_r = (\alpha_r, \beta_r).$$

Taubes’ compactness analysis shows that, after passing to a subsequence, the zero sets of α_r converge to a finite collection

$$\{(C_i, m_i)\},$$

where each C_i is a J -holomorphic curve and $m_i \in \mathbb{Z}_{>0}$ is a multiplicity.

These curves satisfy the homology identity

$$\sum_i m_i [C_i] = PD(e).$$

Hence if

$$SW_X(\mathfrak{s}_e) \neq 0,$$

then the class

$$PD(e)$$

can, in a suitable sense, be represented by a J -holomorphic subvariety.

This conclusion is already very strong: a gauge-theoretic nonvanishing statement forces a certain homology class to admit a pseudoholomorphic representative.

5.4.11 Gromov invariant

For a symplectic 4-manifold, one can try to define an invariant by counting the J -holomorphic curves representing a given homology class.

The corresponding invariant constructed by Taubes is usually denoted

$$Gr_X(e).$$

It is not simply “counting all J -holomorphic curves”.

The reason is that one must deal with:

- the orientation of curves;
- transversality;
- disconnected curves;
- multiple components;
- multiply covered tori;
- different homology splittings.

So a rigorous Gromov invariant requires a rather careful definition.

But its basic idea is still a weighted count of the pseudoholomorphic configurations representing $PD(e)$.

5.4.12 The $SW = Gr$ theorem

Taubes' central theorem says that this symplectic curve count agrees exactly with the Seiberg–Witten invariant.

Theorem 5.4.4 (Taubes, $SW = Gr$). Let (X, ω) be a closed symplectic 4-manifold with

$$b_2^+(X) > 1.$$

For $e \in H^2(X; \mathbb{Z})$, let

$$\mathfrak{s}_e = \mathfrak{s}_\omega \otimes \mathcal{E}, \quad c_1(\mathcal{E}) = e.$$

Under compatible orientation conventions, Taubes' Gromov invariant and the Seiberg–Witten invariant correspond:

$$SW_X(\mathfrak{s}_e) = Gr_X(e).$$

The equality here requires Taubes' precise definition of $Gr_X(e)$, in particular when multiply covered square-zero tori occur.

So this theorem cannot be understood simply as “the SW invariant equals the number of ordinary embedded curves”.

What it really says is: two moduli problems of completely different origins produce the same invariant.

5.4.13 The correspondence mechanism between gauge theory and symplectic geometry

$SW = Gr$ can be understood roughly as the following correspondence:

$$\text{Seiberg–Witten solution} \quad \longleftrightarrow \quad J\text{-holomorphic curve configuration.}$$

The object on the left is a connection–spinor pair

$$(A, \psi)$$

satisfying nonlinear elliptic gauge equations.

The object on the right is a 2-dimensional pseudoholomorphic submanifold.

The key variable connecting the two is the α component of the positive spinor.

As $r \rightarrow \infty$, α tends to a non-zero value away from the curve, and tends to zero near the curve.

Hence one can understand a J -holomorphic curve as the “vortex locus” of the spinor field in the large- r limit.

At the same time, the connection curvature also concentrates around these loci, and records the first Chern class of the line bundle $\mathcal{E}$.

So

$$c_1(\mathcal{E})$$

is at once gauge-theoretic twisting data and determines the homology class of the curve.

5.4.14 Geometric interpretation of canonical class nonvanishing

Take

$$e = 0.$$

Then $\mathfrak{s}_e = \mathfrak{s}_\omega$, and the corresponding homology class is zero.

The basic configuration of the Gromov invariant is the empty curve, whose count is ± 1 .

Hence $SW = Gr$ gives

$$SW_X(\mathfrak{s}_\omega) = \pm 1.$$

So Taubes' nonvanishing theorem can in fact be viewed as the special case of the $SW = Gr$ theorem at $e = 0$.

By conjugation, this in turn shows that the symplectic canonical class K_ω is a basic class.

5.4.15 Constraints of basic classes on the symplectic area

Let K be an arbitrary Seiberg–Witten basic class.

Write it as

$$K = -K_\omega + 2e$$

for some twisting class e .

If the corresponding Seiberg–Witten invariant is non-zero, Taubes' theorem gives a J -holomorphic subvariety representing $PD(e)$.

If $e \neq 0$, then its symplectic area satisfies

$$e \cdot [\omega] = \int_{PD(e)} \omega > 0.$$

Hence

$$(K + K_\omega) \cdot [\omega] = 2e \cdot [\omega] \geq 0.$$

Thus

$$K \cdot [\omega] \geq -K_\omega \cdot [\omega].$$

If one applies the same idea to the conjugate basic class, one obtains strong constraints on the basic classes in the direction of the symplectic class.

In minimal symplectic manifolds, these estimates can be strengthened further.

5.4.16 The symplectic adjunction formula

Let

$$C \subset X$$

be an embedded J -holomorphic curve.

Since J preserves the tangent bundle on C , the normal bundle also carries a natural complex line bundle structure.

From the restriction of the tangent bundle

$$TX|_C = TC \oplus NC$$

one obtains the additivity formula for the first Chern class.

Pairing with $[C]$, we have

$$c_1(TX, J) \cdot [C] = c_1(TC) + c_1(NC).$$

Here

$$c_1(TC) = 2 - 2g(C),$$

while the Chern number of the normal bundle equals the self-intersection:

$$c_1(NC) = C^2.$$

Since $K_\omega = -c_1(TX, J)$, we obtain

$$2g(C) - 2 = C^2 + K_\omega \cdot C.$$

This is the symplectic adjunction formula.

Hence for J -holomorphic curves, the Seiberg–Witten adjunction inequality of the previous section is in fact attained as an equality.

5.4.17 Exceptional spheres and Taubes theory

Let (X, ω) be a symplectic 4-manifold admitting a class

$$E \in H_2(X; \mathbb{Z})$$

with

$$E^2 = -1.$$

If E is a symplectic exceptional class, then for a suitable compatible almost complex structure it can be represented by an embedded J -holomorphic sphere.

The adjunction formula gives

$$-2 = -1 + K_\omega \cdot E,$$

so

$$K_\omega \cdot E = -1.$$

This is exactly the standard relation in a blow-up between the exceptional class and the canonical class.

If

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2,$$

then the symplectic canonical class satisfies

$$K_{\widetilde{X}} = K_X + e$$

under the usual blow-up orientation convention.

This formula is entirely compatible with the preceding Seiberg–Witten blow-up formula.

5.4.18 Minimal symplectic manifolds

A symplectic 4-manifold is called **symplectically minimal** if there is no symplectically embedded self-intersection -1 sphere.

Taubes theory establishes a strong link between Seiberg–Witten basic classes and these exceptional spheres.

In particular, in the symplectic category with $b_2^+ > 1$, a blow-up produces

$$K_\omega \pm e$$

as a new family of basic classes, while the exceptional class itself can be detected by pseudoholomorphic curves.

Hence Seiberg–Witten theory can detect not only the existence of a symplectic structure, but also be used to study whether a symplectic manifold can be blown down.

5.4.19 Example: K3

The K3 surface admits a Kähler structure, and hence in particular is a symplectic manifold.

Its canonical bundle is trivial, so that

$$K_\omega = 0.$$

Taubes' theorem therefore gives

$$SW_{K3}(0) = \pm 1,$$

which is consistent with the earlier gauge-theoretic computation.

On the other hand, for an embedded J -holomorphic curve C , the symplectic adjunction formula gives

$$2g(C) - 2 = C^2.$$

For example, if $C^2 = 0$, then

$$g(C) = 1.$$

This is in complete agreement with the fact that the fiber of an elliptic K3 surface is a torus of self-intersection zero.

5.4.20 Example: Blow-up of K3

Let

$$\widetilde{X} = K3 \# \overline{\mathbb{CP}}^2.$$

Denote the exceptional class by e .

Since the canonical class of the original K3 is zero, the symplectic canonical class after the blow-up is

$$K_{\widetilde{X}} = e.$$

The Seiberg–Witten blow-up formula above (Theorem 5.2.1) then gives the basic classes

$$\pm e.$$

Hence the symplectic canonical basic class predicted by Taubes' theorem appears precisely among the basic classes produced by the blow-up formula.

This is the simplest consistency check between the two theories.

5.4.21 The case $b_2^+ = 1$

Taubes theory also applies to many symplectic 4-manifolds with $b_2^+ = 1$, but in this case the Seiberg–Witten invariant has a chamber dependence.

The symplectic form ω naturally singles out a distinguished chamber, usually called the symplectic chamber.

In this chamber, the canonical Spin^c structure still satisfies the corresponding nonvanishing theorem.

But one cannot, as in the case $b_2^+ > 1$, regard this invariant as a chamber-independent invariant without further comment.

Hence in the basic theory this paper mainly continues to use the case $b_2^+ > 1$.

5.4.22 Summary of this section

A symplectic structure ω selects a compatible almost complex structure J , and thus produces a canonical Spin^c structure $\mathfrak{s}_\omega$.

Under the conventions of this paper, its determinant class is

$$c_1(L_{\mathfrak{s}_\omega}) = -K_\omega,$$

where

$$K_\omega = -c_1(TX, J)$$

is the symplectic canonical class.

Taubes' nonvanishing theorem gives

$$SW_X(\mathfrak{s}_\omega) = \pm 1,$$

so $\pm K_\omega$ are both Seiberg–Witten basic classes.

More generally, if

$$\mathfrak{s}_e = \mathfrak{s}_\omega \otimes \mathcal{E},$$

then Taubes' $SW = Gr$ theorem relates

$$SW_X(\mathfrak{s}_e)$$

to the Gromov count of the pseudoholomorphic curves representing $PD(c_1(\mathcal{E}))$.

Hence there is a direct correspondence between Seiberg–Witten theory and symplectic geometry: large-parameter gauge-theoretic solutions degenerate into pseudoholomorphic curves.

This enables Seiberg–Witten invariants to detect the symplectic canonical class, to constrain embedded surfaces, to study minimality, and to control further the smooth topology of a symplectic 4-manifold.

The next step would be to continue the discussion of the typical applications of Seiberg–Witten theory to the classification of smooth 4-manifolds, in particular exotic smooth structures, and of why one and the same topological 4-manifold can have different sets of Seiberg–Witten basic classes.

5.5 Seiberg–Witten theory and smooth structures on 4-manifolds

One of the most important significances of Seiberg–Witten theory is that it can detect the smooth structure of a 4-manifold.

In dimension four, there is a very striking difference between topological classification and smooth classification. Two closed smooth 4-manifolds may be homeomorphic to each other, yet admit no diffeomorphism.

This phenomenon is called an **exotic smooth structure**.

Seiberg–Witten invariants are precisely one of the main tools for studying phenomena of this kind.

5.5.1 Why the intersection form does not suffice to classify smooth structures

Let X be a closed, simply connected, oriented 4-manifold.

Its intersection form is

$$Q_X : H^2(X; \mathbb{Z}) / \text{Tor} \times H^2(X; \mathbb{Z}) / \text{Tor} \longrightarrow \mathbb{Z}.$$

It is the central topological invariant in the oriented homeomorphism classification.

Freedman theory shows that, in the category of simply connected topological 4-manifolds, the intersection form together with the corresponding additional topological data controls the homeomorphism type to a large extent.

But in the smooth category, the intersection form is far from enough.

There may exist two smooth manifolds X and Y satisfying

$$X \cong Y$$

as oriented topological manifolds, and hence having the same intersection form, but

$$X \not\cong Y$$

as smooth manifolds.

In other words, the intersection form cannot see all the smooth information.

The role of gauge theory is precisely to construct new invariants that are sensitive to diffeomorphism, but not necessarily invariant under homeomorphism.

5.5.2 Basic classes are smooth invariants

Assume that

$$f : X \longrightarrow Y$$

is an orientation-preserving diffeomorphism.

Then f induces on cohomology

$$f^* : H^2(Y; \mathbb{Z}) \longrightarrow H^2(X; \mathbb{Z}),$$

and preserves the intersection form.

At the same time, f pulls the Spin^c structures on Y back to Spin^c structures on X , satisfying

$$c_1(L_{f^*\mathfrak{s}}) = f^*c_1(L_{\mathfrak{s}}).$$

The Seiberg–Witten equations correspond naturally under a diffeomorphism, so the corresponding invariants remain unchanged.

Consequently, if K is a basic class of Y , then f^*K is a basic class of X .

Hence the basic-class set

$$\mathcal{B}(X) = \{c_1(L_{\mathfrak{s}}) \mid SW_X(\mathfrak{s}) \neq 0\}$$

is a smooth invariant, in the sense that it is considered under the automorphisms of $H^2(X; \mathbb{Z})$.

Hence if two homeomorphic 4-manifolds have basic-class data that cannot be matched with each other by an intersection-form isometry, then they cannot be diffeomorphic.

5.5.3 The basic strategy for distinguishing exotic smooth structures

When using Seiberg–Witten invariants to distinguish smooth structures, one usually employs the following mechanism.

First construct two smooth 4-manifolds X and Y , and use topological 4-manifold theory to prove that they are homeomorphic.

Then compute

$$SW_X \quad \text{and} \quad SW_Y.$$

If there exists no intersection-form-preserving cohomology isomorphism that matches the Seiberg–Witten data of the two, then one obtains

$$X \not\cong Y$$

in the smooth category.

Hence the structure of the whole argument is

$$\text{topological classification} + \text{gauge-theoretic smooth invariant}.$$

The former proves that the two manifolds are the same topologically, the latter proves that they are different in the smooth category.

5.5.4 A typical source: knot surgery

Fintushel–Stern knot surgery is one of the classical methods for constructing exotic smooth 4-manifolds.

Its basic starting point is a smooth 4-manifold containing a self-intersection zero torus

$$T \subset X, \quad T^2 = 0.$$

Since the tubular neighborhood of T is homeomorphic to

$$T^2 \times D^2,$$

one can remove this neighborhood, and then use a knot complement to construct a new 4-manifold

$$X_K.$$

Under suitable assumptions on the fundamental group and on the embedding, X_K can have the same homeomorphism type as the original X .

But its Seiberg–Witten invariant records the Alexander polynomial of the knot K .

Hence by choosing knots with different Alexander polynomials, one obtains a large number of smooth 4-manifolds that are homeomorphic but pairwise non-diffeomorphic.

The complete knot surgery formula depends on gluing theory, whose proof we do not develop here.

What matters is its mechanism:

the Alexander polynomial in three-dimensional knot theory enters the four-dimensional Seiberg–Witten invariant through a cut-and-paste construction, thereby producing different smooth structures.

5.5.5 K3 and exotic smooth structures

The K3 surface is an important model for this idea.

The standard K3 satisfies

$$b_2^+ = 3, \quad b_2^- = 19,$$

and its intersection form is

$$Q_{K3} \cong 3H \oplus 2(-E_8).$$

Its standard Seiberg–Witten theory has only a very simple basic-class structure, in which the canonical class is zero.

On the other hand, the self-intersection zero torus fibers in an elliptic K3 allow one to perform knot surgery.

Under suitable conditions, the resulting manifolds are homeomorphic to K3, but their Seiberg–Witten invariants change with the chosen knot.

Thus one obtains smooth manifolds that are homeomorphic but not diffeomorphic to the standard K3.

Hence even when the intersection form is exactly the same, the Seiberg–Witten invariants can still see different smooth structures.

5.5.6 Blow-up also changes the basic-class structure

We have seen above that if K is a basic class of X , then in

$$\widetilde{X} = X \# \overline{\mathbb{CP}}^2$$

there appear

$$K + e \quad \text{and} \quad K - e.$$

Hence the basic-class set can record the blow-up operation.

If one blows up k times in succession, each original basic class produces

$$K \pm e_1 \pm \cdots \pm e_k.$$

So Seiberg–Witten theory can distinguish a manifold from many of its blow-ups, and this has a direct connection with the question of minimality.

In the symplectic category, Taubes theory further relates these exceptional classes to symplectic (-1) -spheres.

5.5.7 Seiberg–Witten invariants are not complete classification invariants

Although the Seiberg–Witten invariants are very strong, they do not completely classify smooth 4-manifolds.

Different smooth manifolds may have exactly the same Seiberg–Witten invariants.

There may even exist a large number of interesting smooth manifolds whose ordinary Seiberg–Witten invariants all vanish.

Hence from

$$SW_X = SW_Y$$

one cannot conclude

$$X \cong Y.$$

The Seiberg–Witten invariant should be understood as a strong constraint on the smooth structure, rather than as a complete encoding of it.

This is also why modern four-dimensional manifold theory still has to use at the same time Donaldson invariants, Floer theories, stable homotopy refinements, symplectic techniques, trisections and other structures.

5.5.8 The global logic of Seiberg–Witten theory

We can now review the chain of logic of the whole Seiberg–Witten construction.

First, a Riemannian metric gives a Clifford algebra, and the representation theory of the Clifford algebra produces half-spinor spaces $\Delta^\pm$.

Afterwards one uses

$$\text{Spin}^c(4)$$

to globalize these pointwise representations into spinor bundles

$$S^\pm.$$

A Spin^c structure also naturally carries a determinant line bundle

$$L = \det S^+.$$

Choosing a unitary connection A on L , and combining it with the Levi–Civita connection, gives a spinor connection ∇_A as well as the Dirac operator

$$D_A : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$

On the other hand, the special representation theory of the four-dimensional Clifford algebra gives a quadratic map

$$q : S^+ \longrightarrow i\Lambda_+^2.$$

Combining these two structures produces the Seiberg–Witten equations

$$D_A\psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

The equations have gauge symmetry, so what is really studied is the solution space modulo gauge.

Linearizing gives the Seiberg–Witten deformation complex. Gauge fixing turns it into an elliptic Fredholm problem, so that one obtains the expected dimension

$$d(\mathfrak{s}) = \frac{c_1(L_{\mathfrak{s}})^2 - (2\chi + 3\sigma)}{4}.$$

Generic perturbation takes care of transversality, the Weitzenböck formula together with Abelian gauge fixing gives compactness, and the homology orientation gives the orientation of the moduli space.

Finally, one extracts an integer from the compact oriented moduli space, and obtains the Seiberg–Witten invariant.

Hence the whole theory can be summarized as

$$\text{Spin}^c \text{ geometry} \longrightarrow \text{elliptic gauge equations} \longrightarrow \text{moduli space} \longrightarrow \text{smooth topology}.$$

5.5.9 Where four-dimensionality shows up

Many key structures of Seiberg–Witten theory depend strongly on dimension four.

First, only in dimension four do 2-forms sit in the middle dimension, so that the Hodge star gives

$$\Lambda^2 = \Lambda_+^2 \oplus \Lambda_-^2.$$

Second, four dimensions enjoy the special Lie group decomposition

$$\text{Spin}(4) \cong SU(2)_+ \times SU(2)_-.$$

This makes for a particularly close representation-theoretic relation between positive spinors and self-dual 2-forms.

It is precisely this relation that allows the quadratic spinor expression $q(\psi)$ to be of the same type as the self-dual part of the curvature F_A^+ .

Hence the second Seiberg–Witten equation can be written as

$$F_A^+ = q(\psi).$$

Together with the special properties of the four-dimensional intersection form, the signature, embedded surfaces and middle-dimensional topology, this makes the connection between gauge theory and manifold topology exceptionally strong in dimension four.

5.5.10 Comparison with Donaldson theory

Donaldson theory and Seiberg–Witten theory both construct smooth invariants by means of four-dimensional gauge theory, but their analytic complexity differs greatly.

Donaldson theory studies the ASD equation

$$F_A^+ = 0$$

satisfied by $SU(2)$ or $SO(3)$ connections.

This is a non-Abelian gauge theory, the curvature contains an $A \wedge A$ term, and the moduli space may exhibit bubbling.

Hence its compactification requires Uhlenbeck theory.

The gauge group of Seiberg–Witten theory is $U(1)$, and the connection part is Abelian. Although the addition of the spinor keeps the equations nonlinear, the Weitzenböck formula provides very strong a priori estimates, which makes compactness considerably simpler.

It is precisely this analytic tractability that makes the Seiberg–Witten invariants easier to compute in practice than the Donaldson invariants in many four-dimensional topological problems.

But Donaldson theory and Seiberg–Witten theory are not two unrelated theories. There are deep connections between them, which are further linked through the Witten conjecture, Floer theory and modern gauge-theoretic constructions.

5.5.11 Final summary

The core of Seiberg–Witten theory is not some isolated formula, but a whole mechanism running from local geometry to global topology.

A Spin^c structure provides spinors and a determinant line bundle;
 a connection together with Clifford multiplication produces the Dirac operator;
 the quadratic spinor map relates spinor geometry to the self-dual curvature;
 the Weitzenböck formula converts the Dirac equation into powerful analytic estimates;
 elliptic deformation theory endows the moduli space with a finite-dimensional smooth structure;
 compactness and orientation allow the moduli space to be counted;
 and the resulting Seiberg–Witten invariant in turn constrains the geometry and topology of a smooth 4-manifold.

It can be used to study

- positive scalar curvature;
- embedded surfaces and minimal genus;
- blow-ups and exceptional spheres;
- symplectic canonical classes;
- symplectic minimality;
- exotic smooth structures.

In particular, Taubes' $SW = Gr$ theorem shows that gauge theory and pseudoholomorphic curve theory are not two mutually independent techniques, but rather two ways of describing the same deep structure in four-dimensional symplectic geometry.

From this point of view, the basic idea of Seiberg–Witten theory can be summarized as follows:

Encoding the differential-geometric data on a smooth 4-manifold into an elliptic gauge-theoretic moduli problem, and then extracting from this moduli space topological information sensitive to the smooth structure.

This is also the fundamental reason why gauge theory occupies a central position in four-dimensional manifold theory.

Appendix A

Analytic preliminaries

The purpose of this appendix is to collect the basic analytic material needed in four-dimensional gauge theory. We do not attempt to develop partial differential equations in full generality. Instead, we focus on the analytic structures that occur repeatedly in the study of the ASD and Seiberg–Witten equations:

$$\begin{aligned} & \text{differential operator} \longrightarrow \text{principal symbol} \longrightarrow \text{ellipticity} \\ & \longrightarrow \text{Sobolev completion} \longrightarrow \text{elliptic regularity} \longrightarrow \text{Fredholm theory.} \end{aligned}$$

Throughout this appendix, unless otherwise stated, X denotes a smooth Riemannian manifold and

$$E \longrightarrow X, \quad F \longrightarrow X$$

denote smooth real or complex vector bundles. When L^2 -inner products or formal adjoints are involved, the bundles are assumed to carry bundle metrics compatible with the relevant connections.

A.1 Linear differential operators and principal symbols

A.1.1 Linear differential operators on Euclidean space

We begin with the local model. Let $U \subset \mathbb{R}^n$ be an open set, and let $u : U \longrightarrow \mathbb{R}$ be a smooth function.

A linear differential operator of order at most m has the form

$$P = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha,$$

where $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$ is a multi-index, $|\alpha| = \alpha_1 + \dots + \alpha_n$, and

$$\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

Thus

$$Pu = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha u.$$

The terms with $|\alpha| = m$ are called the **highest-order terms** of P . The operator

$$P_m = \sum_{|\alpha|=m} a_\alpha(x) \partial^\alpha$$

is called the **principal part** of P .

The principal part plays a distinguished role in the local analytic behavior of the differential equation $Pu = f$.

Why the highest-order part matters. Consider, for example,

$$P = -\frac{d^2}{dx^2} + a(x)\frac{d}{dx} + b(x).$$

For a highly oscillatory function $u(x) = e^{i\lambda x}$, we have

$$u'(x) = i\lambda u(x), \quad u''(x) = -\lambda^2 u(x).$$

Hence

$$Pu = (\lambda^2 + i\lambda a(x) + b(x))u.$$

As $|\lambda| \rightarrow \infty$, the second-order term grows like $|\lambda|^2$, whereas the first-order term grows only like $|\lambda|$, and the zeroth-order term remains bounded with respect to λ .

Thus, at high frequency, the highest-order part dominates the lower-order terms. This is the basic analytic reason why the principal part governs ellipticity and regularity.

A.1.2 Differential operators between vector bundles

Let

$$E \longrightarrow X, \quad F \longrightarrow X$$

be smooth vector bundles.

A linear map $P : \Gamma(E) \rightarrow \Gamma(F)$ is called a **linear differential operator of order at most m** if, after choosing local coordinates on X and local frames for E and F , it can be written as

$$P = \sum_{|\alpha| \leq m} A_\alpha(x) \partial^\alpha,$$

where $A_\alpha(x) : E_x \rightarrow F_x$ is a smoothly varying family of linear maps.

More concretely, suppose

$$\text{rank } E = r, \quad \text{rank } F = s.$$

In local trivializations

$$E|_U \cong U \times \mathbb{K}^r, \quad F|_U \cong U \times \mathbb{K}^s,$$

where

$$\mathbb{K} = \mathbb{R} \quad \text{or} \quad \mathbb{C},$$

a section of E is represented by a vector-valued function $u : U \rightarrow \mathbb{K}^r$, and

$$A_\alpha(x) \in M_{s \times r}(\mathbb{K}).$$

Thus locally

$$Pu(x) = \sum_{|\alpha| \leq m} A_\alpha(x) \partial^\alpha u(x).$$

The principal part is

$$P_m = \sum_{|\alpha|=m} A_\alpha(x) \partial^\alpha.$$

A.1.3 An intrinsic characterization by commutators

The coordinate description above is useful for explicit computations, but the notion of differential operator is intrinsic.

For $f \in C^\infty(X)$, we regard multiplication by f as an operator on sections. Given $P : \Gamma(E) \rightarrow \Gamma(F)$, define the commutator

$$[P, f] = P \circ f - f \circ P.$$

A bundle homomorphism $A : E \rightarrow F$ acts pointwise on sections: $(Au)(x) = A_x u(x)$. Such an operator is called a **zeroth-order differential operator**.

Inductively, P is called a **differential operator of order at most m** if for every $f \in C^\infty(X)$ the commutator $[P, f]$ is a differential operator of order at most $m - 1$.

The key point is that taking a commutator with multiplication by a smooth function lowers the differential order by one.

Example A.1.1. Let $\nabla : \Gamma(E) \rightarrow \Omega^1(X; E)$ be a connection. By the Leibniz rule, $\nabla(fu) = df \otimes u + f\nabla u$. Therefore

$$[\nabla, f]u = \nabla(fu) - f\nabla u = df \otimes u.$$

The right-hand side contains no derivative of u , and is therefore zeroth order in u . Hence ∇ is a first-order differential operator.

Example A.1.2. For the exterior derivative $d : \Omega^k(X) \rightarrow \Omega^{k+1}(X)$, we have $d(f\alpha) = df \wedge \alpha + f d\alpha$. Hence $[d, f]\alpha = df \wedge \alpha$. Again the right-hand side is zeroth order in α , so d is a first-order differential operator.

A.1.4 The principal symbol

Let $P : \Gamma(E) \rightarrow \Gamma(F)$ be a differential operator of order m .

In local coordinates,

$$P = \sum_{|\alpha| \leq m} A_\alpha(x) \partial^\alpha.$$

Definition A.1.3. For

$$x \in X, \quad \xi = \sum_{j=1}^n \xi_j dx^j \in T_x^* X,$$

the **principal symbol** of P at (x, ξ) is the linear map $\sigma_P(x, \xi) : E_x \rightarrow F_x$ defined by

$$\sigma_P(x, \xi) = \sum_{|\alpha|=m} A_\alpha(x) \xi^\alpha,$$

where $\xi^\alpha = \xi_1^{\alpha_1} \cdots \xi_n^{\alpha_n}$.

Thus the principal symbol is obtained formally by keeping only the highest-order terms and replacing ∂_j by the cotangent variable ξ_j .

The principal symbol is homogeneous of degree m in the cotangent variable:

$$\sigma_P(x, t\xi) = t^m \sigma_P(x, \xi).$$

This homogeneity reflects the fact that the symbol records precisely the order- m part of the operator.

A.1.5 The principal symbol as a bundle map

Let $\pi : T^*X \rightarrow X$ denote the cotangent-bundle projection.

Pulling back E and F by π , we obtain vector bundles

$$\pi^*E \rightarrow T^*X, \quad \pi^*F \rightarrow T^*X.$$

Their fibers at $(x, \xi) \in T^*X$ are

$$(\pi^*E)_{(x, \xi)} = E_x, \quad (\pi^*F)_{(x, \xi)} = F_x.$$

Thus the principal symbol can be viewed globally as a bundle morphism $\sigma_P : \pi^*E \rightarrow \pi^*F$ covering the identity map on T^*X .

Explicitly, $((x, \xi), v) \mapsto ((x, \xi), \sigma_P(x, \xi)v)$.

In other words, for every cotangent vector $\xi \in T_x^*X$, the principal symbol gives a linear map $\sigma_P(x, \xi) : E_x \rightarrow F_x$.

This bundle-map point of view is particularly useful because ellipticity will simply mean that this bundle map is invertible away from the zero section of T^*X .

A.1.6 The symbol of a first-order operator

Suppose

$$P = \sum_{j=1}^n A_j(x) \partial_j + B(x)$$

is a first-order differential operator.

For $\xi = \sum_{j=1}^n \xi_j dx^j$, the principal symbol is

$$\sigma_P(x, \xi) = \sum_{j=1}^n \xi_j A_j(x).$$

The zeroth-order term $B(x)$ does not contribute to the principal symbol.

Consequently, if two first-order operators differ only by a zeroth-order term, then they have the same principal symbol.

More generally, for operators of order m , adding an operator of order at most $m - 1$ does not change the principal symbol.

A.1.7 The symbols of basic geometric operators

We now compute the principal symbols of the geometric differential operators that will occur repeatedly in gauge theory.

Proposition A.1.4. Let $\nabla : \Gamma(E) \rightarrow \Omega^1(X; E)$ be a connection. Then

$$\sigma_\nabla(x, \xi) : E_x \rightarrow T_x^*X \otimes E_x$$

is given by

$$\sigma_\nabla(x, \xi)(v) = \xi \otimes v.$$

Proof. Choose $f \in C^\infty(X)$ such that $df_x = \xi$. By the Leibniz rule,

$$[\nabla, f]u = df \otimes u.$$

Evaluating at x ,

$$[\nabla, f]u(x) = \xi \otimes u(x).$$

Thus

$$\sigma_\nabla(x, \xi)(v) = \xi \otimes v.$$

□

Proposition A.1.5. The principal symbol of $d : \Omega^k(X) \rightarrow \Omega^{k+1}(X)$ is $\sigma_d(x, \xi)(\alpha) = \xi \wedge \alpha$.

Proof. For $f \in C^\infty(X)$, we have $[d, f]\alpha = df \wedge \alpha$. Taking $df_x = \xi$ gives $\sigma_d(x, \xi)(\alpha) = \xi \wedge \alpha$. □

A.1.8 Composition of principal symbols

Suppose $P : \Gamma(E_0) \rightarrow \Gamma(E_1)$ has order m , and $Q : \Gamma(E_1) \rightarrow \Gamma(E_2)$ has order l .

Then the composition $Q \circ P$ has order at most $m + l$.

The highest-order part is obtained by composing the highest-order parts of P and Q . Consequently,

$$\sigma_{Q \circ P}(x, \xi) = \sigma_Q(x, \xi) \circ \sigma_P(x, \xi).$$

This identity is one of the most important algebraic properties of the principal symbol. In particular, if D is a first-order operator, then $\sigma_{D^2}(x, \xi) = \sigma_D(x, \xi)^2$.

This observation will later allow us to deduce the ellipticity of Dirac-type operators from the fact that their squares are operators of Laplace type.

A.2 Elliptic operators

A.2.1 Definition of ellipticity

Let $P : \Gamma(E) \rightarrow \Gamma(F)$ be a differential operator of order m .

Definition A.2.1. The operator P is called **elliptic** if for every $x \in X$ and every non-zero cotangent vector $0 \neq \xi \in T_x^*X$, the principal symbol $\sigma_P(x, \xi) : E_x \rightarrow F_x$ is an isomorphism.

Equivalently, if $0_X \subset T^*X$ denotes the zero section, then P is elliptic precisely when

$$\sigma_P : \pi^*E|_{T^*X \setminus 0_X} \rightarrow \pi^*F|_{T^*X \setminus 0_X}$$

is a vector-bundle isomorphism.

The zero section is excluded because for an operator of positive order, homogeneity gives $\sigma_P(x, 0) = 0$.

Thus no positive-order operator can have an invertible symbol at $\xi = 0$.

A.2.2 Geometric meaning of ellipticity

Ellipticity means that the highest-order part of the differential operator controls every cotangent direction.

Consider

$$P = \frac{\partial}{\partial x_1}$$

on $\mathbb{R}^n$. Its principal symbol is $\sigma_P(\xi) = \xi_1$.

If $\xi = dx^2$, then $\sigma_P(dx^2) = 0$. Therefore P is not elliptic.

The failure of ellipticity has a direct analytic interpretation. The equation $\frac{\partial u}{\partial x_1} = 0$ implies only that u is independent of x_1 : $u(x_1, \dots, x_n) = f(x_2, \dots, x_n)$, where f is arbitrary.

Thus the equation provides no differential control in the $x_2, \dots, x_n$ directions.

This example illustrates the basic principle: ellipticity means that no non-zero cotangent direction is invisible to the highest-order part of the equation.

A.2.3 The Euclidean Laplacian

Consider the operator

$$\Delta_0 = - \sum_{j=1}^n \frac{\partial^2}{\partial x_j^2}$$

on $\mathbb{R}^n$.

Its principal symbol is

$$\sigma_{\Delta_0}(\xi) = - \sum_{j=1}^n \xi_j^2 = -|\xi|^2.$$

If $\xi \neq 0$, then $|\xi|^2 > 0$, and therefore $\sigma_{\Delta_0}(\xi) \neq 0$.

Thus Δ_0 is elliptic.

In fact, $\sigma_{\Delta_0}(\xi)^{-1} = -\frac{1}{|\xi|^2}$.

This simple formula already hints at elliptic regularity: inversion of the Laplacian introduces a factor $|\xi|^{-2}$, which corresponds heuristically to gaining two derivatives.

A.2.4 Operators of Laplace type

Let $P : \Gamma(E) \longrightarrow \Gamma(E)$ be a second-order differential operator.

Definition A.2.2. The operator P is called an **operator of Laplace type** if its principal symbol is

$$\sigma_P(x, \xi) = -|\xi|^2 \text{id}_{E_x}$$

for every $(x, \xi) \in T^*X$.

Every operator of Laplace type is elliptic, since for $\xi \neq 0$ the map $-|\xi|^2 \text{id}_{E_x}$ is invertible.

The basic geometric example is the connection Laplacian $\nabla^* \nabla$.

More generally, an operator of the form $\nabla^* \nabla + \mathcal{R}$, where $\mathcal{R} \in \Gamma(\text{End } E)$ is zeroth order, has the same principal symbol as $\nabla^* \nabla$, and is therefore also of Laplace type.

This is exactly the structure that appears in Weitzenböck formulas.

A.2.5 The Hodge Laplacian

Recall from Section 1.4 that the Hodge Laplacian is

$$\Delta = dd^* + d^*d : \Omega^k(X) \longrightarrow \Omega^k(X).$$

Its principal symbol is

$$\sigma_\Delta(x, \xi) = -|\xi|^2 \text{id}_{\Lambda^k T_x^* X}.$$

Hence the Hodge Laplacian is an operator of Laplace type, and in particular it is elliptic.

For every $0 \neq \xi \in T_x^* X$, we have

$$\sigma_\Delta(x, \xi)^{-1} = -\frac{1}{|\xi|^2} \text{id}_{\Lambda^k T_x^* X}.$$

The ellipticity of the Hodge Laplacian is one of the basic analytic inputs in Hodge theory.

A.2.6 Formal adjoints and principal symbols

Suppose that X is Riemannian and that E, F carry bundle metrics.

Let $P : \Gamma(E) \longrightarrow \Gamma(F)$ be a differential operator of order m .

Its **formal adjoint** $P^* : \Gamma(F) \longrightarrow \Gamma(E)$ is characterized by $(Pu, v)_{L^2} = (u, P^*v)_{L^2}$ for compactly supported smooth sections u and v , or for arbitrary smooth sections when X is closed.

To construct it, choose local orthonormal frames and write $P = \sum_{|\alpha| \leq m} a_\alpha(x) \partial^\alpha$ and $d \text{ vol} = w(x) dx$. Repeated integration by parts gives

$$P^*v = w^{-1} \sum_{|\alpha| \leq m} (-1)^{|\alpha|} \partial^\alpha (w a_\alpha^* v).$$

Here a_α^* is the fiberwise adjoint matrix. The defining pairing also proves uniqueness: the difference of two candidates pairs to zero with every compactly supported test section. The local constructions therefore agree on overlaps and define a global differential operator. No boundary term occurs under the support assumptions above.

With the principal-symbol convention used above, $\sigma_{P^*}(x, \xi) = (-1)^m \sigma_P(x, \xi)^*$.

Since taking adjoints preserves invertibility, $\sigma_P(x, \xi)$ is invertible if and only if $\sigma_{P^*}(x, \xi)$ is invertible.

Therefore

$$P \text{ is elliptic} \iff P^* \text{ is elliptic}.$$

A.2.7 The Hodge–de Rham operator

For a tangent vector v , contraction is the map

$$(\iota_v \alpha)(v_1, \dots, v_{q-1}) = \alpha(v, v_1, \dots, v_{q-1}), \quad \alpha \in \Lambda^q T_x^* X;$$

it is zero on scalars. In an orthonormal wedge basis, wedging by a basis covector inserts that covector, whereas contraction by its dual removes it with the same alternating sign. Hence, for a real covector ξ ,

$$(\xi \wedge)^* = \iota_{\xi^\#}, \quad \iota_{\xi^\#}(\xi \wedge \alpha) + \xi \wedge \iota_{\xi^\#} \alpha = |\xi|^2 \alpha.$$

The first identity concerns a fiberwise adjoint. The extra minus sign in $\sigma_{d^*} = -\iota_{\xi\#}$ comes from the first-order formal-adjoint convention above.

Consider the first-order operator

$$D = d + d^* : \Omega^*(X) \longrightarrow \Omega^*(X).$$

Since $d^2 = 0$ and $(d^*)^2 = 0$, we obtain

$$D^2 = dd^* + d^*d = \Delta.$$

Taking principal symbols gives

$$\sigma_D(x, \xi)^2 = \sigma_\Delta(x, \xi) = -|\xi|^2 \text{id}.$$

For $\xi \neq 0$, the right-hand side is invertible. Therefore $\sigma_D(x, \xi)$ is itself invertible.

Indeed, $\sigma_D(x, \xi)^{-1} = -\frac{1}{|\xi|^2} \sigma_D(x, \xi)$.

Consequently, $d + d^*$ is a first-order elliptic differential operator.

This is the simplest geometric model of a Dirac-type operator.

A.2.8 Dirac-type operators

Let $E \longrightarrow X$ be a metric vector bundle and let $D : \Gamma(E) \longrightarrow \Gamma(E)$ be a first-order differential operator.

Definition A.2.3. The operator D is called a **Dirac-type operator** if its square is an operator of Laplace type. Equivalently, $\sigma_D(x, \xi)^2 = -|\xi|^2 \text{id}_{E_x}$.

If $\xi \neq 0$, then $\sigma_D(x, \xi)^{-1} = -\frac{1}{|\xi|^2} \sigma_D(x, \xi)$, so every Dirac-type operator is elliptic.

In geometric applications, one frequently has a Weitzenböck formula of the form

$$D^2 = \nabla^* \nabla + \mathcal{R},$$

where $\mathcal{R} \in \Gamma(\text{End } E)$ is a zeroth-order curvature term.

Since $\mathcal{R}$ does not contribute to the principal symbol, this again shows that D is elliptic.

For a Spin^c structure, it is often preferable to consider the full Dirac operator

$$D_A = \begin{pmatrix} 0 & D_A^- \\ D_A^+ & 0 \end{pmatrix} : \Gamma(S^+ \oplus S^-) \longrightarrow \Gamma(S^+ \oplus S^-).$$

Its principal symbol is Clifford multiplication: $\sigma_{D_A}(x, \xi) = c(\xi)$. The Clifford relation $c(\xi)^2 = -|\xi|^2 \text{id}$ therefore gives ellipticity immediately.

In particular, the chiral operator $D_A^+ : \Gamma(S^+) \longrightarrow \Gamma(S^-)$ is elliptic.

A.2.9 Why d itself is not elliptic

The exterior derivative $d : \Omega^k(X) \longrightarrow \Omega^{k+1}(X)$ has principal symbol $\sigma_d(x, \xi)(\alpha) = \xi \wedge \alpha$.

This map is generally not an isomorphism.

For example, $d : \Omega^0(X) \longrightarrow \Omega^1(X)$ has symbol

$$\sigma_d(x, \xi) : \mathbb{R} \longrightarrow T_x^* X, \quad c \longmapsto c\xi.$$

If $\dim X > 1$, the source has dimension 1, while the target has dimension $\dim X > 1$. Therefore the symbol cannot be an isomorphism.

Thus d by itself is generally not elliptic.

This does not mean that d has no useful elliptic structure. Rather, one has to combine it with its adjoint, as in $d + d^*$, or place it inside an elliptic complex.

A.2.10 The four-dimensional operator $d^* \oplus d^+$

Let X now be an oriented Riemannian 4-manifold.

Recall that $\Omega^2(X) = \Omega^+(X) \oplus \Omega^-(X)$, and let $\pi_+ : \Omega^2(X) \rightarrow \Omega^+(X)$ be the projection onto the self-dual part.

Define

$$d^+ = \pi_+ \circ d : \Omega^1(X) \rightarrow \Omega^+(X).$$

The basic gauge-theoretic first-order operator is

$$D = d^* \oplus d^+ : \Omega^1(X) \rightarrow \Omega^0(X) \oplus \Omega^+(X).$$

We now compute its principal symbol.

The symbol of d is

$$\sigma_d(x, \xi)(a) = \xi \wedge a.$$

Therefore

$$\sigma_{d^+}(x, \xi)(a) = \pi_+(\xi \wedge a).$$

On the other hand, $\sigma_{d^*}(x, \xi)(a) = -\iota_{\xi^\#} a$, where $\xi^\#$ denotes the metric dual of ξ .

Hence

$$\sigma_D(x, \xi)(a) = \left(-\iota_{\xi^\#} a, \pi_+(\xi \wedge a) \right).$$

We now show that this map is an isomorphism for every $\xi \neq 0$.

Suppose $\sigma_D(x, \xi)(a) = 0$. Then $\iota_{\xi^\#} a = 0$ and $\pi_+(\xi \wedge a) = 0$.

Choose an oriented orthonormal basis e^1, e^2, e^3, e^4 of $T_x^* X$ such that $\xi = |\xi|e^1$.

Write $a = a_1e^1 + a_2e^2 + a_3e^3 + a_4e^4$.

The first equation gives

$$0 = \iota_{\xi^\#} a = |\xi|a_1,$$

hence $a_1 = 0$.

Thus $a = a_2e^2 + a_3e^3 + a_4e^4$.

Now

$$\xi \wedge a = |\xi| \left(a_2e^{12} + a_3e^{13} + a_4e^{14} \right).$$

With the chosen orientation, a basis for Λ^+ is given by

$$e^{12} + e^{34}, \quad e^{13} - e^{24}, \quad e^{14} + e^{23}.$$

Therefore

$$\pi_+(\xi \wedge a) = \frac{|\xi|}{2} \left[a_2(e^{12} + e^{34}) + a_3(e^{13} - e^{24}) + a_4(e^{14} + e^{23}) \right].$$

Since these three self-dual forms are linearly independent, the equation $\pi_+(\xi \wedge a) = 0$ implies $a_2 = a_3 = a_4 = 0$.

Hence $a = 0$.

Thus $\sigma_D(x, \xi)$ is injective.

But $\dim \Lambda^1 T_x^* X = 4$, while

$$\dim (\Lambda^0 T_x^* X \oplus \Lambda^+ T_x^* X) = 1 + 3 = 4.$$

Therefore injectivity implies bijectivity.

We have proved:

Proposition A.2.4. On an oriented Riemannian 4-manifold, the operator

$$d^* \oplus d^+ : \Omega^1(X) \longrightarrow \Omega^0(X) \oplus \Omega^+(X)$$

is a first-order elliptic differential operator.

This operator is one of the two basic elliptic operators in four-dimensional gauge theory, the other being the Dirac operator.

A.2.11 The twisted operator $d_A^* \oplus d_A^+$

Let $P \longrightarrow X$ be a principal G -bundle and let A be a connection on P .

The connection induces the covariant exterior derivative

$$d_A : \Omega^k(X; \text{ad } P) \longrightarrow \Omega^{k+1}(X; \text{ad } P).$$

In a local trivialization, $d_A \alpha = d\alpha + [A \wedge \alpha]$.

The term $[A \wedge \alpha]$ contains no derivatives of α , and is therefore zeroth order.

Thus

$$\sigma_{d_A}(x, \xi) = \sigma_d(x, \xi) \otimes \text{id}_{(\text{ad } P)_x}.$$

Explicitly, $\sigma_{d_A}(x, \xi)(\alpha) = \xi \wedge \alpha$.

Define

$$d_A^+ = \pi_+ \circ d_A : \Omega^1(X; \text{ad } P) \longrightarrow \Omega^+(X; \text{ad } P).$$

Its principal symbol is

$$\sigma_{d_A^+}(x, \xi)(a) = \pi_+(\xi \wedge a).$$

Similarly, $\sigma_{d_A^*}(x, \xi)(a) = -\iota_{\xi\#} a$.

Hence

$$\sigma_{d_A^* \oplus d_A^+}(x, \xi)(a) = \left(-\iota_{\xi\#} a, \pi_+(\xi \wedge a) \right),$$

with the identity acting implicitly on the $\text{ad } P$ -factor.

The same linear-algebra argument as in the untwisted case shows that this symbol is invertible whenever $\xi \neq 0$.

Therefore:

Proposition A.2.5. Let A be a connection on a principal G -bundle $P \longrightarrow X$ over an oriented Riemannian 4-manifold. Then

$$d_A^* \oplus d_A^+ : \Omega^1(X; \text{ad } P) \longrightarrow \Omega^0(X; \text{ad } P) \oplus \Omega^+(X; \text{ad } P)$$

is a first-order elliptic differential operator.

This is precisely the gauge-fixed linear operator associated with the ASD equation.

A.2.12 Lower-order perturbations do not change ellipticity

Suppose $P, Q : \Gamma(E) \rightarrow \Gamma(F)$ are differential operators of order m , and suppose $P - Q$ has order at most $m - 1$.

Then P and Q have the same principal part, and therefore $\sigma_P = \sigma_Q$.

It follows immediately that

$$P \text{ is elliptic} \iff Q \text{ is elliptic.}$$

Thus ellipticity is insensitive to lower-order terms.

This observation is fundamental in gauge theory. After linearizing a nonlinear gauge equation, one often obtains an operator of the form $D + \mathcal{R}$, where D is a Dirac-type or Hodge-type first-order operator and $\mathcal{R}$ is zeroth order.

Since

$$\sigma_{D+\mathcal{R}} = \sigma_D,$$

the lower-order term does not affect ellipticity.

For example, changing a connection in a Dirac operator changes the operator only by a zeroth-order Clifford multiplication term. Hence all Dirac operators associated with different smooth connections have the same principal symbol.

A.2.13 Why ellipticity is analytically useful

Ellipticity is defined purely in terms of the principal symbol, but its main consequences concern solutions of differential equations.

Let $P : \Gamma(E) \rightarrow \Gamma(F)$ be an elliptic differential operator of order m .

After passing to Sobolev spaces, one obtains estimates of the form

$$\|u\|_{W^{k,p}} \leq C (\|Pu\|_{W^{k-m,p}} + \|u\|_{L^p}).$$

This is called an **elliptic estimate**.

This is one of the central estimates of the analytic theory, and is one of the few formulas in this appendix that will be emphasized in this way.

The basic interpretation is that Pu controls m derivatives of u .

For example, for a second-order elliptic operator, $Pu \in L^p$ typically implies $u \in W^{2,p}$, whereas for a first-order elliptic operator, $Pu \in L^p$ typically implies $u \in W^{1,p}$.

More generally, $Pu \in W^{k-m,p}$ implies $u \in W^{k,p}$, provided u is initially interpreted as a suitable weak solution.

This phenomenon is called **elliptic regularity**.

The additional term $\|u\|_{L^p}$ in the elliptic estimate is necessary because an elliptic operator need not be injective. For example, on a closed connected manifold, $\Delta c = 0$ for every constant function c .

Thus Pu cannot control the component of u lying in $\ker P$.

After projecting away from the kernel, one obtains a stronger estimate of the form

$$\|u - P_{\ker P} u\|_{W^{k,p}} \leq C \|Pu\|_{W^{k-m,p}}.$$

On a compact manifold, elliptic regularity and compact Sobolev embedding further imply that suitable Sobolev realizations of elliptic operators are Fredholm.

In particular, $\ker P$ and $\operatorname{coker} P$ are finite-dimensional.

This fact is the analytic reason why the deformation spaces and obstruction spaces appearing in gauge theory are finite-dimensional, even though the configuration space and the gauge group themselves are infinite-dimensional.

The Sobolev spaces and elliptic estimates needed to make these statements precise will be developed next.

A.3 L^p and Sobolev spaces

The elliptic estimates discussed above cannot be formulated effectively only in terms of smooth sections. The natural functional-analytic setting is provided by L^p and Sobolev spaces.

There are two basic reasons for passing from smooth sections to Sobolev sections.

First, the space $\Gamma(E)$ of smooth sections is not a Banach space with respect to a single natural norm. In order to apply tools such as the Banach implicit function theorem and Fredholm theory, one needs suitable Banach-space completions.

Second, limits of smooth solutions or approximate solutions need not be smooth a priori. Sobolev spaces provide a class of sections which is large enough to be stable under weak convergence, while elliptic regularity will later show that weak solutions of elliptic equations are often much more regular than their initial definition suggests.

A.3.1 L^p spaces of sections

Let (X, g) be a Riemannian manifold, and let $E \rightarrow X$ be a real or complex vector bundle equipped with a bundle metric.

The Riemannian metric determines the Riemannian volume form $d\mu_g$, while the bundle metric determines a pointwise norm $|u(x)|$ on the fibers of E .

Definition A.3.1. Let $1 \leq p < \infty$. The space $L^p(E)$ consists of measurable sections u of E , defined up to equality almost everywhere, such that $\int_X |u|^p d\mu_g < \infty$. Its norm is

$$\|u\|_{L^p} = \left(\int_X |u|^p d\mu_g \right)^{1/p}.$$

For $p = \infty$, define

$$\|u\|_{L^\infty} = \operatorname{ess\,sup}_{x \in X} |u(x)|.$$

For every $1 \leq p \leq \infty$, the space $L^p(E)$ is a Banach space.

When $p = 2$, the space $L^2(E)$ is a Hilbert space with inner product

$$(u, v)_{L^2} = \int_X \langle u, v \rangle d\mu_g.$$

Thus $\|u\|_{L^2}^2 = (u, u)_{L^2}$.

The Hilbert-space structure of $L^2(E)$ will later be used to define orthogonal complements, adjoints of closed operators, and orthogonal projections onto kernels of elliptic operators.

A.3.2 Hölder inequality

The most basic estimate for products of L^p -sections is Hölder's inequality.

Theorem A.3.2 (Hölder inequality). Suppose $1 \leq p, q, r \leq \infty$ satisfy $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$. Then $fg \in L^r$ whenever

$$f \in L^p, \quad g \in L^q,$$

and

$$\|fg\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}.$$

More generally, suppose $E_1, E_2, E_3 \rightarrow X$ are metric vector bundles and $B : E_1 \otimes E_2 \rightarrow E_3$ is a smooth fiberwise bilinear map satisfying $|B(u, v)| \leq C|u||v|$. Then

$$\|B(u, v)\|_{L^r} \leq C\|u\|_{L^p} \|v\|_{L^q}.$$

This applies, for example, to wedge products of differential forms, matrix multiplication, Lie brackets, and Clifford multiplication.

In particular, $|\alpha \wedge \beta| \leq C|\alpha||\beta|$ implies

$$\|\alpha \wedge \beta\|_{L^r} \leq C\|\alpha\|_{L^p} \|\beta\|_{L^q}.$$

This elementary estimate will later be used repeatedly to control nonlinear terms such as $a \wedge a$ in Yang–Mills theory.

A.3.3 Weak derivatives in Euclidean space

Sobolev spaces are obtained by allowing derivatives in a weak sense.

Let $U \subset \mathbb{R}^n$ be open, and let $u, v \in L^1_{\text{loc}}(U)$.

Definition A.3.3. The function v is called the **weak derivative** of u in the x_j -direction if

$$\int_U u \partial_j \varphi \, dx = - \int_U v \varphi \, dx$$

for every test function $\varphi \in C_c^\infty(U)$. In this case we write $\partial_j u = v$.

The definition is motivated by integration by parts.

If u is classically differentiable, then its classical derivative is also its weak derivative. Thus weak differentiation extends ordinary differentiation.

Example A.3.4. Consider $u(x) = |x|$ on an interval containing the origin. The function u is not classically differentiable at $x = 0$, but its weak derivative is represented by

$$u'(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

The value assigned at $x = 0$ is irrelevant, because L^p -functions are defined only up to sets of measure zero.

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$, higher weak derivatives $\partial^\alpha u$ are defined by iterating the same procedure.

A.3.4 Sobolev spaces on Euclidean domains

Definition A.3.5. Let $U \subset \mathbb{R}^n$ be open,

$$k \in \mathbb{Z}_{\geq 0}, \quad 1 \leq p \leq \infty.$$

The **Sobolev space** $W^{k,p}(U)$ consists of all $u \in L^p(U)$ whose weak derivatives $\partial^\alpha u$ belong to $L^p(U)$ for every multi-index satisfying $|\alpha| \leq k$.

For $1 \leq p < \infty$, one may use the norm

$$\|u\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^p}^p \right)^{1/p}.$$

For $p = \infty$, one uses

$$\|u\|_{W^{k,\infty}} = \max_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^\infty}.$$

With these norms, $W^{k,p}(U)$ is a Banach space.

When $p = 2$, one often writes $H^k(U) = W^{k,2}(U)$.

In these notes we shall generally keep the notation $W^{k,2}$ because it makes the dependence on the Sobolev exponent explicit.

A.3.5 Sobolev spaces of sections of vector bundles

Let $E \rightarrow X$ be a metric vector bundle over a Riemannian manifold.

Choose a metric-compatible connection $\nabla : \Gamma(E) \rightarrow \Omega^1(X; E)$.

For a smooth section $u \in \Gamma(E)$, iterating the covariant derivative gives

$$\nabla^j u \in \Gamma((T^*X)^{\otimes j} \otimes E).$$

The preceding Euclidean definition can be extended intrinsically by interpreting these covariant derivatives weakly.

Definition A.3.6. Let

$$k \in \mathbb{Z}_{\geq 0}, \quad 1 \leq p < \infty.$$

The **Sobolev space** $W^{k,p}(E)$ consists of sections $u \in L^p(E)$ whose weak covariant derivatives satisfy

$$\nabla^j u \in L^p((T^*X)^{\otimes j} \otimes E)$$

for every $0 \leq j \leq k$.

A standard Sobolev norm is

$$\|u\|_{W^{k,p}} = \left(\sum_{j=0}^k \|\nabla^j u\|_{L^p}^p \right)^{1/p}.$$

For $p = \infty$, one uses

$$\|u\|_{W^{k,\infty}} = \max_{0 \leq j \leq k} \|\nabla^j u\|_{L^\infty}.$$

When no confusion can arise, the bundle E will be omitted from the notation.

A.3.6 Weak covariant derivatives

We now make the weak interpretation of the covariant derivative precise.

Suppose $u \in L^1_{\text{loc}}(E)$.

Definition A.3.7. A section $v \in L^1_{\text{loc}}(T^*X \otimes E)$ is called the **weak covariant derivative** of u if

$$\int_X \langle v, \phi \rangle d\mu_g = \int_X \langle u, \nabla^* \phi \rangle d\mu_g$$

for every $\phi \in \Gamma_c(T^*X \otimes E)$.

Here

$$\nabla^* : \Gamma(T^*X \otimes E) \longrightarrow \Gamma(E)$$

is the formal adjoint of ∇ .

The minus sign familiar from the Euclidean formula

$$\int (\partial_j u) \varphi = - \int u \partial_j \varphi$$

is already incorporated into the formal adjoint ∇^* .

Higher weak covariant derivatives are defined in the same way, or equivalently by iterating weak covariant differentiation.

Thus Sobolev regularity does not require pointwise differentiability. Differentiation is encoded through integration against smooth, compactly supported test sections.

A.3.7 Independence of the auxiliary connection

The definition of $W^{k,p}(E)$ appears to depend on the choice of connection. On a compact manifold, however, different smooth connections give equivalent Sobolev norms.

Let

$$\nabla \quad \text{and} \quad \nabla'$$

be two connections on E .

Their difference is tensorial: $\nabla' - \nabla = A$, where $A \in \Omega^1(X; \text{End } E)$.

Hence

$$\nabla' u = \nabla u + Au.$$

Since X is compact and A is smooth, there is a constant $C > 0$ such that $|A(x)| \leq C$ for all $x \in X$. Therefore

$$\|\nabla' u\|_{L^p} \leq \|\nabla u\|_{L^p} + C\|u\|_{L^p}.$$

Conversely, $\nabla = \nabla' - A$, and hence

$$\|\nabla u\|_{L^p} \leq \|\nabla' u\|_{L^p} + C\|u\|_{L^p}.$$

For higher derivatives one obtains expressions of the form

$$(\nabla')^k u = \nabla^k u + \sum_{j=0}^{k-1} B_j \nabla^j u,$$

where the B_j are smooth tensor fields determined by A , its covariant derivatives, and the two connections.

Using compactness of X , one obtains constants $C_1, C_2 > 0$ such that

$$C_1 \|u\|_{W^{k,p},\nabla} \leq \|u\|_{W^{k,p},\nabla'} \leq C_2 \|u\|_{W^{k,p},\nabla}.$$

Thus the two Sobolev norms are equivalent.

The same reasoning shows that changing the Riemannian metric or the bundle metric on a compact manifold produces an equivalent Sobolev norm.

Consequently, although a connection and metrics are used to define the norm, the resulting Sobolev space is intrinsic up to canonical equivalence of Banach norms.

A.3.8 Density of smooth sections

Suppose $1 \leq p < \infty$ and X is compact.

Then $\Gamma(E)$ is dense in $W^{k,p}(E)$.

Thus every Sobolev section can be approximated in the $W^{k,p}$ -norm by smooth sections.

This fact is important because many geometric identities are first proved for smooth sections and are then extended to Sobolev sections by density and continuity.

On a non-compact manifold, local arguments are usually formulated using compactly supported smooth sections $\Gamma_c(E)$ and local Sobolev spaces.

A.3.9 Local Sobolev spaces

Elliptic regularity is fundamentally a local phenomenon, so it is useful to introduce local Sobolev spaces.

Definition A.3.8. A section u belongs to $W_{\text{loc}}^{k,p}(E)$ if $\chi u \in W^{k,p}(E)$ for every $\chi \in C_c^\infty(X)$.

Equivalently, u belongs to $W^{k,p}$ on every relatively compact coordinate neighborhood. The notation $L_{\text{loc}}^p(E)$ is defined similarly.

A.4 Sobolev embedding and compactness

The usefulness of Sobolev spaces comes from the fact that control of weak derivatives implies improved integrability, continuity, and compactness.

The quantity

$$k - \frac{n}{p}, \quad n = \dim X,$$

plays a central role.

It may be regarded as a measure of the effective regularity of $W^{k,p}$: increasing k improves regularity, while decreasing p weakens it.

A.4.1 Sobolev embedding theorem

We use the following form of the Sobolev embedding theorem.

Theorem A.4.1 (Sobolev embedding). Let X be a compact n -dimensional Riemannian manifold and let $E \rightarrow X$ be a smooth vector bundle.

Suppose

$$k_0 \geq k_1, \quad 1 \leq p_0, p_1 < \infty,$$

$$\text{and } k_0 - \frac{n}{p_0} \geq k_1 - \frac{n}{p_1}.$$

Then there is a continuous embedding $W^{k_0, p_0}(E) \hookrightarrow W^{k_1, p_1}(E)$.

Equivalently, there is a constant $C > 0$ such that

$$\|u\|_{W^{k_1, p_1}} \leq C \|u\|_{W^{k_0, p_0}}.$$

The comparison $k_0 - \frac{n}{p_0} \geq k_1 - \frac{n}{p_1}$ is therefore the basic bookkeeping rule behind Sobolev embeddings.

A.4.2 The first-order Sobolev exponent

Suppose $1 \leq p < n$.

Define the **Sobolev conjugate exponent** $p^* = \frac{np}{n-p}$.

Equivalently, $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

The Sobolev embedding theorem gives $W^{1, p}(E) \hookrightarrow L^{p^*}(E)$.

Indeed,

$$1 - \frac{n}{p} = -\frac{n}{p^*}.$$

Thus one weak derivative in L^p produces improved integrability.

A.4.3 The four-dimensional critical embedding

The most important case for four-dimensional gauge theory is

$$n = 4, \quad p = 2.$$

Then

$$p^* = \frac{4 \cdot 2}{4 - 2} = 4.$$

Therefore $W^{1, 2}(E) \hookrightarrow L^4(E)$.

In particular, there is a constant $C > 0$ such that

$$\|u\|_{L^4} \leq C \|u\|_{W^{1, 2}}.$$

The Sobolev indices of the two spaces are equal: $1 - \frac{4}{2} = -1$, while $0 - \frac{4}{4} = -1$.

Thus this is a **critical Sobolev embedding**.

This particular embedding is one of the central analytic facts in four-dimensional gauge theory.

For example, if $a, b \in W^{1, 2}$, then $a, b \in L^4$. Hölder's inequality therefore gives $ab \in L^2$ and

$$\|ab\|_{L^2} \leq \|a\|_{L^4} \|b\|_{L^4} \leq C \|a\|_{W^{1, 2}} \|b\|_{W^{1, 2}}.$$

For differential forms with values in a Lie algebra bundle, the same argument gives

$$\|a \wedge b\|_{L^2} \leq C \|a\|_{W^{1,2}} \|b\|_{W^{1,2}}.$$

Taking $a = b$, we obtain

$$\|a \wedge a\|_{L^2} \leq C \|a\|_{W^{1,2}}^2.$$

This is precisely the estimate needed to control the nonlinear term in the curvature formula

$$F_{A_0+a} = F_{A_0} + d_{A_0}a + a \wedge a.$$

A.4.4 Why dimension four is critical

The preceding estimate is closely tied to the dimension of the underlying manifold.

A connection perturbation is a 1-form $a \in W^{1,2}$.

Its linear derivative term satisfies $d_A a \in L^2$.

In dimension four, $W^{1,2} \hookrightarrow L^4$, so

$$a \wedge a \in L^2$$

as well.

Thus both the linear term $d_A a$ and the quadratic term $a \wedge a$ naturally lie at the same L^2 -regularity level.

This matching of regularities is one manifestation of the critical nature of four-dimensional Yang–Mills theory.

A.4.5 Rellich compactness

Continuous Sobolev embedding alone is not sufficient for compactness arguments. One needs a stronger property.

Definition A.4.2. Let B_0 and B_1 be Banach spaces. A continuous inclusion $B_0 \hookrightarrow B_1$ is called a **compact embedding** if every bounded sequence in B_0 has a subsequence which converges strongly in B_1 .

The fundamental compactness theorem is the Rellich theorem.

Theorem A.4.3 (Rellich compactness). Let X be a compact n -dimensional Riemannian manifold and $E \rightarrow X$ a smooth vector bundle. Suppose $k_0 > k_1$ and $k_0 - \frac{n}{p_0} > k_1 - \frac{n}{p_1}$. Then the embedding $W^{k_0, p_0}(E) \hookrightarrow W^{k_1, p_1}(E)$ is compact.

The strict inequality between the Sobolev regularity indices is essential.

For example, on a compact 4-manifold, $W^{1,2}(E) \hookrightarrow L^q(E)$ is compact for every $1 \leq q < 4$.

By contrast, $W^{1,2}(E) \hookrightarrow L^4(E)$ is in general only continuous, not compact.

This failure of compactness at the critical exponent is closely related to concentration and bubbling phenomena in four-dimensional gauge theory.

A.4.6 Weak and strong convergence

Suppose $\{u_j\}$ is bounded in $W^{1,2}(E)$ over a compact 4-manifold.

Since $W^{1,2}(E)$ is reflexive, after passing to a subsequence there exists $u \in W^{1,2}(E)$ such that $u_j \rightharpoonup u$ weakly in $W^{1,2}(E)$.

Rellich compactness then implies, after passing to a further subsequence if necessary, $u_j \rightarrow u$ strongly in $L^q(E)$ for every $1 \leq q < 4$.

The distinction between weak convergence in the critical Sobolev space and strong convergence in subcritical spaces is fundamental in gauge-theoretic compactness arguments.

A.4.7 Morrey embedding

When the Sobolev regularity index is positive, Sobolev sections acquire continuous representatives.

Theorem A.4.4 (Morrey embedding). Let X be a compact n -dimensional Riemannian manifold and let $E \rightarrow X$ be a smooth vector bundle.

Suppose

$$m - \frac{n}{p} \geq k + \alpha,$$

where

$$m, k \in \mathbb{Z}_{\geq 0}, \quad 0 < \alpha < 1.$$

Then there is a continuous embedding $W^{m,p}(E) \hookrightarrow C^{k,\alpha}(E)$.

If

$$m - \frac{n}{p} > k + \alpha,$$

then the embedding is compact.

Here $C^{k,\alpha}(E)$ denotes the space of C^k -sections whose k -th covariant derivatives are Hölder continuous with exponent α .

In particular, if $m - \frac{n}{p} > 0$, then Sobolev sections in $W^{m,p}(E)$ have continuous representatives.

A.4.8 Important four-dimensional consequences

Let now $\dim X = 4$.

Several consequences will be used repeatedly.

First, $W^{1,2}(E) \hookrightarrow L^4(E)$.

Second, for every $1 \leq q < 4$, the embedding $W^{1,2}(E) \hookrightarrow L^q(E)$ is compact.

Third, $W^{2,2}(E) \hookrightarrow L^q(E)$ for every finite $q < \infty$.

Indeed, $2 - \frac{4}{2} = 0$, whereas $0 - \frac{4}{q} < 0$ for every finite q .

Notice that $W^{2,2}(E)$ lies at the critical threshold for continuity: $2 - \frac{4}{2} = 0$. Thus one does not obtain an embedding $W^{2,2}(E) \hookrightarrow C^0(E)$ from the Sobolev–Morrey theorem.

Fourth, if $p > 4$, then $1 - \frac{4}{p} > 0$. Morrey's theorem therefore gives $W^{1,p}(E) \hookrightarrow C^{0,\alpha}(E)$ for $0 < \alpha \leq 1 - \frac{4}{p}$.

In particular,

$$W^{1,p}(E) \hookrightarrow C^0(E), \quad p > 4.$$

This is frequently used after elliptic estimates have produced $W^{1,p}$ -bounds with $p > 4$. Finally, $W^{3,2}(E) \hookrightarrow C^{0,\alpha}(E)$ for every $0 < \alpha < 1$.

A.5 Sobolev multiplication

Gauge theory is nonlinear. It is therefore not sufficient to know that Sobolev spaces are Banach spaces. One must also understand when nonlinear pointwise operations preserve Sobolev regularity.

The relevant nonlinear operations include wedge products, Lie brackets, matrix multiplication, multiplication of spinors by functions, and Clifford multiplication.

A.5.1 The basic multiplication theorem

Let $E_1, E_2, E_3 \rightarrow X$ be smooth vector bundles and suppose that $B : E_1 \otimes E_2 \rightarrow E_3$ is a smooth fiberwise bilinear map.

Examples include $(f, g) \mapsto fg$, $(\alpha, \beta) \mapsto \alpha \wedge \beta$, $(a, b) \mapsto [a, b]$, and Clifford multiplication.

The basic regularity threshold is

$$kp > n, \quad n = \dim X.$$

Under this condition, $W^{k,p} \hookrightarrow C^0$.

One therefore expects multiplication by a $W^{k,p}$ -section to behave like multiplication by a bounded function.

Theorem A.5.1 (Sobolev multiplication). Let X be a compact n -dimensional manifold and suppose $kp > n$.

Then a smooth fiberwise bilinear map $B : E_1 \otimes E_2 \rightarrow E_3$ induces a continuous bilinear map

$$W^{k,p}(E_1) \times W^{k,p}(E_2) \rightarrow W^{k,p}(E_3).$$

In particular, for scalar functions, $W^{k,p}(X)$ is a Banach algebra:

$$\|fg\|_{W^{k,p}} \leq C\|f\|_{W^{k,p}}\|g\|_{W^{k,p}}.$$

The mechanism behind the theorem is the Leibniz rule. Schematically,

$$\nabla^j B(u, v) = \sum_{a+b=j} B_{a,b}(\nabla^a u, \nabla^b v),$$

where the $B_{a,b}$ are smooth bundlewise bilinear maps.

Sobolev embedding is used to place sufficiently low-order derivatives in L^∞ , while Hölder's inequality controls the remaining factors.

A.5.2 The critical multiplication estimate in dimension four

The condition $kp > n$ is sufficient for $W^{k,p}$ to be a Banach algebra, but it is not necessary for every nonlinear expression occurring in gauge theory.

In dimension four, $W^{1,2}$ is critical: $1 \cdot 2 < 4$. Thus $W^{1,2}$ is not a Banach algebra. Nevertheless, $W^{1,2} \hookrightarrow L^4$, so Hölder's inequality gives a continuous bilinear map

$$W^{1,2} \times W^{1,2} \longrightarrow L^2.$$

More precisely,

$$\|uv\|_{L^2} \leq C\|u\|_{W^{1,2}}\|v\|_{W^{1,2}}.$$

For $\text{ad } P$ -valued 1-forms, $a, b \in W^{1,2}(\Lambda^1 T^* X \otimes \text{ad } P)$, one obtains

$$\|a \wedge b\|_{L^2} \leq C\|a\|_{W^{1,2}}\|b\|_{W^{1,2}}.$$

In particular,

$$\|a \wedge a\|_{L^2} \leq C\|a\|_{W^{1,2}}^2.$$

Therefore, if $a \in W^{1,2}(\Lambda^1 T^* X \otimes \text{ad } P)$, then the curvature expression

$$F_{A_0+a} = F_{A_0} + d_{A_0}a + a \wedge a$$

belongs naturally to L^2 .

This is one of the reasons why $W^{1,2}$ is the natural critical Sobolev space for four-dimensional connections.

A.5.3 Multiplication by smoother sections

Another frequently used principle is that sufficiently regular Sobolev sections act as multipliers on lower-order Sobolev spaces.

Suppose $kp > n$. Then $W^{k,p} \hookrightarrow L^\infty$.

Consequently, if

$$u \in W^{k,p}, \quad v \in L^q,$$

then $uv \in L^q$ and

$$\|uv\|_{L^q} \leq C\|u\|_{W^{k,p}}\|v\|_{L^q}.$$

More generally, appropriate Leibniz estimates imply continuity of multiplication maps of the form

$$W^{k,p} \times W^{l,q} \longrightarrow W^{l,q}$$

when the first factor has sufficiently high Sobolev regularity.

This is used repeatedly when lower-order coefficients in an elliptic operator are themselves Sobolev sections.

A.5.4 Sobolev gauge groups

The multiplication theorem is essential for constructing Sobolev completions of gauge groups.

For example, in Seiberg–Witten theory on a closed 4-manifold, one may consider

$$\mathcal{G}_{k+1} = W^{k+1,2}(X, U(1)).$$

For $k \geq 2$, we have $k+1 > 2 = \frac{\dim X}{2}$.

Equivalently, $2(k+1) > 4$.

Hence $W^{k+1,2} \hookrightarrow C^0$, and $W^{k+1,2}$ is a Banach algebra.

Pointwise multiplication therefore extends continuously to $W^{k+1,2}$ -maps, and composition with the smooth group operations on $U(1)$ is well behaved.

Thus $\mathcal{G}_{k+1}$ can be given the structure of a Banach Lie group.

Correspondingly, one may take the Sobolev configuration space

$$\mathcal{C}_k = (A_0 + W^{k,2}(i\Lambda^1)) \times W^{k,2}(S^+).$$

The gauge action is

$$u \cdot (A, \psi) = (A - 2u^{-1}du, u\psi)$$

with the precise coefficient determined by the convention for the determinant connection used in these notes.

The Sobolev multiplication theorem guarantees, for sufficiently large k , that this action is smooth as a map between Banach manifolds.

This Sobolev completion is necessary before applying the Banach implicit function theorem to the Seiberg–Witten equations.

A.6 Weak solutions and elliptic regularity

We now return to elliptic differential operators.

Let $L : \Gamma(E) \rightarrow \Gamma(F)$ be a linear differential operator of order r , where

$$E \rightarrow X, \quad F \rightarrow X$$

are smooth metric vector bundles over a Riemannian manifold X .

The point of introducing Sobolev spaces is that the equation $Lu = f$ can be interpreted even when u is not classically differentiable.

A.6.1 Weak solutions

Suppose $f \in L^p_{\text{loc}}(F)$ and $u \in L^p_{\text{loc}}(E)$.

Let $L^* : \Gamma(F) \rightarrow \Gamma(E)$ denote the formal adjoint of L .

Definition A.6.1. The section u is called a **weak L^p -solution** of $Lu = f$ if

$$\int_X \langle f, \phi \rangle d\mu_g = \int_X \langle u, L^*\phi \rangle d\mu_g$$

for every $\phi \in \Gamma_c(F)$.

This definition transfers all derivatives from u onto the smooth test section ϕ .

If u is sufficiently regular for Lu to be defined classically, then the weak equation is equivalent to the classical equation.

The importance of the weak formulation is that weak limits of sequences of solutions can still satisfy the equation even before any regularity of the limit has been established.

A.6.2 The fundamental elliptic regularity theorem

We now state the central regularity theorem for elliptic operators.

Theorem A.6.2 (Elliptic regularity). Let X be a compact Riemannian manifold without boundary, and let $L : \Gamma(E) \rightarrow \Gamma(F)$ be an elliptic differential operator of order r .

Let $1 < p < \infty$ and $k \in \mathbb{Z}_{\geq 0}$.

Suppose $u \in L^p(E)$ is a weak solution of $Lu = f$ with $f \in W^{k,p}(F)$.

Then $u \in W^{k+r,p}(E)$.

Moreover, there exists a constant $C = C(L, k, p, g, E, F) > 0$ such that

$$\|u\|_{W^{k+r,p}} \leq C (\|Lu\|_{W^{k,p}} + \|u\|_{L^p}).$$

The estimate

$$\|u\|_{W^{k+r,p}} \leq C (\|Lu\|_{W^{k,p}} + \|u\|_{L^p})$$

is called an **elliptic estimate** or an **elliptic a priori estimate**.

The main content is that an elliptic operator of order r controls r derivatives of the unknown section.

Thus, schematically,

$$Lu \in W^{k,p} \implies u \in W^{k+r,p}.$$

This gain of derivatives is the basic regularity mechanism behind elliptic PDE.

A.6.3 Why the lower-order L^p -term is necessary

The term $\|u\|_{L^p}$ in the elliptic estimate cannot in general be omitted.

The reason is that L may have a non-trivial kernel.

Suppose $u \in \ker L$. Then $Lu = 0$.

An estimate of the form $\|u\|_{W^{r,p}} \leq C\|Lu\|_{L^p}$ would imply $u = 0$, and would therefore force $\ker L = 0$.

This is false for many elliptic operators.

For example, if X is closed and connected, then every constant function satisfies $\Delta u = 0$.

Thus the additional term $\|u\|_{L^p}$ controls precisely the low-frequency part of u which is not detected by Lu .

Later, after projecting away from the kernel, this additional term can be removed.

A.6.4 The Fourier-space model

The reason ellipticity controls derivatives is particularly transparent for constant-coefficient operators on Euclidean space.

Consider first $-\Delta u = f$ on $\mathbb{R}^n$.

Taking the Fourier transform gives

$$|\xi|^2 \widehat{u}(\xi) = \widehat{f}(\xi),$$

up to the Fourier-transform sign convention.

Hence

$$|\xi|^2 |\widehat{u}(\xi)| = |\widehat{f}(\xi)|.$$

Since multiplication of the Fourier transform by $|\xi|^2$ corresponds to taking two derivatives, the equation controls two derivatives of u .

For a general constant-coefficient elliptic operator L of order r , one has

$$\widehat{Lu}(\xi) = \sigma_L(i\xi) \widehat{u}(\xi),$$

where the factor i depends only on the Fourier convention.

Ellipticity implies that $\sigma_L(\xi)$ is invertible for $\xi \neq 0$.

Because the principal symbol is homogeneous of degree r , $\sigma_L(t\xi) = t^r \sigma_L(\xi)$.

Restricting to the unit sphere $|\xi| = 1$, compactness and invertibility imply the existence of a constant $C > 0$ such that $|v| \leq C |\sigma_L(\xi) v|$ for all $|\xi| = 1$.

By homogeneity, $|\xi|^r |v| \leq C |\sigma_L(\xi) v|$ for every $\xi \neq 0$.

Substituting $v = \widehat{u}(\xi)$ gives the high-frequency estimate

$$|\xi|^r |\widehat{u}(\xi)| \leq C |\widehat{Lu}(\xi)|.$$

Thus the invertibility of the principal symbol is precisely what allows the equation to control the highest derivatives of u .

The point $\xi = 0$ is not controlled by this argument. This is the Fourier-space origin of the additional lower-order term $\|u\|_{L^p}$ in the global elliptic estimate.

A.6.5 Local elliptic estimates

Ellipticity is fundamentally a local property.

Let $U' \Subset U \subset X$, where $U' \Subset U$ means that the closure of U' is compact and contained in U .

Then an elliptic operator L of order r satisfies an interior estimate of the form

$$\|u\|_{W^{k+r,p}(U')} \leq C \left(\|Lu\|_{W^{k,p}(U)} + \|u\|_{L^p(U)} \right).$$

The constant C depends on L , the two domains, the metric, and the Sobolev exponents, but not on u .

This form of the estimate is particularly important in gauge theory, because local gauge fixing often produces an elliptic equation only on a coordinate ball.

For example, after imposing Coulomb gauge, an ASD connection $A = d + a$ satisfies locally

$$(d^* \oplus d^+)a = (0, -(a \wedge a)^+).$$

The interior elliptic estimate can then be applied to a on a smaller ball.

A.6.6 From local to global estimates

On a compact manifold without boundary, the global elliptic estimate is obtained by combining local estimates with a finite partition of unity.

Choose coordinate neighborhoods $U_1, \dots, U_N$ covering X , together with smooth functions $\chi_1, \dots, \chi_N$ such that $\sum_{j=1}^N \chi_j = 1$.

Each section can be written as

$$u = \sum_{j=1}^N \chi_j u.$$

The operator applied to a localized section satisfies

$$L(\chi_j u) = \chi_j Lu + [L, \chi_j]u.$$

Since L has order r , the commutator $[L, \chi_j]$ has order at most $r - 1$.

Thus localization introduces only lower-order terms.

Applying the local estimate to each $\chi_j u$ and summing the resulting inequalities produces the global estimate.

This is another manifestation of the fact that the principal symbol controls the analytic behavior, whereas lower-order terms can be treated perturbatively.

A.6.7 Elliptic bootstrapping

Suppose $Lu = f$.

If initially $f \in L^p$, elliptic regularity gives $u \in W^{r,p}$.

If in fact $f \in W^{1,p}$, then $u \in W^{r+1,p}$.

More generally,

$$f \in W^{k,p} \implies u \in W^{k+r,p}.$$

For nonlinear equations the right-hand side may itself depend on u . One then combines Sobolev multiplication with elliptic regularity.

A typical situation is $Lu = N(u)$, where $N(u)$ contains only lower-order nonlinear terms.

One first uses the initial Sobolev regularity of u to control $N(u)$. The elliptic estimate then improves the regularity of u . This improved regularity gives better control of $N(u)$, and the procedure can be repeated.

This iterative mechanism is called **elliptic bootstrapping**.

A.6.8 Bootstrapping in Seiberg–Witten theory

This mechanism occurs directly in the Seiberg–Witten equations.

Write $A = A_0 + a$.

The Dirac equation becomes $D_{A_0}\psi = -\frac{1}{2}c(a)\psi$, while the curvature equation together with Coulomb gauge gives

$$(d^* \oplus d^+)a = (0, q(\psi) + \eta - F_{A_0}^+).$$

Both D_{A_0} and $d^* \oplus d^+$ are first-order elliptic operators.

In the compactness argument, this process starts with the bounds in Sections 4.5.8 and 4.5.9. Once a and ψ have sufficient initial Sobolev regularity, Sobolev multiplication controls the nonlinear terms

$$c(a)\psi \quad \text{and} \quad q(\psi).$$

Applying elliptic regularity to the two equations improves the Sobolev regularity of a and ψ .

Repeating the argument yields $a, \psi \in W^{k,p}$ for arbitrarily large k , and Sobolev embedding then gives smoothness.

A.6.9 Weyl's lemma

The preceding theorem has an important immediate consequence.

Corollary A.6.3 (Weyl's lemma). Let $L : \Gamma(E) \rightarrow \Gamma(F)$ be an elliptic differential operator on a smooth manifold.

If $u \in L^p_{\text{loc}}(E)$ is a weak solution and Lu is smooth, then $u \in \Gamma(E)$.

Indeed, if $Lu \in \Gamma(F)$, then locally $Lu \in W^{k,p}$ for every k .

Elliptic regularity therefore gives $u \in W^{k,p}_{\text{loc}}$ for every k .

Taking k sufficiently large and applying Sobolev embedding gives arbitrarily high classical regularity. Hence u is smooth.

This result explains why one may construct moduli spaces using Sobolev completions without changing the ultimate class of smooth solutions: weak Sobolev solutions of the elliptic gauge-fixed equations are smooth.

A.7 Elliptic estimates and coercive estimates

The elliptic estimate contains a lower-order term because the operator may have a kernel.

On a compact manifold one can improve the estimate by explicitly removing the kernel.

A.7.1 The kernel projection

Let X be compact and without boundary, and let $L : \Gamma(E) \rightarrow \Gamma(F)$ be an elliptic operator of order r .

For the moment we work in the Hilbert-space setting $p = 2$.

Let $P_{\ker L} : L^2(E) \rightarrow \ker L$ denote the L^2 -orthogonal projection onto the kernel.

Elliptic regularity will imply that $\ker L \subset \Gamma(E)$, and, as we shall prove below, the kernel is finite-dimensional.

One has the strengthened estimate

$$\|u - P_{\ker L} u\|_{W^{k+r,2}} \leq C \|Lu\|_{W^{k,2}}.$$

In particular, if $u \perp \ker L$, then

$$\|u\|_{W^{k+r,2}} \leq C \|Lu\|_{W^{k,2}}.$$

Such an estimate is often called a **coercive elliptic estimate**.

A.7.2 Why projection removes the lower-order term

The mechanism is easiest to see for $k = 0$.

Suppose, by contradiction, that there were no constant C such that

$$\|u\|_{W^{r,2}} \leq C \|Lu\|_{L^2}$$

for every $u \perp \ker L$.

Then there would exist a sequence $u_j \perp \ker L$ satisfying $\|u_j\|_{W^{r,2}} = 1$ and $\|Lu_j\|_{L^2} \rightarrow 0$.

Since the sequence is bounded in $W^{r,2}$, Rellich compactness gives, after passing to a subsequence, $u_j \rightarrow u$ strongly in L^2 .

Applying the ordinary elliptic estimate to $u_j - u_l$ gives

$$\|u_j - u_l\|_{W^{r,2}} \leq C (\|L(u_j - u_l)\|_{L^2} + \|u_j - u_l\|_{L^2}).$$

Both terms on the right tend to zero.

Hence $\{u_j\}$ is Cauchy in $W^{r,2}$, and therefore $u_j \rightarrow u$ strongly in $W^{r,2}$.

Continuity of L gives $Lu = 0$, so $u \in \ker L$.

On the other hand, $u_j \perp \ker L$ for every j , hence $u \perp \ker L$.

Therefore $u = 0$.

But strong convergence in $W^{r,2}$ then gives $\|u_j\|_{W^{r,2}} \rightarrow 0$, contradicting $\|u_j\|_{W^{r,2}} = 1$.

Thus the coercive estimate holds.

This argument is a standard example of the interaction between elliptic estimates and Rellich compactness. For its application to $d^* \oplus d^+$, including the L^p version, see the proof of Proposition 2.4.2.

A.8 Fredholm operators and elliptic operators

The next consequence of ellipticity is one of the most important structural facts in gauge theory: on a compact manifold, an elliptic operator behaves like a finite-dimensional linear map up to finite-dimensional errors.

A.8.1 Fredholm operators

Let

$$B_0, \quad B_1$$

be Banach spaces, and let $T : B_0 \rightarrow B_1$ be a bounded linear operator.

Definition A.8.1. The operator T is called a **Fredholm operator** if $\dim \ker T < \infty$, the image $\operatorname{im} T$ is closed in B_1 , and the quotient $B_1 / \operatorname{im} T$ is finite-dimensional.

Definition A.8.2. The quotient space $\operatorname{coker} T = B_1 / \operatorname{im} T$ is called the **cokernel** of T .

Thus Fredholmness means that both the failure of injectivity and the failure of surjectivity are finite-dimensional.

A.8.2 The Fredholm index

Definition A.8.3. Let $T : B_0 \longrightarrow B_1$ be Fredholm. Its **Fredholm index** is

$$\text{ind } T = \dim \ker T - \dim \text{coker } T.$$

The kernel measures infinitesimal non-uniqueness, whereas the cokernel measures the obstruction to solving the equation.

Thus the index may be viewed as the net difference

$$\text{degrees of freedom} - \text{obstructions}.$$

This interpretation will later become the expected dimension of a gauge-theoretic moduli space.

A.8.3 The Sobolev realization of an elliptic operator

Let X be a compact Riemannian manifold without boundary, and let $L : \Gamma(E) \longrightarrow \Gamma(F)$ be an elliptic differential operator of order r .

Since L contains at most r derivatives, it extends continuously to

$$L : W^{r,2}(E) \longrightarrow L^2(F).$$

Equivalently, one may regard L as an unbounded operator $L_a : L^2(E) \longrightarrow L^2(F)$ with domain $\text{Dom}(L_a) = W^{r,2}(E)$.

This is called the **analytic realization** of L .

A.8.4 Fredholm property of elliptic operators

Theorem A.8.4 (Fredholm property of elliptic operators). Let X be a compact Riemannian manifold without boundary and let $L : \Gamma(E) \longrightarrow \Gamma(F)$ be an elliptic differential operator of order r .

Then the Sobolev realization

$$L : W^{r,2}(E) \longrightarrow L^2(F)$$

is Fredholm.

Moreover,

$$\ker L \subset \Gamma(E), \quad \ker L^* \subset \Gamma(F),$$

both kernels are finite-dimensional, $\text{im } L$ is closed, and

$$\text{im } L = (\ker L^*)^\perp.$$

Similarly,

$$\text{im } L^* = (\ker L)^\perp.$$

This theorem is the basic bridge from infinite-dimensional PDE to finite-dimensional topology.

A.8.5 Finite-dimensionality of the kernel

We now explain why ellipticity forces $\ker L$ to be finite-dimensional.

Suppose $u \in \ker L$.

The elliptic estimate gives

$$\|u\|_{W^{r,2}} \leq C\|u\|_{L^2}.$$

Thus the L^2 -unit ball in $\ker L$ is bounded in $W^{r,2}(E)$.

Since $W^{r,2}(E) \hookrightarrow L^2(E)$ is compact for $r > 0$, the L^2 -unit ball in the kernel is compact in L^2 .

But the closed unit ball of an infinite-dimensional normed vector space cannot be compact.

Therefore $\dim \ker L < \infty$.

The same argument applies to the formal adjoint L^* , which is also elliptic.

Thus $\dim \ker L^* < \infty$.

A.8.6 Closedness of the range

Let $f_j = Lu_j$ be a sequence in $\operatorname{im} L$ such that $f_j \rightarrow f$ in $L^2(F)$.

Replace each u_j by $u_j - P_{\ker L} u_j$.

This does not change Lu_j , and now $u_j \perp \ker L$.

The coercive estimate gives

$$\|u_j - u_l\|_{W^{r,2}} \leq C\|L(u_j - u_l)\|_{L^2}.$$

Since $Lu_j = f_j$ is Cauchy in $L^2(F)$, the sequence u_j is Cauchy in $W^{r,2}(E)$.

Hence $u_j \rightarrow u$ for some $u \in W^{r,2}(E)$.

Continuity of L gives $Lu = f$.

Therefore $f \in \operatorname{im} L$.

Thus $\operatorname{im} L$ is closed.

A.8.7 The cokernel and the adjoint kernel

Because the range is closed, Hilbert-space orthogonal decomposition gives

$$L^2(F) = \operatorname{im} L \oplus (\operatorname{im} L)^\perp.$$

Here the identification with the formal adjoint requires a regularity step. If $v \in (\operatorname{im} L)^\perp \subset L^2(F)$, then $(Lu, v)_{L^2} = 0$ for every smooth test section u . Thus $L^*v = 0$ in the distributional sense. Elliptic regularity makes v smooth. Conversely, a smooth solution of $L^*v = 0$ annihilates the range, first on smooth sections and then on $W^{r,2}(E)$ by density. Therefore

$$(\operatorname{im} L)^\perp = \ker L^*.$$

Hence

$$L^2(F) = \operatorname{im} L \oplus \ker L^*.$$

It follows that

$$\operatorname{coker} L \cong \ker L^*.$$

Thus the kernel on the right is the smooth kernel of the formal adjoint. Equivalently, it is the kernel of the Hilbert-space adjoint of L viewed as an unbounded operator on L^2 with domain $W^{r,2}$. It should not be confused with the bounded adjoint of $L : W^{r,2} \rightarrow L^2$, whose definition uses a Sobolev inner product.

Consequently,

$$\text{ind } L = \dim \ker L - \dim \ker L^*.$$

Thus the cokernel can be represented concretely by smooth solutions of the adjoint homogeneous equation $L^*v = 0$.

A.8.8 The Fredholm alternative

The preceding decomposition gives a precise solvability criterion.

Consider $Lu = f$.

Since

$$\text{im } L = (\ker L^*)^\perp,$$

the equation has a solution if and only if $f \perp \ker L^*$.

Equivalently,

$$\int_X \langle f, \phi \rangle d\mu_g = 0$$

for every $\phi \in \ker L^*$.

This is the **Fredholm alternative**.

Thus all obstructions to solving $Lu = f$ are contained in the finite-dimensional vector space

$$\ker L^* \cong \text{coker } L.$$

If a solution exists, then all solutions are of the form

$$u = u_0 + v, \quad v \in \ker L.$$

Hence $\ker L$ measures non-uniqueness, whereas $\text{coker } L$ measures non-existence.

A.8.9 Example: the Laplacian

Let X be a closed connected Riemannian manifold and consider

$$\Delta : W^{2,2}(X) \longrightarrow L^2(X).$$

The Laplacian is self-adjoint: $\Delta^* = \Delta$.

Its kernel consists of constant functions:

$$\ker \Delta = \mathbb{R}.$$

Therefore

$$\text{coker } \Delta \cong \ker \Delta = \mathbb{R}.$$

The equation $\Delta u = f$ is solvable if and only if $f \perp 1$, that is, $\int_X f d\mu_g = 0$.

Thus the familiar zero-average condition for solving the Poisson equation is exactly the Fredholm obstruction.

Since

$$\dim \ker \Delta = \dim \text{coker } \Delta = 1,$$

we have $\text{ind } \Delta = 0$.

A.8.10 Self-adjoint elliptic operators

Suppose $L = L^*$ is an elliptic operator acting on sections of the same bundle.

Then

$$\ker L = \ker L^*.$$

Hence $\text{ind } L = 0$.

In particular, if one can additionally prove $\ker L = 0$, then $\text{coker } L = 0$.

Thus L is both injective and surjective.

This simple observation is frequently useful in nonlinear analysis: for a Fredholm operator of index zero, injectivity implies surjectivity, and conversely.

A.8.11 Compact perturbations

Fredholm operators are stable under compact perturbations.

Theorem A.8.5 (Stability under compact perturbations). Let $T : B_0 \rightarrow B_1$ be Fredholm and let $K : B_0 \rightarrow B_1$ be compact. Then $T + K$ is Fredholm and $\text{ind}(T + K) = \text{ind } T$.

This result is particularly important for differential operators.

Suppose $L_0, L_1 : \Gamma(E) \rightarrow \Gamma(F)$ are elliptic operators of the same order r and have the same principal symbol.

Then $L_1 - L_0$ has order at most $r - 1$.

As a map $W^{r,2}(E) \rightarrow L^2(F)$, the difference factors through

$$W^{r,2}(E) \hookrightarrow W^{r-1,2}(E) \rightarrow L^2(F).$$

The first map is compact by Rellich's theorem, while the second is bounded.

Therefore $L_1 - L_0$ is compact.

It follows that $\text{ind } L_1 = \text{ind } L_0$.

Thus the Fredholm index is insensitive to lower-order terms.

A.8.12 Dependence on the principal symbol

The previous argument shows an important principle:

if two elliptic operators have the same principal symbol, then they have the same Fredholm index.

More generally, the index depends only on the homotopy class of the elliptic symbol.

This is the point at which elliptic PDE begins to interact directly with topology.

The Atiyah–Singer index theorem goes much further: it computes the analytic index

$$\text{ind } L = \dim \ker L - \dim \text{coker } L$$

entirely from the topological data encoded by the principal symbol.

For the purposes of gauge theory, this means that the expected dimension of a moduli problem can eventually be expressed in terms of characteristic classes of the underlying manifold and bundles.

A.8.13 Fredholm index and moduli spaces

Suppose a nonlinear equation is written as $\mathcal{F}(u) = 0$ between suitable Banach spaces or Banach manifolds.

At a solution u , the linearization is $D_u\mathcal{F}$.

If $D_u\mathcal{F}$ is Fredholm, then $\ker D_u\mathcal{F}$ is the space of infinitesimal deformations, while $\operatorname{coker} D_u\mathcal{F}$ is the obstruction space.

If the linearization is surjective, then $\operatorname{coker} D_u\mathcal{F} = 0$, and the Banach implicit function theorem gives a smooth local solution space with tangent space $\ker D_u\mathcal{F}$.

In this regular case,

$$\dim \ker D_u\mathcal{F} = \operatorname{ind} D_u\mathcal{F}.$$

Thus the Fredholm index is the **expected dimension** or **virtual dimension** of the moduli problem.

This is the analytic mechanism underlying the dimension formulas in both Donaldson and Seiberg–Witten theory.

A.9 Elliptic complexes

Many gauge-theoretic deformation problems are naturally described not by a single elliptic operator, but by a complex of differential operators.

A.9.1 Differential complexes

Consider a sequence of vector bundles $E_0, E_1, \dots, E_N$ over X , together with differential operators

$$D_i : \Gamma(E_i) \longrightarrow \Gamma(E_{i+1}).$$

Definition A.9.1. The sequence

$$0 \longrightarrow \Gamma(E_0) \xrightarrow{D_0} \Gamma(E_1) \xrightarrow{D_1} \dots \xrightarrow{D_{N-1}} \Gamma(E_N) \longrightarrow 0$$

is called a **differential complex** if $D_{i+1}D_i = 0$ for every i .

The condition $D_{i+1}D_i = 0$ implies

$$\operatorname{im} D_i \subset \ker D_{i+1}.$$

Hence one can define the cohomology of the complex.

Definition A.9.2. The i -th **cohomology group** of the differential complex is

$$H^i = \frac{\ker D_i}{\operatorname{im} D_{i-1}},$$

with the evident modifications in the first and last degrees.

In a deformation problem, H^1 often represents infinitesimal deformations modulo infinitesimal symmetries.

A.9.2 The symbol complex

Assume for simplicity that the operators D_i are first order.

For every $x \in X$ and every $\xi \in T_x^*X$, the principal symbols give a sequence of finite-dimensional vector spaces

$$0 \longrightarrow (E_0)_x \xrightarrow{\sigma_{D_0}(x,\xi)} (E_1)_x \xrightarrow{\sigma_{D_1}(x,\xi)} \cdots \xrightarrow{\sigma_{D_{N-1}}(x,\xi)} (E_N)_x \longrightarrow 0.$$

Because $D_{i+1}D_i = 0$, the composition rule for principal symbols gives

$$\sigma_{D_{i+1}}(x, \xi) \circ \sigma_{D_i}(x, \xi) = 0.$$

Thus the principal symbols themselves form a complex.

This is called the **symbol complex**.

A.9.3 Definition of an elliptic complex

Definition A.9.3. A differential complex is called an **elliptic complex** if, for every $x \in X$ and every $0 \neq \xi \in T_x^*X$, its symbol complex is exact.

Thus ellipticity of a complex does not mean that each individual operator D_i is elliptic.

Indeed, in many important examples the individual operators cannot be elliptic because the ranks of their source and target bundles differ.

Instead, exactness of the symbol sequence says that the failure of one symbol to be injective is exactly accounted for by the image of the preceding symbol.

A.9.4 The de Rham complex

The basic example is the de Rham complex

$$0 \longrightarrow \Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(X) \longrightarrow 0.$$

The symbol of d is $\sigma_d(x, \xi)(\alpha) = \xi \wedge \alpha$.

For $\xi \neq 0$, the sequence

$$0 \longrightarrow \Lambda^0 T_x^* X \xrightarrow{\xi \wedge} \Lambda^1 T_x^* X \xrightarrow{\xi \wedge} \cdots \xrightarrow{\xi \wedge} \Lambda^n T_x^* X \longrightarrow 0$$

is exact.

Indeed, if $\xi \wedge \alpha = 0$ and $\xi \neq 0$, the contraction identity gives

$$\alpha = \xi \wedge (|\xi|^{-2} \iota_{\xi\#} \alpha).$$

This proves exactness in positive degrees; in degree zero, wedging a scalar with a nonzero covector is injective. Therefore the de Rham complex is elliptic.

Notice that d itself is generally not elliptic. Ellipticity appears only after the entire complex is taken into account.

A.9.5 The Laplacian associated with an elliptic complex

For a first-order differential complex, define the operator in degree i

$$\Delta_i = D_{i-1}D_{i-1}^* + D_i^*D_i.$$

This is a second-order operator

$$\Delta_i : \Gamma(E_i) \longrightarrow \Gamma(E_i).$$

To see ellipticity, put $a = \sigma_{D_{i-1}}(x, \xi)$ and $b = \sigma_{D_i}(x, \xi)$ for $\xi \neq 0$. With our symbol convention,

$$\sigma_{\Delta_i}(x, \xi) = -(aa^* + b^*b).$$

If this symbol kills v , taking its inner product with v gives $a^*v = 0$ and $bv = 0$. Exactness says $\ker b = \operatorname{im} a$, so v belongs to both $\operatorname{im} a$ and its orthogonal complement. Thus $v = 0$, and the symbol is invertible.

A section $u \in \Gamma(E_i)$ satisfies $\Delta_i u = 0$ if and only if $D_i u = 0$ and $D_{i-1}^* u = 0$.

Indeed,

$$(\Delta_i u, u)_{L^2} = \|D_i u\|_{L^2}^2 + \|D_{i-1}^* u\|_{L^2}^2.$$

Thus the **harmonic space**

$$\mathcal{H}^i = \ker \Delta_i$$

can be written as

$$\mathcal{H}^i = \ker D_i \cap \ker D_{i-1}^*.$$

On a compact manifold, elliptic theory implies that $\mathcal{H}^i$ is finite-dimensional.

A.9.6 Hodge decomposition for elliptic complexes

On the closed manifold X , regard $\Delta_i : W^{2,2}(E_i) \rightarrow L^2(E_i)$. Its formal self-adjointness and the Fredholm alternative give a bounded inverse on the orthogonal complement of $\mathcal{H}^i$. Extend this inverse by zero on $\mathcal{H}^i$ and denote it by G_i . If Π_i is harmonic projection, then

$$f = \Pi_i f + \Delta_i G_i f = \Pi_i f + D_{i-1}(D_{i-1}^* G_i f) + D_i^*(D_i G_i f).$$

The two potentials belong to $W^{1,2}$ when $f \in L^2$. The three summands are mutually orthogonal: harmonic sections are killed by the appropriate adjoints, and the other two summands are orthogonal because $D_i D_{i-1} = 0$. Integration by parts extends these identities from smooth sections by density. Hence we obtain the orthogonal decomposition

$$L^2(E_i) = \operatorname{im} D_{i-1} \oplus \mathcal{H}^i \oplus \operatorname{im} D_i^*.$$

Here the images mean the images of the maps with domain $W^{1,2}$ and target L^2 ; the displayed decomposition also shows that they are closed. For smooth f , elliptic regularity makes $G_i f$ smooth. If moreover $D_i f = 0$, then f is orthogonal to $\operatorname{im} D_i^*$, so its coexact component vanishes. Thus f differs from its harmonic part by D_{i-1} of a smooth section. A harmonic exact section has zero norm by integration by parts. Every smooth cohomology class therefore has a unique harmonic representative, and

$$H^i \cong \mathcal{H}^i.$$

Consequently, $\dim H^i < \infty$.

This is the general analytic mechanism behind the finite-dimensionality of deformation and obstruction spaces in gauge theory.

A.9.7 The gauge-fixed operator of a three-term complex

Consider a first-order elliptic complex

$$0 \longrightarrow \Gamma(E_0) \xrightarrow{D_0} \Gamma(E_1) \xrightarrow{D_1} \Gamma(E_2) \longrightarrow 0.$$

The natural gauge-fixed operator in the middle degree is

$$D_0^* \oplus D_1 : \Gamma(E_1) \longrightarrow \Gamma(E_0) \oplus \Gamma(E_2).$$

Its kernel is

$$\ker(D_0^* \oplus D_1) = \ker D_0^* \cap \ker D_1 = \mathcal{H}^1.$$

Hence

$$\ker(D_0^* \oplus D_1) \cong H^1.$$

The formal adjoint is

$$D_0 \oplus D_1^* : \Gamma(E_0) \oplus \Gamma(E_2) \longrightarrow \Gamma(E_1).$$

Its kernel is

$$\ker(D_0 \oplus D_1^*) \cong H^0 \oplus H^2.$$

Therefore

$$\operatorname{coker}(D_0^* \oplus D_1) \cong H^0 \oplus H^2.$$

It follows that

$$\operatorname{ind}(D_0^* \oplus D_1) = \dim H^1 - \dim H^0 - \dim H^2.$$

This is precisely the index formula that appears in gauge-theoretic deformation complexes.

A.9.8 The ASD deformation complex

Let $P \longrightarrow X$ be a principal bundle over an oriented Riemannian 4-manifold, and let A be an ASD connection: $F_A^+ = 0$.

The infinitesimal gauge action is

$$d_A : \Omega^0(X; \operatorname{ad} P) \longrightarrow \Omega^1(X; \operatorname{ad} P),$$

while the linearization of the ASD equation is

$$d_A^+ : \Omega^1(X; \operatorname{ad} P) \longrightarrow \Omega^+(X; \operatorname{ad} P).$$

Since $d_A^2 \xi = [F_A, \xi]$, we have $d_A^+ d_A \xi = [F_A^+, \xi]$.

At an ASD connection, $F_A^+ = 0$, and therefore $d_A^+ d_A = 0$.

Thus one obtains the deformation complex

$$0 \longrightarrow \Omega^0(X; \operatorname{ad} P) \xrightarrow{d_A} \Omega^1(X; \operatorname{ad} P) \xrightarrow{d_A^+} \Omega^+(X; \operatorname{ad} P) \longrightarrow 0.$$

This is an elliptic complex.

The first cohomology group

$$H_A^1 = \frac{\ker d_A^+}{\operatorname{im} d_A}$$

is the space of infinitesimal ASD deformations modulo infinitesimal gauge transformations.

The corresponding gauge-fixed operator is

$$d_A^* \oplus d_A^+ : \Omega^1(X; \operatorname{ad} P) \longrightarrow \Omega^0(X; \operatorname{ad} P) \oplus \Omega^+(X; \operatorname{ad} P).$$

Its ellipticity is exactly the analytic reason why the ASD deformation space and obstruction space are finite-dimensional.

A.9.9 The Seiberg–Witten deformation complex

The same structure appears in Seiberg–Witten theory.

At a solution (A, ψ) , the infinitesimal gauge action gives an operator d_0 , and the linearization of the Seiberg–Witten equations gives an operator d_1 .

Gauge invariance implies $d_1 d_0 = 0$.

Thus one obtains a deformation complex

$$0 \longrightarrow i\Omega^0(X) \xrightarrow{d_0} i\Omega^1(X) \oplus \Gamma(S^+) \xrightarrow{d_1} i\Omega^+(X) \oplus \Gamma(S^-) \longrightarrow 0.$$

Its principal part consists of the operators

$$d, \quad d^+, \quad D_A^+.$$

All terms involving the background spinor ψ are zeroth order and therefore do not affect the symbol complex.

The symbol sequence is exact for every $0 \neq \xi \in T_x^* X$, so the Seiberg–Witten deformation complex is elliptic.

The gauge-fixed operator is $d_0^* \oplus d_1$.

After Sobolev completion it is Fredholm.

Its kernel describes infinitesimal deformations satisfying the gauge condition, while its cokernel is the obstruction space.

The Fredholm index $\operatorname{ind}(d_0^* \oplus d_1)$ is the expected dimension of the Seiberg–Witten moduli problem.

At an irreducible regular solution,

$$H^0 = 0, \quad H^2 = 0,$$

and therefore

$$\dim H^1 = \operatorname{ind}(d_0^* \oplus d_1).$$

Thus at such a point the expected dimension agrees with the actual local dimension of the moduli space.

A.10 Banach manifolds and the implicit function theorem

The spaces which occur naturally in gauge theory are infinite-dimensional. After Sobolev completion, however, they are modeled on Banach or Hilbert spaces, so that one can use infinite-dimensional differential calculus.

The basic principle is the same as in finite-dimensional differential geometry: if a nonlinear equation has a surjective linearization, then its solution set is locally a smooth manifold whose tangent space is the kernel of the linearization.

A.10.1 Banach manifolds

Let B be a real Banach space.

Definition A.10.1. A Hausdorff topological space $\mathcal{X}$ is called a **Banach manifold modeled on B** if every point $x \in \mathcal{X}$ has an open neighborhood $U \subset \mathcal{X}$ together with a homeomorphism $\varphi : U \rightarrow V$, where $V \subset B$ is open, such that the transition maps between overlapping charts are smooth maps between open subsets of Banach spaces.

If the model space B is a Hilbert space, then $\mathcal{X}$ is called a **Hilbert manifold**.

All Sobolev spaces $W^{k,2}(E)$ are Hilbert spaces. Hence many Sobolev configuration spaces used in gauge theory are naturally Hilbert manifolds.

More generally,

$$W^{k,p}(E), \quad 1 < p < \infty,$$

is a Banach space, so $W^{k,p}$ -completions naturally lead to Banach manifolds.

A.10.2 Differentiability between Banach spaces

Let

$$B_0, \quad B_1$$

be Banach spaces and let $U \subset B_0$ be open.

Definition A.10.2. A map $F : U \rightarrow B_1$ is **Fréchet differentiable** at $x \in U$ if there exists a bounded linear map $D_x F : B_0 \rightarrow B_1$ such that

$$\frac{\|F(x+h) - F(x) - D_x F(h)\|_{B_1}}{\|h\|_{B_0}} \rightarrow 0$$

as $\|h\|_{B_0} \rightarrow 0$.

The bounded linear map $D_x F$ is called the **differential** or **linearization** of F at x . Higher differentiability is defined by iterating the Fréchet derivative.

A map between Banach manifolds is called smooth if its expression in local Banach charts is smooth.

A.10.3 Tangent spaces

Let $\mathcal{X}$ be a Banach manifold modeled on B .

Take C^1 curves $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{X}$ with $\gamma(0) = x$. For a chart φ near x , declare

$$\gamma_1 \sim_x \gamma_2 \iff (\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0), \quad T_x \mathcal{X} := \{\gamma : \gamma(0) = x\} / \sim_x.$$

Curves may be restricted to smaller intervals. If $\tilde{\varphi}$ is another chart, the chain rule gives

$$(\tilde{\varphi} \circ \gamma)'(0) = D(\tilde{\varphi} \circ \varphi^{-1})_{\varphi(x)}(\varphi \circ \gamma)'(0).$$

The transition derivative is a bounded linear isomorphism, with inverse the derivative of the inverse transition. Thus the equivalence relation is independent of the chart.

The map $[\gamma]_x \mapsto (\varphi \circ \gamma)'(0)$ is a bijection onto B : a vector $v \in B$ is represented by $\varphi^{-1}(\varphi(x) + tv)$ for sufficiently small t . Transport addition, scalar multiplication, and the norm of B through this map. The resulting norm is complete. A change of chart preserves the vector operations and gives an equivalent complete norm, since its derivative and inverse are bounded. Consequently $T_x \mathcal{X}$ has a well-defined Banach topology, although there is no preferred norm.

For a smooth map $F : \mathcal{X} \rightarrow \mathcal{Y}$, define $D_x F[\gamma]_x = [F \circ \gamma]_{F(x)}$. In charts φ and ψ this is represented by the bounded linear map

$$D(\psi \circ F \circ \varphi^{-1})_{\varphi(x)}.$$

The same transition formula proves that it is well defined and gives $D_x(G \circ F) = D_{F(x)}G \circ D_x F$. Here C^1 in Banach spaces includes continuity of the derivative in operator norm; higher C^r regularity is defined recursively.

A.10.4 Affine Sobolev spaces of connections

The space of connections gives a basic example.

Fix a smooth connection A_0 on a vector bundle or principal bundle. Any other connection can be written as $A = A_0 + a$, where $a \in \Omega^1(X; \text{ad } P)$ in the non-Abelian case, or $a \in i\Omega^1(X)$ for a unitary line bundle.

After Sobolev completion one obtains the affine Hilbert space

$$A_0 + W^{k,2}(\Lambda^1 T^* X \otimes \text{ad } P).$$

Its tangent space at every point is canonically

$$W^{k,2}(\Lambda^1 T^* X \otimes \text{ad } P).$$

Thus the space of connections is not naturally a vector space, but it is an affine space modeled on a Sobolev space.

A.10.5 Sobolev configuration spaces in Seiberg–Witten theory

For a fixed Spin^c structure, choose a smooth reference connection A_0 .

For sufficiently large k , define

$$\mathcal{C}_k = (A_0 + W^{k,2}(i\Lambda^1)) \times W^{k,2}(S^+).$$

This is an affine Hilbert manifold modeled on $W^{k,2}(i\Lambda^1) \oplus W^{k,2}(S^+)$.

Hence

$$T_{(A,\psi)}\mathcal{C}_k \cong W^{k,2}(i\Lambda^1) \oplus W^{k,2}(S^+).$$

The Seiberg–Witten map extends, for suitable Sobolev indices, to a smooth map

$$\mathcal{F} : \mathcal{C}_k \longrightarrow W^{k-1,2}(S^-) \oplus W^{k-1,2}(i\Lambda_+^2).$$

Its zero set is the Sobolev solution space of the Seiberg–Witten equations.

Elliptic regularity later shows that these Sobolev solutions are smooth after an appropriate gauge choice.

A.10.6 Submersions and regular values

Definition A.10.3. A smooth map $F : \mathcal{X} \longrightarrow \mathcal{Y}$ between Banach manifolds is called a **submersion at x** if

$$D_x F : T_x \mathcal{X} \longrightarrow T_{F(x)} \mathcal{Y}$$

is surjective and its kernel is complemented in $T_x \mathcal{X}$.

The complemented-kernel condition means that there exists a closed subspace V such that

$$T_x \mathcal{X} = \ker D_x F \oplus V.$$

In Hilbert spaces every closed subspace has an orthogonal complement. Thus, in the Hilbert-manifold setting, surjectivity is enough.

For Fredholm operators the kernel is finite-dimensional, and every finite-dimensional subspace of a Banach space is complemented. Hence a surjective Fredholm linearization automatically satisfies the splitting condition.

Definition A.10.4. A point $y \in \mathcal{Y}$ is called a **regular value** of F if F is a submersion at every $x \in F^{-1}(y)$.

A point in the target which is not regular is called a **critical value**.

A.10.7 The Banach implicit function theorem

We first state a convenient form of the implicit function theorem.

Theorem A.10.5 (Banach implicit function theorem). Let

$$B_0, \quad B_1, \quad B_2$$

be Banach spaces, and let

$$F : U \subset B_0 \times B_1 \longrightarrow B_2$$

be a smooth map defined on a neighborhood U of (x_0, y_0) .

Suppose $F(x_0, y_0) = 0$ and the partial differential $D_y F(x_0, y_0) : B_1 \rightarrow B_2$ is an isomorphism of Banach spaces.

Then there exist neighborhoods

$$x_0 \in U_0 \subset B_0, \quad y_0 \in U_1 \subset B_1,$$

and a unique smooth map $f : U_0 \rightarrow U_1$ such that $f(x_0) = y_0$ and $F(x, y) = 0$ for $(x, y) \in U_0 \times U_1$ if and only if $y = f(x)$.

Thus the zero set of F is locally the graph of a smooth map.

The theorem is formally identical to the ordinary finite-dimensional implicit function theorem, but invertibility must now be understood in the category of bounded linear maps between Banach spaces.

A.10.8 The regular level-set theorem

The form used most often in gauge theory is the following consequence.

Theorem A.10.6 (Regular level-set theorem). Let $F : \mathcal{X} \rightarrow \mathcal{Y}$ be a smooth map between Banach manifolds, and let $x_0 \in \mathcal{X}$ satisfy $F(x_0) = y_0$.

Suppose $D_{x_0} F$ is surjective and $\ker D_{x_0} F$ is complemented.

Then, in a neighborhood of x_0 , $F^{-1}(y_0)$ is a smooth Banach submanifold of $\mathcal{X}$, and

$$T_{x_0} F^{-1}(y_0) = \ker D_{x_0} F.$$

If the differential is Fredholm and surjective, then the kernel is finite-dimensional. Consequently, the local solution space is a finite-dimensional smooth manifold.

Its dimension is $\dim \ker D_{x_0} F$.

Since surjectivity implies $\operatorname{coker} D_{x_0} F = 0$, one has

$$\dim \ker D_{x_0} F = \operatorname{ind} D_{x_0} F.$$

Thus, for a regular Fredholm problem, the Fredholm index is the actual local dimension of the solution space.

A.10.9 Regular solutions in gauge theory

Suppose a gauge-fixed nonlinear equation is represented locally by a smooth map $F : \mathcal{S} \rightarrow \mathcal{Y}$, where $\mathcal{S}$ is a local slice for the gauge action.

Let $C \in \mathcal{S}$ be a solution: $F(C) = 0$.

The solution is called **regular** if

$$D_C F : T_C \mathcal{S} \rightarrow \mathcal{Y}$$

is surjective.

In a gauge-theoretic deformation complex, $\operatorname{coker} D_C F \cong H_C^2$.

Hence regularity is equivalent to $H_C^2 = 0$.

At an irreducible solution one also has $H_C^0 = 0$.

Therefore, at an irreducible regular solution, $H_C^0 = H_C^2 = 0$.

The regular level-set theorem then gives a smooth finite-dimensional local moduli space with tangent space

$$T_{[C]}\mathcal{M} \cong H_C^1.$$

Moreover,

$$\dim \mathcal{M} = \dim H_C^1 = \text{ind}(d_0^* \oplus d_1).$$

This explains analytically why the expected dimension agrees with the actual dimension at irreducible regular solutions.

A.11 Fredholm maps and finite-dimensional reduction

The implicit function theorem gives a smooth local solution space when the linearization is surjective.

If the linearization is Fredholm but not surjective, the solution space need not be a manifold. Nevertheless, Fredholm theory implies that the failure of regularity is finite-dimensional.

This leads to the Kuranishi model.

A.11.1 Nonlinear Fredholm maps

Definition A.11.1. Let

$$\mathcal{X}, \quad \mathcal{Y}$$

be Banach manifolds. A smooth map $F : \mathcal{X} \rightarrow \mathcal{Y}$ is called a **Fredholm map** if for every $x \in \mathcal{X}$ the differential

$$D_x F : T_x \mathcal{X} \rightarrow T_{F(x)} \mathcal{Y}$$

is a Fredholm operator.

Since the index of a Fredholm operator is locally constant, the function $x \mapsto \text{ind } D_x F$ is locally constant.

Hence, if $\mathcal{X}$ is connected, the index is independent of x .

Definition A.11.2. If $\mathcal{X}$ is connected, the common value $\text{ind } D_x F$ is called the **index of the Fredholm map** F and is denoted $\text{ind } F$.

A.11.2 The basic finite-dimensional splitting

Let $F : B_0 \rightarrow B_1$ be a smooth map between Banach spaces satisfying $F(0) = 0$.

Suppose the linearization $D = D_0 F : B_0 \rightarrow B_1$ is Fredholm.

Set $K = \ker D$.

Since D is Fredholm, $\dim K < \infty$.

Choose a closed complement X' such that $B_0 = K \oplus X'$.

Since $R = \operatorname{im} D$ is closed and has finite codimension, choose a finite-dimensional complement C such that $B_1 = R \oplus C$.

The space C may be identified with $\operatorname{coker} D$.

Let $\pi_R : B_1 \rightarrow R$ and $\pi_C : B_1 \rightarrow C$ denote the corresponding projections.

The restriction $D|_{X'} : X' \rightarrow R$ is an isomorphism.

Indeed, it is injective because $X' \cap \ker D = 0$, and it is surjective because $R = \operatorname{im} D$.

By the bounded inverse theorem, its inverse is bounded.

A.11.3 Solving the infinite-dimensional part

Write a point of B_0 as

$$u + v, \quad u \in K, \quad v \in X'.$$

Consider the equation $\pi_R F(u + v) = 0$.

At $(u, v) = (0, 0)$, the derivative in the v -direction is

$$D_v(\pi_R F)(0, 0) = D|_{X'} : X' \rightarrow R,$$

which is an isomorphism.

The Banach implicit function theorem therefore gives a neighborhood $N \subset K$ of 0 and a unique smooth map $f : N \rightarrow X'$ with $f(0) = 0$ such that $\pi_R F(u + v) = 0$ for sufficiently small u, v if and only if $v = f(u)$.

Thus the infinite-dimensional R -component of the equation has been solved completely.

Every sufficiently small solution of $F = 0$ must therefore have the form

$$u + f(u), \quad u \in N \subset K.$$

A.11.4 The Kuranishi map

The remaining equation is the finite-dimensional C -component.

Definition A.11.3. The smooth map $\kappa : N \rightarrow C$ defined by $\kappa(u) = \pi_C F(u + f(u))$ is called the **Kuranishi map** associated with the solution 0.

Since $K = \ker D$ and $C \cong \operatorname{coker} D$, the Kuranishi map has the form

$$\kappa : \ker D \rightarrow \operatorname{coker} D.$$

Locally, solving the original equation is equivalent to solving $\kappa(u) = 0$ in the finite-dimensional space K .

More precisely, a sufficiently small element $x \in B_0$ satisfies $F(x) = 0$ if and only if $x = u + f(u)$ for some $u \in \kappa^{-1}(0)$.

Thus the local solution space is modeled on

$$\kappa^{-1}(0) \subset \ker D.$$

A.11.5 The differential of the Kuranishi map

The Kuranishi map satisfies $\kappa(0) = 0$.

Moreover, $D_0\kappa = 0$.

Indeed, the linear part of F is D , and D vanishes on $K = \ker D$.

Furthermore, the image of D lies in R , so the projection π_C kills the entire linear part.

Consequently, the Kuranishi map begins at quadratic or higher order.

Thus $\ker D$ is the linearized, or infinitesimal, deformation space. It need not be the tangent space of the actual zero set, which may be singular. Even $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = x^2$, has $\ker D_0F = \mathbb{R}$ although its zero set is the single point 0. The equation $\kappa(u) = 0$ determines which small parameters yield solutions; the vanishing of $D_0\kappa$ alone does not integrate every infinitesimal deformation.

The construction is local and therefore also applies to $F : \mathcal{X} \rightarrow \mathcal{Y}$ near p with $F(p) = y_0$: choose charts sending p, y_0 to zero and apply the Banach-space argument. For a smooth section of a Banach bundle, use a local trivialization to express the section as a map into the model fiber. At a zero, changing the trivialization multiplies its derivative by the fiber isomorphism; the derivative of the transition contributes nothing because the section itself vanishes there. The kernel and cokernel therefore have the expected intrinsic meanings. The particular map κ depends on charts and complements, but each construction identifies its zero set with the same local solution set. This is the version used for the SW section on a gauge slice.

A.11.6 The unobstructed case

If $\text{coker } D = 0$, then $C = 0$.

Hence the Kuranishi map has zero target and is identically zero.

The local solution space is therefore simply a neighborhood of 0 in $\ker D$.

This recovers the regular level-set theorem.

Thus the Kuranishi picture extends the implicit-function-theorem picture to the obstructed case.

A.11.7 Deformation and obstruction spaces

For a three-term elliptic deformation complex

$$0 \longrightarrow E_0 \xrightarrow{d_0} E_1 \xrightarrow{d_1} E_2 \longrightarrow 0,$$

the gauge-fixed linearization is $D = d_0^* \oplus d_1$.

Hodge theory gives $\ker D \cong H^1$, while

$$\text{coker } D \cong H^0 \oplus H^2.$$

At an irreducible solution, $H^0 = 0$, so $\text{coker } D \cong H^2$.

Consequently, near an irreducible solution, the Kuranishi map takes the form

$$\kappa : N \subset H^1 \longrightarrow H^2.$$

This explains the terminology:

H^1 = deformation space,

while

H^2 = obstruction space.

A sufficiently small parameter $u \in H^1$ determines a nearby solution, with the correction supplied by the implicit function theorem, exactly when $\kappa(u) = 0$. To integrate a prescribed infinitesimal vector $v \in H^1$ by a curve of solutions, one needs a curve $u(t)$ in this zero set with $u(0) = 0$ and $u'(0) = v$; this does not follow from $D_0\kappa = 0$.

A.11.8 Kuranishi model for a gauge-theoretic moduli space

Let C be a solution of a gauge-theoretic equation, and choose a local slice $\mathcal{S}_C$ through C .

After identifying the tangent space of the slice with the orthogonal complement of the infinitesimal gauge directions, the nonlinear equation can be written locally as

$$F_C(\dot{C}) = 0, \quad \dot{C} \in T_C\mathcal{S}_C.$$

Its linearization is the gauge-fixed deformation operator.

Suppose first that C is irreducible. Then the stabilizer is trivial, and a neighborhood of $[C]$ in the moduli space is locally homeomorphic to $\kappa^{-1}(0)$, where

$$\kappa : N \subset H_C^1 \longrightarrow H_C^2.$$

If $H_C^2 = 0$, then $\kappa \equiv 0$, and the moduli space is smooth near $[C]$, with

$$T_{[C]}\mathcal{M} \cong H_C^1.$$

If $H_C^2 \neq 0$, then the local moduli space may be singular and is controlled by the finite-dimensional zero set $\kappa^{-1}(0)$.

A.11.9 The role of the stabilizer

If the solution C has a non-trivial stabilizer $\text{Stab}(C)$, then the Kuranishi construction can be chosen $\text{Stab}(C)$ -equivariantly.

The local quotient is then modeled not simply on $\kappa^{-1}(0)$, but on $\kappa^{-1}(0)/\text{Stab}(C)$.

Thus reducible solutions may produce quotient singularities even if the obstruction space vanishes.

This is why two distinct issues must be separated: $H_C^2 \neq 0$ corresponds to analytic obstruction, whereas $H_C^0 \neq 0$ corresponds to a non-trivial infinitesimal stabilizer.

At an irreducible regular solution both disappear: $H_C^0 = H_C^2 = 0$.

A.12 Fredholm maps and the Sard–Smale theorem

The previous discussion describes the local structure near a fixed solution.

A different question is whether one can choose auxiliary parameters so that every solution is regular.

In gauge theory these auxiliary parameters may be perturbing forms, metrics, or other geometric data.

The appropriate infinite-dimensional analogue of Sard's theorem is the Sard–Smale theorem.

A.12.1 Generic subsets

Definition A.12.1. A subset R of a topological space is called **residual** if it contains a countable intersection of open dense subsets.

A property is said to hold **generically** if it holds on a residual subset.

By the Baire category theorem, residual subsets of complete metric spaces are dense.

Hilbert manifolds and the parameter spaces normally used in gauge theory are Baire spaces under the standard hypotheses.

Thus a generic set is, in particular, dense, although it need not contain an open dense subset.

A.12.2 Regular values of Fredholm maps

Let $F : \mathcal{X} \rightarrow \mathcal{Y}$ be a smooth Fredholm map between connected Hilbert manifolds.

If $y \in \mathcal{Y}$ is a regular value, then $F^{-1}(y)$ is a smooth finite-dimensional manifold.

At $x \in F^{-1}(y)$, surjectivity gives $\text{coker } D_x F = 0$, and therefore

$$\dim \ker D_x F = \text{ind } F.$$

Hence

$$\dim F^{-1}(y) = \text{ind } F.$$

If $\text{ind } F < 0$, a regular fiber must be empty, since a smooth manifold cannot have negative dimension.

A.12.3 The Sard–Smale theorem

We use the smooth Hilbert-manifold version of the theorem.

Theorem A.12.2 (Sard–Smale). Let $F : \mathcal{X} \rightarrow \mathcal{Y}$ be a smooth Fredholm map between second-countable Hilbert manifolds modeled on separable spaces, and assume that $\mathcal{X}$ is connected.

Then the following hold.

If $\text{ind } F < 0$, then for a residual subset of $y \in \mathcal{Y}$ one has $F^{-1}(y) = \emptyset$.

If $\text{ind } F \geq 0$, then the regular values of F form a residual subset of $\mathcal{Y}$. For every such regular value y , $F^{-1}(y)$ is a smooth, possibly empty, finite-dimensional manifold of dimension $\text{ind } F$.

This is the infinite-dimensional transversality theorem needed in gauge theory.

A.12.4 Parameterized nonlinear equations

The most useful gauge-theoretic formulation involves a parameter space.

Let

$$\Lambda, \quad \mathcal{X}, \quad \mathcal{Y}$$

be Hilbert manifolds and let

$$\mathcal{F} : \Lambda \times \mathcal{X} \rightarrow \mathcal{Y}$$

be smooth.

Fix $y_0 \in \mathcal{Y}$.

For each $\lambda \in \Lambda$, define

$$\mathcal{F}_\lambda : \mathcal{X} \longrightarrow \mathcal{Y}, \quad \mathcal{F}_\lambda(x) = \mathcal{F}(\lambda, x).$$

We are interested in the solution space

$$\mathcal{S}_\lambda = \{x \in \mathcal{X} \mid \mathcal{F}_\lambda(x) = y_0\}.$$

The corresponding **universal solution space** is

$$\mathcal{S} = \{(\lambda, x) \in \Lambda \times \mathcal{X} \mid \mathcal{F}(\lambda, x) = y_0\}.$$

A.12.5 Universal transversality

Suppose y_0 is a regular value of the universal map

$$\mathcal{F} : \Lambda \times \mathcal{X} \longrightarrow \mathcal{Y}.$$

Then

$$\mathcal{S} = \mathcal{F}^{-1}(y_0)$$

is a smooth Hilbert manifold.

Consider the natural projection

$$\pi : \mathcal{S} \longrightarrow \Lambda, \quad (\lambda, x) \longmapsto \lambda.$$

The fiber over λ is precisely

$$\pi^{-1}(\lambda) = \{\lambda\} \times \mathcal{S}_\lambda.$$

Thus the question of whether $\mathcal{S}_\lambda$ is smooth becomes the question of whether λ is a regular value of π .

Suppose in addition that each map $\mathcal{F}_\lambda$ is Fredholm.

At $(\lambda, x) \in \mathcal{S}$, write $L = D_x \mathcal{F}_\lambda$ and $Q = D_\lambda \mathcal{F}$. The tangent space is

$$T_{(\lambda, x)} \mathcal{S} = \{(\nu, v) : Q\nu + Lv = 0\}, \quad D\pi(\nu, v) = \nu.$$

Surjectivity of the total derivative $Q + L$ yields the exact sequence

$$0 \longrightarrow \ker L \longrightarrow T_{(\lambda, x)} \mathcal{S} \xrightarrow{D\pi} T_\lambda \Lambda \xrightarrow{[Q]} \operatorname{coker} L \longrightarrow 0.$$

Indeed, $[Q]\nu = 0$ precisely when one can solve $Lv = -Q\nu$, and total surjectivity makes $[Q]$ onto. Thus $\ker D\pi \cong \ker L$ and $\operatorname{coker} D\pi \cong \operatorname{coker} L$. The range of $D\pi$ is the closed kernel of $[Q]$. Consequently $D\pi$ is Fredholm; it is surjective exactly when L is. This proves the regular-value equivalence, as well as

$$\operatorname{ind} \pi = \operatorname{ind} \mathcal{F}_\lambda.$$

The Sard–Smale theorem therefore implies that for generic $\lambda \in \Lambda$, the fiber $\mathcal{S}_\lambda$ is a smooth finite-dimensional manifold of dimension $\operatorname{ind} \mathcal{F}_\lambda$, provided the index is non-negative.

This is the standard mechanism of generic transversality.

A.12.6 Why enlarging the parameter space helps

Suppose the unperturbed equation is $F(x) = 0$.

At a solution x , the differential $D_x F$ may fail to be surjective.

One introduces a parameter $\lambda \in \Lambda$ and replaces the equation by $\mathcal{F}(\lambda, x) = 0$.

The universal differential is

$$D_{(\lambda, x)} \mathcal{F} : T_\lambda \Lambda \oplus T_x \mathcal{X} \longrightarrow T_0 \mathcal{Y}.$$

Even if the partial differential in the x -direction $D_x \mathcal{F}_\lambda$ is not surjective, the additional parameter directions $T_\lambda \Lambda$ may fill the missing cokernel directions.

Thus a sufficiently rich parameter space can make the universal linearization surjective even when individual equations are not regular.

Sard–Smale then converts universal surjectivity into generic surjectivity for the individual parameter values.

A.12.7 Application to the Seiberg–Witten equations

For the perturbed Seiberg–Witten equations, take a perturbation $\eta \in i\Omega_+^2(X)$ and consider

$$D_A \psi = 0, \quad F_A^+ = q(\psi) + \eta.$$

After Sobolev completion, the perturbation space is replaced by a suitable Sobolev completion $\mathcal{P}$ of $i\Omega_+^2(X)$.

The universal Seiberg–Witten map has the form

$$\mathcal{F} : \mathcal{P} \times \mathcal{C}_k \longrightarrow W^{k-1,2}(S^-) \oplus W^{k-1,2}(i\Lambda_+^2),$$

with

$$\mathcal{F}(\eta, A, \psi) = (D_A \psi, F_A^+ - q(\psi) - \eta).$$

At a zero (η, A, ψ) , the derivative in the perturbation direction $\nu \in T_\eta \mathcal{P}$ contributes $(0, -\nu)$ to the universal linearization.

Thus the perturbation variables directly provide additional directions in the curvature component of the target.

After passing to the irreducible quotient and establishing universal surjectivity, the universal zero set becomes a smooth Hilbert manifold.

The projection

$$\pi : \mathcal{Z} \longrightarrow \mathcal{P}$$

from the universal moduli space to the perturbation space is Fredholm.

Its index is the expected dimension $d(s)$.

Sard–Smale then implies that the regular values of π form a residual subset of the perturbation space.

Hence, for a generic perturbation η , every irreducible solution is regular.

Equivalently, $H_{(A, \psi)}^2 = 0$ at every irreducible solution.

A.12.8 Generic smoothness of the moduli space

Suppose η is a generic perturbation and suppose, in addition, that there are no reducible solutions.

Then every solution is irreducible and regular.

Consequently,

$$H_{(A,\psi)}^0 = H_{(A,\psi)}^2 = 0.$$

The perturbed moduli space $\mathcal{M}_{s,\eta}$ is therefore a smooth finite-dimensional manifold. Its tangent space at a solution is

$$T_{[(A,\psi)]}\mathcal{M}_{s,\eta} \cong H_{(A,\psi)}^1.$$

Its dimension is

$$\dim \mathcal{M}_{s,\eta} = d(s).$$

Thus the expected dimension computed from the Fredholm index becomes the actual dimension after transversality has been achieved.

A.12.9 Negative expected dimension

Suppose $d(s) < 0$.

For a generic perturbation, Sard–Smale gives a regular moduli problem.

If a solution existed, the local moduli space near that solution would be a smooth manifold of dimension $d(s) < 0$, which is impossible.

Therefore, generically,

$$\mathcal{M}_{s,\eta} = \emptyset.$$

This explains why gauge-theoretic moduli spaces of negative expected dimension are generically empty.

A.12.10 Regularity versus irreducibility

It is important not to confuse regularity with irreducibility.

A solution is **irreducible** when its stabilizer is minimal. In terms of the deformation complex this corresponds, in the standard Seiberg–Witten setting, to $H^0 = 0$.

A solution is **regular** when the linearized equation is surjective, which corresponds to $H^2 = 0$.

These are logically independent conditions.

Generic perturbations are used to achieve $H^2 = 0$, while separate arguments are needed to exclude reducible solutions.

Only when $H^0 = H^2 = 0$ does the ordinary implicit-function-theorem picture give a smooth local moduli space without stabilizer singularities.